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INTERNATIONAL SEMINAR ON EDUCATION

**“Knowledge Integration for Advancing SDGs in the Era
of Global Transformation”**

INTERNATIONAL PROCEEDING

ORGANIZED BY:

**FACULTY OF TARBIYAH AND TEACHER TRAINING
UNIVERSITAS ISLAM NEGERI SULTANAH NAHRASIYAH LHOKSEUMAWE**

in Collaboration with:

Christ College (Autonomous), Irinjalakuda, India; SEGi University & Colleges, Malaysia; Abra State Institute of Science and Technology, Philippines; Indonesian Education Share to Care Volunteers. Universitas Almuslim Bireun, Indonesia; Nursing Academy Kesdam Iskandar Muda Lhokseumawe, Indonesia.



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**Faculty of Tarbiyah and Teacher Training, Universitas Islam Negeri Sultanah
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Faculty of Tarbiyah and Teacher Training, Universitas Islam Negeri Sultanah Nahrasiyah Lhokseumawe is one of the faculties committed to advancing education, research, and academic collaboration in Indonesia. The faculty actively promotes academic excellence through seminars, conferences, community engagement, and international partnerships involving scholars, lecturers, researchers, and practitioners from various countries. Through these academic activities, the faculty aims to strengthen global networking and contribute to the development of education in the era of global transformation.

About the Conference

The Faculty of Tarbiyah and Teacher Training, Universitas Islam Negeri Sultanah Nahrasiyah Lhokseumawe proudly organized the International Seminar on Education and Call for Paper with the theme:

“Knowledge Integration for Advancing SDGs in the Era of Global Transformation.”

The conference was conducted in a hybrid meeting format on August 11, 2025, at Aula Serbaguna UIN Sultanah Nahrasiyah Lhokseumawe, Aceh, Indonesia. The seminar brought together academics, researchers, teachers, education practitioners, and students from different institutions and countries to discuss current issues, innovations, and challenges in education and social transformation.

This international seminar featured keynote speakers and experts from India, Philippines, Malaysia, and Indonesia who shared their insights in the fields of education, linguistics, Islamic law, educational technology, and early childhood education. The conference also provided opportunities for participants to present research papers and publish selected articles in international proceedings and reputable journals.

The seminar aimed to encourage interdisciplinary discussion, strengthen international academic collaboration, and support the achievement of Sustainable Development Goals (SDGs) through knowledge integration and educational innovation.

Topic Areas

1. Language Education
2. Linguistics
3. Technology in ELT
4. Early Childhood Education
5. Young Learners Education
6. Management of Education
7. Science and Psychology
8. Science and Mathematics
9. Culture and Religion
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Preface

On behalf of the Faculty of Tarbiyah and Teacher Training, Universitas Islam Negeri Sultanah Nahrasiyah Lhokseumawe, I would like to extend a warm welcome to all distinguished keynote speakers, presenters, researchers, academicians, students, and participants attending the International Seminar on Education and Call for Paper.

This international seminar is organized as a forum for academic discussion, research dissemination, and international collaboration under the theme **“Knowledge Integration for Advancing SDGs in the Era of Global Transformation.”** Through this conference, scholars and practitioners from different countries and academic backgrounds are invited to exchange ideas, share experiences, and discuss innovations and challenges in education and related fields.

In the era of rapid global transformation, education holds an essential role in promoting sustainable development and improving the quality of human resources. Therefore, academic collaboration and knowledge integration become increasingly important in responding to contemporary educational and social issues. We are honored to have keynote speakers and contributors from India, Philipines, Malaysia, and Indonesia who enrich this seminar with their expertise and perspectives.

The Faculty of Tarbiyah and Teacher Training continuously supports academic excellence, research development, and international networking as part of its commitment to advancing education and strengthening global partnerships. We believe that this seminar will create meaningful academic discussions and encourage future collaborations among scholars, educators, researchers, and institutions worldwide.

I would like to express my sincere appreciation to all organizing committees, collaborating institutions, keynote speakers, presenters, and participants for their valuable contribution and support in making this conference possible. Hopefully, this seminar and its proceedings will provide beneficial contributions to the advancement of education, research, and society.

Dr. Jumat Barus, S.S., M.S.

Dean of Faculty of Tarbiyah and Teacher Training

Universitas Islam Negeri Sultanah Nahrasiyah Lhokseumawe, Aceh, Indonesia

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DIGITAL LITERACY AS A CATALYST EQUIPPING COMMUNITIES WITH 21ST-CENTURY SKILLS FOR SUSTAINABLE DEVELOPMENT

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ABSTRACT

Digital literacy is increasingly recognized as a multidimensional social capability that extends beyond technical skills to encompass critical, ethical, and collaborative engagement with digitally mediated environments. This chapter conceptualizes digital literacy as a catalytic force for equipping communities with 21st-century skills and advancing sustainable development, drawing on contemporary debates in digital competence, community empowerment, and critical media studies. It examines how digital literacy mediates access to knowledge, participation, and power, highlighting its role in strengthening community resilience while also reproducing inequalities when framed in technosolutionist or individualistic terms. By foregrounding issues of inequality, access, and the politics of digital infrastructures, this report argues for a transformative, community-based framework grounded in contextual embeddedness, participation, critical reflection, ethics, and inclusivity. It concludes that treating digital literacy as a strategic social investment, rather than a narrow technical objective, is essential for aligning digital transformation with social justice and long-term sustainable development.

Keywords: *Digital literacy; Sustainable development; Community empowerment; 21st-century skills; Social inequality; Critical digital literacy.*

INTRODUCTION

The rapid expansion of digital technologies over the last decade has profoundly reshaped how societies communicate, work, learn, and participate in public life. However, the growing ubiquity of digital tools has not automatically translated into equitable social progress or sustainable development. While connectivity rates continue to rise globally, disparities in how individuals and communities meaningfully engage with digital environments remain stark. Recent scholarship emphasizes that digital transformation without corresponding growth in human capability risks reinforcing structural inequalities rather than alleviating them (van Dijk, 2020; Ragnedda & Ruiu, 2023). In this context, digital literacy emerges not merely as a technical skill set but as a critical social competence that mediates participation, inclusion, and agency in contemporary societies.

Conventional understandings of digital literacy have long been dominated by instrumental perspectives, focusing on operational skills such as using devices, navigating software, or accessing online information. Although these competencies remain necessary, they are increasingly insufficient in addressing the complex demands of the 21st century. Scholars argue that such narrow conceptualizations overlook the

cognitive, ethical, and socio-cultural dimensions of digital engagement, including critical evaluation of information, awareness of algorithmic influence, and responsible participation in digital publics (Ng, 2021; Sefton-Green et al., 2020). As a result, digital literacy must be reframed as a multidimensional construct that enables individuals and communities not only to use technology, but to understand, question, and shape its role in society.

This reframing becomes particularly urgent when digital literacy is examined through the lens of sustainable development. The United Nations' Sustainable Development Goals (SDGs) highlight education, equity, decent work, and inclusive institutions as interdependent objectives, all of which are increasingly mediated by digital infrastructures. Empirical and conceptual studies suggest that digital literacy functions as a cross-cutting enabler that amplifies progress across multiple SDGs, especially when embedded within community contexts rather than isolated individual interventions (UNESCO, 2021; Hatlevik et al., 2023). Without adequate digital literacy, access to digital technologies risks becoming symbolic rather than transformative, offering connection without empowerment.

At the community level, the stakes of digital literacy extend beyond employability and economic productivity. Communities with limited digital literacy often face restricted access to public services, reduced civic participation, and heightened vulnerability to misinformation and digital exploitation. Research indicates that digital literacy influences how communities mobilize knowledge, sustain local initiatives, and negotiate power relations in digitally mediated environments (Ragnedda, Rosa, & Muschert, 2020; Walton & Donner, 2022). Thus, digital literacy operates as a form of social capital, shaping collective resilience and the capacity for sustainable community development.

Despite its growing importance, digital literacy initiatives frequently adopt top-down, technocratic approaches that prioritize short-term skill acquisition over long-term capacity building. Critical scholars caution that such approaches risk reproducing forms of digital dependency, where communities remain consumers of technology rather than active co-creators of digital futures (Selwyn, 2022; Couldry & Mejias, 2019). This critique underscores the need to move beyond technosolutionist narratives and toward literacy models that are grounded in local contexts, cultural values, and ethical considerations. Digital literacy, in this sense, is inseparable from questions of power, voice, and representation.

Moreover, the assumption that digital literacy is a neutral or universally beneficial construct has been increasingly challenged. Studies in critical digital studies highlight that digital platforms are embedded with ideological, economic, and political logics that shape user behavior and knowledge production (Kellner & Share, 2021; Noble, 2018). Although access to technology may be expanding, the capacity to critically engage with digital systems remains unevenly distributed. This unevenness has significant implications for sustainable development, as communities lacking critical digital literacy are more susceptible to manipulation, exclusion, and environmental or social harm mediated through digital channels.

Against this backdrop, this chapter positions digital literacy as a catalytic force for equipping communities with 21st-century skills essential for sustainable development. Rather than treating digital literacy as an end in itself, the chapter conceptualizes it as a dynamic process that fosters critical thinking, ethical awareness, collaboration, and collective agency. By adopting a conceptual, theoretical, and critically reflective approach, this chapter seeks to contribute to ongoing scholarly debates on how digital literacy can be reimagined as a socially embedded capability that supports inclusive and sustainable futures (Bawden & Robinson, 2020; Livingstone, 2021). In doing so, it argues that digital literacy must be understood not simply as learning to live with technology, but as learning to live well through it.

METHOD

This study adopts a qualitative conceptual approach to examine the role of digital literacy as a catalyst for community empowerment and sustainable development. Rather than relying on empirical data collection, the paper is grounded in an extensive review and critical synthesis of existing literature from interdisciplinary fields, including digital education, sociology, communication studies, and development studies. Relevant scholarly sources such as peer-reviewed journal articles, international policy reports, and theoretical frameworks were systematically selected based on their relevance to digital literacy, 21st-century skills, and sustainability discourse. This approach allows the study to identify key themes, patterns, and conceptual relationships that inform the evolving understanding of digital literacy in contemporary society.

Furthermore, the analysis employs a critically reflective framework to interpret how digital literacy is positioned within broader socio-cultural, political, and ethical contexts. By integrating perspectives from critical digital studies and community development theory, this paper seeks to move beyond instrumental interpretations of digital literacy toward a more transformative and socially embedded perspective. The method emphasizes analytical interpretation and theoretical integration, enabling the development of a comprehensive framework that connects digital literacy with issues of inequality, empowerment, and sustainable development. Through this approach, the study contributes to ongoing academic discussions by offering a nuanced and context-sensitive conceptualization of digital literacy as a multidimensional and dynamic capability.

RESULT AND DISCUSSION

Conceptualizing Digital Literacy in the 21st Century

The concept of digital literacy has undergone significant theoretical expansion over the past two decades, reflecting the increasing complexity of digital environments and their entanglement with social life. Early definitions of digital literacy largely emphasized functional competence, focusing on users' ability to operate digital devices and access online information efficiently. While these definitions were adequate during the early stages of digital adoption, contemporary scholars argue that they fail to capture the epistemic, ethical, and political dimensions of digital

engagement in networked societies (Bawden & Robinson, 2020; Ng, 2021). In the 21st century, digital literacy must therefore be reconceptualized as a dynamic and context-sensitive capability rather than a static set of technical skills.

Recent theoretical models increasingly frame digital literacy as a multilayered construct that integrates cognitive, social, and critical dimensions. Frameworks such as the European Commission's DigComp and UNESCO's Global Framework on Digital Literacy Skills emphasize competencies related not only to information access and communication, but also to content creation, problem-solving, safety, and ethical awareness (Carretero, Vuorikari, & Punie, 2022; UNESCO, 2023). These models reflect a broader scholarly consensus that digital literacy involves understanding how digital systems shape knowledge production, social interaction, and power relations. Consequently, being digitally literate implies the ability to navigate uncertainty, evaluate credibility, and engage responsibly within digitally mediated ecosystems.

A key shift in contemporary theorization is the recognition of digital literacy as inherently social rather than purely individual. Research in sociocultural and critical traditions highlights that digital practices are embedded within communities, institutions, and cultural norms that shape how technologies are interpreted and used (Sefton-Green et al., 2020; Walton & Donner, 2022). From this perspective, digital literacy is co-constructed through participation in shared practices, informal learning networks, and community-based knowledge exchange. This view challenges deficit-oriented approaches that locate digital illiteracy solely within individuals, instead drawing attention to structural conditions such as access, language, cultural relevance, and institutional support.

Another defining feature of 21st-century digital literacy is its critical orientation. Critical digital literacy extends beyond the evaluation of information accuracy to include awareness of algorithmic governance, datafication, and platform economies. Scholars argue that users must be able to recognize how digital platforms shape visibility, influence behavior, and monetize attention, often in opaque ways (Selwyn, 2022; Zuboff, 2019). Although not all users need technical knowledge of algorithms, critical literacy entails an understanding of their social consequences. Without this critical dimension, digital literacy risks becoming a form of compliance that enables participation without agency.

Ethical considerations further complicate contemporary conceptualizations of digital literacy. Issues such as data privacy, surveillance, online harm, and digital well-being have become central concerns in both academic discourse and public policy. Studies suggest that ethical digital literacy involves moral judgment, empathy, and responsibility in digital interactions, particularly within diverse and unequal societies (Floridi et al., 2020; Pangrazio & Cardozo-Gaibisso, 2021). This ethical turn underscores that digital literacy cannot be separated from broader discussions of citizenship, rights, and social responsibility. In this sense, digital literacy intersects with moral education and civic competence.

Importantly, digital literacy in the 21st century is increasingly framed as a lifelong and adaptive process. Rapid technological change means that specific tools and platforms quickly become obsolete, while underlying principles of critical thinking, learning-to-learn, and adaptability retain their relevance. Research highlights that sustainable digital literacy depends less on mastery of current technologies and more on the capacity to continuously update one's understanding and practices in response to evolving digital landscapes (Hatlevik et al., 2023; Redecker, 2020). This orientation aligns digital literacy with broader educational goals related to resilience and future readiness.

From a global perspective, conceptual debates on digital literacy have also begun to challenge Western-centric assumptions embedded in dominant frameworks. Scholars from the Global South argue that many prevailing models overlook local knowledge systems, linguistic diversity, and socio-economic realities, thereby limiting their applicability across contexts (Ragnedda & Rui, 2023; Adam & Carter, 2021). A more inclusive conceptualization of digital literacy recognizes plurality, acknowledging that meaningful digital engagement may take different forms depending on cultural values, community needs, and developmental priorities. Such perspectives caution against universalized benchmarks that ignore contextual specificity.

In community contexts, digital literacy takes on an explicitly transformative dimension. Rather than focusing solely on individual competence, community-oriented models emphasize collective learning, shared problem-solving, and participatory knowledge creation. Empirical studies demonstrate that when digital literacy initiatives are embedded within community development efforts, they are more likely to foster empowerment, social cohesion, and sustainable outcomes (Gonzales, 2020; Toyama, 2021). This approach positions digital literacy as a relational practice that strengthens community agency and supports locally grounded innovation.

Taken together, these theoretical developments suggest that digital literacy in the 21st century should be understood as a meta-competence that integrates technical proficiency, critical awareness, ethical judgment, and social participation. It is neither neutral nor value-free, but shaped by ideological, economic, and cultural forces that influence how digital technologies are designed and used. Recognizing this complexity is essential for aligning digital literacy with sustainable development goals, as simplistic or instrumental approaches risk overlooking the deeper social transformations required for long-term impact (Livingstone, 2021; Selwyn, 2022).

This chapter adopts such an expanded conceptualization of digital literacy as the foundation for examining its catalytic role in equipping communities with 21st-century skills. By situating digital literacy within broader socio-cultural and ethical frameworks, the discussion moves beyond questions of access and skills acquisition toward issues of agency, justice, and sustainability. In doing so, it prepares the conceptual ground for subsequent sections that explore how digital literacy intersects with community empowerment and sustainable development in practice.

21st-Century Skills and Community Empowerment

The discourse on 21st-century skills has gained significant prominence in global education and development agendas, often framed as a response to rapid technological change, economic uncertainty, and shifting labor markets. These skills commonly including critical thinking, creativity, collaboration, communication, and problem-solving are increasingly viewed as essential for navigating complex social and professional environments. However, much of the existing literature has focused predominantly on individual employability, positioning 21st-century skills as personal assets rather than collective resources (Ananiadou & Claro, 2009; Voogt & Roblin, 2012). Recent studies argue that such an individualistic framing limits the transformative potential of these skills, particularly in community and development contexts (Trilling & Fadel, 2021; OECD, 2023).

Digital literacy plays a pivotal role in mediating how 21st-century skills are acquired, practiced, and mobilized within communities. Research suggests that digital environments increasingly function as primary spaces where critical thinking, collaboration, and creativity are enacted, whether through online problem-solving, civic engagement platforms, or community-based digital initiatives (Hatlevik et al., 2023; Ng & Tan, 2024). In this sense, digital literacy does not merely complement 21st-century skills but actively shapes their expression. Communities with higher levels of digital literacy are better positioned to leverage digital tools for collective learning, innovation, and social coordination, thereby strengthening communal capacity rather than isolated individual success.

Community empowerment literature emphasizes that empowerment is fundamentally a relational and collective process, involving increased control over resources, decision-making, and knowledge production. Scholars argue that empowerment cannot be reduced to skill acquisition alone; it requires opportunities for participation, voice, and agency within social structures (Perkins & Zimmerman, 1995; Rappaport, 2020). When digital literacy is integrated into community settings, it can facilitate these processes by enabling communities to access information, articulate local concerns, and engage with broader networks of influence (Gonzales, 2020; Walton & Donner, 2022). Thus, digital literacy becomes a conduit through which 21st-century skills are translated into meaningful social action.

Critical thinking, often cited as a cornerstone of 21st-century skills, acquires particular significance in digitally mediated communities. The proliferation of misinformation, algorithmic filtering, and persuasive digital content demands heightened evaluative capacities at both individual and collective levels. Studies indicate that communities equipped with critical digital literacy are more resilient to disinformation and better able to engage in deliberative dialogue around shared concerns (Kahne & Bowyer, 2020; Nyhan & Reifler, 2021). This collective critical capacity supports democratic participation and informed decision-making, which are central to sustainable community development.

Collaboration and communication, two other core 21st-century skills, are similarly transformed through digital literacy. Digital platforms enable new forms of horizontal collaboration that transcend geographic boundaries, allowing communities to share resources, co-create knowledge, and build solidarity across contexts. However, effective participation in such networks requires more than access; it depends on communicative competence, cultural sensitivity, and ethical awareness in digital interaction (Pangrazio & Cardozo-Gaibisso, 2021; Redecker, 2020). Digital literacy supports these competencies by fostering reflective communication practices and an understanding of digital norms, thereby enhancing the quality of collective engagement.

Creativity and innovation, frequently associated with economic growth, also hold social significance at the community level. Research on grassroots innovation highlights how digitally literate communities can creatively adapt technologies to local needs, generating contextually relevant solutions rather than adopting externally imposed models (Toyama, 2021; Smith et al., 2020). Such community-driven innovation challenges deficit narratives that portray marginalized communities as passive recipients of technology. Instead, digital literacy enables communities to become active designers of their own developmental pathways, aligning creative practices with local values and sustainability goals.

Importantly, the relationship between digital literacy, 21st-century skills, and community empowerment is not inherently equitable. Power asymmetries related to gender, age, socioeconomic status, and language continue to shape who benefits from digital skill development initiatives. Studies caution that without intentional inclusivity, programs aimed at fostering 21st-century skills may inadvertently reinforce existing hierarchies within communities (Ragnedda & Ruiu, 2023; Selwyn, 2022). This insight underscores the need for community-based approaches that foreground participation, reflexivity, and social justice as integral components of digital literacy development.

From a sustainability perspective, empowered communities equipped with 21st-century skills are better positioned to engage with long-term challenges such as environmental management, public health, and economic resilience. Digital literacy supports access to data, coordination of local action, and dissemination of indigenous and experiential knowledge, all of which contribute to sustainable practices (Leal Filho et al., 2021; Sterling, 2020). When communities can critically interpret information and collaborate effectively, they are more capable of negotiating sustainable solutions that balance local priorities with global imperatives.

In synthesizing these perspectives, it becomes evident that digital literacy serves as a crucial bridge between 21st-century skills and community empowerment. Rather than treating skills as abstract competencies detached from social contexts, this chapter conceptualizes them as practices embedded within community life and mediated by digital engagement. Such a framing highlights the importance of shifting from individual-centered skill development toward collective capacity building, where digital literacy functions as a catalyst for empowerment, resilience, and sustainable

development. This conceptual alignment sets the stage for a deeper examination of how digital literacy intersects with sustainability agendas, which is explored in the following section.

Digital Literacy and Sustainable Development: A Critical Alignment

The relationship between digital literacy and sustainable development has become increasingly prominent within global policy and academic discourses, particularly as digital technologies are positioned as key drivers of social and economic transformation. Sustainable development, as articulated in the Sustainable Development Goals (SDGs), emphasizes the integration of economic growth, social inclusion, and environmental protection. Recent scholarship argues that digital literacy functions as a transversal enabler across these dimensions, shaping how individuals and communities engage with development processes in digitally mediated contexts (Leal Filho et al., 2021; UNESCO, 2021). However, this alignment is neither automatic nor neutral, requiring critical examination of how digital competencies are cultivated and mobilized.

From an educational and social perspective, digital literacy supports sustainable development by enhancing access to knowledge, facilitating lifelong learning, and enabling informed participation in civic life. Studies indicate that digitally literate communities are better equipped to access public services, engage in democratic processes, and respond adaptively to social and environmental challenges (Hatlevik et al., 2023; Livingstone, 2021). These capacities are closely aligned with SDG 4 (Quality Education) and SDG 16 (Peace, Justice, and Strong Institutions), illustrating how digital literacy contributes to sustainability by strengthening institutional trust and participatory governance.

Economically, digital literacy has been linked to inclusive growth and decent work, particularly in contexts where digital platforms mediate access to employment and entrepreneurship opportunities. Research suggests that individuals and communities with higher levels of digital literacy are more capable of leveraging digital economies while mitigating risks such as precarious labor and platform exploitation (van Dijk, 2020; Ragnedda & Rui, 2023). In this sense, digital literacy supports SDG 8 (Decent Work and Economic Growth) not merely by enhancing productivity, but by enabling critical engagement with digital labor systems and fostering more equitable participation in emerging economic models.

Environmental sustainability further underscores the importance of digital literacy, as environmental data, climate information, and sustainability practices are increasingly communicated through digital channels. Scholars argue that digital literacy enables communities to interpret environmental data, participate in digital environmental activism, and adopt sustainable practices informed by evidence-based knowledge (Leal Filho et al., 2021; Sterling, 2020). However, this potential is contingent upon critical literacy that allows communities to distinguish credible information from misleading or politicized narratives, particularly in the context of climate change communication.

Despite these contributions, the alignment between digital literacy and sustainable development is often framed in overly optimistic or technosolutionist terms. Technological solutionism assumes that digital tools and skills alone can resolve complex social and environmental problems, overlooking structural inequalities and contextual constraints (Selwyn, 2022; Morozov, 2019). Critical scholars caution that such narratives risk depoliticizing development by shifting attention away from power relations, governance, and resource distribution. Without a critical orientation, digital literacy initiatives may inadvertently support unsustainable practices or reinforce dominant interests under the guise of innovation.

Moreover, digital development itself carries sustainability risks that must be acknowledged within discussions of digital literacy. The environmental footprint of digital infrastructures, including energy consumption, electronic waste, and extractive supply chains, presents significant challenges to sustainable development goals (Belkhir & Elmeligi, 2018; Hilty & Aebischer, 2020). Digital literacy that neglects these material dimensions risks promoting uncritical consumption of technology. A sustainability-oriented conception of digital literacy therefore includes awareness of the environmental costs of digitalization and encourages responsible and informed digital practices.

Social sustainability also demands attention to issues of equity and inclusion in digital literacy development. Persistent digital divides shaped by socioeconomic status, gender, geography, and language limit who can benefit from digitally enabled development initiatives. Research highlights that unequal access to meaningful digital literacy undermines the inclusive ethos of the SDGs, particularly SDG 10 (Reduced Inequalities) (Ragnedda & Ruiu, 2023; Gonzales, 2020). Aligning digital literacy with sustainable development thus requires intentional strategies that prioritize marginalized communities and address structural barriers to participation.

At the policy level, scholars emphasize the need for coherence between digital literacy frameworks and sustainability agendas. Fragmented policies that treat digitalization and sustainable development as separate domains often fail to realize their synergistic potential (OECD, 2023; Redecker, 2020). Integrative approaches that embed digital literacy within sustainability-oriented education, community development, and governance structures are more likely to generate long-term impact. Such alignment shifts digital literacy from a technical policy objective to a strategic investment in sustainable social transformation.

In sum, while digital literacy holds significant promise as an enabler of sustainable development, its effectiveness depends on how it is conceptualized, implemented, and governed. A critical alignment between digital literacy and sustainability requires moving beyond access and skills toward ethical awareness, critical reflection, and contextual sensitivity. By situating digital literacy within broader debates on equity, environmental responsibility, and social justice, this chapter argues for a sustainability-oriented approach that recognizes digital literacy as both a resource and a responsibility. This critical perspective provides a foundation for

examining issues of inequality and power in digital literacy development, which are addressed in the following section.

Inequality, Access, and the Politics of Digital Literacy

Discussions of digital literacy are inseparable from broader questions of inequality and power that shape access to, and use of, digital technologies. While global connectivity has expanded significantly, scholars caution that the persistence of digital inequality reflects deep-rooted social, economic, and political structures rather than temporary technological gaps. Research demonstrates that disparities in digital literacy mirror existing forms of social stratification, including class, gender, geography, and linguistic capital (van Dijk, 2020; Ragnedda & Ruiu, 2023). As such, digital literacy cannot be treated as a neutral or universally accessible competence, but must be understood within the politics of inclusion and exclusion that characterize digital societies.

Early policy responses to digital inequality largely focused on physical access to devices and internet connectivity. Although access remains a prerequisite for participation, contemporary studies emphasize that access alone is insufficient to ensure meaningful engagement. The concept of the “second-level digital divide” highlights differences in skills and usage patterns, while the “third-level digital divide” draws attention to unequal outcomes and benefits derived from digital engagement (Scheerder, van Deursen, & van Dijk, 2017; van Deursen & Helsper, 2020). Digital literacy sits at the center of these divides, mediating whether access translates into empowerment or reproduces disadvantage.

Power relations embedded within digital platforms further complicate the politics of digital literacy. Digital environments are not passive tools but are shaped by commercial, political, and ideological interests that influence what information is visible, whose voices are amplified, and which forms of knowledge are valued. Critical scholarship reveals how algorithmic systems and data-driven economies privilege certain actors while marginalizing others, often in ways that are opaque to users (Noble, 2018; Zuboff, 2019). Digital literacy, therefore, requires an understanding of these power structures in order to foster agency rather than passive consumption.

Language and culture play a crucial yet often overlooked role in shaping digital literacy inequalities. Much of the digital content and educational resources available online are produced in dominant global languages, limiting meaningful participation for linguistically marginalized communities. Studies indicate that linguistic exclusion in digital spaces constrains knowledge production and reinforces epistemic inequalities, particularly in the Global South (Adam & Carter, 2021; Ragnedda & Ruiu, 2023). A politically informed approach to digital literacy must therefore recognize multilingualism and cultural diversity as central, rather than peripheral, considerations.

Gendered dimensions of digital literacy further illustrate how social norms and power relations influence digital participation. Research consistently shows that women and girls in many contexts face barriers to digital access and skill development, shaped by cultural expectations, safety concerns, and unequal educational opportunities

(Antonio & Tuffley, 2014; Hilbert, 2020). These gendered inequalities are not simply technical issues but reflect broader patterns of social exclusion. Digital literacy initiatives that fail to address these dynamics risk reinforcing gender disparities under the guise of technological inclusion.

Community-based perspectives highlight that digital literacy inequalities are often experienced collectively rather than individually. Marginalized communities may face compounded barriers related to infrastructure, education, and institutional support, limiting their capacity to engage with digital systems on their own terms. Empirical studies suggest that community-driven digital literacy initiatives, which emphasize local participation and shared learning, are more effective in addressing structural inequalities than top-down interventions (Gonzales, 2020; Toyama, 2021). Such approaches reposition communities as active agents in shaping digital inclusion strategies.

The political dimension of digital literacy also extends to governance and policy-making. Decisions regarding digital infrastructure, education curricula, and platform regulation are often made without meaningful community participation, leading to misalignment between policy objectives and local needs. Scholars argue that inclusive digital governance requires participatory approaches that integrate community voices into decision-making processes (Couldry & Mejias, 2019; OECD, 2023). Digital literacy, in this sense, supports not only individual competence but also collective capacity to engage with and influence digital policy environments.

Critically, the framing of digital literacy as a personal responsibility can obscure structural accountability. When individuals are expected to adapt to digital systems without corresponding institutional reform, responsibility for exclusion is shifted away from policymakers and technology providers. This narrative risks normalizing inequality and legitimizing unequal outcomes as individual failure rather than systemic injustice (Selwyn, 2022; van Dijk, 2020). A critical politics of digital literacy challenges this framing by emphasizing shared responsibility and structural change.

So these perspectives, it becomes evident that digital literacy is deeply entangled with questions of inequality, power, and justice. Addressing digital divides requires more than expanding access or offering skills training; it demands a politically informed approach that confronts structural barriers and centers marginalized voices. By situating digital literacy within the politics of inclusion and exclusion, this chapter underscores the necessity of aligning digital literacy initiatives with broader commitments to social equity and sustainable development. This critical lens prepares the groundwork for the following section, which proposes a transformative framework for community-based digital literacy.

Toward a Transformative Framework for Community-Based Digital Literacy

In response to the structural inequalities and power asymmetries embedded in contemporary digital environments, there is an increasing call for transformative approaches to digital literacy that move beyond instrumental skill development. Transformative digital literacy frameworks emphasize empowerment, critical

consciousness, and social change rather than adaptation to existing technological systems. Scholars argue that such approaches are essential for ensuring that digital literacy contributes meaningfully to sustainable development, particularly in marginalized and under-resourced communities (Freire, 1970; Selwyn, 2022). Within this perspective, digital literacy becomes a process of collective learning and reflection aimed at reshaping social relations mediated by technology.

A central principle of a community-based digital literacy framework is contextual embeddedness. Digital literacy initiatives that are disconnected from local realities often fail to resonate with community needs and values, resulting in limited uptake and sustainability. Research highlights that effective programs are those that integrate local knowledge, cultural practices, and lived experiences into digital learning processes (Gonzales, 2020; Adam & Carter, 2021). Contextualization ensures that digital literacy is not perceived as an externally imposed agenda but as a locally meaningful resource that supports community-defined goals.

Participation and co-creation constitute another foundational element of transformative digital literacy. Rather than positioning communities as passive recipients of training, participatory approaches involve community members in the design, implementation, and evaluation of digital literacy initiatives. Studies in community informatics demonstrate that co-created digital projects foster a sense of ownership and agency, which are critical for long-term impact (Walton & Donner, 2022; Toyama, 2021). Through collaborative learning and shared problem-solving, digital literacy becomes a collective practice that strengthens social cohesion and mutual accountability.

Critical reflection is integral to a transformative framework, enabling communities to interrogate the social, economic, and political dimensions of digital technologies. Drawing on traditions of critical pedagogy, scholars emphasize the importance of dialogue and reflection in developing critical digital consciousness (Pangrazio & Cardozo-Gaibisso, 2021; Kellner & Share, 2021). This reflective dimension encourages communities to question dominant technological narratives, recognize power structures, and envision alternative digital futures aligned with sustainability and social justice.

Ethical orientation further distinguishes transformative digital literacy from technocratic models. Ethical digital literacy encompasses considerations of privacy, data ownership, environmental impact, and digital well-being. As digital technologies increasingly permeate everyday life, scholars argue that ethical reasoning must be embedded within digital literacy practices to support responsible and sustainable engagement (Floridi et al., 2020; Hilty & Aebischer, 2020). Community-based approaches are particularly well-suited to fostering ethical dialogue, as they allow collective negotiation of norms and values in digital contexts.

Capacity building within a transformative framework extends beyond individual skill acquisition to include institutional and structural dimensions. Sustainable digital literacy initiatives often involve partnerships between community organizations, educational institutions, civil society, and local governments. Research

indicates that such multi-stakeholder collaborations enhance resource sharing, policy alignment, and institutional support, increasing the likelihood of long-term sustainability (Leal Filho et al., 2021; OECD, 2023). By embedding digital literacy within broader development ecosystems, communities gain greater leverage to influence digital governance and policy decisions.

Inclusivity is a non-negotiable principle of transformative digital literacy. Programs that fail to address intersecting forms of exclusion such as those based on gender, age, disability, or language risk reproducing inequalities even as they expand access. Scholars emphasize the importance of inclusive design and targeted support to ensure that marginalized groups are not left behind in digital literacy initiatives (Ragnedda & Ruiu, 2023; Hilbert, 2020). Community-based frameworks provide opportunities to identify and address such exclusions through locally informed strategies.

Importantly, a transformative digital literacy framework recognizes learning as an ongoing and adaptive process. Rapid technological change necessitates continuous learning, experimentation, and reflection rather than fixed curricula. Studies highlight that communities that cultivate learning cultures characterized by openness, adaptability, and shared inquiry are better positioned to navigate digital uncertainty and pursue sustainable development pathways (Hatlevik et al., 2023; Redecker, 2020). This orientation aligns digital literacy with broader notions of resilience and future readiness.

Finally, these principles, a transformative framework for community-based digital literacy positions technology as a means rather than an end. By centering context, participation, critical reflection, ethics, and inclusivity, such a framework aligns digital literacy with the broader goals of empowerment and sustainability. This approach challenges dominant narratives that equate digital progress with technological adoption, instead emphasizing social transformation and collective agency. The following section builds on this framework by examining its implications for policy, education, and community development in diverse contexts.

Implications for Policy, Education, and Community Development

The reconceptualization of digital literacy as a transformative, community-based capability carries significant implications for public policy. Policymakers are increasingly urged to move beyond infrastructure-centric digital agendas toward holistic strategies that integrate digital literacy with social equity and sustainability goals. Research indicates that policies focusing solely on connectivity and device provision often fail to address deeper capability gaps and may inadvertently widen social inequalities (van Dijk, 2020; Selwyn, 2022). A policy orientation grounded in transformative digital literacy emphasizes long-term capacity building, participatory governance, and alignment with sustainable development priorities rather than short-term technological targets.

From an educational perspective, the expanded understanding of digital literacy challenges traditional curricular approaches that isolate digital skills within discrete

subjects or technical training modules. Scholars argue that digital literacy should be embedded across curricula as a transversal competence linked to critical thinking, ethical reasoning, and civic engagement (Redecker, 2020; Livingstone, 2021). This integration requires pedagogical shifts that prioritize inquiry-based learning, dialogic teaching, and reflective practice. Such approaches position learners not merely as users of digital tools, but as critical participants in digitally mediated knowledge ecosystems.

Teacher education and professional development represent a crucial leverage point in operationalizing transformative digital literacy. Studies consistently highlight that educators' own digital literacy, including critical and ethical dimensions, significantly influences how digital practices are enacted in learning environments (Hatlevik et al., 2023; Instefjord & Munthe, 2020). Professional development programs that foreground reflective pedagogy, contextual sensitivity, and ethical awareness are more likely to foster meaningful digital learning experiences. This implication underscores that digital literacy is not only a student outcome, but a systemic educational capacity.

In community development contexts, the implications of transformative digital literacy extend beyond formal education systems. Community-based organizations, libraries, and civil society actors play a vital role in facilitating informal and non-formal digital learning spaces. Research demonstrates that such spaces are particularly effective in reaching marginalized populations who may be excluded from institutional education pathways (Gonzales, 2020; Walton & Donner, 2022). Supporting these actors through policy recognition, funding, and partnerships enhances the reach and sustainability of digital literacy initiatives grounded in community realities.

The alignment between digital literacy and sustainable development also necessitates rethinking evaluation and impact measurement. Conventional metrics that emphasize participation rates or skill acquisition provide limited insight into whether digital literacy initiatives contribute to empowerment or sustainability. Scholars advocate for evaluative frameworks that capture qualitative outcomes such as agency, collective problem-solving, and community resilience (Toyama, 2021; Leal Filho et al., 2021). Such measures better reflect the transformative aspirations of community-based digital literacy and support evidence-informed policy refinement.

At the governance level, transformative digital literacy implies greater community involvement in digital decision-making processes. Participatory governance models that include community voices in discussions on data use, platform regulation, and digital service design are increasingly recognized as essential for democratic accountability (Coudry & Mejias, 2019; OECD, 2023). Digital literacy equips communities with the knowledge and confidence needed to engage meaningfully in these processes, thereby strengthening institutional legitimacy and social trust.

Equity considerations remain central across policy, education, and community development implications. Targeted strategies are required to address intersecting forms of exclusion related to gender, disability, age, and linguistic diversity. Research suggests that inclusive digital literacy policies are most effective when they are co-

designed with marginalized groups and grounded in local knowledge systems (Ragnedda & Ruiu, 2023; Hilbert, 2020). This approach shifts digital inclusion from a compensatory model toward one based on recognition and empowerment.

Finally, these implications point toward the need for cross-sector collaboration. Sustainable digital literacy development cannot be achieved by isolated actors working in silos. Partnerships among governments, educational institutions, community organizations, and the private sector are essential for aligning resources, expertise, and values (OECD, 2023; Leal Filho et al., 2021). When guided by a transformative framework, such collaborations can support digital ecosystems that are not only innovative, but also socially just and environmentally responsible.

In summary, the implications of reconceptualizing digital literacy as a catalyst for sustainable development extend across policy, education, and community practice. Rather than offering prescriptive solutions, this chapter highlights strategic orientations that prioritize critical engagement, inclusivity, and long-term social impact. These implications reinforce the argument that digital literacy is a collective investment in sustainable futures, setting the stage for the concluding section that synthesizes the chapter's key insights.

CONCLUSION

This chapter has argued that digital literacy must be understood as far more than a set of technical competencies required to navigate digital tools. In the context of accelerating digitalization and global sustainability challenges, digital literacy emerges as a socially embedded capability that shapes how communities learn, participate, and exercise agency. Drawing on contemporary scholarship, the chapter has demonstrated that instrumental approaches to digital literacy are insufficient for addressing the structural inequalities and complex socio-ecological problems that characterize the 21st century (Selwyn, 2022; Livingstone, 2021). Instead, digital literacy must be reconceptualized as a critical, ethical, and collective practice.

By situating digital literacy within the broader discourse of 21st-century skills, this chapter highlights its role in fostering critical thinking, collaboration, creativity, and civic engagement at the community level. These skills, when mediated through digital literacy, function not merely as individual assets but as collective resources that strengthen social cohesion and resilience. Research consistently shows that communities equipped with such capacities are better positioned to respond to uncertainty, negotiate change, and pursue sustainable development pathways aligned with local priorities (Hatlevik et al., 2023; Gonzales, 2020).

The critical alignment between digital literacy and sustainable development underscores both opportunity and risk. While digital literacy can act as a cross-cutting enabler of the Sustainable Development Goals, uncritical or techno solutionist approaches risk reproducing inequality and environmental harm. As discussed, digital infrastructures themselves are embedded within political economies that shape access, visibility, and outcomes (Ragnedda & Ruiu, 2023; Hilty & Aebischer, 2020).

Recognizing these tensions is essential if digital literacy initiatives are to contribute meaningfully to sustainability rather than serve as symbolic gestures of progress.

This chapter has also foregrounded the politics of digital literacy, emphasizing that access to technology does not equate to access to power. Inequalities related to socioeconomic status, gender, language, and geography continue to shape who benefits from digital transformation. A critical digital literacy perspective challenges narratives that individualize responsibility for digital exclusion and instead call for structural accountability and inclusive governance (van Dijk, 2020; Couldry & Mejias, 2019). In this sense, digital literacy is inseparable from broader struggles for social justice and democratic participation.

In response to these challenges, the chapter proposed a transformative, community-based framework for digital literacy. Centred on contextual relevance, participation, critical reflection, ethical awareness, and inclusivity, this framework positions communities as active agents rather than passive recipients of digital interventions. Such an approach aligns digital literacy with principles of sustainable development by prioritizing long-term capacity building, collective learning, and locally grounded innovation (Toyama, 2021; Leal Filho et al., 2021).

The implications outlined for policy, education, and community development further reinforce the need to treat digital literacy as a strategic social investment. Policies that integrate digital literacy with sustainability agendas, educational practices that embed critical and ethical dimensions, and community initiatives that foreground participation and equity are more likely to yield enduring impact. These orientations shift the focus from short-term technological adoption toward the cultivation of resilient, empowered communities capable of shaping their digital futures (OECD, 2023; Redecker, 2020).

In conclusion, digital literacy should be understood not simply as learning to use digital technologies, but as learning to live, decide, and act well within digitally mediated societies. As communities confront intertwined challenges of inequality, environmental degradation, and rapid technological change, digital literacy represents a critical lever for sustainable development when grounded in critical awareness and collective agency. Future research and practice should continue to explore how context-sensitive, community-driven digital literacy initiatives can contribute to more just, inclusive, and sustainable futures.

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TECHNOLOGY ENHANCED LANGUAGE LEARNING THROUGH ICT INTEGRATION TO SUPPORT SUSTAINABLE DEVELOPMENT GOALS (SDGS)

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ABSTRACT

The integration of Information and Communication Technology (ICT) has increasingly transformed language education in response to global digital development and sustainable educational demands. However, the implementation of technology in language learning often emphasizes technological utilization rather than its contribution to sustainable education aligned with the Sustainable Development Goals (SDGs). This study aims to examine the role of Technology Enhanced Language Learning through ICT integration in supporting sustainable language education, particularly in achieving SDG 4 on quality education. This research employed a library METHOD by synthesizing theoretical perspectives, empirical studies, and international educational policy documents related to ICT-based language learning and sustainability frameworks. The findings reveal that ICT integration promotes learner-centered learning environments, enhances accessibility and flexibility of language education, supports lifelong learning practices, and fosters global communication competence. Furthermore, technology-enhanced language learning contributes to inclusive and equitable educational opportunities while addressing sustainability challenges in digital education. Despite its potential benefits, several challenges such as digital inequality, limited technological competence, and institutional readiness remain significant concerns for sustainable implementation. This study highlights that aligning ICT integration with sustainability-oriented educational strategies provides a strategic pathway toward resilient and future-oriented language education systems in the digital era.

Keywords: *Technology Enhanced Language Learning, ICT Integration, Sustainable Development Goals (SDGs), Digital Language Education, Library Research*

INTRODUCTION

The rapid development of digital technology has significantly transformed educational practices across various disciplines, including language learning. The integration of technology into educational environments has shifted traditional teacher-centered instruction toward more interactive and learner-centered approaches (Yilmaz, 2017). Technology Enhanced Language Learning enables learners to access authentic learning materials, collaborate digitally, and engage in flexible learning environments beyond classroom boundaries (Zhonghao Zhou, 2018). These technological advancements indicate that digital innovation plays a crucial role in shaping modern educational systems (Sapri Saputra, 2024).

Information and Communication Technology (ICT) has become an essential component in improving the effectiveness of language learning processes (Benini, 2014). The use of ICT tools such as online platforms, multimedia applications, and virtual communication environments enhances students' engagement and motivation in learning foreign languages (Cuculic, 2021). Previous studies have demonstrated that ICT integration supports collaborative learning, autonomous learning, and improved language proficiency outcomes (Herwiana et al., 2025). Therefore, ICT-based instruction contributes to more dynamic and meaningful learning experiences.

Furthermore, education plays a fundamental role in achieving sustainable global development agendas promoted by the United Nations (Abera, 2023). Sustainable Development Goal 4 emphasizes inclusive and equitable quality education and promotes lifelong learning opportunities for all (Visioli & Stapleton, 2024). The integration of ICT in education has been recognized as a strategic approach to improving educational accessibility and quality worldwide, particularly in developing countries (Muhammad Muhammad et al., 2021). Reports from UNESCO highlight that digital technology facilitates inclusive education and supports sustainable learning ecosystems (Saini, 2020).

Although numerous studies have examined the effectiveness of ICT in language learning environments (Zhao, 2004), most research primarily focuses on technological performance and learning outcomes rather than its contribution to sustainable development perspectives. Limited attention has been given to how Technology Enhanced Language Learning supports broader educational sustainability aligned with SDGs frameworks. This gap indicates the need for a comprehensive literature-based investigation connecting ICT integration with sustainable educational development. Thus, understanding the relationship between ICT integration and sustainable education is essential in responding to global educational challenges in the digital era (Dong et al., 2025). Educational institutions are increasingly required to adopt innovative learning strategies that not only improve academic achievement but also contribute to long-term sustainable development goals (GRANO & PRIETO, 2020).

Based on these considerations, this study aims to explore Technology Enhanced Language Learning through ICT integration in supporting Sustainable Development Goals (SDGs) using theoretical perspective approach.

The concept of Technology Enhanced Language Learning has been widely discussed as an approach that combines pedagogical innovation with technological advancement to support meaningful learning experiences. Scholars argue that ICT integration enables multimodal learning environments where learners interact through digital platforms, online collaboration, and authentic communication practices (Thibaut & Curwood, 2018). Such integration shifts language learning from teacher-centered instruction toward learner-centered and technology-mediated knowledge construction processes (Li & Swanson, 2014).

From a pedagogical perspective, ICT-supported language learning aligns with constructivist learning theory, which emphasizes active participation and knowledge co-construction through interaction and experience (Diyal, 2024). Digital tools such as

learning management systems, online communication platforms, and multimedia resources allow learners to engage in authentic language use beyond classroom limitations (Yurieva et al., 2021). Consequently, technology-enhanced environments contribute to improving learner autonomy, critical thinking skills, and collaborative learning competencies required in contemporary education (Paethrangsri et al., 2024).

In relation to sustainable education, integrating ICT into language learning supports inclusive and equitable access to educational resources, particularly for learners facing geographical or institutional limitations (Aradhana & Bhattacharya, 2024). Sustainable learning practices emphasize long-term accessibility, adaptability, and continuous knowledge development, which are facilitated through digital learning ecosystems (Radenkova, 2024). Therefore, ICT integration functions not only as an instructional medium but also as a mechanism for promoting sustainable educational transformation.

Recent academic discussions further highlight that digital education contributes to achieving Sustainable Development Goals promoted by the United Nations by promoting lifelong learning opportunities and reducing educational inequality (CHEN et al., 2024). Language learning supported by technology enables global communication competence and intercultural awareness, both of which are essential components of sustainable global citizenship (Handayani, 2024). However, the alignment between technology-enhanced pedagogy and sustainability-oriented educational frameworks remains underexplored within language education research.

Accordingly, synthesizing existing literature becomes essential to understand how ICT integration can systematically support sustainable development within language learning contexts (Sultana et al., 2023). Through a comprehensive examination of previous theoretical and empirical studies, this research positions Technology Enhanced Language Learning as a strategic educational response to global transformation challenges. This integrative perspective strengthens the conceptual foundation for examining the intersection between ICT innovation and Sustainable Development Goals in language education.

METHOD

This study employed a library METHOD to examine the role of Technology Enhanced Language Learning through ICT integration in supporting Sustainable Development Goals (SDGs) within language education contexts (Azizah et al., 2025). Library research focuses on collecting and analyzing theoretical perspectives, empirical findings, and scholarly discussions derived from existing academic sources rather than primary field data (Abdurrahman, 2024). This approach enables researchers to synthesize knowledge from diverse studies in order to construct comprehensive conceptual understanding (Schick-Makaroff et al., 2016).

The data sources of this study consisted of peer-reviewed journal articles, academic books, conference proceedings, and international policy documents related to ICT integration, technology-enhanced language learning, and sustainable education (González-Zamar et al., 2020). Relevant literature was selected based on its alignment

with the research topic, theoretical relevance, and contribution to discussions on digital learning and sustainable development in education (Hamadi & El-Den, 2023). The selected references primarily addressed technology-supported language learning practices and their relationship with educational sustainability principle.

The data collection process involved identifying, reviewing, and categorizing scholarly publications that discuss ICT implementation in language education and its contribution to Sustainable Development Goals, particularly SDG 4 on quality education promoted by the United Nations (Costa et al., 2024). The researcher systematically organized the literature according to thematic relevance, including technological integration in pedagogy, digital learning environments, and sustainability-oriented educational practices (Huda & Sariyatun, 2025). This categorization facilitated a structured analysis of recurring concepts and theoretical patterns across studies (Grodal, 2020).

Data analysis was conducted using qualitative content analysis by interpreting and synthesizing key arguments, findings, and theoretical frameworks presented in the selected literature (Assarroudi et al., 2018). The analysis emphasized identifying conceptual relationships between technology-enhanced learning and sustainable educational development (Orozco-Messana et al., 2020). Through interpretative synthesis, this study generated an integrated understanding of how ICT integration supports sustainable language learning practices in the digital era.

RESULTS AND DISCUSSION

ICT Integration in Technology Enhanced Language Learning

The synthesis of reviewed literature indicates that ICT integration has fundamentally transformed the implementation of Technology Enhanced Language Learning (TELL) in contemporary educational contexts (Zainuddin, 2023). Language learning is no longer limited to conventional classroom interaction but has evolved into digitally mediated environments that enable continuous learning engagement (Safae EL GAZI et al., 2025). The integration of technological tools allows learners to access authentic linguistic resources, thereby expanding opportunities for meaningful language exposure. Additionally, ICT integration facilitates the transition from traditional instructional delivery toward technology-supported learning ecosystems (Barak, 2007). Digital platforms such as online learning environments, multimedia applications, and communication technologies provide interactive learning experiences that enhance learner participation (Hosseini pour Kamali, 2021). The reviewed studies consistently demonstrate that technology functions not merely as a supplementary teaching aid but as an integral component shaping instructional design and learning interaction.

Furthermore, the integration of ICT supports multimodal language learning processes in which learners simultaneously engage with visual, auditory, and textual inputs (Al-Muttairi & Al-Alusi, 2025). This multimodality strengthens comprehension and language retention by accommodating diverse learning preferences (Dean, 2016). As identified across multiple scholarly discussions, technology-enhanced

environments promote deeper cognitive engagement compared to single-mode instructional approaches commonly found in traditional classrooms.

Another significant finding emerging from the literature is the role of ICT in increasing learning flexibility (Noronha, 2025). Technology enables asynchronous and synchronous learning interactions that allow learners to manage learning activities according to individual pace and learning needs (Frank, 2008). This flexibility contributes to sustained learner engagement and supports personalized learning pathways, which are increasingly emphasized in modern educational frameworks.

In addition, ICT integration enhances opportunities for authentic communication practices in language learning (Sirghi & Sevciuc, 2025). Digital communication tools enable learners to interact with global communities through online discussions, collaborative projects, and virtual exchanges (Pongen, 2024). Such interaction exposes learners to real-world language use and intercultural communication experiences, strengthening communicative competence beyond textbook-based learning contexts.

The analysis also reveals that effective ICT integration reshapes the role of educators within Technology Enhanced Language Learning environments (Nurmala et al., 2023). Teachers increasingly function as facilitators, learning designers, and digital mediators rather than sole knowledge transmitters (Afriani et al., 2025). This pedagogical transformation demonstrates that technology integration influences not only learning tools but also instructional roles and classroom dynamics, reinforcing the evolving nature of language education in the digital era.

Learner-Centered Transformation in Digital Language Learning

The integration of ICT within Technology Enhanced Language Learning has significantly shifted language education toward learner-centered pedagogical practices (Mai, 2020). The reviewed literature consistently highlights that digital technologies empower learners to actively participate in constructing knowledge rather than passively receiving instructional content (Ferdiansyah et al., 2025). This transformation reflects a broader movement in education that prioritizes learner autonomy, engagement, and individualized learning experiences. ICT-supported learning environments enable students to regulate their own learning processes through access to diverse digital resources (Carneiro & Steffens, 2013). Learners are able to select materials, manage learning pace, and revisit instructional content according to personal needs and proficiency levels (Stary & Totter, 2006). Such flexibility encourages self-directed learning behavior, which plays an essential role in developing independent language learners capable of sustaining learning beyond formal educational settings.

Another important finding emerging from the literature concerns the enhancement of collaborative learning through digital platforms (Susiana & Priyatin, 2025). Online discussion forums, virtual group projects, and interactive communication tools facilitate peer interaction and cooperative knowledge construction (Hew & Cheung, 2010). These collaborative activities promote meaningful language use as learners negotiate meaning, exchange ideas, and engage in

communicative problem-solving tasks within digital environments. Moreover, learner-centered digital learning environments contribute to increased learner motivation and engagement (Crown et al., n.d.). Interactive multimedia content, gamified learning applications, and real-time feedback mechanisms create dynamic learning experiences that sustain learner interest (V & Reddy, 2025). Studies indicate that when learners actively interact with technological tools, they demonstrate higher participation levels and stronger commitment to language learning activities. The literature synthesis further reveals that ICT integration supports differentiated learning approaches in language education (Andrew, 2025). Technology enables educators to accommodate diverse learner abilities, learning styles, and linguistic backgrounds through adaptive learning resources (Chauhan, 2025). As a result, instruction becomes more inclusive, allowing learners with varying competencies to participate effectively in the learning process without experiencing significant learning barriers.

Finally, the transformation toward learner-centered digital language learning reshapes educational responsibility by positioning learners as active agents of knowledge development (Premawardhena, 2020). Technology-enhanced environments encourage critical thinking, creativity, and reflective learning practices that extend beyond linguistic competence (Xerri, 2025). This shift demonstrates that ICT integration contributes not only to language acquisition but also to the development of broader cognitive and lifelong learning skills required in sustainable educational systems.

ICT Integration and Sustainable Education toward SDG 4

The synthesis of reviewed literature demonstrates that ICT integration in Technology Enhanced Language Learning plays a significant role in advancing sustainable education aligned with Sustainable Development Goal 4 promoted by the United Nations, which emphasizes inclusive and equitable quality education (Sagir et al., 2025). Digital technologies expand access to educational resources by minimizing geographical and institutional barriers that traditionally limit learning participation (Hushin, 2025). Through technology-supported learning environments, language education becomes more accessible to diverse learner populations across different socio-economic contexts. ICT integration contributes to educational sustainability by promoting equal learning opportunities through digital accessibility (Baena-Navarro et al., 2024). Online platforms, open educational resources, and virtual learning systems allow learners to engage in educational activities regardless of physical location (Deichakivska et al., 2024). This expanded access reflects one of the primary objectives of SDG 4, namely ensuring lifelong learning opportunities that are inclusive and adaptable to changing global conditions.

Furthermore, technology-enhanced language learning supports sustainable knowledge development by enabling continuous learning beyond formal classroom settings (Lai, 2019). Digital learning environments encourage learners to independently explore linguistic resources, participate in global communication networks, and engage in ongoing skill development (Terhune, 2013). Such practices reinforce lifelong

learning principles, which are central to sustainable education frameworks promoted within global educational policies. Another important finding emerging from the literature is the role of ICT in reducing educational inequality (Masana, 2020). Technology-mediated instruction enables marginalized learners, including those in remote or under-resourced areas, to access learning materials that were previously unavailable (Wajnah et al., 2025). As a result, ICT integration contributes to narrowing educational gaps while promoting equitable participation in language education, thereby supporting social sustainability within educational systems.

In addition, ICT-supported language learning fosters global competence and intercultural awareness among learners (Arias Cifuentes & Solano Cahuana, 2025). Digital communication tools facilitate interaction across cultural and linguistic boundaries, enabling learners to develop communication skills necessary for global collaboration (Pongen, 2024). These competencies align with sustainability education goals that emphasize responsible global citizenship, mutual understanding, and cross-cultural cooperation in an interconnected world.

However, the literature also suggests that achieving sustainable education through ICT integration requires alignment between technological innovation, pedagogical strategies, and institutional support (Baena-Navarro et al., 2024). Sustainable implementation depends not only on technology availability but also on digital literacy development, teacher preparedness, and supportive educational policies (Firda Nurjihan Salsabila et al., 2024). Therefore, ICT integration should be viewed as a comprehensive educational strategy that connects technological advancement with long-term sustainability objectives in language learning.

Challenges in Implementing Sustainable ICT-Based Language Learning

The literature analysis reveals that despite the significant potential of ICT integration in Technology Enhanced Language Learning, several challenges continue to hinder its sustainable implementation (Sari & Abrar, 2024). One of the most frequently identified issues concerns unequal access to technological infrastructure across educational contexts (Waqar et al., 2024). Limited internet connectivity, inadequate digital devices, and insufficient technological resources remain major barriers, particularly in developing regions and under-resourced educational institutions.

Another critical challenge highlighted in previous studies relates to disparities in digital literacy among educators and learners (Buthelezi & Jiyane, 2026). Effective integration of ICT requires not only access to technology but also the ability to utilize digital tools pedagogically (Joshi, 2022). Many educators experience difficulties in adapting instructional strategies to technology-enhanced environments, resulting in technology being used merely as a presentation tool rather than as a transformative learning medium.

Furthermore, the literature indicates that institutional readiness significantly influences the success of ICT-based language learning implementation (Rafiee & Abbasian-Naghneh, 2021). Educational institutions often face limitations in providing

continuous professional development programs that support teachers in integrating technology effectively (A Dysart & Weckerle, 2015). Without adequate training and institutional support, sustainable ICT implementation becomes difficult to achieve despite technological availability.

Pedagogical adaptation also emerges as a substantial challenge in technology-enhanced language education (Lizamuddin et al., 2025). Transitioning from traditional teacher-centered instruction to digitally mediated learner-centered approaches requires changes in teaching philosophy and classroom management practices (Y. W. Li, 2016). Resistance to pedagogical change among educators may slow the adoption of innovative learning models, thereby limiting the effectiveness of ICT integration.

In addition, concerns regarding digital inequality extend beyond infrastructure to include differences in learners' socio-economic backgrounds (Qaribilla et al., 2024). Students with limited technological exposure or inadequate digital learning environments at home may experience learning disadvantages in technology-dependent educational systems. This condition highlights that sustainable ICT integration must consider social and economic dimensions to prevent the widening of educational disparities.

The reviewed literature ultimately suggests that sustainable ICT-based language learning cannot rely solely on technological advancement. Successful implementation requires coordinated efforts involving policy development, institutional commitment, teacher capacity building, and equitable resource distribution (Imbong et al., 2025). Addressing these challenges is essential to ensure that technology-enhanced language learning genuinely contributes to sustainable education rather than reinforcing existing educational inequalities.

Educational Implications of Technology Enhanced Language Learning in the Digital Era

The synthesis of existing literature indicates that the integration of ICT within Technology Enhanced Language Learning carries significant implications for future educational practices. Language education in the digital era requires instructional models that are adaptive, flexible, and responsive to technological development (Moroz et al., 2025). The findings suggest that technology-enhanced learning environments encourage educators to redesign instructional approaches that prioritize meaningful interaction, learner engagement, and sustainable knowledge development.

One important educational implication concerns the transformation of teaching roles in digitally mediated learning environments (Pavlik, 2015). Educators are increasingly required to function as facilitators, mentors, and learning designers who guide students in navigating digital learning resources (Afriani et al., 2025). This shift emphasizes pedagogical competence alongside technological proficiency, highlighting the necessity for continuous professional development to support sustainable teaching practices.

Furthermore, ICT integration promotes curriculum innovation within language education (Rozak & Ayaz, 2023). Digital technologies enable the incorporation of

authentic learning materials, global communication activities, and collaborative online tasks that reflect real-world language use (Fuat & Malbert, 2020). As a result, language curricula become more relevant to contemporary communication demands and better aligned with global educational transformation trends.

Another implication emerging from the literature relates to the development of twenty-first century skills among learners (Shobirin, 2025). Technology-enhanced language learning environments foster critical thinking, creativity, digital literacy, and intercultural communication competence (Chen & Yang, 2014). These competencies are essential for preparing learners to participate effectively in global knowledge societies while supporting sustainable human development objectives aligned with agendas promoted by the United Nations.

The findings also suggest that educational policymakers play a crucial role in ensuring sustainable ICT integration (Msambwa et al., 2024). Institutional policies must support equitable access to technological resources, teacher training initiatives, and infrastructure development to maximize the benefits of digital learning environments (Sabrina Nur Syifa et al., 2024). Without systemic policy alignment, technological innovation may fail to produce long-term educational impact.

Ultimately, Technology Enhanced Language Learning supported by ICT integration represents a strategic pathway toward resilient and future-oriented education systems (Kadek Adyatna Wedananta et al., 2025). By aligning technological innovation with sustainability principles, language education can contribute to inclusive learning opportunities and lifelong education practices (Mahdir Muhammad et al., 2025). These implications demonstrate that technology-enhanced pedagogy is not merely a response to digital transformation but a foundational component of sustainable education in the modern era.

CONCLUSION

This study explored the role of Technology Enhanced Language Learning through ICT integration in supporting Sustainable Development Goals within language education contexts. Based on the synthesis of existing literature, the findings indicate that ICT integration significantly transforms language learning practices by promoting accessibility, flexibility, and learner-centered instructional environments. Technology-enhanced learning enables learners to actively engage in knowledge construction while facilitating meaningful communication and collaborative learning experiences beyond traditional classroom limitations.

The analysis further demonstrates that ICT integration contributes substantially to sustainable education, particularly in supporting SDG 4, which emphasizes inclusive and equitable quality education and lifelong learning opportunities. Digital learning environments expand educational access, reduce geographical barriers, and promote continuous learning engagement. Consequently, technology-enhanced language learning serves as an effective framework for aligning pedagogical innovation with global sustainability objectives.

Despite its potential benefits, the study also highlights several challenges affecting sustainable ICT implementation, including unequal technological infrastructure, limited digital literacy, and insufficient institutional readiness. These challenges indicate that successful integration of ICT requires comprehensive educational strategies involving teacher capacity development, policy support, and equitable resource distribution. Sustainable technology integration therefore depends on coordinated efforts among educators, institutions, and policymakers.

From an educational perspective, the findings imply that future language education should prioritize adaptive pedagogical models that integrate technological innovation with sustainability principles. Educators are encouraged to redesign instructional practices that foster learner autonomy, digital competence, and global communication skills. Additionally, educational institutions should strengthen professional development programs to ensure effective and sustainable use of technology in teaching and learning processes.

This study recommends that future research expand the discussion by examining empirical evidence of ICT-based language learning implementation across diverse educational contexts. Further investigations may explore the effectiveness of technology-enhanced learning in promoting sustainable educational outcomes through mixed or empirical research approaches. Overall, aligning Technology Enhanced Language Learning with Sustainable Development Goals provides a strategic pathway toward resilient, inclusive, and future-oriented language education systems in the digital era.

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QUALITY EDUCATION IN EARLY CHILDHOOD: THE FOUNDATION FOR GLOBAL TRANSFORMATION

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ABSTRACT

Sustainable Development Goal 4 (SDG 4) calls for inclusive and equitable quality education and the promotion of lifelong learning opportunities for all. Within this global agenda, early childhood development (ECD) has emerged as a critical foundation for achieving long-term social, economic, and environmental transformation. This article examines the central role of early childhood education (ECE) in advancing SDG 4, particularly Target 4.2, which emphasizes universal access to quality early childhood development, care, and pre-primary education by 2030. Drawing on global data, neuroscientific evidence, and policy initiatives, this paper discusses the developmental importance of the early years and explores contemporary challenges such as inequitable access, the digital divide, teacher shortages, and climate-related disruptions. The discussion is supported by examples from Malaysia, including the C-HAT health initiative and the Future Skills for All (FS4A) digital programme. The findings highlight that investing in early childhood education not only supports academic readiness but also strengthens health systems, reduces inequality, and contributes to sustainable development across sectors. The paper concludes by emphasizing the urgent need for collaborative and evidence-based action to ensure that quality early childhood education becomes a global priority before 2030.

Keywords: *Early Childhood Education; Sustainable Development Goal 4; Global Transformation*

INTRODUCTION

In 2015, the United Nations adopted the 17 Sustainable Development Goals (SDGs) as a universal framework to address global challenges and promote sustainable development by 2030. Among these goals, SDG 4 focuses specifically on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all. Education is widely acknowledged as a transformative force capable of reducing poverty, improving health outcomes, fostering gender equality, and driving economic growth. However, the foundation of quality education does not begin in primary school or secondary education; it begins in early childhood.

Target 4.2 of SDG 4 calls for universal access to quality early childhood development, care, and pre-primary education. This emphasis reflects a growing global recognition that the earliest years of life are the most critical period for shaping cognitive abilities, social competencies, emotional regulation, and physical health. Early childhood development encompasses the holistic growth of children from the prenatal stage to approximately eight years of age, a developmental window that significantly influences lifelong trajectories.

The importance of early childhood has been repeatedly highlighted by researchers and global leaders. As noted by Bill Gates, the first five years of life profoundly influence the remaining decades of an individual's existence. This statement aligns with neuroscientific findings demonstrating that more than one million neural connections are formed every second during the early years. These early neural pathways shape future learning capacity, behavior, and health outcomes.

Despite the overwhelming evidence supporting early investment, global disparities remain deeply concerning. According to UNICEF (2023), approximately 43% of children under the age of five are not developmentally on track. Furthermore, more than 175 million children worldwide are not enrolled in preschool (UNICEF, 2019). These figures reveal significant inequalities in access to quality early learning opportunities and underscore the urgency of systemic intervention.

This article explores the role of quality early childhood education as a strategic foundation for global transformation. It analyzes scientific evidence, global challenges, and practical case studies to demonstrate how early childhood education functions as a central driver of sustainable development.

METHOD

This study employs a qualitative descriptive approach based on document analysis. The analysis draws upon global policy frameworks related to SDG 4, international statistical reports from UNICEF, and documented educational initiatives implemented in Malaysia. The methodological approach integrates theoretical perspectives on early childhood development, empirical data from global reports, and applied examples of educational innovation.

By synthesizing conceptual frameworks, developmental science, and national case studies, this article aims to provide a comprehensive and multidimensional understanding of how early childhood education contributes to achieving SDG 4 and supports broader sustainable development goals. The discussion integrates global policy perspectives with neuroscientific evidence and practical implementation strategies to bridge theory and practice. Through this integrative approach, the article not only highlights the foundational importance of early childhood development for lifelong learning outcomes but also demonstrates how early educational investment generates cross-sectoral benefits in health, economic productivity, and social equity. Ultimately, this analysis seeks to position early childhood education not merely as a preparatory stage of schooling but as a strategic driver of sustainable and inclusive global transformation.

RESULTS AND DISCUSSION

Early childhood development is widely recognized as a multidimensional process encompassing cognitive, physical, social, and emotional growth. These domains do not develop independently; rather, they are deeply interconnected and mutually reinforcing. Cognitive development influences language acquisition and problem-solving skills, while social and emotional development shape self-regulation,

resilience, and interpersonal competence. Physical health and nutrition further support neurological development and overall well-being.

Neuroscientific research provides compelling evidence for the importance of early experiences in shaping brain architecture. During the first few years of life, the brain develops at an extraordinary pace. More than one million neural connections are formed every second, creating the structural foundation for future learning. Although neural plasticity continues throughout life, early childhood represents the most sensitive period for establishing these foundational connections. Positive experiences, such as responsive caregiving and stimulating learning environments, strengthen neural circuits through what researchers describe as “serve and return” interactions. Conversely, exposure to chronic stress, neglect, malnutrition, or adversity can disrupt neural development and produce long-lasting consequences.

These findings highlight that investment in early childhood education must extend beyond academic instruction. Young children require nurturing care that includes access to health services, adequate nutrition, emotional security, physical safety, and meaningful early learning experiences. When these components are integrated, children are more likely to enter primary school ready to learn and to sustain positive developmental trajectories.

Importantly, early childhood education does not function in isolation within the Sustainable Development Goals framework. Rather, it operates as a central hub connecting multiple SDGs. By improving early learning outcomes, societies can reduce intergenerational poverty, thereby contributing to SDG 1 (No Poverty). By integrating health and nutrition interventions within early childhood programmes, communities advance SDG 3 (Good Health and Well-being). Expanding access to quality childcare also supports maternal workforce participation, promoting SDG 5 (Gender Equality) and SDG 8 (Decent Work and Economic Growth). In this way, early childhood education serves as a foundational strategy for sustainable and inclusive development.

Despite its recognized importance, achieving universal access to quality early childhood education remains challenging. Inequitable access continues to disproportionately affect children from marginalized communities. Economic disparities, geographic isolation, and limited infrastructure often prevent families from accessing quality preschool programmes. Additionally, the rapid digital transformation of education has exposed a persistent digital divide. While technology offers new opportunities for learning, children in under-resourced communities frequently lack access to devices, internet connectivity, and digital literacy support.

Teacher shortages and inadequate professional training further complicate the situation. High-quality early childhood education depends on skilled educators who understand child development and employ evidence-based pedagogical practices. However, many educators face limited training opportunities, low compensation, and insufficient professional support. Climate-related disruptions, including natural disasters and environmental instability, increasingly interrupt schooling and disproportionately affect vulnerable populations.

In response to these challenges, innovative and transformative practices have emerged. Nature-based learning approaches have gained attention for their ability to foster resilience, creativity, and environmental awareness. By engaging children with natural environments, educators cultivate not only cognitive skills but also emotional connection to sustainability. Similarly, play-based STEAM education integrates science, technology, engineering, arts, and mathematics through joyful exploration. This approach nurtures curiosity, creativity, and problem-solving skills that are essential in a rapidly evolving global landscape.

Culturally responsive pedagogy also plays a vital role in promoting inclusion and equity. When educational practices reflect children's identities, languages, and cultural backgrounds, learners experience a stronger sense of belonging and engagement. Professional teacher networks further enhance educational quality by enabling educators to collaborate, share best practices, and continuously refine their teaching approaches.

Malaysia provides illustrative examples of how integrated initiatives can advance early childhood education within the SDG framework. One significant concern in Malaysia is the increasing prevalence of childhood overweight and obesity, partly driven by poor dietary habits and excessive screen time. To address this issue, the Ministry of Health, in collaboration with UNICEF, launched the C-HAT (Cara Hidup Anda Terbaik) programme. Implemented across multiple states, this initiative promotes healthy eating and physical activity among school-aged children. Although focused on health outcomes, the programme supports holistic child development and reinforces the interconnected nature of SDG 3 and SDG 4.

Another important initiative is the Future Skills for All (FS4A) programme, which supports Malaysia's Digital Educational Learning Initiative. Through partnerships with major technology companies such as Google, Microsoft, and Apple, the programme enhances digital literacy among students and teachers. By reaching thousands of schools, FS4A addresses the digital divide and equips learners with future-ready skills. Such initiatives demonstrate how strategic collaboration between the government and private sectors can strengthen educational equity and innovation.

CONCLUSION

Quality early childhood education represents the cornerstone of sustainable global transformation. Scientific research confirms that the early years are the most critical period for brain development and lifelong learning. Global statistics reveal persistent inequalities that threaten progress toward SDG 4, yet innovative practices and collaborative initiatives provide promising pathways forward.

Investing in early childhood is not merely a matter of educational reform; it is a strategic decision that influences health systems, economic productivity, social cohesion, and environmental sustainability. As the 2030 deadline approaches, stakeholders at every level must prioritize early childhood development. Policymakers must allocate sufficient resources, educators must continue to innovate and collaborate,

researchers must generate evidence-based solutions, and communities must actively participate in creating nurturing environments for children.

The urgency of action cannot be overstated. The future of global transformation depends on the quality of experiences provided to the youngest members of society today. Decisions made in the present—regarding policy priorities, funding allocation, teacher preparation, and community engagement will shape not only educational systems but the social and economic landscape of future generations. Failure to invest adequately in early childhood development risks perpetuating cycles of inequality, poverty, and limited opportunity. Conversely, sustained and strategic investment in early childhood education has the potential to generate long-term returns that extend far beyond individual learners, strengthening national resilience, promoting inclusive growth, and advancing the collective aspirations embedded within the Sustainable Development Goals. In this sense, early childhood education is not simply the beginning of formal schooling; it is the beginning of sustainable development itself.

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THE FUTURE OF TEACHING IN A GLOBALIZED WORLD: ADAPTING EDUCATION FOR TOMORROW'S LEARNERS

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ABSTRACT

Processes of globalization, rapid technological innovation, and complex global challenges are reshaping what it means to teach and learn in the 21st century. This paper examines the future of teaching in a globalized world, focusing on the competencies required by tomorrow's learners, the evolving role of teachers, and the strategic use of educational technologies. Drawing on recent research on 21st-century skills, global competence, AI in education, and teacher professional transformation, it argues that education systems must move toward flexible, inclusive, and globally oriented models that remain rooted in local values and contexts. The paper outlines major drivers of change, analyzes key skills such as critical thinking, digital literacy, and global citizenship, and explores emerging challenges including the digital divide, ethical use of AI, and the tension between global and local identities. It concludes by proposing strategic directions for teachers as guides, designers, and equity advocates in a world where education is increasingly seen as a passport not only to individual opportunity but also to shared global responsibility.

Keywords: *Globalization, 21st-Century Skills, Educational Technology, Teacher Role, Global Citizenship, Digital Literacy*

INTRODUCTION

Globalization has fundamentally reshaped contemporary education systems by intensifying the movement of people, ideas, capital, and technologies across national borders (Bernal, 2025). Learners today are embedded in transnational networks of information and influence, where local experiences are continuously shaped by global economic, cultural, and environmental dynamics. For teachers, this context generates unprecedented opportunities to connect classrooms, access diverse resources, and foster intercultural understanding, while also creating new demands to prepare students for uncertain labor markets, rapid technological change, and complex planetary challenges (Komalasari, 2025; Rachmawati & Sari, 2024).

Educational research and policy discourse increasingly emphasize that traditional, content-heavy, transmission-oriented models of schooling are no longer sufficient (OECD, 2018a; Rivas et al., 2025). Instead, systems are called to cultivate 21st-century competencies such as critical thinking, creativity, collaboration, digital literacy, and global citizenship, alongside strong foundational knowledge. Teachers are thus expected to act not merely as lecturers but as designers of learning environments, mentors of lifelong learners, and mediators between global trends and local realities (OECD, 2018b; Salzburg Global, n.d.).

This paper explores four guiding questions: (1) What are the main drivers of change shaping the future of teaching? (2) Which key skills and competencies are required by tomorrow's learners? (3) How is technology transforming teaching and learning in globalized contexts? (4) What challenges and strategic responses characterize the evolving role of teachers? By integrating conceptual frameworks with recent empirical studies, the paper aims to offer a comprehensive yet practice-relevant account of how teaching can adapt to global realities in ethical and inclusive ways.

METHOD

This study adopts a qualitative approach using a literature review and conceptual analysis to examine the future of teaching in a globalized world. The data are derived from secondary sources, including peer-reviewed journal articles, policy reports (e.g., OECD and UNESCO), and recent publications related to globalization, educational technology, and 21st-century skills. Sources were selected based on their relevance and credibility.

The analysis was conducted through a thematic approach, guided by four research questions: (1) drivers of change in teaching, (2) key competencies for learners, (3) the role of technology, and (4) challenges and strategies for teachers. Relevant findings from the literature were categorized and synthesized to provide an integrated discussion. This approach enables a comprehensive understanding of current educational transformations and offers conceptual insights into adapting teaching practices in a globalized context.

RESULT AND DISCUSSION

Globalization and the Changing Educational Landscape

Globalization is often defined as the intensification of worldwide social relations linking distant localities in such a way that local happenings are shaped by events occurring many miles away (OECD, 2018a). In education, globalization manifests through international policy borrowing, global learning metrics, transnational curriculum trends, and the expansion of cross-border education markets. Institutions are evaluated against international benchmarks, and students increasingly compete and collaborate in global arenas (Bernal, 2025; OECD, 2024). Fadhillah (2025) underscores that knowledge and cultures are now deeply interconnected, and that teachers face "new demands and opportunities." These demands include preparing students to navigate multicultural environments, to critically engage with global media and information flows, and to understand how local issues such as employment or environmental degradation are tied to global systems. At the same time, opportunities arise for schools to participate in international collaborations, exchange programs, and global problem-solving initiatives (Parekh, 2024; Rachmawati & Sari, 2024).

Global challenges such as climate change, pandemics, widening inequalities, and large-scale migration have further intensified the imperative for transformative education (Bernal, 2025; OECD, 2024). The United Nations' Sustainable Development Goals (SDGs) explicitly position education as both a goal in itself (SDG 4) and a driver

of progress across other goals, including poverty reduction, gender equality, and climate action. UNESCO and OECD frameworks highlight that learners must develop the capacity to understand complex systems, appreciate interdependence, and act responsibly at local and global levels (OECD, 2018a, 2024). These developments shift the focus of education from solely preparing individuals for employment toward preparing them for responsible global citizenship and sustainable development (Inggriyani, 2024; Syahputra, 2024). For teachers, this entails integrating global issues into curricula, facilitating critical dialogue on controversial topics, and encouraging students to link classroom learning with community and global action (Mudinillah, 2024).

Drivers of Change in the Future of Teaching

Fadhillah (2025) identifies four key drivers of change: technological growth, cross-border mobility, global challenges, and lifelong learning requirements. These align closely with current research and policy analyses.

The rapid proliferation of digital technologies (including artificial intelligence (AI), virtual and augmented reality (VR/AR), learning analytics, and cloud platforms) has significantly altered the landscape of teaching and learning (Bernal, 2025; Intel, 2023). Educational technology is no longer a peripheral support but a central component of instructional design, assessment, and school management. AI-enabled tools offer new possibilities for personalized learning, adaptive assessment, and automated feedback, while VR/AR environments facilitate immersive simulations and virtual field trips (Rachmawati & Sari, 2024). However, scholars emphasize that technological innovation must be pedagogically purposeful and ethically grounded (Inggriyani, 2024). Without thoughtful integration, digital tools can amplify existing inequities, overload teachers, or reduce learning to algorithmic optimization rather than human development (Syahputra, 2024).

Increased cross-border mobility through migration, international study, and digital communication has diversified student populations and expanded the circulation of curricular models (Parekh, 2024; Rahman, 2025). Classrooms are now characterized by linguistic, cultural, and religious plurality, which can enrich learning but also generate tensions if not accompanied by inclusive, culturally responsive pedagogies (Komalasari, 2025). Digital communication technologies also enable ideas and educational innovations to spread rapidly across contexts, sometimes leading to uncritical adoption of global "best practices" that are not always suited to local realities (Lisnawati, 2025). Teachers thus need competencies to critically evaluate external models and adapt them to their own learners' needs and cultures.

Global challenges function as powerful drivers of educational reform by highlighting the limitations of narrow, discipline-bound, and exam-oriented schooling (Bernal, 2025; OECD, 2024). Issues such as climate change, public health, and social justice require systems thinking, ethical reasoning, and cross-sector collaboration. Curriculum frameworks increasingly emphasize interdisciplinary, project-based

approaches that allow learners to investigate real-world problems and envision transformative solutions (Mudinillah, 2024; Rachmawati & Sari, 2024).

Automation, AI, and the changing nature of work have shortened the shelf-life of many skills and created demand for continuous reskilling and upskilling (Bernal, 2025; Komalasari, 2025). Higher education and vocational institutions are expected to develop flexible learning pathways that support learners throughout their lives, not just during initial schooling. Teachers are called to model lifelong learning themselves, engaging in ongoing professional development and reflective practice (Lisnawati, 2025; OECD, 2018b).

Key Skills and Competencies for Tomorrow's Learners

International frameworks such as the OECD Learning Compass 2030 identify a cluster of knowledge, skills, attitudes, and values that learners require to thrive in rapidly changing societies (OECD, 2018a, 2024). These include cognitive skills (e.g., critical thinking, problem-solving), social and emotional skills (e.g., empathy, collaboration, resilience), and values oriented toward human dignity, sustainability, and justice. Fadhillah (2025) highlights four key skill domains: (a) critical thinking and problem-solving, (b) digital literacy and AI collaboration, (c) intercultural competence and global citizenship, and (d) creativity, adaptability, and resilience. These domains are strongly supported by recent empirical and theoretical work on future-ready education (Rachmawati & Sari, 2024; Rivas et al., 2025).

Critical thinking and complex problem-solving enable learners to interrogate information, challenge assumptions, and design responses to novel situations (Rivas et al., 2025). Education systems must prioritize higher-order thinking by integrating inquiry-based tasks, argumentative writing, and collaborative problem-solving into daily practice. Project-based learning has been shown to foster these competencies by engaging students with real-world issues and requiring them to apply knowledge in authentic contexts (Mudinillah, 2024; Rachmawati & Sari, 2024).

Digital literacy now extends beyond basic technical skills to include critical evaluation of online information, understanding of algorithms, and the ability to collaborate productively with AI systems (Bernal, 2025; Intel, 2023). Inggriyani (2024) emphasizes that AI-powered education should help learners understand how data are collected and used, recognize biases, and participate in shaping ethical digital futures rather than being passive users. This aligns with broader calls for "digital citizenship" and "data literacy" as core school outcomes (Syahputra, 2024).

Global competence includes the capacity to examine local, global, and intercultural issues; understand and appreciate the perspectives of others; engage in open, appropriate interactions; and take responsible action for sustainability and well-being (OECD, 2018a, 2024). Fadhillah's (2025) emphasis on intercultural competence and global citizenship resonates with emerging work on AI-supported global citizenship education, which explores how digital tools can facilitate cross-cultural dialogue and ethical reflection (Inggriyani, 2024; Syahputra, 2024).

Creativity, adaptability, and resilience help learners navigate rapidly changing environments, cope with failure, and generate novel solutions (Rahman, 2025; Rivas et al., 2025). Studies of 21st-century learning profiles in the era of global challenges show that students who develop these dispositions are better equipped to handle uncertainty and to contribute to innovation in their communities. Such dispositions are nurtured through pedagogies that encourage experimentation, reflection, and meaningful risk-taking rather than high-stakes rote testing (Mudinillah, 2024; Rachmawati & Sari, 2024).

The Role of Technology in Teaching and Learning

Online and blended learning models have moved from emergency responses to structural features of many education systems (Bernal, 2025; Intel, 2023). These models combine synchronous and asynchronous interaction, enabling flexibility in pace, place, and mode of learning. When well designed, blended approaches can support differentiated instruction, extend learning beyond classroom walls, and facilitate international collaboration through virtual exchanges (Parekh, 2024; Rachmawati & Sari, 2024).

AI-driven platforms can analyze learner data to provide adaptive content, personalized feedback, and targeted interventions (Ingriyani, 2024). Such systems hold promise for embedding global citizenship and SDG-related content into personalized pathways, but must be governed by robust ethical frameworks regarding bias, privacy, and accountability. Research on AI for global citizenship education highlights the need to balance efficiency gains with humanistic and justice-oriented educational goals (Syahputra, 2024).

Virtual classrooms and collaboration platforms enable students from different countries to work together on shared projects, engage in dialogue, and co-create knowledge (Bernal, 2025; Parekh, 2024). Programs such as international project-based learning, global MOOCs, and online exchange initiatives demonstrate how technology can facilitate "internationalization at home" for institutions that may lack resources for physical mobility (Rachmawati & Sari, 2024; "Teaching Tomorrow," 2025).

Gamified learning environments and immersive VR/AR applications have shown potential to increase motivation, deepen conceptual understanding, and enable safe experimentation with complex systems (Rachmawati & Sari, 2024). For example, VR simulations can allow students to explore climate scenarios or historical events from multiple perspectives, thereby fostering systems thinking and empathy. However, such tools require significant teacher support and critical framing to avoid superficial engagement (Intel, 2023).

Challenges and Ethical Tensions in a Globalized, Digital Future

One of the most pressing challenges is the persistence of digital divides in infrastructure, access, and skills (Bernal, 2025; Komalasari, 2025). Studies show that learners from low-income, rural, or marginalized communities often lack reliable connectivity, devices, or supportive learning environments, which can exacerbate

educational inequalities when technology-intensive models are adopted. Teachers must therefore advocate for inclusive policies and adopt low-tech or hybrid strategies that do not leave any learners behind (Rachmawati & Sari, 2024). The expectation that teachers continuously adapt to new technologies, curricula, and accountability regimes can generate significant workload and professional identity stress (Lisnawati, 2025; OECD, 2018b). The transformation of the educator's role in the digital era is both empowering and demanding, requiring teachers to serve as designers, facilitators, and data interpreters while maintaining relational and pastoral responsibilities. Without adequate support, professional learning, and recognition, these demands risk leading to burnout and resistance to innovation (Lisnawati, 2025).

Globalized curricula and English-dominated digital resources may overshadow local languages, cultures, and knowledge systems (Syahputra, 2024). Scholars caution that uncritical adoption of global templates can contribute to cultural homogenization and marginalization of indigenous or minority perspectives. Teachers thus face the delicate task of fostering global citizenship while affirming local identities and epistemologies (Fadhillah, 2025; OECD, 2024). The expansion of AI and data-driven systems raises ethical questions about surveillance, algorithmic bias, and the commercialization of student data (Inggriyani, 2024; Syahputra, 2024). Research on AI and global citizenship education calls for critical data literacy, participatory governance, and explicit attention to social justice in the design and use of educational technologies. Teachers play a crucial role in helping students interrogate how AI systems work and whose interests they serve (Inggriyani, 2024; Syahputra, 2024).

Strategies for Teachers in a Globalized World

Building on Fadhillah's (2025) framework, recent literature suggests several strategic directions for teachers navigating the future of teaching. Teachers are encouraged to begin with clear learning goals and then select technologies that genuinely enhance engagement, collaboration, and understanding (Mudinillah, 2024; Rachmawati & Sari, 2024). Innovative practices are most effective when technology is embedded within coherent pedagogical models such as project-based learning, inquiry-based instruction, and formative assessment (Intel, 2023). Culturally responsive pedagogy involves recognizing and valuing students' cultural backgrounds as assets for learning, and addressing power relations within curriculum and classroom discourse (Komalasari, 2025). In globalized settings, this includes critically examining whose knowledge is represented in textbooks and digital resources, and creating spaces for students to share their own experiences and perspectives (Rachmawati & Sari, 2024). Teachers can facilitate international projects, virtual exchanges, and participation in global MOOCs or online communities to expand students' horizons (Bernal, 2025; Parekh, 2024). Such initiatives develop language skills, intercultural competence, and collaborative problem-solving, while also modeling how to work across difference toward shared goals ("Teaching Tomorrow," 2025).

Integrating sustainability, human rights, and global citizenship into subject teaching allows students to connect curriculum content with real-world responsibilities

(Inggriyani, 2024; OECD, 2024). AI-mediated simulations and data visualizations can help learners explore the consequences of different choices and policies, but must be accompanied by critical discussion and action-oriented projects (Syahputra, 2024). The teacher of the future is often portrayed as a lifelong learner who engages in continuous professional development, reflective practice, and collaborative inquiry (Lisnawati, 2025; OECD, 2018b). OECD's work on "Teaching for the Future" stresses the importance of teacher agency (teachers' capacity to influence curriculum, policy, and innovation) as a key driver of sustainable educational change. Professional learning communities, mentoring, and practitioner research can support teachers in experimenting with new approaches while maintaining a strong sense of professional identity and purpose (Rachmawati & Sari, 2024; Salzburg Global, n.d.).

As guides and mentors, teachers scaffold students' learning journeys, support socio-emotional development, and foster reflective and ethical decision-making (OECD, 2018b; Salzburg Global, n.d.). As lifelong learners, they model curiosity, adaptability, and resilience that qualities that are essential for learners facing an uncertain future (Lisnawati, 2025; Rachmawati & Sari, 2024). As global thinkers and local advocates, they mediate between global agendas and local realities, ensuring that global competencies are contextualized and culturally meaningful (OECD, 2024; Parekh, 2024). Finally, as equity promoters, they actively work to reduce educational disparities, challenge discriminatory practices, and ensure inclusive participation in both physical and digital learning spaces (Inggriyani, 2024; Komalasari, 2025; Syahputra, 2024).

CONCLUSION

The future of teaching in a globalized world is characterized by intertwined processes of technological acceleration, deepening interdependence, and persistent inequality (Bernal, 2025; Komalasari, 2025; OECD, 2024). To prepare tomorrow's learners, education systems must adapt through innovation, inclusivity, and flexibility while remaining anchored in humanistic values and local knowledge. Teachers are central to this transformation as designers of learning environments, mentors of global citizens, and advocates for justice in digital and physical classrooms (Fadhillah, 2025; OECD, 2018b; Salzburg Global, n.d.). Realizing this vision requires sustained investment in teacher professional learning, equitable digital infrastructure, ethical governance of AI, and genuine partnerships among educators, communities, policymakers, and technology developers (Bernal, 2025; Inggriyani, 2024; Syahputra, 2024).

Echoing Malcolm X's assertion that "education is the passport to the future," Fadhillah's (2025) perspective and the broader literature suggest that in a globalized era this passport must equip learners not only to succeed individually but also to contribute responsibly to shared global futures. The future of teaching, therefore, is not merely about adapting to change but about actively shaping a more just, sustainable, and interconnected world through education.

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CHAPTER 1: EDUCATION

OPTIMIZATION OF SIMPLE PROJECTS TO IMPROVE STUDENT LEARNING OUTCOMES ON THE DIGESTIVE SYSTEM MATERIAL AT SMPN 3 KUTABLANG

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ABSTRACT

At junior high school, science learning still faces challenges in delivering abstract concepts, such as the human digestive system. The lack of concrete visual media makes it difficult for students to understand the structure and process of digestion. This study aims to develop and implement a simple teaching aid made from local materials as a learning medium to improve students' learning outcomes. This study used a quantitative approach with a quasi-experimental one-group pretest-posttest design involving 28 eighth-grade students at SMP Negeri 3 Kutablang. The digestive system teaching aid was designed in a two-dimensional format and used during learning. The results showed an increase in students' average scores from 55% on the pretest to 83% on the posttest. Additionally, students demonstrated active engagement during the activity, and teachers gave positive feedback on the effectiveness of the media. It can be concluded that the use of simple teaching aids can enhance students' conceptual understanding and learning outcomes in the digestive system material.

Keywords: *Simple Teaching Aid, Digestive System, Learning Outcomes, Science Learning*

INTRODUCTION

Science learning at the junior high school level is important in shaping students' initial understanding of complex and abstract scientific concepts, such as the human digestive system. This material is often difficult for students to grasp due to limited visualization and a lack of hands-on experience. To address this, the use of simple teaching aids is an effective solution for transitioning abstract concepts into more concrete and understandable concepts (Dewi et al., 2020; Saraswati & Ananda, 2023).

According to Nghitoolwa et al. (2024) the application of visual aids in science classes significantly improved students' learning outcomes and interest. Vidak et al. (2023) also reported similar findings, revealing that visualization using aids, including technology-based tools like AR, strengthened student engagement and understanding of human body structures and biological processes. Meanwhile, research by Riswanto and Mahfud (2022) found that interactive visual media positively impacted science learning outcomes.

However, it is important to note that not all teaching aids need to be high-tech. Simple, affordable learning media based on local materials are more applicable in many

schools, especially in areas with limited facilities, such as SMP Negeri 3 Kutablang. Research by Ramdani and Nurhasanah (2023) and Pramita et al. (2022) supports this. They found that simple teaching aids made from recycled materials can improve learning outcomes and foster students' environmental awareness.

Previous observations at SMP Negeri 3 Kutablang indicate that learning about the digestive system is still delivered conventionally without adequate teaching aids. Interviews with teachers and grade analysis indicate low student achievement in this material. Therefore, innovation is needed through the development and optimization of simple teaching aids that are appropriate to the school's needs and conditions. Based on this background, this study aims to develop simple teaching aids as learning media for the digestive system at SMP Negeri 3 Kutablang and to evaluate their effectiveness in improving student learning outcomes. This approach is not only locally relevant but also reflects global trends in educational innovation based on visual and practical learning experiences.

METHOD

This study used a quantitative descriptive approach with a quasi-experimental method and a one-group pretest-posttest design. This design was chosen to determine the impact of simple teaching aids on improving student learning outcomes in the digestive system. This approach allows for evaluation of changes that occur before and after the intervention in the same group (Nugroho & Suryani, 2021). This design is also commonly used in educational research with a limited scope, including in schools with limited facilities (Maulana et al., 2021).

This study was conducted at SMP Negeri 3 Kutablang, Bireuen Regency, Aceh Province, from November to December 2024. The subjects were eighth-grade students, selected purposively based on their learning needs and willingness to participate. A total of 28 students participated in the learning and evaluation process.

The initial stage of the activity began with observations and interviews with science teachers to identify the main challenges in the learning process, particularly related to the topic of the human digestive system. The needs analysis revealed that the learning methods used were still conventional and lacked adequate visual aids, resulting in students experiencing difficulties in understanding the digestive process. Based on these findings, the researchers designed and developed a simple demonstration tool in the form of a human digestive system model. The tool was constructed using locally available materials such as used bottles, balloons, funnels, small tubes, and food coloring, which were selected based on considerations of cost efficiency, safety, and ease of replication. Prior to implementation, the content of the demonstration tool was validated by science teachers to ensure its suitability and alignment with the learning material, as the validity of learning media plays a crucial role in effectively conveying scientific concepts (Fauziah et al., 2021).

The implementation of the learning activities began with a pretest administered to students in the form of 20 multiple-choice questions to measure their initial understanding of the digestive system. This was followed by the learning process,

during which the teacher presented the digestive system material using the prepared demonstration tool, with assistance from a group of students to support the learning activities. After the learning session was completed, a posttest with questions similar to those in the pretest was administered to assess changes in students' learning outcomes and to evaluate the effectiveness of the implemented learning media.

The test instruments were developed and validated beforehand to ensure the material's appropriateness and difficulty level. Pretest and posttest data were analyzed using paired sample *t*-tests to determine the significance of differences in scores (Yuliana & Prasetyo, 2020). According to Kandukoori et al. (2024), this method effectively measures innovative media's influence on students' cognitive achievement in limited classroom contexts.

During the learning process, observations of student engagement and reflective interviews with teachers were also conducted to evaluate the effectiveness of teaching aids. Photo documentation and field notes were included as supporting qualitative data. This method is supported by international studies showing that simple teaching aids can significantly improve student understanding through experiential learning (Nghitoolwa et al., 2024). Local research by Ramdani and Nurhasanah (2023) also confirmed that learning media based on recycled materials are effective in improving student understanding in the context of limited school facilities.

RESULTS AND DISCUSSION

Pretest and Post-test Results

The activity began with a pretest administered to eighth-grade students at SMP Negeri 3 Kutablang. The pretest consisted of 20 multiple-choice questions about the human digestive system. Based on the evaluation results, the average pretest score was 55%, or approximately 11 questions answered correctly. This result indicates that most students still had a low basic understanding of the material.

After the learning process was conducted using a two-dimensional cardboard model of the digestive system that was used directly by each student, a posttest was administered with equivalent questions. The posttest results showed an increase in the average score to 83%, or equivalent to 17 correct questions out of 20. The 28% difference in scores indicates a significant improvement following the instructional media intervention.

Table 1. Percentage Data of Pretest and Posttest Scores

Test Types	Average Value (%)	Average Correct
Pretest	55%	11 of 20 questions
Post-test	83%	17 of 20 questions

The data indicate that the use of interactive and visual aids has a positive impact on improving students understanding. This finding supports research by Nghitoolwa et

al. (2024), which showed that the direct use of concrete visual aids can strengthen students' retention and understanding of science concepts.

Student Response and Engagement

Throughout the lesson, students demonstrated high levels of enthusiasm and engagement. They actively held, pointed to, and explained parts of the digestive system using their respective props. This activity created a fun learning environment, encouraged interaction between students and the teacher, and built students' confidence in explaining the material.

Individually distributed cardboard props were effective in stimulating visual and kinesthetic learning styles. This aligns with the results of a study by Ramdani and Nurhasanah (2023) and is also supported by Susanti and Widodo (2021), who showed that the use of context-based props can build students' emotional and cognitive engagement during the lesson.



Figure 1. Making Simple Material-Based Digestive System Teaching Aid



Figure 2. Final Two-Dimensional Digestive System Model

Figure.1 shows the process of making a human digestive system demonstration tool by researcher and junior high school students using simple materials such as cardboard, colored paper, and glue. This activity aims to introduce the concept of the digestive system visually and creatively. Figure. 2 shows the final two-dimensional human digestive system demonstration made from cardboard and colored paper, displaying the main digestive organs arranged anatomically to support students' visual understanding of digestive system structure and function.



Figure 3. Students Presenting a Human Digestive System Model

Students present the final results of a cardboard human digestive system demonstration kit, complete with organ labels such as the mouth, esophagus, stomach, small intestine, large intestine, and anus. This activity aims to train students to recognize and explain the parts of the digestive system independently through a visual approach and hands-on practice.

Teacher Evaluation

A science teacher at SMP Negeri 3 Kutablang responded positively to the visual aids used. The teacher stated that these media were very helpful in conveying digestive system material, which is often difficult for students to understand due to its abstract nature. The teacher added that the tools were easy to replicate, economical, and reusable for other classes. This opinion is supported by studies by Dewi et al. (2020) and Siregar and Lubis (2022), which stated that contextual and simple visual aids significantly assist teachers in explaining complex material, particularly in biology and science, based on local wisdom.

Data-Problem Relationship

The main problem identified in this activity was low student learning outcomes due to a lack of concrete learning media. The pretest results confirmed this condition. However, after learning with visual aids, there was a significant increase in posttest scores and student engagement. This demonstrates that the use of appropriate learning media can be a practical solution to the limitations of facilities and conventional learning methods.

CONCLUSION

Based on the research results, it can be concluded that the use of a simple two-dimensional model of the digestive system significantly improved students' learning outcomes in the digestive system at SMP Negeri 3 Kutablang. This was evident in the increase in students' average scores from 55% in the pretest to 83% in the posttest after

the intervention. Furthermore, this model also increased student engagement and active participation during the learning process. Positive responses from teachers indicate that this model can be an effective and applicable solution to address the limited availability of science learning resources in schools.

Using simple, locally sourced, and easily replicable models supports learning outcomes and provides teachers with an innovative alternative in delivering abstract material. Therefore, it is recommended that similar models be further developed and integrated into other science learning environments to encourage improved educational quality in schools with limited facilities.

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ANALYSIS OF ELEMENTARY SCHOOL STUDENTS' PROBLEM-SOLVING ABILITIES IN SOLVING FRACTION OPERATIONS

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ABSTRACT

This study aims to analyze the mathematical problem-solving abilities of elementary school students in solving fraction operations with the same denominator. The study used a descriptive qualitative method with 20 fourth-grade students in one of the elementary schools in Bireuen Regency, Aceh as subjects. Data were collected through essay tests and interviews developed based on four problem-solving indicators according to Polya, namely understanding the problem, planning, implementing, and re-checking. The results showed that the ability to understand the problem was classified as high (80%), while the planning (55%), implementing (60%), and re-checking (35%) stages were still classified as moderate to low. Common student errors lie in misunderstanding the concept of similar denominators and the application of fraction addition operations. Based on these results, learning efforts are needed that emphasize the development of conceptual understanding through visual representation activities, the use of real-world contexts, and reflective guidance so that students can improve their critical and strategic thinking skills in solving mathematical problems.

Keywords: *Ability, Problem Solving, Arithmetic Operations, Fraction*

INTRODUCTION

Mathematical problem-solving ability is the process by which an individual analyzes or interprets a problem in order to find a solution (Sagita & Ermawati, 2023). Problem-solving ability is an essential skill that every mathematics teacher needs to nurture so that students can develop their mathematical competence (Hensberry & Jacobbe, 2012). The NCTM emphasizes that learning standards require students to possess five core skills: problem-solving, communication, connections, reasoning, and representation.

However, the reality on the ground shows that students' problem-solving abilities remain low. This is confirmed by the results of two international studies: the Trends in International Mathematics and Science Study (TIMSS) and the Programme for International Student Assessment (PISA). According to the 2015 TIMSS report, Indonesia ranked 42nd out of 49 countries with an average score of 397. This data indicates that Indonesian students' abilities to understand complex information, master theories, analyze and solve problems, use tools and procedures, and conduct investigations are still relatively low. This condition aligns with the results of the 2022 PISA, published by the OECD in 2023, where Indonesia only scored 366, or around

69% of the global average. This fact further confirms that the quality of Indonesian students' mathematical abilities still falls short of expectations.

According to research by Damayanti & Kartini (2022), Polya explains that mathematical problem-solving skills consist of four main aspects. First, students must be able to understand the given problem. Second, they must design an appropriate strategy to solve it. Third, they must implement the solution according to the planned strategy. Fourth, they must evaluate or double-check the obtained solution. Thus, the low achievement of Indonesian students in international studies can be attributed to their suboptimal mastery of these four aspects of problem-solving.

Fractions are a crucial topic in elementary school mathematics that is closely related to problem-solving skills. This material forms the foundation for mastering advanced mathematical concepts such as ratio, proportion, algebra, and probability. However, various studies have shown that fractions are one of the most difficult topics for students to understand in various countries (Syukra, Adrias, & Syam, 2025; Putri, Syam, & Chandra, 2025). In Indonesia, the results of the 2023 National Assessment reported that more than 40% of elementary school students still experience difficulties in solving simple fraction operations, particularly addition and subtraction with the same denominator. This situation indicates the urgency of examining students' problem-solving abilities in the context of fractions.

Students' difficulties in understanding fraction operations are often influenced by whole number bias, which is the tendency for students to use inappropriate thinking strategies. One form of this is adding denominators directly even though they are the same, resulting in incorrect answers (Rahayungsih & Jaafar, 2025). A similar phenomenon has also been found in the educational context in South Korea, where *Responsiveness to Intervention -based learning interventions* have proven effective in improving students' abilities in fraction operations and reducing achievement gaps between student groups (Yoon & Kim, 2025). Meanwhile, several studies have shown that elementary school students still experience difficulties in operations with fractions with unlike denominators (Nuraini, 2016; Setyaningrum & Riswari, 2023; Anggraeni, Darta & Rohimah, 2022). These obstacles arise primarily from a weak understanding of the concept of fractional equivalence and the procedure for equalizing denominators, while studies on operations with fractions with like denominators are still relatively limited. Yet, mastery of this material provides the foundation before students learn more complex fraction concepts. This shows that there is a research gap *that* needs to be filled through this study.

Furthermore, several international studies confirm that conceptual understanding of fractions plays a more important role than mere procedural skills (Lee & Choi, 2024). However, at the elementary school level, procedural mastery in operations with fractions with like denominators remains a crucial prerequisite for students to develop deeper conceptual understanding. Therefore, research examining students' problem-solving abilities in the context of operations with fractions with like denominators is highly relevant.

Based on this background, this study aims to analyze elementary school students' problem-solving abilities in solving fractions with like denominators. The study focuses on identifying common errors made by students, the problem-solving strategies used, and the cognitive factors influencing their success or failure. Theoretically, this research is expected to enrich the literature on mathematics learning, particularly on fractions, and support the *constructivist theoretical framework*, where students construct understanding through meaningful problem-solving experiences (Kilpatrick et al., 2021).

METHODS

The METHOD used is descriptive qualitative. The research population included all fifth-grade students, while the sample was selected using *purposive sampling* of 20 students aged 10-11 years. Data were collected through tests and interviews. The research was conducted in an elementary school located in Bireuen, Aceh with 20 sixth grade students as research subjects. The research instrument was a 70-minute essay test consisting of two questions. The questions were arranged based on indicators of mathematical problem-solving abilities which, include: (1) understanding the problem, (2) making a plan, (3) implementing the plan and (3) checking back.

RESULTS AND DISCUSSION

Based on the problem-solving ability test conducted, a picture was obtained of the extent to which fourth-grade students were able to solve problems related to fractions with the same denominator. Analysis was conducted on each problem-solving ability indicator through descriptive questions to assess students' ability to use problem-solving indicators.

Table 1. Results of Analysis of Students' Problem-Solving Ability Based on Indicators

Polya Indicator	Student Activity Description	Key Research Findings	Success Percentage
Understanding the Problem	Students can mention important information from the problem, such as the value of fractions (for example $\frac{3}{6}$ and $\frac{2}{6}$), and describe it in the form of a simple visual model.	Most students were able to understand the context of the story and correctly identify relevant information. They were also able to visualize fractions correctly.	80%
Planning the Settlement	Students try to determine the appropriate mathematical operation to solve the problem.	Many students immediately perform addition operations without first reviewing	55%

		the meaning of denominators or the relationships between fractions. This indicates that some students still lack a grasp of the concept of common denominators.	
Executing the Plan	Students draw visual models of fractions and calculate the results of operations based on the plans they make.	Students' visual representations were quite good, but errors were still found in combining two fractions into a coherent whole. Some students also had difficulty interpreting calculation results conceptually.	60%
Checking Again	Students are asked to evaluate the final results and match them with the picture or context of the question.	Most students don't reflect on or double-check their answers. They tend to end their work as soon as they get the result without ensuring it fits the context of the problem.	35%

To clarify students' abilities in each indicator, the following presents two examples of the work results of fourth grade students in solving problems with fractions with the same denominator.

Question 1 .

A cake is cut into 6 equal pieces. Ani eats 3 pieces, while Budi eats 2 pieces.

- Describe using the area of a rectangle that shows the part of the cake that Ani and Budi ate!*
- How much of the cake is left uneaten? Solve by estimating the area of the following rectangle!*
- Is there another way to determine the number of cakes Ani and Budi ate? If so, write your answer.*

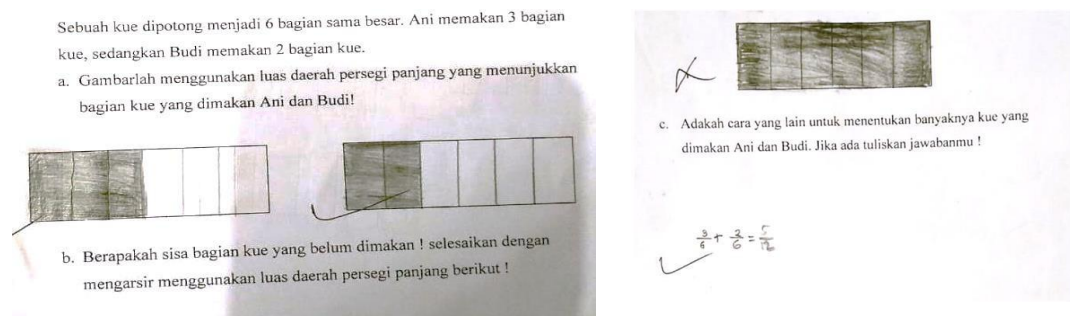


Figure 1. Student Answer Sheet F1

The figure illustrates that the student was able to understand the basic information of the problem well. He successfully illustrated the division of the cake into six equal parts and shaded the parts that Ani and Budi ate correctly. However, in the planning stage of the solution, the student made a conceptual error in the operation of adding fractions. It can be seen from the notes on the answer sheet that the student wrote the calculation $\frac{3}{6} + \frac{2}{6} = \frac{5}{12}$. This error indicates that the student added the denominators of the fractions when both fractions have the same denominator. The correct addition result should be $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

Thus, the main error occurs at the stage of implementing the plan, because students do not yet understand that the denominator does not need to be changed when both fractions have the same denominator.

Question 2

Roni has $\frac{1}{5}$ liter orange juice. Then he mixed in

$\frac{2}{5}$ a liter of apple juice. Describe the total amount of juice Roni has using the area of a rectangle!

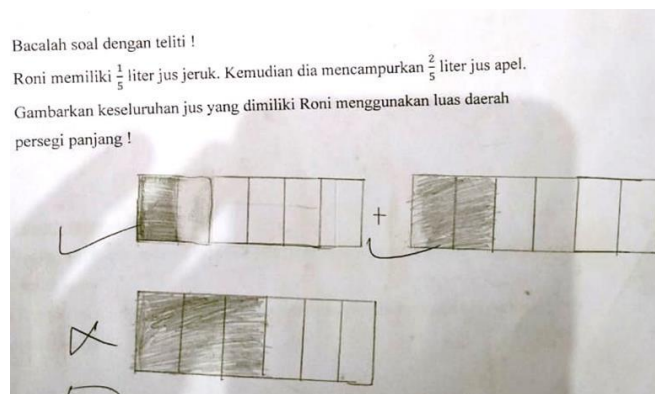


Figure 2. Student Answer Sheet F2

Based on the answer sheet of student F2, it appears that the student has been able to draw two rectangles, each divided into five equal parts. One part is shaded to show $\frac{1}{5}$ and two parts are shaded to show $\frac{2}{5}$. Conceptually, this step is correct and shows the ability to understand the problem and carry out the plan well. However, the student has not combined the two fraction models into one complete representation that shows the total result $\frac{3}{5}$. This means that the student has understood the concept of parts of a whole, but still has difficulty uniting the two fractions in the same visual model. This shows that the student's conceptual understanding of the meaning of fraction operations is still partial.

The results of this study indicate that fourth-grade students' problem-solving abilities are considered high (80%), but their ability to plan, implement, and review solutions is still in the moderate to low category. This finding aligns with Polya's (1973) assertion that the problem-solving process relies not only on mathematical skills but also on the ability to understand context, develop strategies, and reflect on the results. The results of this study are in line with the findings of Althamiranda, Fitria, & Nurrohmat (2025) who showed that elementary school students still experience difficulties, especially in the planning and implementation stages of problem-solving strategies as described by George Polya. They noted that many students are not yet accustomed to translating problems into appropriate mathematical models and tend to rely on algorithmic procedures without understanding the meaning of the problem context. This condition shows that students' thought processes are still mechanical and have not yet developed into the reflective reasoning needed in mathematical problem solving.

In line with this, research by Wijayanti & Amin (2019) revealed that in solving contextual mathematics problems, high-ability students use visual, symbolic, and verbal representations simultaneously, but students with lower ability levels only rely on symbolic or verbal representations. This confirms that the skill of connecting visual and symbolic representations is very important for strengthening the mathematical thinking process. Furthermore, research by Nuraini, Lubis & Wandini (2024) shows that students who have good visual representation skills tend to be able to demonstrate a more systematic thinking process in understanding and solving mathematical problems, while students who are less, often only write the final answer without describing the complete representation stage. This finding is consistent with the view of Novianti, Karim, and Jasmaniah (2023), who stated that students are not yet accustomed to directing and regulating themselves in the learning process, making it difficult for them to plan and manage appropriate strategies when engaging in problem-solving activities.

Thus, it can be concluded that elementary school students' mathematical problem-solving abilities are still focused on the initial stage, namely understanding the problem, while the stages of strategy planning, procedure implementation, and review of results are not optimal. To address this, mathematics learning needs to be designed to integrate the use of multiple representations (concrete, visual, and

symbolic), reflective discussions, and thinking scaffolding so that students are able to develop more meaningful conceptual understanding and problem-solving strategies.

CONCLUSION

The results of this study indicate that elementary school students' mathematical problem-solving abilities in solving fractions with like denominators vary at each stage of the solution. While most students were able to understand the problem well, they still experienced difficulties in planning a solution strategy, implementing the procedure correctly, and rechecking the results. Common errors that emerged were related to understanding the concept of like denominators and incorrect application of addition operations.

Based on these findings, it is recommended that the mathematics learning process in elementary schools emphasize conceptual understanding through exploratory activities and the use of visual representations, rather than solely on procedural application. Teachers need to provide step-by-step guidance (scaffolding) so that students can reflect on each step of the solution. Furthermore, exercises involving everyday contexts need to be increased to help students understand the meaning of fractions more concretely. Further research is recommended to develop problem-based learning models or conceptual visual approaches that can improve students' abilities in the planning and reflection stages of mathematical problem-solving.

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THE IMPLEMENTATION OF THE INQUIRY LEARNING MODEL ASSISTED BY THE POWTOON APPLICATION TO IMPROVE STUDENTS' PROBLEM-SOLVING SKILLS AT SMP NEGERI 2 PEUSANGAN SIBLAH KRUENG

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ABSTRACT

The objective of this study is to examine the improvement in students' problem-solving skills through the implementation of the Inquiry Learning model assisted by the Powtoon application on the topic of probability in Grade VII at SMP Negeri 2 Peusangan Siblah Krueng. It also aims to determine students' problem-solving skills when taught using conventional learning methods on the same topic and grade level. The approach used in this study is quantitative, and the research design is quasi-experimental. The population and sample consisted of 36 seventh-grade students at SMP Negeri 2 Peusangan Siblah Krueng. Data collection techniques included tests and observation sheets. Data analysis was conducted using hypothesis testing and percentage analysis. The results of the study indicated that the obtained t-value (ttest) was 3.867, which is greater than the t-table value of 1.303 at a 5% significance level ($t_{table} = 1.303 < t_{count} = 3.867$), leading to the acceptance of the alternative hypothesis (H_a) and the rejection of the null hypothesis (H_0). This implies that the improvement in students' problem-solving skills using the Inquiry Learning model assisted by the Powtoon application is significantly better than that of students taught through using conventional learning methods assisted by instructional videos on the topic of probability in Grade VII at SMP Negeri 2 Peusangan Siblah Krueng.

Keywords: *Inquiry-Based Learning, Powtoon, Problem Solving, Probability*

INTRODUCTION

Education is one of the essential aspects of human life. Through quality education, individuals are guided to develop their potential and capabilities to solve various problems they encounter. Education also contributes to shaping individuals to become qualified and morally upright. According to Law No. 20 of 2003 on the National Education System, education is a conscious and deliberate effort to create a learning environment and learning process so that learners actively develop their potential to possess spiritual-religious strength, self-control, personality, intelligence, noble character, and skills needed by themselves, society, the nation, and the state. The goal of education as stated in the Preamble of the 1945 Constitution is to educate the nation's life. The intended intelligence is not limited to intellectual intelligence alone but encompasses a broader, holistic concept. This educational objective ultimately determines the success of shaping quality human beings without overlooking the role

of other elements in mathematics education.

Mathematics plays various roles that make it a crucial field of study. One of its key roles is serving as a tool for thinking, helping students understand the mathematical concepts they are learning. In learning mathematics, students need the ability to understand mathematical concepts, solve problems, and think critically one of which is the problem-solving ability. Problem-solving ability is a crucial skill that students must possess to resolve mathematical problems. According to Nurhayati (2017), mathematical problem-solving ability is a process through which students learn and solve problems, enabling them to acquire and apply knowledge they already have in problem-solving.

Based on interviews conducted by the researcher at SMP Negeri 2 Peusangan Siblah Krueng, it was revealed by a mathematics teacher that most seventh-grade students had difficulty mastering the concepts of probability. These difficulties include an inability to identify known elements, what is being asked, and the sufficiency of the required elements. Students also struggled to formulate mathematical problems or develop mathematical models. They often became confused when attempting to apply strategies to solve both mathematical and real-world problems. This indicates that students lack adequate problem-solving skills, particularly in identifying the known and unknown components of a problem and determining the sufficiency of the given information essential part of the problem-solving process.

Problem-solving ability is a necessary skill that must be instilled in students to help them navigate the challenges they will face throughout life. With this skill, they will be able to solve problems accurately and apply the knowledge acquired in school to real-life situations. To improve students' problem-solving skills, a conducive learning environment and effective instructional methods are essential. Learning approaches should encourage students to be independent, creative, and actively involved in classroom activities. One such instructional model that promotes creativity, critical thinking, and active learning is the Inquiry-Based Learning model assisted by Powtoon.

Inquiry-Based Learning is a model that encourages student engagement in the learning process. According to Kelana (2021:7), Inquiry-Based Learning is a systematic process of thinking and discovery in which knowledge is constructed through personal discovery rather than memorization. According to Kurniawati (2022), a learning model is a pattern or plan used by teachers to organize learning materials and student activities, serving as a guide for implementing instructional processes. The advantages of Inquiry-Based Learning include: (a) development of cognitive, affective, and psychomotor domains, (b) accommodating various learning styles, (c) encouraging behavioral change through experience, and (d) promoting student autonomy.

One of useful tool to support the implementation of Inquiry-Based Learning is the Powtoon application, which allows for the creation of interactive learning media. Powtoon enables educators to incorporate various animations into presentations, making learning more engaging and reducing student boredom. According to Awalia et al. (2019), the use of animated content in Powtoon can help students stay focused,

engaged, and motivated during lessons. As an audiovisual learning medium, Powtoon enhances classroom instruction, making it more dynamic and enjoyable for students. Research by Ulandari et al. (2019) showed that the Inquiry-Based Learning model significantly improved student outcomes, as evidenced by higher average scores in the experimental group compared to the control group on various problem-solving indicators.

The Powtoon application can be effectively used by teachers and students to create engaging and interactive presentations. It also serves as a powerful medium for enhancing student comprehension of mathematical concepts and other subjects. Powtoon's advantages include ease of video creation, a wide selection of engaging animations, support for images and audio, replayability, interactivity through feedback, accessibility anytime and anywhere, and short video durations that help maintain student motivation. Based on the above considerations, the researcher is interested in conducting a study titled: "The Implementation of the Inquiry-Based Learning Model Assisted by the Powtoon Application to Improve Students' Problem-Solving Skills at SMP Negeri 2 Peusangan Siblah Krueng."

METHOD

The approach used in this study is quantitative, and the type of research employed is quasi-experimental. The research design applied is the nonequivalent pretest–posttest control group design. The study was conducted at SMP Negeri 2 Peusangan Siblah Krueng. The population and sample consisted of 36 seventh-grade students from two classes at SMP Negeri 2 Peusangan Siblah Krueng. The data collection technique used in this study was a test, and the data analysis technique involved hypothesis testing.

RESULTS AND DISCUSSION

The Independent Sample T-test was conducted to determine whether the research hypothesis is accepted or rejected. The hypothesis tested is as follows:

Table 1. Independent Sample T-test Results

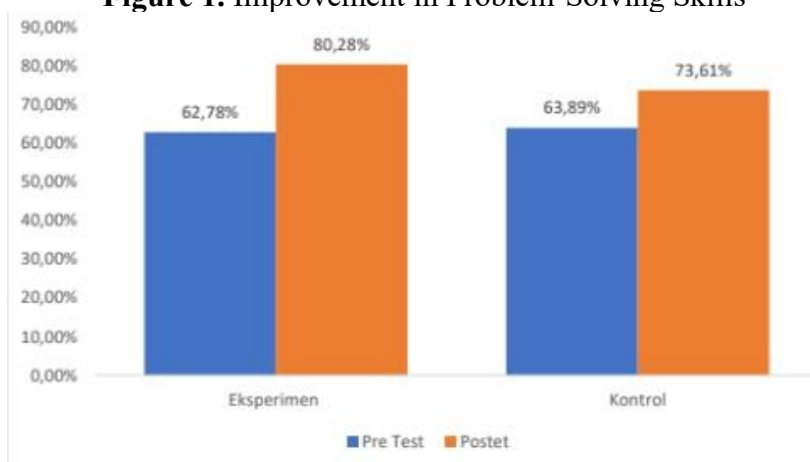
		t-test for Equality of Means				
		t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference
Students' Problem Solving	Equal variances assumed	3,867	34	0,039	3,611	4,164

	Equal variances not assumed	3,867	33,389	0,039	3,611	4,164
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Based on the results of the Independent Sample T-test on the post-test scores and the analysis of Levene's Test, it was found that the significance value was 0.039. Since $0.039 < 0.05$, the alternative hypothesis (H_a) is accepted. This indicates that the implementation of the Problem Based Learning model has a significant effect on the problem-solving ability of Grade VIII students at SMP Negeri 2 Peusangan Siblah Krueng. Furthermore, the t-table value at the 5% significance level is 1.303, while the t-count value is 3.867 ($t_{table} = 1.303 < t_{count} = 3.867$), which confirms the acceptance of H_a related to students' problem-solving skills. Based on these results, it can be concluded that the improvement in students' problem-solving ability taught using the Inquiry-Based Learning model assisted by the Powtoon application is better than those taught using conventional learning methods assisted by instructional videos, particularly in the Probability topic for Grade VIII students at SMP Negeri 2 Peusangan Siblah Krueng.

The improvement in students' problem-solving skills taught using the Inquiry-Based Learning method assisted by the Powtoon application is better than that of students taught using conventional learning assisted by instructional videos on the topic of Probability in Grade VIII of SMP Negeri 2 Peusangan Siblah Krueng. For further illustration, refer to the following figure:

Figure 1. Improvement in Problem-Solving Skills



Based on the explanation of the graph above, it shows that Grade VIII students of SMP Negeri 2 Peusangan Siblah Krueng enjoyed learning through the Inquiry Learning model, which led to an improvement in their mathematics learning, particularly in the topic of probability. The results of this study are consistent with the findings of Nurhalimah (2022), who stated that the Inquiry Learning model is a method that can assist teachers in achieving the goals of learning. The Inquiry Learning model has several advantages in improving students' abilities in various areas.

In its implementation, the Inquiry Learning model engages students actively in the learning process. At the beginning of the lesson, students are asked to reread their textbooks. By using Inquiry Learning, students are encouraged to express their opinions, which in turn helps them retain the material more effectively. Based on observations conducted by two teacher observers on the teacher's activities during the learning process, the average percentage score obtained was 93%, which falls into the "very good" category of teacher performance. Similarly, the observation of students' activities by the same two observers yielded an average score of 91%, also categorized as "very good."

The Inquiry Learning model is an effective method that can help teachers deliver lessons more effectively to achieve learning objectives. This is due to the many advantages offered by the model. Some of these advantages include increasing students' self-confidence, improving learning outcomes, enhancing students' mathematical communication skills, developing students' problem-solving abilities in learning materials, improving students' social skills, and more. However, the Inquiry Learning model also has certain drawbacks, such as a noisier classroom environment than usual and students feeling anxious about being called on to answer questions. Nonetheless, these drawbacks can be managed by the teacher. One way to address them is for the teacher to guide students to behave more orderly during the learning process.

CONCLUSION

Based on the research findings, it can be concluded that the obtained t-value is as follows: the t-table value at the 5% significance level is 1.303, which is lower than the t-count of 3.867. This indicates that the alternative hypothesis (H_a) is accepted and the null hypothesis (H_0) is rejected. These results imply that the improvement in students' problem-solving skills using the Inquiry Learning method assisted by the Powtoon application is better than that of students taught through conventional learning using instructional videos on the topic of probability in Grade VIII of SMP Negeri 2 Peusangan Siblah Krueng.

Based on these findings, the following suggestions are proposed. For teachers, the Inquiry Learning method assisted by the Powtoon application can be used in teaching probability by integrating it with other learning media. For students, the Inquiry Learning method assisted by the Powtoon application can have a positive impact on developing mathematical problem-solving skills, particularly in the topic of probability. For future researchers, further studies may be conducted on different mathematical topics to serve as comparative research and to improve the quality and standard of education.

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TEACHERS' PERCEPTIONS OF KNOWLEDGE INTEGRATION FOR GLOBAL CITIZENSHIP EDUCATION: A QUALITATIVE STUDY FROM KERALA COLLEGES OF INDIA

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ABSTRACT

This qualitative study explores how college teachers in Kerala understand and practice knowledge integration as part of Global Citizenship Education (GCE). It seeks to uncover teachers' interpretations of global citizenship and the strategies they use to bring global perspectives into their classrooms. The research also examines the challenges teachers face in linking subject knowledge with social responsibility and global awareness. A qualitative methodology was adopted, involving semi structured interviews with ten college lecturers from disciplines including English, Education, and Social Sciences. The interview data were analysed thematically. The findings reveal that while most teachers value the idea of global awareness, they have varying levels of familiarity with the formal concept of GCE. Many participants naturally attempt to connect their teaching to global issues such as climate change, gender equality, and cultural diversity etc. often by relating these issues to local contexts. However, they also reported significant constraints, including an exam focused curriculum, limited time, and lack of institutional support or training in GCE. Despite these hurdles, some innovative pedagogical approaches were noted, for example, using literature to discuss global issues or involving students in community projects tied to the Sustainable Development Goals. The study concludes that clearer guidelines, professional development programmes, and curriculum reforms are needed to empower teachers to effectively integrate knowledge for global citizenship. By doing so, higher education in Kerala can better prepare students as informed, responsible global citizens.

Keywords: *Global Citizenship Education, Knowledge Integration, Critical Pedagogy, Higher Education, Kerala, Qualitative Research*

INTRODUCTION

Global Citizenship Education (GCE) has emerged as a significant concept in international education over the past two decades, reflecting growing concerns about global challenges such as inequality, environmental sustainability, and social conflict. GCE refers to educational approaches that aim to equip learners with the knowledge, values, and skills needed to engage responsibly with global issues and to contribute to a more just, inclusive, and peaceful world (Banks, 2017). Rather than implying citizenship beyond the nation-state, GCE emphasises a sense of shared humanity, global interdependence, and respect for diversity. UNESCO (2015) conceptualises GCE through three interrelated dimensions: cognitive (understanding global issues and systems), socio-emotional (empathy, values, and attitudes), and behavioural (responsible action at local and global levels).

At the policy level, GCE has gained strong recognition worldwide, notably through its inclusion in the United Nations Sustainable Development Goals (Target 4.7). In India, the National Education Policy 2020 echoes GCE principles by advocating ethical, social, and universal human values, as well as multidisciplinary learning. Although elements of global awareness have long existed in Indian

education through peace education and environmental studies, GCE as an explicit educational framework remains relatively new, particularly in higher education.

A key pedagogical approach closely linked to GCE is knowledge integration. Knowledge integration involves connecting learning across disciplines and relating academic content to real-life contexts, enabling students to develop a holistic understanding of complex global issues (Beane, 1997). Global challenges such as climate change, migration, or social inequality cannot be adequately understood through a single disciplinary lens. Therefore, effective GCE often requires teachers to move beyond subject boundaries and integrate perspectives from history, economics, ethics, science, and culture. Research suggests that integrated learning enhances critical thinking, problem-solving skills, and the ability to apply knowledge meaningfully.

Teachers play a central role in implementing both GCE and knowledge integration. Their beliefs, perceptions, and professional experiences strongly influence classroom practice (Borg, 2003). Studies indicate that while many teachers support holistic and value-based education in principle, they often face constraints in practice, including rigid curricula, exam-oriented systems, limited time, and lack of institutional support. Toh and Ng (2019) found that teachers in the Asia-Pacific region generally recognise the importance of global awareness but differ in their confidence and preparedness to integrate GCE due to limited training and resources. In India, formal exposure to GCE and interdisciplinary pedagogy in pre-service and in-service teacher education remains limited, resulting in uneven implementation largely driven by individual initiative.

The theoretical foundation of this study is informed by Paulo Freire's critical pedagogy. Freire (1970) argued against the "banking" model of education and proposed a dialogic, problem-posing approach that encourages critical consciousness and social transformation. Freirean pedagogy aligns closely with GCE, as both emphasise critical thinking, dialogue, and the integration of knowledge with lived experiences. From this perspective, education should not only raise awareness of global issues but also empower learners to reflect and act upon them, linking reflection with action (praxis).

This study is situated within the higher education context of Kerala, a state known for its high literacy rate, strong social awareness, and emphasis on education as a tool for social development. Kerala's sociocultural values, including ideas such as *vasudhaiva kutumbakam* ("the world is one family"), and its global exposure through migration and digital media, provide a favourable context for GCE. Students and teachers in Kerala are generally aware of international events and global issues, offering a strong foundation for integrating global perspectives into higher education.

However, despite this conducive environment, challenges remain. Higher education in Kerala is largely examination-driven, with content-heavy syllabi that leave limited space for interdisciplinary and value-based learning. While recent curricular developments, such as elective courses on environmental studies and human rights, indicate progress, the explicit integration of GCE is still inconsistent. Teachers often lack clear guidelines, training, and institutional support to systematically incorporate global citizenship themes into their teaching.

In this context, the present study focuses on knowledge integration as a key mechanism for implementing GCE in higher education. By examining college teachers' awareness, perceptions, and classroom practices in Kerala, the study seeks to address a gap in existing research and to understand how global citizenship ideals are interpreted and enacted at the classroom level. Teachers' perspectives are crucial, as they mediate curriculum policy and shape students' learning experiences. Understanding both the opportunities and constraints they face can inform future policy, curriculum design, and teacher professional development aimed at strengthening GCE in higher education.

METHOD

This study employed a qualitative research design to capture teachers' perspectives in their own words. Given the exploratory nature of the research questions, a phenomenological approach was used to understand teachers' lived experiences and the meanings they attach to them. Rather than testing hypotheses or measuring variables, the study aimed to provide a rich and descriptive account of teachers' perceptions and practices related to knowledge integration and global citizenship.

The participants were ten college lecturers from various institutions across Kerala. They represented different academic disciplines, including the Humanities, Social Sciences, and Education, and had teaching experience ranging from five to twenty years. The group included lecturers from both government and private colleges located in different districts. Participants were selected through purposive sampling, focusing on teachers with an interest in innovative teaching or involvement in co-curricular activities related to social issues. Participation was voluntary, and pseudonyms were used to ensure confidentiality.

Data were collected through semi-structured interviews conducted over a two-month period. The interviews used open-ended questions that encouraged participants to reflect on global citizenship education, describe their teaching practices, and discuss challenges they faced. Each interview lasted approximately 45–60 minutes. Seven interviews were conducted in person, while three were carried out online for convenience. Interviews were mainly conducted in English, with occasional use of Malayalam expressions, which were translated where necessary. All interviews were audio-recorded with participants' consent, and observational notes were taken to provide additional context.

The interview recordings were transcribed verbatim and analysed using thematic analysis following Braun and Clarke (2006). The researcher first familiarised herself with the data and then generated initial codes related to the research objectives. These codes were grouped into broader themes, resulting in three main themes: teachers' understanding of global citizenship education, strategies for integrating global and local issues in teaching, and challenges in knowledge integration. To ensure trustworthiness, triangulation, peer review of the coding process, and member checking were employed.

Ethical considerations were carefully observed throughout the study. Participants were informed about the purpose of the research and their rights, including the right to withdraw at any time. Written informed consent was obtained,

and all data were stored securely. The identities of participants and institutions were anonymised to protect confidentiality and encourage honest and open participation.

RESEARCH RESULTS

Awareness and Understanding of GCE

The first theme centres on how aware the teachers were of the concept of global citizenship education and how they interpreted its meaning. Overall, the interviews revealed a spectrum of awareness. A few participants had a relatively clear idea of what GCE entails, often describing it in terms of fostering open mindedness and understanding global interconnections even if they did not use the exact term “GCE”. However, several teachers admitted that they were not previously familiar with the formal terminology. They associated the idea with more familiar concepts like moral education, environmental awareness, or simply “teaching students to be good human beings.” For instance, one senior English lecturer expressed that while she had not heard the term *Global Citizenship Education* explicitly, she believed it meant “teaching beyond the textbook” and encouraging students to learn about the wider world. She said:

“Our job is not just to finish the syllabus. I try to give students exposure to what is happening around the globe. I didn’t know there is a name for it, but I always felt it is part of educating a person.”

Another participant, a lecturer in the Education department, initially conflated GCE with community service, reflecting a common perception that global citizenship is mainly about humanitarian work or social service. He noted:

“Global citizenship... perhaps it means doing service for society, like charity or helping people in need. As teachers, we often guide students in social service activities. Maybe that is what it is.”

These responses show that while the terminology of GCE is not yet widely recognised among the teachers, the underlying principles are not alien to them. Most respondents valued the idea that education should shape socially conscious individuals. They frequently mentioned concepts like empathy, cultural tolerance, and sustainability as important educational outcomes. The difference lay in how systematically these ideas were understood. Only two out of the ten teachers could reference global frameworks (for example, one teacher mentioned the United Nations SDGs in passing), suggesting that formal discourse on GCE has not deeply penetrated faculty development in these colleges. Nonetheless, there was a general openness to the idea once it was discussed. Teachers often remarked that, upon reflection, linking their subject to broader issues was “common sense” and necessary for students’ holistic growth.

Integrating Global and Local Issues in Teaching

The second theme highlights the strategies and examples teachers provided of how they integrate global topics and social issues into their everyday teaching. Despite working within a prescribed syllabus, many teachers found creative ways to make connections between their subject matter and the wider world. An important observation is that teachers frequently used local examples as a bridge to global themes, a pedagogy that can be termed “glocal” (think globally, act locally).

By situating global issues in the local context, they made abstract concepts more concrete for students. For example, an English literature professor described how she used a novel about migration to spark discussion on both historical migration from Kerala to the Gulf countries and contemporary refugee crises globally. According to her, this dual perspective helped students see parallels between the characters' experiences in the novel and real stories they had heard in their own communities. She explained:

“When we read a story about people fleeing their homeland, I ask students to think about our own neighbours or relatives who have worked abroad or the news about refugees in other countries. Suddenly, the literature is not just fiction it is connected to real life events and choices.”

In another instance, a Political Science lecturer incorporated discussions of climate change policy into a unit on the Indian constitution and citizens' rights. He asked his class to debate the right to a clean environment as a fundamental right, linking local environmental issues like the recurring floods in Kerala to global climate challenges. This approach turned a civics lesson into an exploration of environmental citizenship. He shared his experience:

“I gave my students a task to investigate how climate change is affecting Kerala like the floods we had and then we connected it to what they study in newspapers about COP climate conferences. It became a lively discussion on what responsibility each citizen has, not only the government.”

A notable example of integration came from a Botany lecturer who took a very hands-on approach. During our interview at her college campus, she proudly pointed out a vegetable garden maintained by students. She had initiated this as part of a unit on sustainable agriculture. The students learned scientific concepts about soil, plants, and nutrition in class, and simultaneously applied those concepts by growing organic vegetables on campus. This project allowed her to naturally introduce global issues of food security and sustainable farming practices. Students, she noted, began to talk about organic farming trends worldwide and even drew comparisons with kitchen gardens in other countries. This story illustrates how practical activities can link local action to global awareness. Across the interviews, other integration strategies mentioned included

Project work

Several teachers assign projects that require students to research a global issue and examine its local impacts. For instance, a Sociology teacher had groups of students' study topics like “gender equality in our village and in the world” or “local waste management and global environmental goals,” leading students to present findings that connect home and abroad.

Multidisciplinary Seminars

In one college, a lecturer collaborated with colleagues from different departments to organise a seminar on sustainable development, allowing the topic to be explored from multiple disciplinary perspectives. The seminar included a range of speakers, such as a local environmental activist who shared practical

community-based experiences related to sustainability. An international scholar also participated via video call, which helped broaden students' understanding by providing a global perspective on the issue.

Using media and technology

Teachers also mentioned showing short documentaries or TED talks on global topics and then moderating discussions. One teacher recounted using a TED talk on the importance of kindness in a general English class to trigger conversation on empathy across cultures. These examples demonstrate that even without an explicit mandate, some educators are proactively blending global content into their teaching. Importantly, they do so by rooting it in the students' context connecting textbook knowledge to issues in Kerala, and then expanding outward. The result, according to many, is increased student engagement. Teachers observed that students often became more curious and participative when such connections were made, asking questions and sometimes even initiating discussions about news events or social problems. It suggests that knowledge integration for global citizenship can make learning more dynamic and relevant, although it relies heavily on the individual teacher's initiative.

Challenges and Support Mechanisms

The third theme encompasses the various challenges teachers reported, as well as the kinds of support (or lack thereof) that affect their ability to integrate knowledge for GCE. A predominant sentiment was that the current higher education structure in Kerala, being quite traditional and exam oriented, poses practical constraints. Teachers consistently mentioned the pressure to "complete the syllabus" as a major hurdle. With a fixed curriculum and a tight academic calendar, there is little flexibility to diverge into open ended discussions or projects that are not directly tied to examination topics. One lecturer summed it up:

"I have only a limited number of hours with each class. If I spend too much time on tangents even valuable ones like discussing global issues I fear we will fall behind and students might not be prepared for the university exam. It is a constant tension for me."

Another significant challenge is the lack of institutional encouragement or policy support for GCE initiatives. None of the participants recalled any official guidelines from their college or affiliating university about incorporating global citizenship or related themes into teaching. Whatever integration happened was largely due to personal passion. An education faculty member pointed out that while there are broad visions in the national policy, day to day teaching is guided by syllabi that make no mention of GCE. He said:

"We hear big terms like global competence in conferences, but at the college level, nobody tells us how to do it or checks if we do. As long as our results are good, the system is happy. So, many teachers just stick to the usual method."

Moreover, the teachers indicated a need for capacity building. Some felt that they themselves needed more exposure to global issues and pedagogical methods to confidently bring those into their teaching. One relatively new lecturer confessed

that she avoided topics like international politics or complex global economic issues in class because she felt unsure about handling questions or debates that might arise, and she did not want to inadvertently provide incorrect information. This highlights a professional development gap: even well-intentioned teachers might hold back if they feel out of their depth or worry about controversy.

Cultural and political sensitivities can also be a constraint. A few respondents noted that discussing certain global issues can be tricky in the current climate. For example, conversations about human rights or international conflicts might be seen as political, and teachers have to tread carefully to avoid misunderstanding or conflict with prevailing sentiments. One teacher shared that when she tried to discuss the global movement in a class on media studies, some students were enthusiastic but a couple of others reacted defensively, saying it was “against our culture” to talk openly about such matters. She reflected that without guidance on how to navigate these discussions, teachers can feel caught off guard.

On the positive side, teachers identified some supportive factors that enable knowledge integration when it does happen. Personal motivation and commitment to student welfare was one many said they felt a personal responsibility to prepare students for more than exams, which drove them to include global and societal topics. Peer support was another: in colleges where a few teachers were like minded, they formed informal groups to share resources (like interesting articles or case studies) and sometimes collaborated on interdisciplinary events. Support from the college leadership was mentioned in two cases; for instance, a principal who encouraged outreach activities or seminars indirectly created an environment where GCE related efforts were valued.

In summary, the challenges are significant and largely systemic: rigid curricula, exam pressures, and lack of formal support or training. Yet, the presence of even small support networks and the intrinsic dedication of teachers play a crucial role in the few successes noted. These findings underscore that without addressing structural issues, the integration of knowledge for global citizenship will remain patchy and dependent on individual educators, rather than being a standard practice across higher education.

DISCUSSION

The findings of this study reveal both promising developments and notable gaps in the implementation of Global Citizenship Education (GCE) at the college level in Kerala. Overall, teachers demonstrate a positive orientation toward the core values of GCE, such as fostering social awareness, responsibility, and learning beyond rote memorisation. Even when unfamiliar with the formal concept of GCE, many teachers believe that education should extend beyond academic content to address real-world and social issues. This aligns with previous studies, including Toh and Ng (2019), which show that educators across the Asia-Pacific region value global awareness despite uncertainty about implementation.

However, the study also highlights a clear gap between policy ideals and classroom practice. Although national and international frameworks, such as India’s National Education Policy 2020 and UNESCO’s guidelines, promote global citizenship and sustainability, structural constraints within higher education hinder

their realisation. Exam-oriented curricula, rigid syllabi, and limited time restrict teachers' ability to engage in exploratory and critical pedagogies. This situation reflects Freire's notion of the "banking" model of education, where content delivery and assessment take precedence over dialogue and critical engagement. While many teachers aspire to more meaningful learning, institutional expectations often limit their capacity to do so consistently.

Despite these constraints, several teachers employed pragmatic and creative strategies to integrate global and social issues into their teaching, such as linking literary texts to migration or using sustainability projects to connect theory with practice. These approaches support Beane's (1997) argument that integrated curricula enhance relevance and student engagement. Students responded positively when learning was connected to real-life contexts, suggesting that knowledge integration can strengthen critical thinking and participation. Notably, many teachers adopted a "glocal" approach by connecting local experiences with global issues, making abstract global concepts more accessible to students.

Nevertheless, the scope of GCE implementation remained limited. Most global themes addressed were relatively non-controversial, focusing on environmental awareness, empathy, or cultural understanding. More critical discussions of global inequality or power relations were less common. This pattern reflects Andreotti's (2011) distinction between "soft" and "critical" GCE, with teachers tending toward the former. Factors such as sociopolitical sensitivity and lack of confidence or training likely contribute to this cautious approach.

Finally, the findings underscore the importance of teacher training and institutional support. The integration of GCE largely depends on individual initiative rather than systematic planning. Teachers expressed a need for clearer guidance, professional development opportunities, and supportive institutional frameworks. In the Kerala context where students are generally aware of global issues and receptive to such discussions there is strong potential for more meaningful implementation. With curriculum flexibility, professional development, and supportive leadership, GCE can be integrated in ways that complement academic goals rather than compete with them.

CONCLUSION

This qualitative study examined how college teachers in Kerala perceive and practise knowledge integration within the context of Global Citizenship Education (GCE). The findings indicate that teachers generally support the principles of GCE, valuing education that goes beyond rote learning to foster globally aware and responsible individuals. Many teachers have already incorporated global and social perspectives into their teaching, linking curriculum content with real-world issues and enhancing student engagement. However, the study also highlights significant challenges, including exam-oriented curricula, limited policy support, and insufficient training, which restrict the widespread implementation of such practices. These findings suggest that the successful integration of GCE cannot depend solely on individual teacher initiative but requires strong institutional support, curriculum reform, and sustained professional development.

For Kerala, a state with a strong educational tradition, integrating knowledge for global citizenship represents an important step forward. By supporting teachers and embedding global perspectives within curricula, higher education institutions can produce graduates who are academically competent and socially responsible. Overall, this study underscores both the potential and the challenges of implementing GCE and highlights the central role of teachers in realising the goals of Global Citizenship Education in higher education.

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THE EFFECTIVENESS OF PUZZLE GAMES ON FINE MOTOR DEVELOPMENT IN PRESCHOOL-AGE CHILDREN IN PAUD AMAL BIREUEN

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ABSTRACT

Fine motor development is a better indicator than gross motor skills in diagnosing motor disorders in preschoolers. If there is a fine motor disorder, it can have an impact on other developments. Fine motor development disorders can be overcome by providing a stimulus to children, namely by playing puzzles. The purpose of this study was to determine the effectiveness of puzzle games on the fine motor development of preschool children through case studies. The case study research design used is scientific writing, which is designed to solve a problem based on theory and scientific method. With inclusion criteria. Based on the research results on the effectiveness of puzzle games in promoting fine motor development in preschoolers, the findings suggest that puzzle games are effective in supporting fine motor development. Based on the information above, fine motor development carried out with puzzle games is very effective. It can be seen from Tables I and II that the subject can complete puzzle games at the right time without the guidance of parents and teachers. Based on the data in the sentence Most of the development of preschool age children before being given the puzzle playing method was normal (59%). And almost all of the development of preschool-age children after being given the puzzle playing method was normal (88.4%)

Keywords: *Puzzle Games, Fine Motor, Preschool Age Children*

INTRODUCTION

Growth and development during the preschool years are a very influential foundation stage that lays the groundwork for further development (Adriana, 2013). This period is short, so it is referred to as a critical period or golden period. Even the slightest disturbance in growth and development that occurs in preschool-aged children, if not detected and intervened upon as early as possible, will reduce the quality of human resources in the future (Febrikaharisma, 2013).

Fine motor development is the improvement of body movement coordination involving much smaller muscles and nerves. Good motor control allows children to engage in active play. When playing, sensory-motor activities are the largest component used by children, and active play is very important for muscle function development. For example, toys used by babies develop sensory- motor skills, and toys for toddlers and preschoolers greatly help the development of fine motor activities. Research findings indicate that puzzles have a positive impact on the development of fine motor skills in children (Harmila, 2023).

Based on global prevalence data, the incidence of fine motor development disorders in preschool children in the United States in 2017 was 17.8% (Zablotsky et al, 2019). Data on fine motor development disorders in preschool children in Indonesia in 2018 was 7.51% (Riskesdas, 2018). Play is one of the most influential factors in motor development. Play is an activity in which children can practice

skills and express their thoughts creatively. Play provides opportunities to practice skills and develop ideas according to their own ways and abilities. For children aged 0-6 years, playing provides pleasure and develops imagination. Physical movements are not only important for developing physical skills, but also have a positive effect on growth, self-esteem, and even cognitive development. One type of game that is suitable for preschoolers is puzzles (Adriana, 2011).

Puzzles involve arranging pieces of an image to form a complete picture. Images are something that is visualized in two dimensions as an expression of feelings and thoughts. Puzzles are fun educational tools that can be used to develop fine motor skills. Puzzle games can improve fine motor skills, such as classifying objects based on color, shape, and size, as well as enhancing and training creative thinking skills to solve simple problems (Madyastuti, 2016). When children put together thin puzzle pieces made of wood or cardboard, it can hone their fine motor skills (Saraswati, 2011).

Based on the research title, numerous studies have been conducted and have demonstrated consistent findings related to this topic. One of these studies was conducted by Erni Yuniati (2018) and entitled *The Effectiveness of Puzzle Games in Improving Fine Motor Development in Preschool Children*. The study revealed that puzzle games had a significant effect on the fine motor development of preschool children, as indicated by a p-value of 0.0001 (Erni, 2018).

Based on the preliminary study conducted by the researcher at PAUD AMAL Wanita, there were 23 children divided into one class. The results of interviews conducted with kindergarten teachers revealed that the problem faced was the underdevelopment of fine motor skills in 4-5-year-old students at PAUD. This was evident in the assessment of the students' work, which showed that the average student was still classified as lacking. After identification, it was found that there were several problems arising from the learning system implemented by teachers towards students, namely in the delivery of lesson material in the classroom, teachers still used the old pattern dominated by less innovative lecture methods, the use of learning media was also not optimal in the teaching and learning process carried out by teachers so that student motivation and interest in learning was still lacking, and learning activities were more teacher-centered with students only as passive listeners. The objectives of this research are to determine the effectiveness of puzzle games on the fine motor development of preschool children.

Fine Motor Development in Children Aged 3-6 Years

Fine motor development is a critical aspect of early childhood growth, as it plays an essential role in supporting children's overall developmental processes. Fine motor skills refer to a child's ability to control small muscles, particularly those in the hands, fingers, and wrists, which are necessary for precision-based activities such as drawing, writing, cutting, and manipulating small objects (Hurlock, 2013). These skills enable children to perform daily tasks independently and contribute to their readiness for formal learning activities.

Between the ages of 3 and 6, children demonstrate significant progress in their fine motor abilities. At age 3, children are generally able to grasp and manipulate simple objects, such as holding a pencil or crayon to make scribbles on

paper. By age 4, their fine motor skills become more refined, enabling them to imitate simple shapes such as circles, crosses, or squares. At ages 5–6, finger and hand coordination are further developed. Children can write letters and numbers with greater clarity, produce more detailed drawings, and perform daily tasks that require precision, such as buttoning clothes or using utensils properly (Santrock, 2011).

The development of fine motor skills during the preschool years is not only crucial for academic readiness but also supports independence and self-confidence. Children with well-developed fine motor abilities tend to be more prepared for formal learning activities, such as writing and drawing in primary school. Therefore, appropriate stimulation during early childhood plays a vital role in supporting this developmental stage.

Puzzle Games and Fine Motor Development

One effective form of stimulation for fine motor development in early childhood is through educational games, such as puzzles. A puzzle is a game composed of pieces that must be arranged to form a complete picture or pattern. Puzzles come in various forms, ranging from simple ones with large pieces for children aged 3–4, to puzzles featuring numbers, letters, or geometric shapes designed for children aged 5–6 (Mayke, 2001).

Puzzle activities provide both enjoyable and challenging learning experiences. While working on puzzles, children practice shape-matching, memory recall, and problem-solving strategies. These activities demand concentration, patience, and well-coordinated hand–eye movements.

From a fine motor development perspective, puzzles play an important role in enhancing finger dexterity and wrist movement. Children use their fingers to grasp, rotate, and place puzzle pieces in the correct position. These actions indirectly strengthen the small muscles of the hands while also improving manipulative skills. Additionally, puzzles help refine hand–eye coordination, as children must align each piece with the intended space. Such skills form the foundation for later abilities, including writing and drawing (Beaty, 2014).

Problem-Solving Skills

In addition to motor development, puzzles contribute significantly to cognitive growth, particularly problem-solving skills. While completing a puzzle, children are encouraged to think logically, analyze spatial relationships, and experiment with different solutions. This process strengthens their ability to solve problems independently, fostering critical thinking and reasoning skills that are foundational for later learning (Fisher, 2017).

The Effectiveness of Puzzle Games for Preschool Children

1. *Child Development Theories*

- a. **Piaget’s Theory:** According to Jean Piaget, preschool-aged children are in the preoperational stage, where they begin to develop logical thinking but still rely heavily on concrete, hands-on experiences. Puzzles support this

development by allowing children to use practical experiences to solve problems, organize shapes, and understand cause-and-effect relationships (Piaget, 1952).

- b. **Vygotsky's Theory:** Lev Vygotsky emphasized the importance of social interaction and environmental tools in cognitive development. Puzzle activities serve as effective learning tools within this framework, as children can engage in guided play with adults or peers. Interaction through giving hints, collaborative problem-solving, or observing others' strategies enhances cognitive and social skills while scaffolding the child's learning process (Vygotsky, 1978).

2. *Factors Affecting Puzzle Effectiveness*

- a. **Type of Puzzle:** The design and complexity of a puzzle directly influence its effectiveness. Puzzles with pieces that are too small or complex may lead to frustration, whereas puzzles appropriately matched to a child's age and skill level encourage success, build confidence, and maintain motivation.
- b. **Frequency of Play:** Regular engagement with puzzles strengthens their impact. Consistent practice allows children to refine fine motor skills, improve spatial awareness, and develop more advanced problem-solving strategies over time.
- c. **Parental or Caregiver Support:** Children who play puzzles under the guidance of parents or caregivers tend to show greater improvements in fine motor skills. Supportive interactions provide scaffolding, motivation, and reinforcement, helping children navigate challenges and consolidate learning experiences effectively.

Puzzle activities represent a multifaceted tool for preschool education. They simultaneously promote fine motor development, cognitive growth, problem-solving abilities, and social interaction. The effectiveness of puzzles is enhanced when factors such as puzzle type, frequency of engagement, and adult support are carefully considered. Consequently, puzzles can be regarded as an engaging and impactful method to support the holistic development of preschool children aged 3–6 years.

METHOD

This study is descriptive research employing a case study design with two respondents as the research subjects. Data were collected using observation sheets and puzzle games as research instruments. The purpose of this study is to explore the characteristics and behaviors of children who exhibit reluctance to attend school and identify the underlying factors influencing their attitudes toward learning and social interaction in a school setting.

The first subject is a quiet and shy child who refuses to study unless accompanied by a parent. He expresses reluctance to go to school, stating that he has no friends to play with and does not know anyone there, indicating social adjustment difficulties. The second subject is also a quiet child; however, unlike the first subject, he does not require parental accompaniment. Instead, he refuses to attend school due to the absence of a play area, suggesting that the availability of

recreational facilities influences his willingness to attend school. These findings highlight different motivational and social factors affecting each child's readiness for school.

RESULT

The first stage in research involving subjects is assessment. In this case study, the initial assessment focused on puzzle games as a stimulation activity. The purpose of this assessment was to observe the initial responses of both subjects toward puzzle play and to identify their level of independence in completing the activity. The process was carried out with the assistance of parents and teachers to understand the children's behavior during the initial stage and after a brief intervention.

During the initial assessment, Subject I participated in a puzzle activity on Monday at 09:00 AM. At this stage, the subject appeared passive, remained silent, and showed no initiative to assemble the puzzle. He tended to wait for guidance from parents or the teacher before starting the activity. However, after 30 minutes, at 09:30 AM, a follow-up observation was conducted. The results indicated positive progress; the subject began assembling the puzzle independently without requiring assistance from parents or the teacher. This demonstrated an improvement in motivation and independence within a short period after receiving initial stimulation.

A similar pattern was observed in Subject II, who took part in the assessment on Wednesday. At 09:00 AM, the subject exhibited the same passive behavior as Subject I during the initial stage—remaining quiet and waiting for directions from parents or the teacher without attempting to assemble the puzzle. However, by 09:30 AM, the subject was able to work on the puzzle independently without any assistance. This change suggests that both subjects required an initial adaptation period and support, but were able to demonstrate independence once given the opportunity to practice and understand the activity.

Overall, the assessment results indicate that although both subjects were initially passive and reluctant to engage in the activity without guidance, they were able to adapt and gradually increase their independence during puzzle play. This improvement suggests that repeated exposure to puzzle activities can help children become more confident and self-directed in completing tasks. These findings further indicate that educational games such as puzzles can effectively stimulate concentration, initiative, and independence in children, particularly when introduced through supportive and non-coercive approaches.

For both Subject I and Subject II, the researcher observed significant changes when the puzzle activities were conducted within the same session structure. Initially, both subjects showed similar patterns of behavior during the pre-assessment phase, where they appeared passive and refrained from engaging in puzzle assembly without guidance. However, during the post-assessment phase, both subjects demonstrated the ability to complete the activity independently, without requiring assistance from parents or teachers.

For Subject I, the session took place on Tuesday at 09:00 AM. At the beginning of the activity, the subject remained silent and did not attempt to

assemble the puzzle, preferring to wait for guidance from either the parent or the teacher. After 30 minutes, at 09:30 AM, a second observation was conducted. This time, the subject displayed notable progress by independently engaging in puzzle assembly without any external assistance. This outcome suggests that the subject required a brief adjustment period before developing confidence and autonomy in the task.

Similarly, Subject II underwent the same process on Thursday. During the initial observation at 09:00 AM, the subject also showed reluctance to start the activity, remaining quiet and passive while waiting for instructions or encouragement. However, by 09:30 AM, the subject demonstrated the ability to work on the puzzle independently without any help from parents or teachers. This behavioral shift highlights the subject's growing confidence and willingness to engage once familiarized with the activity.

The findings indicate that although both subjects initially required time to adapt, they were capable of transitioning from dependence to independence within a relatively short period. This transition demonstrates that children can develop autonomy when provided with consistent opportunities to engage in structured learning activities. The results further suggest that structured activities such as puzzle games can serve as effective tools to promote problem-solving skills, self-reliance, and active engagement, particularly when facilitated within a supportive learning environment.

DISCUSSION

Based on the findings of this study on the effectiveness of puzzle games in enhancing fine motor development in preschool-aged children, it was concluded that puzzle games are highly effective because they provide substantial support for fine motor skill improvement. Puzzle games involve arranging disassembled picture pieces correctly by recognizing their shapes, colors, and sizes. These activities require children to use both their intelligence and instinct as they dismantle and reassemble pieces according to patterns or colors. Through this process, children learn to search for, rearrange, and rotate pieces to form a complete and meaningful image. Observations of Subjects I and II, as shown in Tables I and II, demonstrated notable changes in fine motor development between the first and second sessions after puzzle therapy was introduced. These changes were evident when the subjects completed the puzzles independently within an appropriate time-frame, without assistance from parents or teachers.

Supporting these findings, previous studies have also highlighted the effectiveness of puzzle games in promoting fine motor development. According to Lilis Maghfuroh (2018), most preschool children (59%) demonstrated normal fine motor development before receiving puzzle play therapy. However, after receiving the therapy, this figure increased significantly to 88.4%. Similarly, research by Fransiska (2012) compared two groups: a control group without puzzle therapy and an experimental group with puzzle therapy. The results showed that 52% of children in the control group exhibited normal fine motor development, while 88% of children in the experimental group achieved normal development, indicating a significant positive impact of puzzle games.

In contrast, Mila's (2012) research explored the use of Play-Doh as an alternative play therapy. Her findings revealed that in the first cycle, before Play-Doh therapy was applied, 59% of children demonstrated good fine motor development, 35% had adequate development, and 7% showed poor development. After the second cycle of Play-Doh therapy, the proportion of children with good development increased to 83%, while 17% remained in the adequate category. Additionally, Haryani (2012) conducted a study using a one-group pretest and post-test design, concluding that ball play therapy significantly improved gross motor development, with a p-value of 0.041.

From these studies, it can be concluded that both puzzle games and Play-Doh activities are effective in enhancing fine motor development in children. While puzzle games primarily train children to improve hand-eye coordination and problem-solving skills through pattern recognition, Play-Doh encourages creativity and muscle strength in the fingers and hands. These findings emphasize the importance of incorporating various play-based therapies in early childhood education to foster holistic motor development.

CONCLUSION

Based on the results of the study, it can be concluded that puzzle games are effective in enhancing fine motor development in preschool-aged children. After conducting two sessions regularly, both subjects demonstrated noticeable improvements in their ability to complete puzzle activities independently without guidance from parents or teachers. This indicates that puzzle play not only engages children cognitively but also supports the refinement of their fine motor skills, which are essential for early learning and daily tasks.

Furthermore, the observations revealed significant differences in the fine motor abilities of both subjects before and after the intervention. Initially, the children exhibited hesitation and required assistance to begin the activity. However, after the implementation of puzzle play therapy, they showed increased confidence and independence in performing the tasks. These findings highlight the role of puzzle games not only in improving fine motor development but also in supporting gross motor skills by promoting hand-eye coordination, spatial awareness, and controlled movements. Consequently, puzzle games can serve as an effective educational tool to foster overall motor development in young children.

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LEARNING MADE EASY: A GUIDED REINVENTION APPROACH FOR STUDENTS

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ABSTRACT

This study aims to explore the implementation of the Guided Reinvention approach in mathematics learning at the elementary school level and its impact on students' conceptual understanding. This approach is grounded in the principle that students can "reinvent" mathematical concepts through contextual experiences and guided teacher support. Using a qualitative case study method, the research was conducted in a fourth-grade classroom of a public elementary school that has adopted the *Merdeka Curriculum*. Data were collected through participatory observation, in-depth interviews, and documentation over five mathematics lessons on fractions. The findings reveal that Guided Reinvention fosters students' active engagement in the learning process, enhances conceptual understanding through concrete, visual, and symbolic representations, and transforms students' perceptions of mathematics into a more positive and enjoyable experience. Rather than merely memorizing procedures, students construct meaning through experience and social interaction. The teacher acts as a facilitator by employing open-ended questions, contextual problems, and reflective activities to guide students' thinking. Moreover, the classroom atmosphere becomes more collaborative and supportive. The study concludes that Guided Reinvention is an effective and humanistic approach to mathematics learning, aligned with the direction of the *Merdeka Curriculum*. The implications highlight the need for teacher training, the development of contextual teaching materials, and the implementation of student-centered learning to create meaningful, enjoyable, and empowering learning experiences. This approach provides new hope that mathematics can become a subject that is both comprehensible and loved by children from an early age.

Keywords: *Guided Reinvention, mathematics, elementary school, constructivism, meaningful learning*

INTRODUCTION

Mathematics is one of the core subjects that plays a crucial role in shaping students' reasoning, logic, and critical thinking skills from the early stages of education (Méndez-Parra et al., 2022). Nevertheless, it cannot be denied that mathematics is still perceived as a daunting subject by many elementary school students. This phenomenon occurs not only in remote areas but also in educational centres with adequate facilities. Many students experience anxiety, fear of failure, or even refusal to actively participate in mathematics learning (Yuliana et al., 2022).

High levels of mathematics anxiety and low motivation have become persistent issues that reemerge year after year. Empirical data from various national

and international educational surveys, such as TIMSS (*Trends in International Mathematics and Science Study*) and PISA (*Programme for International Student Assessment*), show that Indonesian students' mathematics achievement remains below the international average (Karjiyati et al., 2022; Sri Hartatik, 2020). For instance, PISA results indicate that Indonesian students have very weak problem-solving skills in mathematics, particularly in real-world contexts. This highlights a gap between classroom instruction and the demands of 21st-century competencies, which require understanding rather than rote memorization (Stoeffler et al., 2020; Amalina & Vidákovich, 2023). Such evidence points to the failure of traditional teaching approaches in developing conceptual understanding and practical applications of mathematics at the elementary school level (Krishan & Al-Rsa'I, 2023).

One of the key causes of the low quality of mathematics learning in elementary schools is the continued reliance on teacher-centered pedagogy, which emphasizes mechanical problem-solving (Fitriani et al., 2022; Laurens et al., 2018). Teachers often act as knowledge transmitters, while students remain passive recipients. This learning pattern undermines students' critical and exploratory thinking, leading them to view mathematics merely as a collection of formulas to be memorized rather than a meaningful tool for reasoning (Amalina & Vidákovich, 2023). In fact, mathematics is essentially a human construct that has developed historically and contextually, a dimension rarely highlighted in actual classroom practices.

In this context, the Guided Reinvention approach becomes highly relevant and significant to examine and implement. Guided Reinvention is a pedagogical approach in mathematics education that positions students as active discoverers of mathematical concepts under systematic teacher guidance (Prahmana, 2022). Rather than receiving concepts directly from the teacher, students reconstruct them gradually through contextual problem-solving that mirrors the historical development of the concepts themselves. Thus, Guided Reinvention emphasizes constructive, participatory, and experience-based learning.

However, this approach has not yet been widely adopted in mathematics classrooms at the elementary level. The gap between classroom realities and the potential of Guided Reinvention is evident. On one hand, students need meaningful learning experiences that stimulate curiosity. On the other hand, many teachers lack sufficient understanding or training to design reinvention-oriented lessons. Research in mathematics education has also tended to focus more on learning outcomes rather than on students' thinking processes and active role in constructing meaning.

From a theoretical perspective, the dominance of behaviorist approaches in elementary mathematics teaching has created a significant theoretical gap (Shahbana et al., 2020). Behaviorism, with its emphasis on stimulus-response, does little to support exploration of ideas or student creativity (Asfar et al., 2019). Conversely, constructivist approaches to the foundation of Guided Reinvention have not yet fully permeated classroom practices. Modern learning theories such as Vygotsky's social constructivism, Piaget's cognitive development theory, and Freudenthal's *Realistic Mathematics Education (RME)* strongly support the

principles of Guided Reinvention (Smagorinsky, 2018; Wardani et al., 2023). Yet in practice, teachers remain trapped in outdated paradigms that stress memorization and mechanical procedures.

From a research standpoint, there is still limited exploration of the effectiveness of Guided Reinvention in elementary mathematics education in Indonesia. Most studies have focused on secondary or higher education, and much of the research has been conducted abroad. Existing studies in Indonesia are mostly descriptive, lacking in-depth analysis of students' thought processes and failing to address the design of systematic learning models. This indicates a research gap that must be bridged through the development of Guided Reinvention-based instructional designs suited to the characteristics of Indonesian elementary school students.

The novelty of this study lies in its effort to formulate and implement Guided Reinvention specifically for Indonesian elementary schools. This approach not only draws upon contemporary learning theories but also adapts them to local learning cultures and students' social contexts. Moreover, this study seeks to bridge theory and practice through structured, student-centered instructional designs that facilitate students in "Reinventing" fundamental mathematical concepts in meaningful and enjoyable ways. By focusing on students' thinking processes, learning shifts from a passive, monotonous routine to a dynamic and motivating experience.

Specifically, this research and development of Guided Reinvention in elementary mathematics aims to: (1) identify the challenges faced by students and teachers in learning mathematics at the elementary level; (2) design a Guided Reinvention-based instructional model tailored to the characteristics of elementary school students; (3) implement the model in real classroom contexts to observe its effectiveness and students' responses; and (4) evaluate the outcomes to provide feedback for continuous improvement.

Thus, Guided Reinvention is not only an alternative solution to rigid, procedural mathematics instruction but also a concrete effort to develop a generation of students capable of thinking critically, creatively, and reflectively from an early age. Through guided rediscovery of mathematical concepts, students gain not only knowledge of *what* and *how* in mathematics but also an understanding of *why* concepts exist and how they apply in real life. This aligns with the spirit of the *Merdeka Curriculum*, which emphasizes contextual, meaningful, and liberating learning.

Furthermore, the implementation of this approach is expected to create a more humanistic, inclusive, and participatory mathematics learning ecosystem. Teachers are no longer mere instructors but facilitators who help students uncover meaning. Students are not passive participants but active subjects in constructing knowledge. Within such an ecosystem, mathematics learning ceases to be a burden and instead becomes a growth space for thought, imagination, and problem-solving.

In short, through the study and implementation of Guided Reinvention, this article proposes a new paradigm in elementary mathematics education—one that values process over outcomes, understanding over memorization, and exploration over one-way instruction. In this way, "*Mathematics Made Easy*" becomes not merely a slogan but a reality built through appropriate approaches, contextual

instructional designs, and the courage to innovate in basic education. Through this step, we are not only teaching mathematics to children but also nurturing mathematical thinking that will shape them into problem-solvers of the future.

METHOD

This study employed a qualitative approach with a case study method. The qualitative approach was chosen because the research aimed to gain an in-depth understanding of the implementation of the Guided Reinvention approach in elementary mathematics learning, as well as its influence on students' conceptual understanding and participation. An intrinsic case study design was applied, as the researcher sought to explore a specific educational phenomenon—namely the application of Guided Reinvention—within a real and contextual setting at a selected elementary school.

The research was conducted at a public elementary school that had progressively adopted thematic learning and the *Merdeka Curriculum*. The location was selected purposively, based on the availability of teachers who were open to innovation and the presence of heterogeneous student backgrounds. The participants consisted of fourth-grade students as the learners engaged in mathematics through Guided Reinvention, the classroom teacher as the main facilitator of the reinvention process, and the researcher who acted as a moderate participant observer in classroom activities.

The focus of the research was directed toward four main aspects: (1) the process of implementing Guided Reinvention in elementary mathematics learning; (2) the ways in which students constructed meaning of mathematical concepts through guided reinvention facilitated by the teacher; (3) students' responses to the strategies and activities employed in the classroom; and (4) the challenges encountered by teachers and the adaptive strategies they employed in facilitating Guided Reinvention.

To address these research focuses, data were collected through participatory observation, in-depth interviews, and documentation. Observations were conducted by recording teacher–student and student–student interactions, spontaneous student responses when facing contextual mathematical problems, and the strategies teachers used to guide the reinvention process. Observation sheets were designed based on indicators of mathematical thinking, affective engagement, and social interaction. Semi-structured interviews were conducted with teachers, students, and several parents to obtain a comprehensive picture of teaching strategies, learning experiences, and perceptions of Guided Reinvention. Documentation included students' worksheets, teachers' reflective notes, photographs and video recordings of classroom activities, and students' learning products such as discussion outputs, drafts, or concrete models.

Data were analyzed interactively and iteratively following the framework of Miles and Huberman, which includes data reduction, data display, and conclusion drawing. During data reduction, findings from observations, interviews, and documentation were selected and categorized according to the research focus. Relevant data extracts—such as student statements, teacher strategies, or significant interactions—were identified for deeper analysis. Reduced data were then presented

in descriptive narratives, matrices, and visual models to highlight patterns of student engagement and their development of mathematical thinking. Conclusions were drawn through interpretive analysis, with validity ensured through source triangulation, member checking, and peer debriefing with experts or fellow researchers.

To ensure trustworthiness, the study adopted Lincoln and Guba's four criteria. Credibility was achieved through triangulation and member checking, transferability was supported by providing thick contextual descriptions to allow replication in similar contexts, dependability was maintained through a transparent audit trail of the research process, and confirmability was ensured by linking data and interpretations back to their original sources.

In qualitative research, the researcher functions as the primary instrument. The researcher's presence in the classroom allowed direct interaction with participants throughout the learning activities. Nevertheless, a critical distance and objectivity were maintained in interpreting data. The researcher acted as an active observer who occasionally provided technical support when requested by the teacher. Through this methodological design, the study offers a comprehensive account of the dynamics of implementing Guided Reinvention in elementary mathematics classrooms. By emphasizing process rather than outcomes alone, the study provides richer insights into how students develop understanding and mathematical thinking skills through contextual, meaningful, and guided learning experiences.

RESULT

This study aims to examine in depth the implementation of the Guided Reinvention approach in mathematics learning in Grade IV of elementary school. Using a qualitative case study method, the researcher explores the processes, responses, and changes experienced by students, as well as the strategies employed by the teacher in guiding students to reinvent mathematical concepts. Data were collected through participatory observation, in-depth interviews, and documentation across five learning sessions on the topic "*fractions as parts of a whole.*"

Learning Dynamics in the Guided Reinvention Approach

Observations revealed that mathematics learning with the Guided Reinvention approach took place in a more open, dialogic, and collaborative atmosphere compared to conventional teaching. Each session began with a contextual problem, such as dividing fruits, bread, or other concrete objects fairly. These problems were designed to naturally trigger students' exploration of fraction concepts.

Students actively responded through group discussions, drawing sketches, folding paper, and manipulating concrete objects to represent fractions. The learning process progressed gradually from observing real-world phenomena to abstracting mathematical concepts. The teacher provided only minimal direction and mostly posed open-ended questions that stimulated students' reasoning.

For example, in one session, the teacher presented a situation where four cakes had to be divided among eight children. Students were asked to find a fair strategy for division and explain their results. Some discovered that *one-half of one-half equals one-fourth* and demonstrated this by folding a square sheet of paper into four equal parts. This process showed that students were not only learning procedures but also constructing the meaning of fractions through concrete experiences.

Changes in Students' Roles and Attitudes

An important finding was the shift in students' attitudes toward mathematics learning. Initially, students showed dependence on the teacher and fear of making mistakes. However, after several sessions with the Guided Reinvention approach, they began to display courage in trying, discussing, and even questioning mathematical ideas.

Previously quiet students, such as R and L, started actively sharing their opinions within their groups. They also began to correct each other constructively. Learning became a positive social experience that strengthened mathematical communication skills. In interviews, students expressed that they felt *"happier learning mathematics because they could find the answers themselves, not just be told."*

Teacher's Strategies as a Facilitator of Reinvention

The teacher played a key role in the success of this approach. Interviews showed that the teacher experienced a role shift from being a "Concept Transmitter" to a "Facilitator of Thinking". The teacher consciously designed contextual problems relevant to students' experiences, arranged heterogeneous learning groups, and prepared manipulatives as learning aids. The teacher skillfully used guiding questions such as:

1. "Can you divide this into equal parts without using a ruler?"
2. "Can you explain to your friend why this part is called one-fourth?"

Rather than rushing to provide correct answers, the teacher encouraged students to prove their ideas through concrete actions such as folding paper, drawing representations, or demonstrating their reasoning in front of peers.

In reflection, the teacher stated, *"I realized that children have their own ways of thinking. We just need to spark and guide them, not constantly tell them what to do or think."*

Evidence of Students' Understanding Through Learning Products

Documentation of students' work—such as drawings, fraction sketches, and written notes—showed that they could represent fractions in multiple ways. They demonstrated understanding that fractions are parts of a whole, not just numbers to memorize. Students used varied representations, including visual models, concrete objects, and contextual stories. In one activity, students were asked to create a story about "Pizza Sharing." One student wrote:

"My three friends and I shared 2 pizzas. Each of us got half of a pizza, then divided it again into two parts. So I got $1/4$ and $1/4 = 2/4$, which is the same as $1/2$."

This illustrates students' ability to combine verbal, visual, and numerical representations. Students used these representations together to explain their understanding. This shows meaningful learning.

Challenges in Implementation and Coping Strategies

Despite many positive outcomes, the study also identified several challenges:

1. Time requirements: The reinvention process demands more time for exploration. Teachers must manage time effectively while still allowing natural student discovery.
2. Discussion management skills: Not all teachers are accustomed to open-ended discussions and reflective questioning, requiring specific training.
3. Varied student abilities: Students with lower understanding need extra support to keep up in group activities.

The teacher addressed these challenges through several strategies. These included organizing heterogeneous groups, using concrete learning media, and providing additional support during activities. Joint reflections were also conducted at the end of each session to strengthen student understanding.

Emotional and Social Effects on Students

The Guided Reinvention approach also had positive emotional impacts. Students felt more confident, heard, and able to learn without pressure. The classroom atmosphere became lighter and filled with laughter, yet remained focused on learning goals. Social interaction improved as many activities were carried out collaboratively. One student, K, shared:

"Learning like this feels more enjoyable. We can work with friends, try things out, and not be afraid of being wrong."

Conclusion of the Case Study

The findings from this case study highlight that the Guided Reinvention approach in mathematics learning in Grade IV of elementary school:

1. Transforms the learning process from mechanistic to dialogic and constructive.
2. Enables students to build mathematical concepts from concrete, relevant experiences.
3. Enhances students' courage and active engagement in mathematical reasoning.
4. Encourages teachers to transform into facilitators of exploration and reflection.
5. Positively influences students' attitudes, confidence, and social interaction.

Overall, this case study shows that Guided Reinvention is not merely a teaching strategy. Rather, it is a pedagogical approach that humanizes the learning process. It makes mathematics logical, tangible, and meaningful in children's lives.

DISCUSSION

The findings of this study provide several important insights into the implementation of the Guided Reinvention approach in elementary mathematics learning. First, the results demonstrate that mathematics learning can be transformed from a teacher-centered and mechanistic process into an interactive,

dialogic, and constructive activity. This transformation is consistent with the principles of Realistic Mathematics Education (RME), which emphasize the importance of contextual problems and student reinvention of concepts rather than direct transmission (Son et al., 2020; Chong et al., 2019). By engaging with real-life contexts such as food sharing, students were able to construct fraction concepts through concrete experiences, thereby strengthening their conceptual understanding.

Second, the study highlights the positive shift in students' roles and attitudes. Initially, students exhibited dependence on the teacher and fear of making mistakes. However, through the reinvention process, they gradually developed confidence, courage, and active participation in discussions. This aligns with constructivist perspectives, which argue that meaningful learning occurs when learners are given opportunities to explore, test ideas, and negotiate meaning collaboratively (Silalahi, 2019; Lev et al., 2024). The observed social interactions, peer explanations, and constructive corrections illustrate how mathematical learning can also foster communication skills and collaborative competencies.

Third, the role of the teacher in Guided Reinvention is shown to be crucial. Rather than acting as a transmitter of knowledge, the teacher functioned as a facilitator who guided students' thinking through carefully crafted contextual problems and probing questions. Such a shift in teacher roles reflects a deeper pedagogical transformation, where teachers are required to master facilitation skills, inquiry techniques, and reflective questioning strategies. This finding resonates with previous studies emphasizing that effective mathematics teachers are those who can scaffold learning without diminishing student autonomy (Dominguez & Svihla, 2023; Radford et al., 2015).

Furthermore, students' written products and representations provide evidence of deeper conceptual development. The ability to express fractions in multiple forms, concrete, visual, verbal, and symbolic, demonstrates flexibility in mathematical thinking, which is a hallmark of genuine understanding. This supports the argument that learning environments encouraging representation and reflection are more likely to produce long-term conceptual retention.

Nevertheless, the study also revealed several challenges, particularly in terms of time management, teacher readiness, and diverse student abilities. These challenges highlight the necessity for systemic support, including professional development for teachers and curriculum flexibility that accommodates exploratory learning. Without adequate training, teachers may find it difficult to manage open-ended discussions or to balance guided discovery with curriculum demands.

Finally, beyond cognitive outcomes, the Guided Reinvention approach also showed emotional and social benefits. Students expressed enjoyment, reduced anxiety, and increased confidence in learning mathematics. These findings indicate that the approach not only addresses intellectual development but also contributes to affective and social dimensions of learning. This is particularly important in the elementary context, where students' early experiences with mathematics can shape their long-term attitudes toward the subject.

Taken together, these findings underscore the potential of Guided Reinvention as a pedagogical model that humanizes mathematics learning. By

positioning students as active discoverers and the teacher as a facilitator, mathematics becomes more meaningful, enjoyable, and empowering. Future research may focus on scaling this approach to different mathematical topics, examining its long-term effects on learning outcomes, and exploring strategies to overcome the practical challenges identified in this study.

CONCLUSION

This study concludes that the implementation of the Guided Reinvention approach in elementary mathematics learning effectively supports students' conceptual understanding, active engagement, and positive attitudes toward mathematics. By allowing students to "reinvent" mathematical ideas through contextual experiences and teacher-guided exploration, learning becomes more meaningful and less procedural. The findings highlight that Guided Reinvention not only strengthens students' ability to represent concepts in concrete, visual, and symbolic forms but also fosters collaboration, communication, and reflective thinking within the classroom. The role of the teacher as a facilitator, who skillfully guides the process through open-ended questions, contextual problems, and reflective discussions, proves to be essential in enabling students to construct their own knowledge. Furthermore, the approach helps to reduce students' anxiety toward mathematics and transform it into an enjoyable and empowering learning experience. In relation to the research objective, it can be affirmed that Guided Reinvention is an effective and humanistic pedagogical model aligned with the vision of the Merdeka Curriculum. Future studies may explore its application to other mathematical topics, assess its long-term impact on learning outcomes, and identify strategies to support teachers in adopting this approach more broadly.

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EMBEDDING LOCAL CULTURAL VALUES THROUGH SCHOOL CULTURE: A CASE STUDY AT SMP NEGERI 3 MAKMUR, BIREUEN REGENCY

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ABSTRACT

This study aims to explore the local wisdom values developed at UPTD SMP Negeri 3 Makmur, Bireuen Regency. The research employed a qualitative design with a descriptive approach. The data sources consisted of primary data, obtained directly from informants in the field, and secondary data as supporting information derived from school documents. Data were collected through observation, interviews, and documentation. The research informants included the principal, teachers, and students. Data analysis was carried out through data reduction, data display, and conclusion drawing/verification. The findings indicate that several forms of local wisdom are integrated into the school culture of UPTD SMP Negeri 3 Makmur, namely *khanduri maulid* (communal feast commemorating the Prophet's birthday) and traditional Acehese dances such as *tari Ranup Lampuan*, *tari Ratoh Jaroe*, and *tari Laweut*.

Keywords: *integration; local cultural; values; school culture*

INTRODUCTION

The cultural diversity of Indonesia, spread across the entire archipelago, represents a national treasure that must be preserved and sustained. Each region possesses its own distinct characteristics in practicing local wisdom values, thereby reflecting the identity of the local community. According to Gerda (2024), cultural identity is a set of unique values, traditions, languages, and worldviews that characterize a particular group or society. The dimensions of cultural identity include both visible elements, such as language and clothing, as well as intangible aspects such as beliefs and values (Mohyeddin, 2024).

Social changes occurring in society as a result of globalization also affect cultural aspects. The influx of foreign cultures inevitably influences indigenous traditions, making it necessary to undertake efforts to ensure that local culture is not lost amid the dynamics of modernization. Research conducted by Chukwuemeka et al. (2024) explains that language and value systems have become increasingly distorted by globalization. Similarly, Taptiani et al. (2024) argue that globalization has brought significant transformations to local cultural traditions; therefore, the role of the younger generation is crucial in preserving culture.

Schools, as institutions for transmitting cultural values, play a crucial role in introducing local culture to students through various programs developed within the educational setting. Local culture represents the uniqueness of a particular region shaped within a specific geographical context (Sun, 2025). According to Farhaeni and Martini (2023), the inclusion of cultural values in education is

essential for preserving Indonesia's local cultural heritage. One way of fostering local cultural values in schools is through the cultivation of school culture. As Kend.D and Deal (1999) argue, schools possess their own culture consisting of a set of rituals, habits, and morals that shape the behavior of the school community. School culture functions in forming norms, beliefs, the personal development of teachers and students, as well as the overall vision and objectives of the school (Liu, 2023). Furthermore, it reflects the entirety of school life (Iyabode, 2021).

School culture not only reflects the overall activities of the school community but also has a positive impact on student character formation, school quality, student achievement, and institutional development. Bayar and Karaduman (2021) explained that school culture positively influences student achievement. Similarly, Družinec (2019) found that school culture affects school quality and effectiveness. Furthermore, Y. Wang (2023) highlighted the positive correlation between school culture and student motivation. In addition, Hermansyah et al. (2023) emphasized that school programs can foster the development of good student character. In summary, school culture plays a significant role in school development and students' academic achievements. Therefore, schools must be able to cultivate a positive school culture to generate meaningful impacts on institutional growth.

The integration of local cultural values in schools represents an effort to instill cultural values in students in order to foster good character and to contribute to the preservation of cultural heritage. Previous studies (Fitrah et al., 2020; Alawiyah et al., 2023; Fauzah et al., 2023) indicate that the development of local culture in schools can shape students' character. Meanwhile, Litaay et al. (2025) explain that by integrating cultural values into the curriculum, schools can serve as spaces where students learn to live together harmoniously, even amidst diversity.

SMP Negeri 3 Makmur is one of the schools in Bireuen Regency that has incorporated elements of local culture into its school programs. It is well known that the province of Aceh possesses a wide variety of cultural traditions that continue to be preserved by the local community. These are reflected in its arts, language, culinary practices, customs, and other cultural expressions.

METHOD

This study employed a qualitative research design using a descriptive approach. Qualitative research is an approach that begins with certain assumptions and uses a specific interpretive framework or theory that influences the way we view research problems, especially those related to the meanings of social problem contexts (Creswell, 2015). The data sources consisted of: (1) primary data, obtained directly from informants in the field through interviews; and (2) secondary data, serving as supporting information, derived from school documents such as activity photos, school programs, and video recordings. Data were collected through: (1) direct observation to examine phenomena occurring in the field; (2) interviews with informants, for which the researcher had prepared an interview guide containing the list of research questions; and (3) documentation of school activities. The research informants included the principal, teachers, and students. Data were analyzed

through data reduction, data display, and conclusion drawing/verification. To ensure the validity of the data, source triangulation techniques were applied.

RESULT AND DISCUSSION

Based on the findings of this study, it was identified that several forms of local culture have been developed at UPTD SMP Negeri 3 Makmur, Bireuen Regency, namely:

1. *Khanduri Maulid*

Khanduri Maulid is one of the traditions practiced by the Acehese community to commemorate the birthday of the Prophet Muhammad (peace be upon him). The celebration of the Prophet's birthday reflects the implementation of Islamic law, customs, culture, and social practices within Acehese society (Ishak et al., 2022). The commemoration of the Prophet's birth represents an integration between Islam and local culture, observed continuously for four months—Rabiul Awal, Rabi'ul Tsani, Jumadil Awal, and Jumadil Tsani (M. Idris, 2022). Furthermore, Hanafiah (2023) emphasizes that the *maulid* celebration embodies several values, namely socio-religious values, communal festivities, kinship and social bonding, unity, and social harmony. The highest achievement of *maulidurrasul* for the wider community is the collective reflection on the life and history of the Prophet.

The implementation of the *maulid* tradition in schools significantly contributes to character development among students, as it embodies numerous positive values. At UPTD SMP Negeri 3 Makmur, *Khanduri Maulid* is celebrated by involving the entire school community. The incorporation of such local cultural practices serves as a distinctive characteristic of the school. Research by Alawiyah et al. (2023) explains that the character values formed through the *Khanduri Maulid* tradition in schools include *ukhuwah islamiyah* (Islamic brotherhood), exemplary conduct, religious values, and social values.

2. Traditional Dances

Traditional dance is a cultural expression of each region that reflects the identity of a society, where every movement in the dance contains philosophical meaning. In addition to the movements, other aspects that make traditional dances unique are the costumes worn by the dancers and the musical instruments that accompany the performance. Aceh is a province that has many unique dances, which have already gained international recognition. Based on the information obtained, several forms of traditional dances have been developed at UPTD SMP Negeri 3 Makmur, namely *Tari Ranup Lampuan*, *Tari Ratoh Jaroe*, and *Tari Laweut*.

According to J. Wang (2024), dance not only develops children's artistic abilities but also serves as an effective way to instill values in them. The findings of Guo and Li (2025) indicate that learning cultural dances can strengthen students' sense of local cultural identity. Meanwhile, Ramadani (2023) explains that there is a significant correlation between character formation and traditional dance practice.

From these various perspectives, it can be concluded that traditional dances are able to shape students' character and strengthen their love for culture.

CONCLUSION

Based on the findings, it can be concluded that the integration of local cultural values developed in schools represents an effort to preserve cultural heritage while simultaneously shaping students' character. At UPTD SMP Negeri 3 Makmur, several forms of local culture have been developed and incorporated as part of school culture, namely the *Khanduri Maulid* tradition and traditional dances such as *Tari Ranup Lampuan*, *Tari Ratoh Jaroe*, and *Tari Laweut*. This study implies that schools should strengthen the role of cultural education programs as effective platforms for transmitting local wisdom to younger generations. Furthermore, teachers and policymakers are encouraged to design learning strategies and curricula that integrate cultural practices, thereby fostering students' identity, discipline, and appreciation for cultural diversity.

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IMPLEMENTATION OF THE P5 MERDEKA CURRICULUM IN GEOGRAPHY LEARNING AT PEUSANGAN STATE SENIOR HIGH SCHOOL

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ABSTRACT

The problem in this study is that the implementation of the Merdeka Curriculum in the learning process provides teachers with the freedom to choose various teaching methods that can be adapted to the needs and specific interests of their students (differentiated learning). However, due to the changes brought by the new curriculum, this policy also has weaknesses, namely that not all teachers understand differentiated learning because the curriculum is still relatively new. The approach used in this research is a qualitative approach with a descriptive type of research. Qualitative research employs a qualitative approach and presents comprehensive descriptions related to the problems studied. This research is planned to be carried out at SMA Negeri Peusangan during the even semester of the 2024/2025 academic year. The population and sample of the study consist of the Principal, 3 Geography subject teachers, and 10 students. The results of the study conclude that: 1) The implementation of P5 in the Merdeka Curriculum within geography lessons at SMA Negeri 1 Peusangan has been good. 2) The supporting factors for the implementation of the *Projek Penguatan Profil Pelajar Pancasila* (P5) in the Merdeka Curriculum include teacher competence, availability of facilities and infrastructure, student activeness, parental support, well-planned P5 programs, as well as monitoring and evaluation of P5 implementation. 3) There are several inhibiting factors for the optimal implementation of P5 in the Merdeka Curriculum for geography lessons, such as the need to improve the quality of human resources, particularly geography teachers.

Keywords: *Implementation, P5 Merdeka Curriculum, Geography Learning*

INTRODUCTION

Education is a conscious and well-planned effort to create a learning atmosphere and learning process that enables students to actively develop their potential to develop spiritual and religious strength, self-control, personality, intelligence, noble character, and skills needed by themselves, society, the nation, and the state (Law No. 20 of 2003 on the National Education System). Education is a process in human life that serves as a means to acquire knowledge, which will later be useful in supporting life in the future (Azka, 2022).

Educational objectives can be achieved if they are supported by a well-organized curriculum that is aligned with students' needs at school. However, in practice, many students still feel burdened by curriculum changes because they experience difficulties adapting to the new curriculum that has been implemented, namely the Merdeka Curriculum (Kemdikbudristek, 2022). The Merdeka Curriculum, as a form of improvement of the 2013 Curriculum, has certainly received various responses from teachers, students, and parents. While some

support it, many others complain that the curriculum change occurred too quickly to replace the 2013 Curriculum (Saputra & Hadi, 2022). To date, the concept of the Merdeka Curriculum has received diverse responses from various educational institutions that facilitate student learning at the elementary, secondary, and higher education levels (Abidah et al., 2020). In fact, the evaluation of its implementation is described in the Guidebook for the Development of the Pancasila Student Profile Strengthening Project. This section explains that the evaluation of the implementation of the Merdeka Curriculum, especially the strengthening of the Pancasila student profile, essentially must involve students (Kemdikbudristek, 2022).

Since the launch of the Merdeka Curriculum, educators have also faced the challenges of 21st-century education, which require competencies and qualifications of complex human resources capable of responding to educational challenges worldwide. Education will always move forward and be renewed; it is dynamic and able to adapt to keep pace with ongoing changes, along with the development of knowledge. The implementation of the Merdeka Curriculum in schools influences the field of education. Currently, education bears a significant responsibility to respond to the challenges of 21st-century education, in which students are required to develop competitive skills needed in this era. These skills focus on the development of 21st-century competencies, such as critical thinking, problem solving, communication skills, digital literacy, information and communication technology (ICT) literacy, information literacy, and media literacy (Yulianti & Nurvita, 2024).

In order to achieve 21st-century learning through the Merdeka Curriculum, teachers are given the freedom to choose various teaching tools that align with students' learning needs and interests in each school implementing the Merdeka Curriculum. The Pancasila Student Profile Strengthening Project (P5) is a program designed by the Ministry of Education, Culture, Research, and Technology (Kemdikbudristek) as part of the Merdeka Curriculum, with the aim of encouraging the achievement of the Pancasila Student Profile. This program is implemented using a new paradigm, namely project-based learning.

Theoretically and conceptually, project-based learning is supported by constructivist learning theory. Constructivist learning theory provides individuals with the freedom to learn or seek their needs with facilitation from others. This theory argues that individuals construct knowledge and meaning based on their own experiences. Vygotsky's social constructivist theory states that children's learning development is influenced by culture and interactions with both social and physical environments (Suyono, 2020). This theory provides opportunities for learners to actively construct knowledge by comparing new information with prior understanding, learning from real-life experiences in the field, and important part of learning, particularly within the social learning environment.

The Pancasila Student Profile Strengthening Project (P5) is a planned activity aimed at strengthening the character of the Pancasila Student Profile among Indonesian students. This project is integrated with intracurricular programs conducted in the classroom. P5 represents interdisciplinary learning within the Merdeka Curriculum, with the goal of observing and identifying solutions to

problems that occur in the surrounding environment. Through project-based learning, students are provided with many opportunities to learn in both formal conditions and more flexible learning structures, which helps make learning more fun and active. This is because students are directly involved with their surrounding environment as a means of strengthening various competencies contained in the Pancasila Student Profile. Therefore, the implementation of character values through the Pancasila Student Profile Strengthening Project becomes an important area of study.

Based on interviews with teachers at Peusangan 1 Senior High School, the implementation of the Merdeka Curriculum in the independent learning process gives teachers the freedom to choose various teaching methods that can be tailored to the specific needs and interests of their students (differentiated learning). However, due to the new curriculum, this policy also has a weakness in that not all teachers understand differentiated learning because the curriculum change is still new. Based on observations made regarding the implementation of the Merdeka Curriculum, the learning process is blocked by technical obstacles related to the creation of contextual lesson materials and the application of learning strategies or methods, especially by geography teachers, both due to the teachers' own skill factors and office and framework limitations. Not all students can master the use of the Merdeka Curriculum education plan to the fullest because not all teachers have been trained.

As a result, learning management, especially for productive learning, is far from ideal. Students are not fully engaged in learning activities because they are still accustomed to the old model, where students wait for material to be delivered by the teacher, so they just sit quietly. The implementation of the Merdeka Curriculum in the learning management system, especially for geography, is still far from ideal. This is partly because most teachers who teach this subject have not received the Merdeka Curriculum, resulting in a clear difference in perception among teachers.

Therefore, to ensure the success of the Merdeka Curriculum at the senior high school level, this study was conducted to examine the level of understanding and readiness of geography teachers at SMA Negeri 1 Peusangan in using the P5 Merdeka Curriculum in teaching. The results of this study can be used as information to improve teachers' abilities in using the Merdeka Curriculum, especially geography teachers. Similar studies related to the use of P5 in the Merdeka Curriculum in geography learning have been conducted by several researchers, including Wardani (2022). This study found that the mechanism of cooperation with external parties still encounters many obstacles, including incomplete evaluations, the process of adjusting the KKNi curriculum to the MBKM program, which will have an impact on students and lecturers, technical difficulties in implementing the MBKM program, and internship programs.

Furthermore, Sinomi (2022) found that one of the factors that hinders teachers in implementing the independent learning system is that they are comfortable with standards but still struggle with technology. These factors include the low quality of teacher human resources, lack of facilities and learning resources, inadequate infrastructure, age, and lack of previous independent learning

experience. The creativity of students is positively influenced by the use of differentiated learning and assessment as methods of implementing the Merdeka Curriculum pattern. The quality and significance, as students who use independent learning, acquire critical thinking skills, emotional sensitivity, talent, and vivid imagination (Hasnawati, 2021).

METHOD

The approach used in this study is a qualitative approach with a descriptive research type. Qualitative research is research that uses a qualitative approach and presents comprehensive descriptions related to the issues being studied. Specifically, descriptive research can be explained as a type of research that aims to present a complete picture of the social setting or research conducted to explore and clarify a social phenomenon or event, namely by describing the variables related to the problem or unit being studied in relation to the phenomenon being tested (Roosinda, 2021).

The first step when the researcher arrives at the research location is to act as an interviewer. Therefore, the researcher's direct presence in the field is a measure of success in understanding the case being studied. The research was conducted at SMA Negeri 1 Peusangan, Bireuen Regency. The data used in this study were interview guidelines. The data sources in this study were the principal, geography teachers, and students who were respondents to obtain accurate data. The data collection techniques in this study were interviews, observation, and documentation.

RESULTS

SMA Negeri 1 Peusangan, located on Jalan Banda Aceh Medan Matangglumpangdua, Blang Asan, Peusangan District, Bireuen Regency, Aceh Province, is a public senior high school that was established in 1981. With a land area of 14,105 square meters, this school has adequate facilities to support a quality learning and teaching process. State Senior High School 1 Peusangan has received an A accreditation based on Decision No. 842/BAN-SM/SK/2019, issued on October 7, 2019. This demonstrates the school's commitment to providing quality education and producing high-quality graduates.

SMA Negeri 1 Peusangan is also supported by internet access and electricity from the national power company (PLN) to support teaching and learning activities. As a public school, SMA Negeri 1 Peusangan operates under the authority of the Regional Government and is committed to producing young generations who are intelligent, have noble character, and are ready to face future challenges. The results of the research on the implementation of P5 in the Merdeka Curriculum were obtained from interviews with the school principal, Dialita, S.Pd., three teachers, and ten students. Based on the research conducted, the researcher obtained information regarding the involvement of the principal in implementing P5 within the Merdeka Curriculum at SMA Negeri 1 Peusangan. This research involved the use of several methods, including direct field observation to collect data, as well as interviews with various parties, such as the principal. Another method used was documentation. After collecting data through observation, interviews, and

documentation, the researcher then presented the findings regarding the role of the principal in the implementation of the Merdeka Curriculum at SMA Negeri 1 Peusangan.

The principal's main role in determining the success of learning in schools through effective instructional leadership is crucial. The principal must focus on the vision and mission that have been established at the school to maintain leadership capacity and provide exemplary practices in effective learning, so that the school can operate in accordance with the established directions and objectives. Especially in implementing the Merdeka Curriculum, the principal's role from planning to evaluation has a significant impact. Therefore, the principal employs various strategies to develop the Merdeka Curriculum.

The results of interviews with geography teachers at SMA Negeri 1 Peusangan—namely three teachers—regarding the implementation of P5 in the Merdeka Curriculum in geography learning were analyzed by describing the responses that had been answered and tabulated in the form of tables, then analyzed by describing the responses given by the teachers, namely Rani Yulidar, S.Pd., Lisfuaidah, S.Pd., and Nurlaili, S.Pd. From the interview results, it was found that the implementation of P5 in the Merdeka Curriculum in geography learning at SMA Negeri 1 Peusangan has been good. Teachers already understand what P5 is within the Merdeka Curriculum and are familiar with its implementation because they have participated in training. In addition, teachers also understand the important components that must be applied in P5 to enhance student creativity, especially in improving student learning outcomes.

The implementation of P5 in the Merdeka Curriculum in geography learning at SMA Negeri 1 Peusangan has been good, where teachers already understand what P5 is in the Merdeka Curriculum and are familiar with it because they have attended training. Furthermore, teachers are aware of the important components that must be implemented to enhance student creativity, particularly in improving student learning outcomes. In the Merdeka Curriculum, the Project for Strengthening the Profile of Pancasila Students (P5) places students as the main actors. The role of students in P5 includes several important aspects such as being actively involved in project planning, carrying out project activities, collaborating with peers and teachers, and evaluating and reflecting on their learning experiences. P5 is designed to develop students' potential holistically, not only in cognitive aspects, but also affective and psychomotor aspects, as well as to build the character of the Pancasila Student Profile.

DISCUSSION

Based on the results of interviews with the principal and geography teachers at SMA Negeri 1 Peusangan, the implementation of P5 within the Merdeka Learning Curriculum has been very good. However, its implementation needs to consider the learning situations and conditions of students, especially whether the school is located in an urban or rural area. Not all aspects can run smoothly, as there are many factors that may become obstacles, such as students' learning facilities. The implementation of learning is closely related to the role of teachers in classroom instruction, which determines whether learning objectives are achieved or not.

Learning implementation includes the organization of learning activities and the teacher's leadership in the classroom learning process. The organization of learning carried out by geography teachers includes the distribution of tasks to students regarding what must be done during the learning process and the objectives that will and must be achieved through the learning activities. In the learning process, the teacher as a leader plays a role in influencing or motivating students to carry out the expected tasks, so that the teaching process runs smoothly, students can easily and effectively master the subject matter, and learning objectives can be achieved.

Teachers must always strive to strengthen students' motivation to learn. This can be achieved through engaging lesson presentations and pleasant personal relationships, both in learning activities inside and outside the classroom. Based on the research findings, the implementation of the Project for Strengthening the Profile of Pancasila Students (P5) with the theme of a sustainable lifestyle through the utilization of used or recycled materials at SMA Negeri 1 Peusangan shows that students care about their surrounding environment, both at home and at school. In addition, students have become more independent, creative, innovative, and socially responsible.

This is consistent and similar to the research (Yulastuti, 2022), which shows that students also become more independent, creative, innovative, and magnanimous after the implementation of P5 activities. Additionally, research by Sulistyaningrum (2023) showed that the implementation of P5 activities fostered Pancasila values and improved student character. Students must be accustomed to taking the right actions and strive to apply the six character elements in the Pancasila Student Profile, Pancasila Student Profile both at school and at home. The research results are similar to the Implementation can become graduates with the Pancasila Student Profile (Jupriyanto, 2023).

The Pancasila Student Profile Strengthening Project (P5) activity in the Merdeka Curriculum at SMA Negeri 1 Peusangan shows that the implementation of P5 activities also fosters Pancasila values with the dimensions of piety, faith, and morality towards nature, which makes students more protective of the natural environment. And the creative dimension enables students to produce innovative works of economic value from used items around them, thereby developing students' character for the better.

According to Khirurrijal et al (2022), its implementation differs from curriculum design and planning. The Merdeka curriculum requires complex cooperation between teachers and students. Furthermore, in the field of education, the implementation of curriculum development itself has become an urgent matter. Curriculum development is carried out with the aim of simplifying the existing curriculum. The Merdeka Curriculum is implemented to achieve learning outcomes without emphasizing reading and writing, thus not burdening students. However, changes are observed in natural and social sciences.

Based on the results of interviews conducted by researchers at SMA Negeri 1 Peusangan about the P5 independent learning curriculum from geography teachers who have been interviewed by researchers, there are those who have different views in responding to the independent learning curriculum, some of whom are positive. Geography teachers at SMA Negeri 1 Prusangan have different

understandings of the P5 independent learning curriculum, as stated by a geography teacher who said that the P5 independent learning curriculum should be accepted and not questioned. This is because the independent learning curriculum is an idea from the government, and the government has certainly considered its positive and negative impacts. So, if it has been instructed, then it must be implemented because it has become a decision, and everyone must think positively. Therefore, it must be implemented because it has already been decided, and everyone must be optimistic, because it has been carefully considered by many people and experts.

The results of interviews with students regarding the implementation of P5 (Pancasila Student Profile Strengthening Project) in Geography learning at SMA Negeri 1 Peusangan show that students feel enthusiastic and gain many benefits. They consider P5 to be a fun activity that is relevant to their daily lives and helps them develop various skills. Students generally welcome the application of the P5 Merdeka Curriculum in Geography learning. They feel more interested and motivated to learn because the Merdeka Curriculum P5 provides a learning experience that is different from conventional methods. Overall, the interview results show that the Merdeka Curriculum P5 has a positive impact on Geography learning in high school. Students feel more enthusiastic and motivated, and gain a more meaningful learning experience through P5. Although there are some challenges, the benefits felt by students are far greater.

The term “*Merdeka Belajar*” consists of two words, namely ‘*merdeka*’ and ‘*belajar*.’ According to the Big Indonesian Dictionary (KBBD), the word “*merdeka*” has several meanings: (1) free (from slavery, colonialism, etc.), independent; (2) not affected by or free from demands; (3) not bound, not dependent on a particular person or party, free. Thus, the implementation of *merdeka belajar* implies freedom and independence in the learning process, where students have the space to take initiative and actively participate in building their knowledge. The implementation of the Merdeka Curriculum at SMA Negeri 1 Peusangan demonstrates efforts to apply these guidelines in everyday educational practices.

Furthermore, Sinomi (2022) found that one of the factors that hinders teachers in implementing the Merdeka learning system is that they are comfortable with standards but still struggle with technology. These factors include the low quality of teacher human resources, lack of facilities and learning resources, inadequate infrastructure, advanced age, and lack of previous experience with independent learning. Hasnawati (2021) found that student creativity is positively influenced by the use of differentiated learning and assessment as methods of implementing the Merdeka Curriculum. In terms of quality and significance, students who use independent learning acquire critical thinking skills, emotional sensitivity, talent, and vivid imagination.

The main problem faced by teachers in implementing the Merdeka Curriculum is their low level of competence and understanding in designing modules, as well as the suboptimal use of information technology in the learning process and the creation of interactive modules to create a generation of lifelong learners with a learner personality (Astuti et al., 2023). Learning in pioneering schools requires project activities aimed at strengthening the character of students with the Pancasila learner profile. These projects must be designed as effectively as

possible to provide an excellent experience for students, thus requiring reinforcement of the Pancasila learner profile project (Palupi Sri Wijayant et al., 2022). The following are the results of interviews with the project coordinator regarding the obstacles encountered in implementing the Pancasila student profile strengthening project.

Here's how constructivist learning theory supports the implementation of the Merdeka Curriculum:

1. Selection of learning materials and methods
The Merdeka Curriculum gives teachers the freedom to choose and arrange the sequence of learning materials based on the developmental phase of students. This is in line with the principle of constructivism, where learning is seen as an active process that involves the construction of knowledge by students themselves.
2. Differentiated learning
The concept of differentiation in the Merdeka Curriculum allows teachers to accommodate different learning styles among students. Constructivist theory emphasizes the importance of understanding individual diversity and providing relevant and meaningful learning experiences for each student.
3. P5 activities and contextual learning
P5 activities, which involve introducing students to entrepreneurship through field and contextual activities, enable students to build their own understanding of these concepts in a real-world context. This supports the constructivist approach to learning, where learning is more effective when students are actively engaged in meaningful situations.
4. Collaboration between teachers, students, and parents
Constructivism also emphasizes the importance of social interaction in learning. Strong collaboration between teachers, students, and parents is essential in learning. Supporting the learning process at Peusangan 1 Public High School creates an environment that supports better knowledge construction.

By applying constructivist learning theory in the implementation of the Merdeka Curriculum, schools can improve learning effectiveness by emphasizing students' active experiences in constructing knowledge. This approach supports deeper understanding and better achievement of learning objectives. In addition, it helps prepare students to become independent, reflective, and lifelong learners.

CONCLUSION

Based on the results of research related to the implementation of the Merdeka Curriculum at Peusangan 1 Public High School in geography lessons, it can be concluded that: The implementation of the P5 Merdeka Curriculum in geography lessons at SMA Negeri 1 Peusangan has been good, where teachers have used innovative learning such as project-based learning, teachers also help students develop soft skills, such as cooperation, communication and creativity.

The implementation of the P5 Merdeka Curriculum in geography lessons at Peusangan 1 Public High School has been good, where teachers have used

innovative learning methods such as project-based learning, and teachers have also helped students develop soft skills, such as cooperation, communication, and creativity.

There are several factors that hinder the optimal implementation of P5 in the Merdeka Curriculum in geography lessons, including the quality of human resources, particularly geography teachers, which needs to be improved, especially their ability to conduct learning in line with the times, such as the use of learning technology and increased creativity and innovation in learning activities. In addition, the lack of facilities and learning resources or inadequate infrastructure also poses a challenge in the implementation of P5 in the Merdeka Curriculum.

The following are the recommendations that the researcher will provide after conducting research at SMA Negeri 1 Peusangan regarding the implementation of the Pancasila student profile strengthening project:

1. The principal is expected to continue to guide and provide direction to educators on the implementation of the Pancasila student profile strengthening project, as well as to continue to improve the quality of learning at the school, such as the facilities and infrastructure to support project learning activities.
2. Teachers are expected to continue to improve their understanding and increase training related to the Pancasila student profile strengthening project, because in the Merdeka Curriculum, teachers are required to be more creative and innovative facilitators in implementing project-based learning so that it can develop better in the future.
3. Parents are expected to participate in supporting project activities carried out at school and to work together with teachers in monitoring students while they are at home so that the instilling of character values does not only occur at school.
4. For future researchers who will conduct research, it is recommended that the research focus more on the implementation of P5 from the perspective of the Pancasila Student Profile, which is global diversity, mutual cooperation, and critical thinking

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IMPROVING STUDENTS' CREATIVE MATHEMATICAL THINKING ABILITIES ON SPLDV MATERIAL USING *THE CREATIVE PROBLEM SOLVING* (CPS) METHOD USING *FLIPBOOK MEDIA* IN GRADE X OF SMAN 2 KUTACANE

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ABSTRACT

The Creative Problem Solving (CPS) learning model offers a solution to improve student learning outcomes in solving SPLDV problems by training students to think creatively, logically, and systematically, and increasing their confidence in articulating their findings through the Creative Problem Solving (CPS) learning process. In this learning model, researchers used flipbooks to facilitate students' problem-solving. Students solve SPLDV problems because applying this model trains them to think creatively, logically, and systematically, and increases their confidence in articulating their findings through the Creative Problem Solving (CPS) learning process. In this learning model, researchers used flipbooks to facilitate students' problem-solving.

Keywords: *Mathematical Creative Thinking Ability, SPLDV Material, Creative Problem Solving (CPS) Method, Flipbook*

INTRODUCTION

Mathematics has a vital role in shaping students' creative thinking skills. This ability is essential in today's era of globalization and information technology. One of the subjects in the independent curriculum, Mathematics, is a discipline with distinct characteristics compared with other disciplines. The mathematical material that requires creative thinking skills is the Two-Variable Linear Equation System (SPLDV). This material requires a good understanding of concepts and problem-solving skills.

According to the National Council of Instructors of Arithmetic (NCTM), as conveyed by Rahmani J (2018), states, "There are three basic skills in mathematics: *standard problem solving, reasoning and proof, and communication*. In this study, the researcher concluded that creative thinking skills, one of the basic standards of mathematics ability, correspond to the three NCTM ability standards. Creative thinking skills influence students' self-confidence in various problem situations; they help students learn facts, concepts, and mathematical principles and their applications; they help students identify the steps taken when solving problems; and they help determine the priority of strategic steps. Find answers to questions, not just find them in the answers themselves. To be effective creative thinkers, students need ample opportunities to create and solve problems in mathematics and in real-world contexts. A problem usually involves a situation that prompts a person to solve it.

According to (Fatur Rahman et al., 2020), The cause of students' low levels of mathematical creative thinking during mathematics lessons is a suboptimal teaching and learning process. Creative thinking is a higher-order thinking (HOT) skill. HOT is one of the independent curriculum goals that students must achieve. According to the definition (Febrianingsih, 2022), Creative thinking is the students' ability to generate novel, previously undiscovered solutions. Furthermore, many teachers fail to fully utilize the context provided by the real world, even though context can shape students' knowledge and skills through real-world experiences. Therefore, students have difficulty applying their mathematical knowledge in real life.

Based on the results of observations and interviews with the mathematics education teacher, Mrs. Rosmawati S.Pd on Thursday, October 5, 2023, which was conducted at SMA Negeri 2 Kutacane, stated that students' mathematical creative thinking abilities on SPLDV material were still relatively low, as seen from the teaching and learning process in class X which still used the lecture method and did not prioritize a good learning process. This is evident in the test results, in which only 40% of students solved SPLDV problems correctly. Because various kinds of teaching materials are used in mathematics education, teachers need to develop an effective and efficient learning system to help students better understand the lesson. One approach to creating an effective learning system is to use SPLDV-based learning media in conjunction with the Creative Problem-Solving method.

To address this issue, this study will implement the CPS method using flipbooks in SPLDV learning. CPS is a learning method that encourages students to think creatively when solving problems, whereas flipbooks are an interactive and engaging learning medium. According to Fatur Rahman et al. (2020) and Pepkin (2004) describes Creative Problem Solving as a model that trains students to think creatively. This model provides students with the freedom to be active participants in the problem-solving process. The syntax of the Creative Problem-Solving learning model includes problem clarification, idea expression, evaluation and selection, and implementation. It is hoped that students can be trained to reason, construct, and think creatively when solving problems.

Previously, this learning model was also investigated by Tarlina & Afriansyah (2016), who successfully applied it to the material on lines and angles. In this study, researchers aim to use the Creative Problem-Solving model to SPLDV material. The Creative Problem Solving (CPS) learning model provides solutions that can help improve student learning outcomes in solving SPLDV problems because the application of this model can train students to think creatively, logically, and systematically, and will increase their confidence in expressing what they have found through the Creative Problem Solving (CPS) learning process. In this learning model, researchers use flipbook media to facilitate students' solution of SPLDV problems.

As noted above, this study uses flipbooks as teaching materials as needed. A flipbook is a learning tool that systematically and engagingly presents materials, methods, limitations, and assessment options to achieve desired competencies. According to Khafidhoh et al. (2022), a flipbook is a technology-based learning medium used in the flipbook learning process. Flipbooks are interactive media in

Flip PDF format that support the insertion of videos, images, audio, links, and animations.

It is hoped that implementing the CPS method, supported by flipbook media, will improve students' mathematical creative thinking skills in SPLDV materials. In addition, this research is also expected to contribute to the Development of effective and enjoyable mathematics learning methods for students. How can students' mathematical creative thinking skills on SPLDV material be improved using the creative problem solving (CPS) method assisted by flipbook media? This research examined whether the use of the CPS method is efficacious in improving students' creative thinking skills for solving mathematical problems.

METHOD

The approach used in this research is quantitative, meaning that all data are collected and analyzed using quantitative methods (Murtinugraha et al., 2021). Stated quantitative research information refers to data that are present or expressed as numbers obtained from the field; it can also be called quantitative data, expressed as numbers obtained by converting qualitative values into quantitative ones. Next, the experimental class group received treatment using the CPS learning model, whereas the control group received conventional instruction typically used by teachers. After receiving the treatment, both groups were given a posttest to determine the students' final condition. Sugiyono (2019) as follows:

Description of Control Class Research Results

In this study, the data obtained by the researcher is the critical thinking ability of students in mathematics learning for class X of SMA Negeri 2 Kutacane with the application of the Creative Problem Solving (CPS) method assisted by Flipbook Media on SPLDV material, as well as the description of the results of the pre-test and post-test scores. Data ini digunakan untuk menguji efektivitas metode CPS yang didukung oleh Flipbook Media dalam meningkatkan kemampuan berpikir kritis siswa dalam menyelesaikan sistem persamaan linear dua variabel. The control class can be seen in Table 4.1 below:

Table 4.1 Recapitulation of Critical Thinking Skills Values in the Control Class

No	Initials	Pre-test Score	Post-test Score
1	AN	25	75
2	AA	31	56
3	AZ	38	88
4	AU	38	81
5	AD	31	75
6	BS	13	75
7	CN	25	63
8	CV	25	75
9	DW	31	75
10	HS	25	69
11	HK	31	75
12	MA	13	81

13	MR	25	69
14	MS	44	69
15	MZ	13	75
16	MD	38	63
17	NA	13	75
18	OS	25	56
19	RA	31	88
20	SM	38	81
21	ZR	31	63
22	ZA	25	75
Amount		609	1602
Average		27.68	72.82

Based on the 2024 research findings, the average pre-test score for students' critical thinking skills in the control class was 27.68, whereas the average post-test score increased to 72.82. The difference between the pre-test and post-test scores indicates an improvement in students' critical thinking skills in the control class. These data provide an overview of students' initial and final abilities, which serve as a reference for the subsequent analysis.

Description of Experimental Class Research Results

In this study, the data obtained by the researcher is the critical thinking ability of students in mathematics learning for class X of SMA Negeri 2 Kutacane with the application of the Creative Problem Solving (CPS) method assisted by Flipbook Media on SPLDV material, as well as the description of the results of the pre-test and post-test scores. The comparison between pre-test and post-test results provides empirical evidence of changes in students' critical thinking abilities after the implementation of the CPS-based learning intervention. The experimental class can be seen in Table 4.2 below:

Table 4.2 Recapitulation of Critical Thinking Skills Values in the Experimental Class

No	Initials	Pre-test Score	Post-test Score
1	AN	31	88
2	AA	44	88
3	AZ	38	88
4	AU	38	88
5	AD	31	75
6	BS	38	88
7	CN	31	88
8	CV	31	88
9	DW	38	94
10	HS	31	88
11	HK	44	81
12	MA	38	88
13	MR	50	88
14	MS	50	94

15	MZ	38	75
16	MD	25	88
17	NA	31	81
18	OS	38	81
19	RA	25	88
20	SM	25	75
21	ZR	38	94
22	ZA	25	81
Amount		609	775
Average		27.68	35.23

Referring to the 2024 research data presented in the Table above, the average pre-test score of the experimental class was 35.23, while the average post-test score increased to 85.5. The observed increase reflects a change in students' critical thinking performance. This information represents students' initial and final abilities in the experimental class.

Pre-test Score Results

Pre-test results were obtained for both the control and experimental classes. These results provide an overview of students' initial creative mathematical thinking abilities prior to the implementation of various learning treatments. The pre-test results for the control class are presented in Table 4.3 below:

Table 4.3 Pre-test Results for the Control Class

Creative Thinking Ability Indicators							
No	Initials	Smoothness	Authenticity	Elaboration	Flexibility	Score	
		Question 1	Question 2	Question 3	Question 4		
1	AN	0	2	2	0	4	25
2	AA	2	1	2	0	5	31
3	AZ	2	2	2	0	6	38
4	AU	0	2	2	2	6	38
5	AD	2	2	0	1	5	31

RESULT AND DISCUSSION

Definition of Mathematics

Mathematics is an important science in helping our daily lives. According to the (Gayatri, 2022) Establishment that mathematics is a fundamental science that is the foundation for the Development and progress of science, technology, and philosophy. Mathematics is foundational to many fields, and numerous studies demonstrate its utility in everyday life (Sidabutar, 2018). Therefore, everyone who knows this basic science, especially those with a mathematics education, should be able to understand students (Sidabutar, 2018).

FlipBook Media

Remembering that flipbooks are a form of presenting learning media in a virtual format (Cholifah & Muslihasari, 2022). According to Rusnilawati & Gustiana (2017), the use of flipbooks can help teachers present electronic teaching materials more effectively, as they can display images, videos, text, animations, and other content. According to Wibowo & Pratiwi (2018), flipbooks incorporate animations, videos, audio, and other elements, making them interactive learning media that are less monotonous and more engaging.

Previous research on the use of flipbooks in education has shown that they can increase student interest and influence learning outcomes. Students can more easily understand the material, thereby achieving higher learning outcomes (Hadayatullah & Rakhmawati, 2016). Flipbooks are designed to enable students not only to practice the steps presented in the media but also to read, view images, listen, and watch videos (Mafula et al., 2017). The advantages of this medium are that it supports files in PDF, image, video, and animation formats and includes features such as template designs, backgrounds, control buttons, navigation bars, hyperlinks, and background sound. This makes it more engaging. Flipbooks help students better master abstract material and events that cannot be presented in class (Andarini et al., 2013).

Understanding Mathematical Creative Thinking Ability

According to (Faturrohman et al., 2020) The ability to think creatively is one of the most important skills for students because it is a process that arises from innovative ideas derived from data and information analysis, and from gaining insight into an object through creative thinking. The ability to think creatively is a skill within the category of high-order thinking (HOT). HOT is one of the goals of the independent curriculum that students must achieve (Gais & Afriansyah, 2017). Creative thinking is a learning process in which teachers employ various methods and strategies, such as group work, role-playing, and problem-solving, to foster and develop students' creativity.

Based on the definitions of creativity provided by the experts above, it can be concluded that creativity is the discovery of new ideas and concepts that arise from combining existing ones. Creative thinking is an activity that aims to generate ideas and concepts, solve problems, identify patterns, and connect ideas to something meaningful. Think creatively, bring benefit to life, add knowledge, create new knowledge, and develop solutions to problems. Someone who always thinks creatively will have an impact on their personality in planning and deciding an action, and creative people are usually more organized in their actions. In education, creative thinking can help students deepen their understanding of the material by evaluating arguments from textbooks, journals, discussion partners, argumentative partners, and teachers during learning activities.

Understanding the Creative Problem Solving (CPS) Learning Model

According to (Ajeng Pramestika et al., 2020) The CPS Learning Model is a logical approach to learning that builds students' ways of thinking and enables them to develop and demonstrate creative, productive, critical, independent,

collaborative, and communicative thinking and acting skills. Through creative thinking, the full potential of the self can be realized, particularly through imaginative, deviant, and horizontal expressions of ideas (Syahrin et al., 2019). The CPS model provides students with opportunities to actively explore problems, generate ideas, and evaluate solutions. Furthermore, CPS encourages students to be more flexible and open-minded in the learning process, thereby supporting the Development of higher-order thinking skills.

Steps of the Creative Problem Solving (CPS) Learning Model

The implementation of the Creative Problem Solving (CPS) learning model in this study followed several structured stages as proposed by Aris (2014) and presented by Yulita (2022). The first stage involved problem clarification, in which the teacher guided students in understanding the problems presented and identifying the types of solutions required. During this stage, students were encouraged to analyze the problems carefully, which helped them focus on the core issues and supported the Development of their initial critical-thinking skills.

The next stage emphasized the expression of opinions, in which students were free to share ideas and propose various strategies for solving the problems. This stage fostered active participation and creative thinking, as students explored multiple perspectives without immediate judgment. Subsequently, in the evaluation and selection stage, students worked collaboratively in groups to discuss and assess the proposed strategies, selecting the most appropriate solution based on logical reasoning and feasibility. Finally, in the implementation stage, students applied the selected strategy to solve the problem systematically until a final solution was obtained. Through these stages, the CPS learning model encouraged students to think critically, collaborate effectively, and apply problem-solving strategies systematically.

Definition of SPLDV

Defining a linear equation (SPLDV) is a topic in mathematics that concerns solving two linear equations in two variables. Several methods can be used to solve a linear equation (SPLDV), including elimination, substitution, and graphical methods. (Septiani, et al., 2020.). This topic requires students to understand the relationships between variables and apply appropriate problem-solving strategies to obtain accurate solutions. Therefore, mastering SPLDV is essential for developing students' analytical and creative mathematical thinking skills.

CONCLUSION

Based on the results and discussion, it can be concluded that implementing *the Creative Problem Solving* (CPS) method, supported by Flipbook media, can improve students' mathematical creative thinking skills in class X at SMAN 2 Kuta. The findings show that this learning approach helps students develop their creativity while learning mathematics. The CPS method, when used alongside Flipbook media, can serve as an alternative instructional strategy to foster a more engaging, student-centred learning environment. These results can inform the Development of innovative mathematics learning strategies by teachers and researchers.

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ANALYZING FACTORS AFFECTING STUDENT PERFORMANCE IN 3D ANIMATION EDUCATION

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ABSTRACT

Digital technology is evolving rapidly and has transformed higher education, particularly in industry-focused vocational programs that prepare graduates for entry into the global creative industry. This research identifies the factors affecting student learning outcomes in 3D animation courses at Politeknik Negeri Lhokseumawe, from the technical, pedagogical, and psychological perspectives, as a gateway to achieving all phases of Sustainable Development Goal 4 (Quality Education). The research employed a quantitative descriptive method and was administered to 36 students using a structured Likert-scale questionnaire. Descriptive statistics and correlation analyses were also conducted. The technical constraints ($M = 2.72$, $SD = 1.41$), allocated practice time ($M = 3.03$, $SD = 1.12$), and pedagogical support ($M = 3.14$, $SD = 1.08$) emerged as the most significant factors that shaped performance. The strength of correlation was high for hardware constraints and software proficiency difficulties ($r = 0.68$, $p < 0.01$) and moderate for the interval time for practice and concept knowledge ($r = 0.56$, $p < 0.05$). These results suggest that improving digital facilities, flexible practice class schedules, and efficient collaborative teaching methods is necessary to achieve better learning outcomes in animation courses. What is new or original in this study is its thorough analysis of both technical, pedagogical, and psychological dimensions, providing a unified model for digital VET. This research provides theoretical and practical implications for enhancing institutional policies and pedagogical strategies to prepare students worldwide for the global transformation of the creative industries.

Keywords: *3D animation; digital infrastructure; knowledge integration; learning outcomes; SDGs*

INTRODUCTION

The digital transformation of post-secondary education, a movement intensified by the Fourth Industrial Revolution, has significantly altered the skills demanded by the 21st-century workforce. 3D animation is increasingly an industry within the creative arts and is growing rapidly across film, marketing, game Development, architecture, and digital media (Patete & Marquez, 2022). With employers seeking graduates who can blend technical skills with creativity in a global digital economy, universities and other educational institutions are challenged to prepare graduates to meet these demands. This vision is also closely linked to the explicit text of Sustainable Development Goal 4 (SDG 4), which promotes inclusive, quality education for all and is a key enabler of sustainable

Development (UNESCO, 2020; United Nations, 2015).

In vocational education, 3D animation courses are fundamental for students to master animation, modeling, rendering, simulation, digital composition, and computer-based storytelling. However, the learning outcomes of such courses are influenced by several interacting factors. Previous studies show that 3D animation can significantly enhance motivation and cognitive engagement among students (Teplá et al., 2022) and that immersive technologies, such as VR, improve practical competence and learner satisfaction (Conrad et al., 2024; Long et al., 2025). However, technical issues remain the primary barriers, including outdated hardware, limited pedagogical support, and psychological barriers, such as a lack of confidence (Nugroho et al., 2023; Othman et al., 2024). But even the light of such evidence fractures. Most research studies focus on only one side (technical, pedagogical, or psychological). Few studies integrate these dimensions to guide digital professional training, aiming to produce educational outcomes through digital media.

Moreover, studies are insufficiently conducted at Indonesian polytechnics, owing to inadequate laboratory facilities, limited software licenses, and the need to allocate many hours of practice (Imaduddin & Firdaus, 2025; Santosa & Jazuli, 2022; Sri Rahayu, 2023). The value and novelty of the present work is that it takes the "big picture" approach and investigates both technical, pedagogical, and psychological factors that may affect learning 3D animation. Through the lens of an SDG orientation, with a particular focus on SDG 4, this study makes a theoretical and cross-country practitioner contribution regarding what vocational education policies might usefully look like in lower-socioeconomic-developed contexts.

METHOD

This study employed a quantitative descriptive approach with purposive sampling, involving 36 students enrolled in a 3D animation course. A Likert-scale questionnaire (1–5) was used to measure six variables: technical constraints, understanding of course material, time allocated for practice, pedagogical support, self-confidence, and learning environment. Data analysis included descriptive statistics (mean, median, standard deviation) to identify trends and correlation analysis to examine relationships among the variables.

RESULT AND DISCUSSION

The descriptive analysis revealed three significant challenges that significantly affected student learning outcomes in 3D animation courses. The first challenge relates to technical constraints, with a mean score of 2.72 (SD = 1.41), primarily attributable to inadequate computer hardware and outdated software. These limitations hindered students' ability to run animation programs optimally, thereby affecting their productivity and learning effectiveness. Another challenge identified was limited practice time, indicated by a mean score of 3.03 (SD = 1.12). Insufficient opportunities for hands-on practice were associated with weaker conceptual understanding, as students struggled to apply theoretical knowledge to practical animation tasks. The third challenge, insufficient pedagogical support, had the highest mean score of 3.14 (SD = 1.08). This factor contributed to lower student

confidence and reduced engagement during the learning process, suggesting that instructional guidance and feedback are crucial to supporting students' performance in 3D animation education.

Correlation analysis confirmed strong relationships between hardware limitations and difficulties in mastering animation software ($r = 0.68$, $p < 0.01$) and between limited practice time and conceptual understanding ($r = 0.56$, $p < 0.05$). These findings highlight the interplay among infrastructure, time allocation, and teaching support in shaping student performance. The results demonstrate that technical constraints are a central barrier in 3D animation education. Insufficient hardware and outdated software hinder students' ability to engage with complex rendering and simulation tasks, echoing findings that digital access barriers undermine ICT-enabled education (Othman et al., 2024). From a cognitive perspective, this aligns with Cognitive Load Theory, which explains that students expend additional cognitive effort on technical difficulties rather than meaningful learning, thereby reducing their capacity for knowledge construction (Sweller, 1999).

The significant correlation between practice time and conceptual understanding reinforces the importance of sustained hands-on engagement. Without adequate practice, students risk achieving only surface-level comprehension. This is consistent with Multimedia Learning Theory, which emphasizes that meaningful learning occurs when learners have sufficient opportunities to integrate visual and verbal representations through repeated practice (Meyer, 2009). Furthermore, systematic reviews confirm that immersive and interactive technologies are most effective when students have extended opportunities for practice, leading to deeper cognitive processing and greater skill acquisition (Bödding et al., 2025; Liu et al., 2024).

Pedagogical support also emerged as another crucial factor. Insufficient guidance and feedback reduce students' confidence, thereby undermining their motivation to persist with challenging animation tasks. According to Self-Determination Theory, a supportive learning environment fosters intrinsic motivation by satisfying learners' psychological needs for autonomy, competence, and relatedness (Ryan & Deci, 2000, 2020). In the absence of such support, students may disengage or experience lower creative confidence. Recent reviews further confirm that pedagogical scaffolding, when combined with technology, significantly enhances both affective and cognitive outcomes (Sviridova et al., 2023; Thomann et al., 2024).

Taken together, these findings extend prior studies by highlighting the interdependence of technical, pedagogical, and psychological factors in vocational education. While technical infrastructure forms the foundation, it is insufficient without complementary pedagogical strategies and motivational scaffolding. This integrated perspective contributes to advancing the discourse on digital vocational education, particularly in developing-country contexts such as Indonesia. The novelty of this research lies in framing these three dimensions as a holistic model, thereby offering not only empirical validation but also a practical roadmap for policymakers and educators seeking to align vocational education with the demands of the global creative industry.

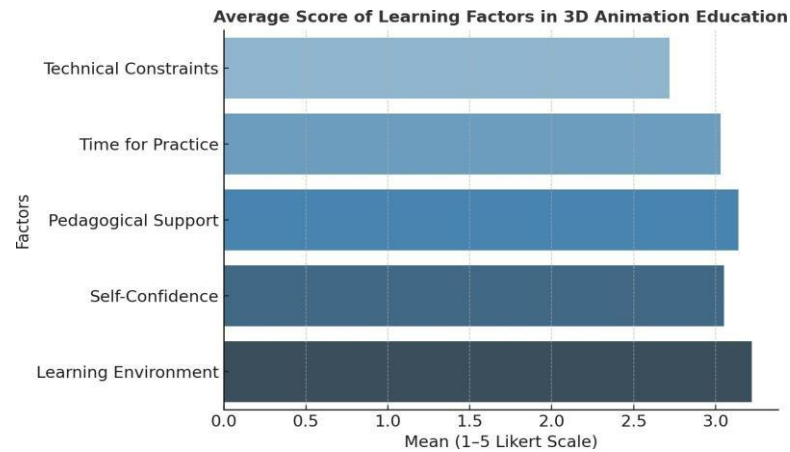


Figure 1. Average Score of Learning Factors in 3D Animation Education

This bar chart illustrates the mean scores for five learning factors, measured using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). Among the factors, Learning Environment obtained the highest mean ($M = 3.22$), suggesting relatively positive perceptions. In contrast, Technical Constraints received the lowest mean ($M = 2.72$), indicating significant barriers related to outdated hardware and software. These results emphasize the importance of addressing infrastructural limitations as a prerequisite for improving learning outcomes.

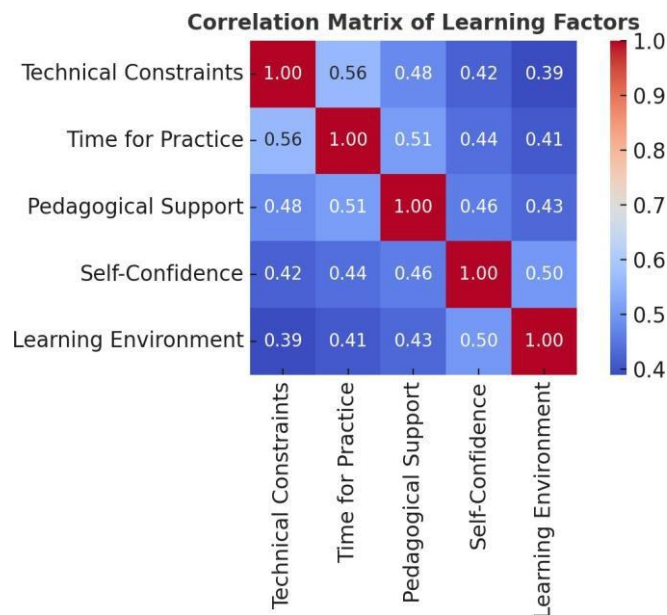


Figure 2. Correlation Matrix of Learning Factors

This heatmap presents the correlation coefficients among the five measured learning factors. The strongest positive correlation was found between Technical Constraints and Time for Practice ($r = 0.68$, $p < 0.01$), followed by Time for Practice and Conceptual Understanding ($r = 0.56$, $p < 0.05$). These relationships indicate that

technical challenges are closely linked to students' ability to engage in practice, which, in turn, affects their mastery of animation concepts. The results highlight the interdependent nature of technical, pedagogical, and psychological dimensions in vocational education.

CONCLUSION

This research finds that the key factors affecting the learning outcomes of 3D animation courses are technical barriers, the pace of practice completion, and insufficient pedagogical support. The findings indicate that inadequate hardware and outdated software not only lead to a lack of technological understanding but also to declines in students' confidence and motivation. Fewer opportunities to practice are associated with weaker conceptual knowledge, and inadequate pedagogical support impairs students' ability to engage in the course.

These challenges require an all-encompassing and integrated response. Strengthening the digital infrastructure and providing flexible, extended practice sessions, along with collaborative learning platforms, are sine qua non for improving the quality of vocational education. These suggestions align with the objectives of SDG 4 (Quality Education) and with a broader restructuring of vocational schools to address global needs in the creative industries.

Theoretically, this research offers insight by developing a model that integrates techno-pedagogical and psychological dimensions into the analysis of student performance. Operationally, it provides actionable guidance for policymakers, educators, and institutional leaders to develop a more dynamic learning environment within digital creative fields. Future research should increase sample sizes across institutions, apply advanced statistical models, such as regression and SEM, and integrate emerging technologies, such as VR and AI platforms. Such attempts will offer better evidence to enhance the practice and theory of digital VET.

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THE INFLUENCE OF CONTEXTUAL TEACHING AND LEARNING (CTL) MODEL ASSISTED BY LOCAL WISDOM-BASED LEARNING VIDEOS ON THE MATHEMATICAL CONNECTIONS OF JUNIOR HIGH SCHOOL STUDENTS

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ABSTRACT

Abstract: This research aims to evaluate the effectiveness of the Contextual Teaching and Learning (CTL) model supported by local wisdom-based learning videos on students' mathematical connection abilities in social arithmetic material. The background of this research is students' limited ability to relate mathematical concepts to everyday contexts. This study employs a quantitative, quasi-experimental design with two seventh-grade classes at SMP Negeri 4 Bireuen: one experimental and one control. The instruments used are mathematical connection tests and observation sheets. Data analysis was conducted using SPSS 30, including normality and homogeneity tests, t-tests, and correlation analyses. The results show that learning with the CTL model, which is assisted by video-based local wisdom, significantly improves students' mathematical connection abilities compared with conventional methods. This model provides a more meaningful and relevant learning experience for students' lives.

Keywords: *CTL, local wisdom, learning videos, mathematical connections, social arithmetic*

INTRODUCTION

Education plays a crucial role in shaping high-quality human resources and is fundamental to the Development of individuals and society as a whole. In this context, mathematics is a core discipline essential for developing logical, analytical, and critical thinking skills (Fauhah & Rosy, 2020). Mathematics education encompasses a range of interconnected topics that require students to not only understand concepts in isolation but also to relate them across different contexts (Utaminingsih & Shufa, 2019). One competency that supports such understanding is mathematical connection, defined as the ability to relate mathematical ideas both within mathematics and across real-world applications (Lubis et al., 2019). However, in practice, many students—especially at the junior high school level—struggle to connect mathematical concepts with real-life situations.

A study conducted at SMP Negeri 4 Bireuen found that students had difficulty applying mathematical knowledge, particularly in social arithmetic, including concepts such as profit and loss. These difficulties include difficulties in

relating different mathematical procedures, interpreting various representations of the same concept, and understanding how mathematical ideas are relevant to everyday scenarios. The learning process is often teacher-centered, relying on direct instruction and rote memorization, which can result in superficial understanding and low student engagement. To address this issue, there is a pressing need for innovative instructional models that promote active student participation and contextual understanding.

Contextual Teaching and Learning (CTL) is an approach that connects academic content to real-world contexts, enabling students to construct knowledge through meaningful experiences. CTL emphasizes student involvement through exploration, discussion, and reflection, making mathematics more engaging and relevant (Hasudungan, 2022). In addition to selecting an appropriate learning model, the use of instructional media is vital for enhancing student engagement and understanding. Video-based learning media, particularly those developed on digital platforms such as Canva, can help visualize mathematical problems in familiar contexts. For example, videos depicting local market transactions can effectively illustrate profit-and-loss scenarios, helping students grasp abstract mathematical ideas more concretely. Moreover, integrating local wisdom into video content—such as traditional Acehnese trading practices—can further enhance the relevance of learning by connecting it to students' cultural backgrounds.

Several studies have highlighted the effectiveness of CTL and video-based learning media in improving students' mathematical skills. Negeri et al. (2010) found that students taught using a contextual approach demonstrated significantly stronger mathematical connection abilities than those taught conventionally. Similarly, Sinaga (2022) and Zulfah (2017) reported that video-assisted instruction—particularly when aligned with students' real-life experiences—can foster more profound understanding, increase motivation, and support the development of problem-solving and critical-thinking skills. Given these considerations, this study aims to examine the effect of the Contextual Teaching and Learning (CTL) model, supported by video-based learning media grounded in local wisdom, on students' mathematical connection ability in social arithmetic. The integration of contextual problems with culturally relevant video content is expected to create a more meaningful and practical learning experience for students. Thus, the research seeks to answer the following question: Does the CTL model supported by local wisdom-based video learning significantly influence students' mathematical connection abilities in social arithmetic topics at SMP Negeri 4 Bireuen?

METHOD

This research uses a quantitative approach with a quasi-experimental design. The research design is a pretest-posttest control Group Design, with two groups: experimental (CTL assisted by locally wisdom-based video learning) and control (conventional methods). The study participants are Grade VII students at SMP Negeri 4 Bireuen. Data collection instruments: Mathematical connection ability test (pretest and posttest), observation of teacher and student activities. Data analysis was conducted using SPSS version 30, including: Normality test (Kolmogorov-

Smirnov), Homogeneity test (Levene's Test), Independent t-test, and Product-moment correlation test.

RESULT AND DISCUSSION

This research was conducted to evaluate the effect of the Contextual Teaching and Learning (CTL) model assisted by local wisdom-based instructional videos on students' mathematical connection abilities in social arithmetic material. The study included two seventh-grade classes at SMP Negeri 4 Bireuen: one experimental class using the CTL model and one control class using conventional instruction. Students' mathematical connection abilities were measured using pre- and post-tests to assess the effectiveness of the CTL model supported by local wisdom-based learning videos.

Result

Table 1. Descriptive Statistics

Descriptive Statistics					
	N	Minimum	Maximum	Mean	Std. Deviation
Pretest experiment class	30	40	70	54.47	7.722
Posttest experiment class	30	77	91	85.73	4.842
Pretest control class	30	44	66	54.27	6.097
Posttest control class	30	71	86	80.00	4.969
Valid N (listwise)	30				

Based on the descriptive statistics, the mean pre-test score for students in the experimental class is 54.47 (SD = 7.722), whereas the control class has a mean pre-test score of 54.27 (SD = 6.097). This indicates that the initial abilities of both classes are relatively balanced. After treatment, the average post-test score of the experimental class increased significantly to 85.73, with a narrower distribution (SD = 4.842), whereas the control class also increased to 80.00 (SD = 4.969). The greater increase in scores and greater consistency of results in the experimental class indicate that the treatment in this study had a greater impact on learning outcomes than the control class.

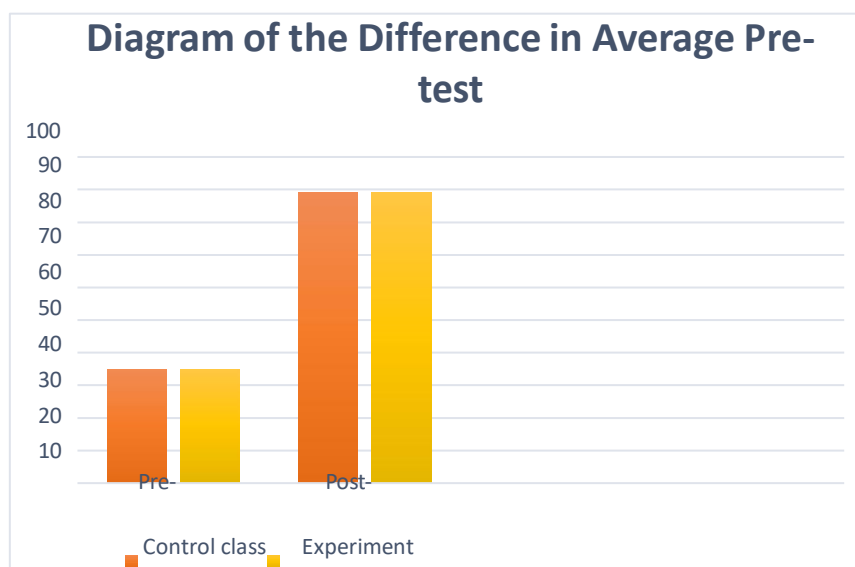


Figure 1: Average Diagram of Student Pretest and Posttest Results

Prerequisite Tests for Analysis

Before conducting further analysis, several prerequisite tests are performed to ensure the validity of the data and its conformity with the necessary statistical assumptions. These tests include normality, homogeneity, hypothesis, and product-moment correlation tests. The results of these prerequisite tests indicated that the data met the assumptions required for subsequent statistical analysis.

1) Normality Test

The normality test is conducted to determine whether the research data are typically distributed. If the data is normally distributed, the following statistical tests can be performed. Data are considered normally distributed if the significance value (sig.) is greater than 0.05; otherwise, the data are not normally distributed. The p-value is less than 0.05; the data are not normally distributed. The results of the normality tests on the pretest and posttest values for the experimental class and the control class are presented as follows:

Table 2. Test of Normality
Test of Normality

Class		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
learning outcomes	Experimental Pre-Test Outcomes	Class .091	30	.200*	.975	30	.693
	Experimental Post-Test Outcomes	Class .114	30	.200*	.957	30	.255

Control Class Pre-Test Learning Outcomes	.129	30	.200*	.941	30	.096
Control Class Post-Test Learning Outcomes	.128	30	.200*	.943	30	.110

*: lower limit of true significance
a.: significant correction Liliefors

Since all Kolmogorov-Smirnov significance values exceed 0.05, it can be concluded that all data are typically distributed. Thus, the data meet the normality assumption, and parametric statistical tests, such as the Independent-Samples T-Test, can be conducted validly. This condition ensures that the comparison of students' mathematical connection abilities between the experimental and control groups is statistically reliable.

2) Homogeneity Test

The homogeneity test is conducted to determine whether the variances of the experimental and control groups are equal. Based on the results of Levene's Test, the $p\text{-value} > 0.05$, indicating that the variances of the two groups are homogeneous and that the data meet the requirements for an independent t-test. This result confirms that differences in post-test scores between the two groups are attributable to the treatment rather than to variance inequality.

Table 3. Homogeneity Test Result

Test of Homogeneity of Variances					
		Levene Statistic	df1	df2	Sig.
learning outcomes	Based on Mean	.865	3	116	.461
	Based on Median	.577	3	116	.631
	Based on Median and with adjusted df	.577	3	34.672	.634
	Based on the trimmed mean	.425	3	116	.736

Based on Levene's test of homogeneity, the p-value (based on the mean) was 0.461, which exceeds the significance level of 0.05. This indicates that the variance of the data across groups is homogeneous or not significantly different; thus, the assumption of homogeneity is satisfied. Therefore, further analysis using parametric tests, such as the Independent-Samples T-Test, can be conducted legitimately.

3) Independent Test

Based on the results of the independent t-test conducted on the posttest data between the experimental class and the control class, a significance value (2-tailed) of < 0.05 was obtained. It can be concluded that there is a significant difference between the two classes. Therefore, the CTL learning model, supported by local

wisdom-based video learning, improves students' mathematical connection skills.

Table 4. Result of the Independent Test

Independent Samples Test										
		Levene's Test for Equality of Variance s		T-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tail ed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
learning outcomes	Equal variances Assumed	13.348	.001	-8.505	58	.000	-19.033	2.238	-23.513	-14.554
	Equal variances not assumed			-8.505	42.974	.000	-19.033	2.238	-23.546	-14.520

Based on the results of the Independent Samples t-test, Levene's test yielded $F = 13.348$ ($p < 0.05$), indicating that the variances are not homogeneous; therefore, the interpretation of the t-test results refers to the 'Equal variances not assumed' row. The t-test results show a t-value of -8.505 ($df = 48$; $p < 0.05$), indicating a significant difference in learning outcomes between students in the experimental and control classes. The average difference in scores between the two groups is -19.033, with a 95% confidence interval ranging from -23.546 to -14.520. Therefore, it can be concluded that the use of the Contextual Teaching and Learning (CTL) model, supported by local wisdom-based learning videos, has a significant effect on improving student learning outcomes compared with conventional learning.

4) Product-Moment Correlation Test

The results of the product-moment correlation test between the pretest and posttest scores show that the $Sig. < 0.05$, and the correlation coefficient value is 0.056. This indicates a strong, significant relationship between initial ability and students' final results: the higher the students' initial ability, the greater their mathematical connection ability after the treatment. Therefore, students' initial mathematical understanding plays an important role in shaping their learning outcomes following the implementation of the CTL model, supported by local wisdom-based learning videos.

Table 1: Uji Korelasi Product Moment

learning outcomes	Pearson Correlation	learning outcomes	class
		1	.615**
class	Sig. (2-tailed)		.000
	N	120	120
	Pearson Correlation	.615**	1
	Sig. (2-tailed)	.000	
	N	120	120

** . Correlation is significant at the 0.01 level (2-tailed).

Based on the Pearson product-moment correlation test, a correlation coefficient of 0.615 was obtained at the significance level $\alpha < 0.05$. This indicates a positive and significant relationship between the learning treatment (experimental and control classes) and student learning outcomes. Therefore, the more appropriate the learning model employed, the higher the students' learning outcomes.

Discussion

This research aims to examine the effect of using the Contextual Teaching and Learning (CTL) model supported by local wisdom-based learning videos on students' mathematical connection abilities. The pre- and posttest data showed a significant improvement in learning outcomes for the experimental group compared with the control group. This indicates that applying the CTL model, combined with video media that highlight local cultural elements, can encourage students to connect mathematical concepts to real-life experiences in their daily lives. Descriptive statistical test results show that the average pretest score of the experimental class is 54.47 and that of the control class is 54.27. This comparable score indicates that students' initial abilities across the two classes are similar prior to receiving different treatments. This serves as a fair basis for comparing the effects of each treatment.

After the treatment, there was a significant increase in the students' posttest scores. The experimental class, which was taught using the CTL model assisted by learning videos, experienced an average increase of 85.73, while the control class, which was taught conventionally, increased to 80.00. The 5.73-point improvement difference indicates that CTL learning, which integrates local context, can provide a more meaningful and relevant learning experience for students. The normality test indicates that the pretest and posttest data from both the experimental and control classes are not significantly nonnormal. This means the data is usually distributed. Furthermore, Levene's test of homogeneity of variance yielded a p-value of 0.461 (> 0.05), indicating that the variances between groups are homogeneous. With the assumptions of normality and homogeneity met, parametric tests can be used validly.

The independent sample t-test showed a significance value (2-tailed), far below the threshold of 0.05. This indicates a significant difference in learning

outcomes between students in the experimental and control classes. The mean difference of -19.033, with a 95% confidence interval from -23.546 to -14.520520, further supports the conclusion that the CTL treatment contributes more to improved learning outcomes. The product-moment correlation test resulted in an r value of 0.615. This coefficient indicates a positive and significant association between the learning model type (experimental versus control) and students' learning outcomes. In other words, the more appropriate the applied learning model is to students' needs and the local context, the higher the results.

The CTL model encourages active student engagement in learning through reflection activities, inquiry, questioning, and group work. Thus, students not only receive information from the teacher but also build their own knowledge through real learning experiences. This aligns with the principles of constructivism, which underpin the CTL model. Such meaningful learning processes enable students to relate various mathematical concepts to real problems they encounter. The use of local wisdom-based learning videos also enriches students' learning experiences. Videos showcasing the economic activities of the Bireuen community, such as the trade of 'kue bhoi' and 'timphan', help students understand the concepts of profit and loss in a context that is close and familiar. This is consistent with findings (Anggraini, 2021) that learning media grounded in local culture facilitates students' understanding of the material because they are linked to their personal experiences.

The product-moment correlation between the pretest and posttest results is 0.615, indicating a moderate-to-strong, significant correlation. This indicates a positive relationship between students' initial abilities and their final results. The higher the students' initial abilities, the greater the improvement they can achieve after participating in CTL-based learning and local wisdom videos. This finding reinforces the assumption that contextual learning can optimize students' individual potential. During implementation, students appeared more enthusiastic and active in the experimental class. This was also evident in the student and teacher activity sheets, which indicated increased student participation and engagement during the learning process. Learning activities do not focus solely on achieving results but also on the process of building deep, meaningful understanding. With this approach, students find it easier to remember concepts and apply them in real life.

In addition, the CTL approach, using locally based cultural media, plays an important role in preserving local wisdom for the younger generation. Through learning that incorporates regional cultural values, students not only gain academic knowledge but also develop a love for their surrounding social and cultural environment. This is a strategic step in developing education that is not only academic but also humanistic and rooted in the nation's culture.

The results of this study align with previous research by Musriliani et al. (2015), which concluded that the CTL model significantly improves students' mathematical connections. Similarly, Guspitarsari et al. (2024) stated that CTL is effective in enhancing understanding of mathematical concepts when combined with visual media. The use of learning videos in this study enhanced the appeal of learning, particularly by fostering a concrete understanding of abstract topics such as mathematics.

Overall, the application of the CTL model, supported by local wisdom-based

video learning, has significantly improved students' mathematical connection skills. This approach not only enhances learning outcomes but also fosters cultural awareness and instills local values in the educational process. Therefore, this learning strategy is highly recommended for implementation in the Independent Curriculum, which emphasizes meaningfulness and diversity in learning.

CONCLUSION

Based on the study results, the implementation of the Contextual Teaching and Learning (CTL) model, supported by video-based instruction grounded in local wisdom, has a significant positive effect on students' mathematical connection abilities in social arithmetic. The CTL model, which emphasizes learning through real-life contexts and is supported by culturally relevant videos, creates a more meaningful, engaging, and effective learning experience. This is evidenced by the significant difference in posttest scores between the experimental and control groups, with the experimental group showing greater improvement. The independent t-test results indicate a statistically significant difference in mathematical connection ability, favoring the CTL approach. In addition, the strong correlation between pretest and posttest results demonstrates that students with higher initial abilities also achieved higher learning outcomes, and that the applied learning model effectively supports student Development. In conclusion, integrating contextual learning and video media grounded in local culture has been shown to enhance students' ability to relate mathematical concepts both within the subject and in real-life situations. This strategy is recommended for mathematics learning, particularly for topics that require real-world application, such as social arithmetic.

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ELEMENTARY STUDENTS' MATHEMATICAL THINKING IN GEOMETRY SUBJECT THROUGH 4C-BASED PROBLEM-BASED LEARNING

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ABSTRACT

This study aims to investigate the impact of problem-based learning (PBL) that focusing on 21st-century skills (4C: critical thinking, communication, collaboration, and creativity) on mathematical thinking skills of fourth-grade elementary students in geometry. A quantitative approach was employed using a quasi-experimental method with a non-equivalent control group design, involving two parallel classes as the experimental and control groups, each consisting of 25 students. The research instruments were a mathematical thinking skills test and an affective questionnaire. The results showed a statistically significant difference between the experimental and control groups in posttest scores of mathematical thinking skills ($p < 0.05$). The experimental group obtained a significantly higher mean gain score (21.80) compared to the control group (11.60), with a t-test result of $t = 4.75$ and $p = 0.000$. Additionally, the questionnaire results showed that the 4C-based PBL increased students' motivation, active participation, and confidence. These findings indicate that integrating PBL with 4C skills effectively improves the mathematical thinking skills of elementary students in learning geometry.

Keywords: *Mathematical Thinking, Problem-Based Learning 4C, geometry.*

INTRODUCTION

Mathematical thinking skills serve as an important foundation for developing students' competencies in the 21st century, particularly for elementary school students. Both national and international curricula emphasize the development of critical thinking, problem-solving, and reflective thinking skills as integral components of early mathematics education (NCTM, 2020). Geometry, in particular, is the subject in which students are required to connect visual representations with abstract mathematical concepts, a process that demands both logical thinking and creativity. However, empirical evidence indicates that many elementary students still struggle to develop their mathematical thinking skills, especially in understanding and conceptually solving geometry problems (Ministry of Education and Culture, 2023).

The urgency for more meaningful geometry learning has increased in response to global demands for improving the quality of Indonesian students' scores in mathematical literacy. The 2022 Programme for International Student Assessment (PISA) results show that Indonesian students' score in mathematical literacy (mean score = 379) remains below the OECD average score of 489 (OECD, 2023). One of the factors contributing to this underperformance is inadequate teaching methods that develop deep conceptual understanding and build students' self-efficacy in mathematics, which are closely interconnected with mathematical thinking.

One relevant approach in addressing these challenges is the 4C-based Problem-Based Learning model, a student-centered learning framework emphasizing real-life problem-solving. 4C-based PBL is a learning model that focuses on authentic problem-solving, identifying and reflecting on ideas, expanding solution applicability, and drawing analogies to similar cases. However, the implementation of PBL at the elementary level requires systematic adaptation in accordance with the cognitive and affective characteristics of elementary school students. The integration of PBL and 4C skills development in elementary school geometry learning provides more meaningful learning experiences for students. This aligns with the findings of Akram et al. (2023), which show that PBL is effective in developing students' 4C competencies in mathematics education. Through collaborative communication, critical thinking, and creativity in solving problems, Akram et al. (2023) confirmed that PBL is effective in developing students' 4C competencies in mathematics education.

While numerous studies have shown that PBL is effective in improving conceptual understanding and students' mathematical learning abilities, few have specifically investigated the impact of 4C-based PBL on elementary students' mathematical thinking skills, which is also the gap this study aims to address. In the contemporary educational context, the integrated 4C-based PBL model has emerged as a promising solution for the development of relevant skills in the 21st century. This study implements the 4C-based PBL for fourth-grade students of AlFatih Integrated Islamic Elementary School in Bireuen District, Aceh, Indonesia. It is expected to facilitate the development of mathematical thinking skills through collaborative and explorative group activities in geometry learning. The 4C-based PBL in geometry learning encourages students to study and actively enrich knowledge through contextual learning experiences. Several existing studies show that educators still face challenges in effectively designing and implementing the 4C-based PBL (Sarnoko et al., 2023).

The issue that initiated this study is the underdeveloped mathematical thinking skills among fourth-grade elementary students in geometry learning, where current learning practices tend to be procedural and lack innovative breakthroughs, thus leading to the inadequate mathematical skills of fourth-grade students. The existing gap between students' mathematical competency development needs and learning activities implemented in the classroom is particularly notable.

Practically, this research provides guidance for educators in designing and implementing 4C-based Problem-Based Learning in geometry instruction to improve students' mathematical thinking skills. However, current teaching methods are still insufficient to foster deep conceptual understanding and students' self-efficacy in mathematics, which consequently hinders the development of mathematical thinking skills in geometry learning. Based on this problem, this study seeks to examine whether there is a difference in students' mathematical thinking skills between those taught using 4C-based Problem-Based Learning and those taught using Problem-Based Learning without 4C integration. In addition, this research also aims to explore students' responses toward geometry learning through the implementation of the 4C-based Problem-Based Learning model in grade IV of SD IT Al Fatih Pusangan Bireuen.

METHOD

This study employs a quantitative approach using a quasi-experimental non-equivalent control group design. This design was chosen due to practical constraints preventing full randomization of research subjects, but external variables remain controllable by comparing two equivalent groups (the experimental and control groups). Research population comprises all fourth-grade students of Integrated Islamic Elementary School Al-Fatih Peusangan consisting of four parallel classes. The research sample consists of 25 students from each class (Class IV & Class IV). The selection criteria for the sample required relatively balanced academic skills based on previous scores, and the selection was based on class characteristics equivalence (homogeneity). The sampling technique used was purposive sampling, where samples were deliberately chosen based on certain considerations to ensure relevance to the research objectives.

In experimental design, the experimental group was taught with the 4C-based (communication, collaboration, critical thinking, and creativity) Problem-Based Learning model. Meanwhile, the control group was taught with the conventional PBL model without explicit integration of 4C skills. Research instruments: the researchers used several instruments: 1) a test to assess students' mathematical thinking skills; 2) a questionnaire: support the quantitative data and evaluate students' affective aspects; and 3) interviews.

Data Collection Procedure: A pre-test was conducted on both groups to determine their initial abilities. The experimental group received geometry instruction using the PBL (Problem-Based Learning) approach based on the 4C framework, while the control group received geometry instruction using conventional PBL. The pre-test was administered before the treatment, and the posttest was administered after the treatment over several sessions. The data were analysed to compare the results of both groups.

The data analysis was carried out in several phases, beginning with normality and homogeneity tests to verify the suitability of the data for parametric analysis. This step was followed by the calculation of gain scores to assess improvements in learning outcomes. An independent samples t-test was then performed to examine the mean differences between the experimental and control groups. Statistical significance was determined at the 95% confidence level ($\alpha = 0.05$) to identify any significant differences between the two groups. In addition, the analysis of the mathematical thinking questionnaire instrument of the two groups and interviews were conducted to provide deeper insights into student performance based on specific indicators of mathematical thinking ability. This series of procedures was designed to produce results that are both valid and reliable.

A closed-ended questionnaire with a five-point Likert scale was utilized to measure students' mathematical thinking skills in geometry through the 4C Problem-Based Learning (PBL) approach (Communication, Collaboration, Critical Thinking, and Creativity). There were 8 statement items arranged based on 4C indicators, such as the ability to work together in groups, explain solutions, think critically, use various strategies, and relate mathematical concepts to everyday life situations. Each item is given a response option from Strongly Agree, Agree,

Neutral, Disagree, and Strongly Disagree. The questionnaire items refer to the theory of 21st century competencies and students' mathematical thinking. All statements were developed by the researchers by considering the context of geometry material, student characteristics, and learning objectives based on 4C Problem-Based Learning (PBL) approach.

Null Hypothesis (H_0): There is no significant difference in students' mathematical thinking skills between the students taught with the 4C-based PBL model and the students taught with PBL without 4C.

Alternative Hypothesis (H_1): There is a significant difference in students' mathematical thinking skills between the students taught with the 4C-based PBL model and the students taught with PBL without 4C.

RESULT AND DISCUSSION

Data description, this research involved two groups of students. Pre-tests and post-tests were conducted to assess their mathematical thinking abilities, specifically in relation to the topic of perimeter and area of two-dimensional geometric shapes.

Table 1. Mean and Standard Deviation of Pre-test and Post-test Score

Groups	Pre-test (Mean $\pm$ SD)	Post-test (Mean $\pm$ SD)	Gain Score (Mean)
Experimental (PBL4C)	62.40 $\pm$ 7.85	84.20 $\pm$ 6.43	21.80
Control (Standard PBL)	63.00 $\pm$ 8.12	74.60 $\pm$ 7.51	11.60

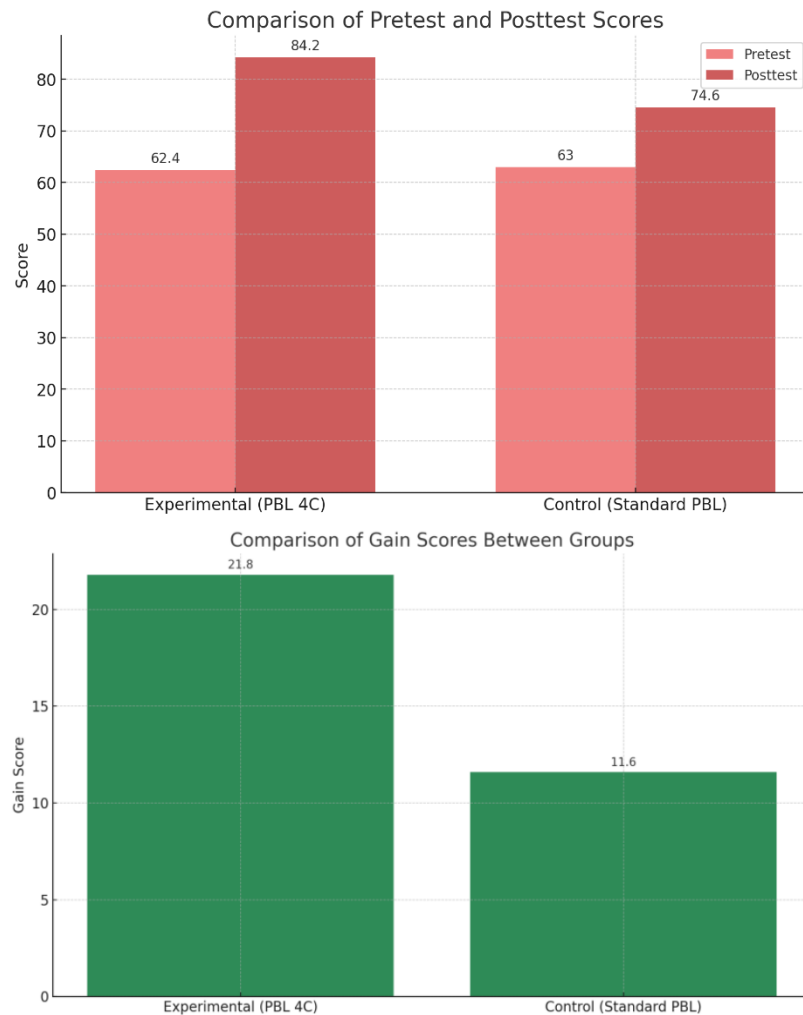
Based on the table above, it is evident that both groups had relatively similar mean pre-test scores. However, in the post-test, the experimental group demonstrated a greater improvement compared to the control group.

Normality Test, the normality test was conducted using the Shapiro-Wilk, resulting in the following outcomes:

Experimental group: $p = 0.212$. Control group: $p = 0.308$

Since the value of $p > 0.05$, the data followed a normal distribution

Homogeneity Test, the homogeneity test was performed using Levene's Test, which resulted in $p = 0.137$. This indicated that the variance of both groups is homogenous. ($p > 0.05$).



This chart illustrates that the experimental group experienced a greater improvement compared to the control group. Meanwhile, gain score chart highlights that the experimental group had a substantially higher mean gain score compared to the control group. Hypothesis Testing (t-Test) was used to examine whether there is a significant difference in the mathematical thinking abilities of students between those who received 4C-based PBL instruction and those who received PBL without the 4C approach. An independent samples t-test was conducted on the post-test scores of both groups.

Table 2. Descriptive Statistics of Post-test Scores for Mathematical Thinking Ability

G roups	N	Me an	Standard Devi at i on (SD)
Experimental(PBL4C)	25	84.20	6.43
Control (Standard PBL)	25	74.60	7.51

t-Statistic (t-calculated): 4.75. Degrees of freedom (df): 48. P-value: 0.000. Significance level (α): 0.05. Since the p-value is less than 0.05, the null hypothesis

(H_0) is rejected, indicating a statistically significant difference between the experimental and control groups in terms of mathematical thinking ability after the intervention. Therefore, Problem-Based Learning (PBL) integrated with 4C skills is proven to be more effective in enhancing mathematical thinking ability of the students compared to standard PBL.

The questionnaire results indicated that students in the experimental group were more active in expressing ideas, collaborating, thinking critically, and demonstrating creativity throughout the learning process. The average questionnaire score for the experimental group was classified as “High” (mean = 4.35 on a 1–5 scale), while the control group only reached the “Moderate” category (mean = 3.70).

Table 3. The list of statements from the questionnaire along with a description of the 4C indicators.

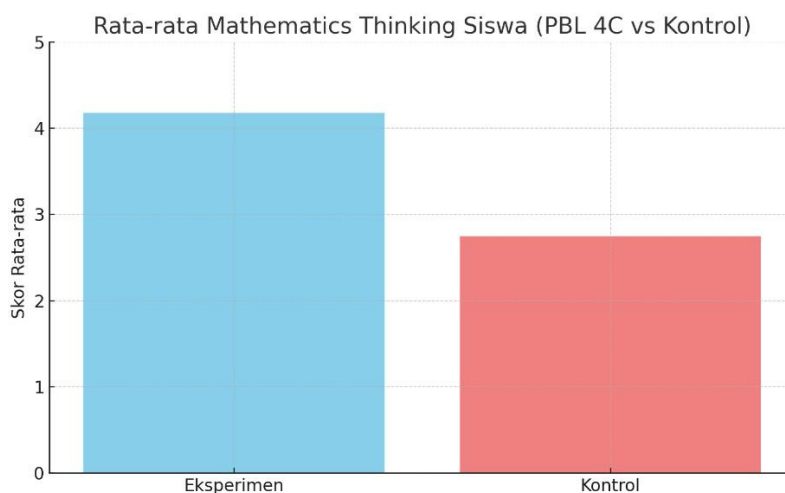
No	Statements	Indicator 4C	Description
1	I try various ways to solve problems on the area and perimeter of flat shapes such as squares, rectangles, and triangles.	<i>Creativity</i>	Measuring the extent to which students use creative ideas and alternative strategies in solving mathematical problems.
2	I am able to explain how I solve problems to friends.	<i>Communication</i>	Measuring students' ability to convey ideas or problem solving procedures verbally to others.
3	I enjoy discussing and working in groups when learning geometry.	<i>Collaboration</i>	Assess students' attitudes in working together, discussing, and interacting socially when completing group assignments.
4	I use logic and reason when solving a problem.	<i>Critical Thinking</i>	Measuring students' critical thinking skills in assessing and selecting problem-solving strategies based on logical reasons.
5	I find it easier to understand mathematical concepts when they are related to everyday life.	<i>Critical Thinking</i>	Assess students' ability to relate mathematical concepts to real contexts to enhance their understanding.
6	I make pictures or sketches to help solve math problems.	<i>Creativity</i>	Measures the extents to which students use visualization or creative representation in the problem-solving process.
7	I am used to double checking my answers and thinking whether my method is correct.	<i>Critical Thinking</i>	Assessing students' reflective habits in evaluating their own solutions to mathematical problems.
8	I am motivated to think		deeper when solving math problems.

Describes
students'
internal
drive
to
think
deeply
and
not
just
accept
answers
immediately
.

The average results of questionnaire distribution for each class:

Class	Average	Median	Standard Deviation	Minimum	Maximum
Experiment	±4.20	~4.25	Low (~0.22)	3.75	4.50
Control	±2.75	~2.75	Medium (~0.30)	2.12	3.38

Students in the experimental class showed positive response of their mathematical thinking skills after 4C-based PBL learning. The distribution of scores in the experimental class was homogeneous. There was a significant difference between students' perceptions in 4C-based PBL learning and PBL learning without 4 C. Hence, the 4C approach helps foster elementary school students' self-confidence and critical, creative, communicative, and collaborative thinking habits in geometry.



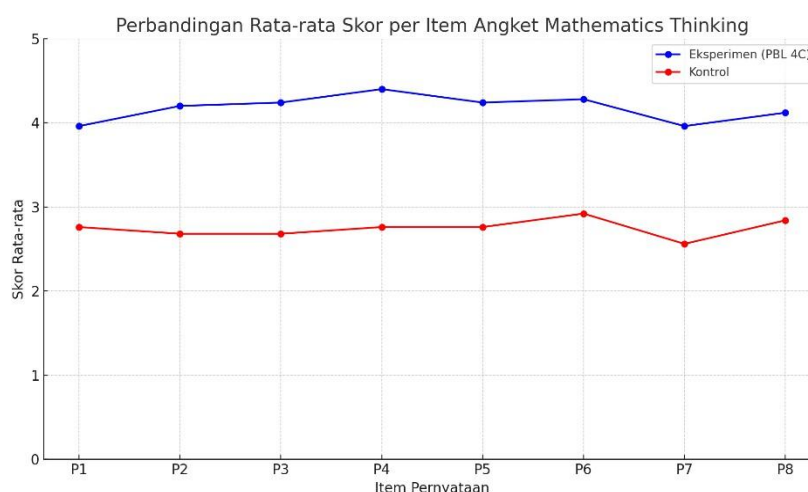
The experimental class had an average score of around 4.20, indicating a high level of mathematical thinking. The control class had an average score of around 2.74, which was lower than the experimental class. The PBL-based learning approach with 4C integration significantly improved students' mathematical thinking skills compared to the PBL learning approach without 4C.

Here is a summary of the results:

Table 4. The average Score per Item

Item	Experiment	Control
P1	~3.92	~2.72
P2	~4.12	~2.76
P3	~4.24	~2.76
P4	~4.28	~2.76
P5	~4.24	~2.84
P6	~4.28	~2.92
P7	~4.08	~2.64
P8	~4.04	~2.88

All items showed significant score differences between the experimental and control classes. Items with differences P4 (logic & reasoning) and P6 (drawing/sketching) showed that the 4C-based PBL approach greatly encouraged students' critical thinking and creativity. The consistently high experimental score pattern strengthened the effectiveness of 4C-based PBL learning on improving elementary school students' mathematical thinking skills in geometry material. More detailed diagram per indicator (P1-P8)



The graph shows a comparison of the average scores for each questionnaire item between the experimental class and the control class.

The results of the study indicated that the implementation of 4C-based Problem-Based Learning (PBL) had a significant positive impact on improving the mathematical thinking abilities of students. The pre-test data showed that both groups had equivalent initial skill levels. However, after the treatment, the experimental group demonstrated greater improvements in scores, both in absolute terms and in gain scores. This improvement aligned with the characteristics of 4C-based PBL, which emphasized collaborative and contextual learning, where students were not only solving problems procedurally but are also encouraged to think critically, express ideas openly, work in teams, and find creative solutions. This was supported by questionnaire results indicating students' positive perceptions of the learning process, as well as interview results that revealed increased motivation and self-confidence. These findings reinforce previous studies (Akram et al., 2023; Sarnoko et al., 2023) which concluded that integrating PBL with 4C skills can meaningfully enhance mathematics learning outcomes.

CONCLUSION

A significant difference exists in the mathematical thinking abilities of students between those who participated in learning using the 4C-based PBL model and those who followed PBL without the 4C approach. The 4C-based PBL model has proven to be more effective in enhancing the mathematical reasoning of students, particularly in understanding perimeter and area of two-dimensional

geometric shapes. Based on this research findings, educators are encouraged to adopt the 4C-based PBL model as a strategic alternative to enhance the quality of mathematics education in elementary schools.

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CHAPTER 2: ENGLISH LANGUAGE TEACHING

THE UTILIZATION OF SPOTIFY APPLICATION TO INCREASE STUDENTS' SKILL IN LISTENING THROUGH ENGLISH PODCAST

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ABSTRACT

This study investigates the effectiveness of using Spotify podcasts as a teaching medium to enhance listening skills in students of senior high school among Class X IPA II students at SMAN 1 Samalanga. The purpose of this research is to determine whether students' mean listening comprehension scores improve significantly after the use of Spotify podcasts as a teaching tool and to examine the effect of this medium on enhancing students' listening skills. Employing a pre-experimental design with a pre-test and post-test, the study involved 32 students selected through purposive sampling. The findings reveal a significant improvement in students' listening comprehension, with the average score increasing from 61.14 to 81.46 and an N-Gain score of 0.6164, which is categorized as moderate. Statistical analysis using a paired-sample t-test indicated a significant difference between pre-test and post-test scores, while the reduction in standard deviation from 11.381 to 6.104 reflects improved score consistency. The results indicate that Spotify podcasts provide engaging and high-quality audio content that effectively addresses challenges in traditional teaching methods and improves learning outcomes. This study highlights the potential of integrating modern technology into English language education by offering an innovative approach to enhancing students' listening skills and overall language proficiency.

Keywords: *Spotify podcasts; listening comprehension; EFL students;*

INTRODUCTION

Listening is a crucial language skill that involves the ability to accurately receive, interpret, and understand spoken messages through active engagement with both verbal and non-verbal cues. In English language learning, listening plays a fundamental role in helping students comprehend conversations, lectures, and multimedia content, which in turn enhances their confidence in speaking and interacting in real-life situations. Effective listening enables learners to process information meaningfully and respond appropriately, making it an essential component of successful communication and language mastery.

In the context of the *Merdeka* Curriculum implemented at SMA Negeri 1 Samalanga, students are expected to actively engage in listening activities by identifying main ideas and supporting details in various spoken texts. The curriculum emphasizes the use of effective listening strategies, such as note-taking, summarizing, and predicting, to improve comprehension and critical thinking. Furthermore, students are encouraged to apply their listening skills in authentic situations, including conversations and multimedia-based learning, to support both academic achievement and social communication.

However, classroom observations at SMA Negeri 1 Samalanga reveal that many students still demonstrate low achievement in listening skills. Students often experience difficulties in understanding spoken information, following oral instructions, and comprehending dialogues involving multiple speakers. These challenges are influenced by several factors, including limited vocabulary mastery, difficulties in recognizing pronunciation and intonation, lack of regular listening practice, unfamiliarity with different accents and speaking styles, limited contextual understanding, and insufficient feedback during learning activities. As a result, students' listening comprehension remains underdeveloped, indicating the need for more effective and engaging instructional approaches.

Listening skills are widely recognized as a fundamental component of effective communication and one of the four core language competencies alongside speaking, reading, and writing (Bodie, 2023). Effective listening requires active attention, cognitive processing, and appropriate responses rather than passive hearing (Raharjo & Arifin, 2024). In educational settings, strong listening abilities are essential for understanding instructional content, participating in discussions, and retaining information, which support deeper comprehension and critical thinking (Şendağ et al., 2024). Beyond academic contexts, listening skills contribute significantly to interpersonal communication, collaboration, and leadership by fostering empathy, trust, and effective decision-making (Drigas et al., 2021; Manusov et al., 2020; Larkin, 2020).

To address students' listening difficulties, the integration of digital technology into language learning has been increasingly recommended. One promising platform is Spotify, a global digital streaming service that provides access to music and podcasts through a user-friendly and flexible system (Ilmi & Nurjati, 2023). Spotify offers millions of audio resources that can be accessed across devices, allowing learners to engage with authentic listening materials anytime and anywhere (Purba et al., 2023). Its advanced recommendation system, powered by artificial intelligence, personalizes content based on users' preferences, while features such as adjustable playback speed and offline listening support autonomous and flexible learning (Hamsia & Roifah, 2023; Terroso-Saenz et al., 2023).

Podcasts, as episodic digital audio programs, offer rich opportunities for language learning by exposing students to authentic speech, varied accents, and diverse topics (Raharjo & Arifin, 2024). Unlike traditional audio materials, podcasts allow learners to pause, replay, and select content according to their learning needs, making them particularly effective for developing listening comprehension and vocabulary acquisition (Masbirorotni et al., 2023; Suwarni et al., 2023). The rapid growth of educational podcasts highlights their potential as an engaging medium for enhancing students' listening skills and motivation in language learning (Şendağ et al., 2024).

Therefore, integrating Spotify podcasts into English language learning at SMA Negeri 1 Samalanga is expected to provide an innovative solution to improve students' listening skills. By offering exposure to native speakers and diverse speaking styles, Spotify can help students develop better comprehension, pronunciation awareness, and confidence in listening. Based on this rationale, the

present study aims to examine the effectiveness of using Spotify podcasts as a teaching tool in improving students' listening skills. Specifically, this research investigates whether there is a significant difference in students' listening comprehension scores before and after the implementation of Spotify podcasts and examines the extent to which Spotify podcasts influence listening skill improvement among X IPA II students at SMA Negeri 1 Samalanga.

METHOD

This research utilizes a Pre-Experimental research design, specifically the one-group pre-test post-test method. Pre-experimental design is chosen because it provides a straightforward approach to observe the effect of a treatment on a single group (Zhu et al. 2025). In this study, there is no control group for comparison, but the same group of students would be assessed before and after the treatment is given. The goal is to determine the effectiveness of using *Spotify Podcasts* in improving students' English listening skills.

According to (Ahmed 2024), a population is the entire group of people or objects relevant to the study. In this research, the population consists of all first-grade students at SMA Negeri 1 Samalanga, focusing on their listening skill. The sample in this research was selected through purposive sampling, meaning the researcher chose the sample based on specific characteristics relevant to the study (Dawis et al. 2023). The selected sample consists of 32 students from Class X IPA II at SMA Negeri 1 Samalanga, chosen based on their initial listening skills and availability to participate in the study.

The primary research instrument utilized in this study is a listening test specifically designed to assess students' comprehension of spoken English based on the content of the Spotify Podcast episodes featured during the treatment. This test consists of multiple-choice questions that provide students with several options from which they must select the correct answer, allowing for the evaluation of their understanding of key concepts and details discussed in the podcasts. Additionally, the test includes essay questions that prompt students to articulate their thoughts and insights regarding the topics presented in the podcasts.

This combination of question types enables a comprehensive assessment of students' listening skills and their ability to engage critically with the podcast content, as they must synthesize information and express their understanding in their own words. Descriptive analysis is a method used to derive general conclusions by elucidating the collected data based on its characteristics. Its primary aim is to create a visual representation or description of each dataset, encompassing measures such as the mean, standard deviation, variance, maximum and minimum values, and the range. This analysis is performed using IBM SPSS Statistics software version 27.

RESULTS AND DISCUSSION

Based on the pre-test and post-test administered to the students, who were taught using Spotify podcasts as a teaching medium, the data regarding students'

pre-test and post-test results were analyzed. The pre-test and post-test scores of students' listening comprehension can be seen in the following table:

Table 1. ListeningComprehensionTestScoresofXIPAIStudents

No	Name	Pre-Test Score	Post-Test Score
1.	AS	75	90
2.	AF	55	80
3.	AZ	65	80
4.	CD	50	80
5.	FM	50	80
6.	HR	70	85
7.	IA	40	65
8.	TA	65	75
9.	KA	35	90
10.	MI	45	89
11.	MA	75	80
12.	MR	55	78
13.	MF	60	80
14.	ND	60	85
15.	NU	60	85
16.	NK	75	90
17.	NR	70	80
18.	NF	70	85
19.	NU	70	80
20.	NR	65	75
21.	PA	75	85
22.	PL	65	80
23.	PR	60	80
24.	RF	60	80
25.	RN	65	90
26.	RA	35	80
27.	RR	45	80
28.	SN	75	80
29.	SY	55	80
30.	SF	60	85
31.	US	60	65
32.	NS	60	75
33.	SZ	75	90
34.	RF	70	89
35.	RH	70	80
Mean Score		61.14	81.46

Based on Table 1 above, it is known that the average pre-test score of students' listening comprehension before being taught using Spotify podcasts as a teaching medium was 61.14, while the post-test results increased significantly to 81.46 after the treatment. The descriptive analysis of the test results of students' listening comprehension can be seen in the following table.

Table 2. Descriptive Analysis of Listening Comprehension Test Results

	N	Minimum	Maximum	Mea	Std.
				n	Deviation
Pre-Test Score	35	35	75	61.14	11.381
Post-Test Score	35	65	90	81.46	6.104
Valid N (listwise)	35			6	

Based on the descriptive analysis above, the average pre-test score was 61.14 with a standard deviation of 11.381, indicating a moderate spread in the students' listening comprehension scores before the treatment. After the intervention, the average post-test score increased to 81.46, with a reduced standard deviation of 6.104, showing both a significant improvement in performance and greater consistency in students' listening comprehension post-treatment.

This improvement of 20.32 points between the pre-test and post-test demonstrates the effectiveness of using Spotify podcasts as a teaching medium in enhancing students' listening comprehension. The narrower spread of scores in the post-test suggests that students performed more uniformly after the intervention, emphasizing the positive impact of the media on their learning outcomes.

The following graph illustrates the comparison between the students' pre-test and post-test scores, showcasing the significant improvement in their listening comprehension after being taught using Spotify podcasts. The visual representation highlights the overall increase in performance from the pre-test to the post-test across the student group.

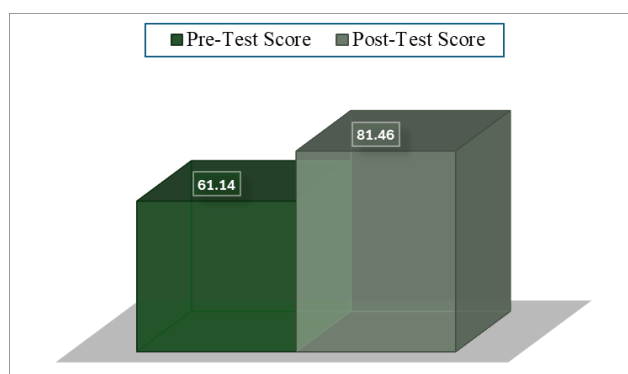


Figure 1. The Difference of Pretest and Post-test scores

Based on the analysis of the mean scores, there was a notable improvement in students' listening comprehension after being taught using Spotify podcasts as a teaching medium. The average pre-test score was 61.14, which increased significantly to 81.46 in the post-test, with a difference of 20.32 points. This improvement demonstrates that the use of podcasts as a teaching tool had a positive impact on the students' ability to comprehend spoken English. The significant increase in scores highlights the effectiveness of the treatment in enhancing students' listening comprehension skills.

Normality Test.

The normality test in this research was conducted using the Shapiro-Wilk test, as the sample size was below 50 students. This test aims to verify whether the data distribution adheres to the assumption of normality, which is essential for conducting parametric statistical analyses. The criteria for determining normality are based on the significance value (p): if the p-value is greater than 0.05, the data is considered to have a normal distribution; conversely, if the p-value is less than 0.05, the data does not meet the assumption of normality. The results of the normality test are presented in Table 4.3 below:

Table 3. Normality Test Results

Table 5. Normality Test Results							
Kelas		Kolmogorov-Smirnov			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	d	Sig.
Hasil	Pre-Test Score	0.117	35	0.200*	0.941	f	
						3	0.06
	Post-Test Score	0.124	35	0.192	0.956	5	2
						3	0.16
						5	7

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

From Table. 3, the significance values for the Shapiro-Wilk test are 0.062 for the pre-test and 0.167 for the post-test, both exceeding the 0.05 threshold. This indicates that the data for both pre-test and post-test scores are normally distributed, confirming compliance with the assumption of normality required for subsequent analyses. The following plot shows the Q-Q plot of the students' pre-test scores, which is used to evaluate the data distribution.

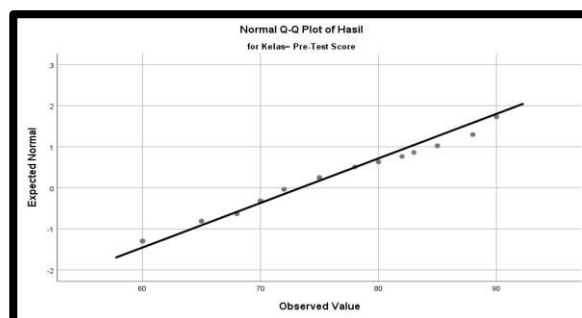


Figure 2. Plot Q-Q

The Q-Q plot presented in the image shows the comparison between observed and expected normal values for the pre-test scores. In a Q-Q plot, if the data points align closely along the diagonal line, it suggests that the data follows a normal distribution. In this plot, the majority of the data points closely follow the diagonal line, indicating that the pre-test scores are approximately normally distributed. Although there are slight deviations at the lower and upper ends of the plot, the overall pattern suggests that the assumption of normality is met. This supports the validity of applying statistical tests that require a normal distribution of data, such as t-tests or ANOVA.

The results of this study indicate a significant improvement in the listening skills of Class X IPA II students at SMAN 1 Samalanga after implementing the teaching method using Spotify podcasts. Based on the pre-test and post-test analysis, the average listening comprehension score increased from 61.14 to 81.46, showing an improvement of 20.32 points. This demonstrates the effectiveness of the podcast-based teaching method in enhancing students' listening comprehension. In other words, using technological media like podcasts can positively contribute to learning English, particularly in the area of listening comprehension skills.

In the descriptive analysis, it is observed that before the treatment, the pre-test scores showed a wide distribution with a standard deviation of 11.381. However, after the treatment, the distribution became more uniform, with a standard deviation decreasing to 6.104. This suggests that after using Spotify podcasts, not only did individual scores improve, but there was also greater consistency across the group's performance. This consistency highlights that the podcast-based teaching method can create an effective and stable learning experience for all students, regardless of individual differences.

The normality test results using Shapiro-Wilk show that both the pre-test and post-test scores meet the assumption of normal distribution ($p\text{-value} > 0.05$). This indicates that the data is valid and can be used for statistical analysis requiring normal distribution assumptions, such as the Paired Sample T-Test. With high data validity, this study can draw robust and reliable conclusions about the effectiveness of the teaching method using Spotify podcasts.

The test of data homogeneity also yielded positive results, with a significant value of 0.428, indicating that the variance distribution across groups is consistent. With this assumption of homogeneity met, the Paired Sample T-Test can be conducted without any issues. The Paired Sample T-Test results show a $p\text{-value}$ of 0.000, which means there is a significant difference between pre-test and post-test scores. Therefore, it can be concluded that the Spotify podcast-based teaching method not only improves students' listening skills but also has a significant impact on the learning process.

Moreover, the N-Gain test results show an average N-Gain of 0.6164, which falls into the moderate category. This indicates that the podcast-based teaching method offers meaningful progress in students' listening comprehension, although there is still room for further improvement. N-Gain values ranging from 20% to 85.71% show that most students experienced significant progress. However, this study also suggests potential enhancements to the method through adjustments in

teaching techniques, materials, or the integration of additional media alongside Spotify podcasts.

The use of Spotify podcasts as a teaching medium highlights the relevance of modern technological methods in education. This media not only makes the learning process more engaging but also provides access to high-quality audio content that can be tailored to students' specific learning needs. It also addresses challenges present in traditional teaching methods, such as limited resources and teacher constraints, while increasing student engagement in learning activities. Overall, the findings of this study indicate that the implementation of Spotify podcasts as a teaching medium was successful in improving the listening skills of Class X IPA II students at SMAN 1 Samalanga. The significant increase in average scores, the improved consistency in students' performance, and the statistical results confirming normality, homogeneity, and meaningful differences between pre-test and post-test scores all support the effectiveness of this method. Additionally, the moderate N-Gain score further demonstrates that the podcast-based teaching approach provided measurable progress in students' listening comprehension. These outcomes confirm the success of the research in achieving its objectives, proving that Spotify podcasts are a valuable and effective tool for enhancing listening skills, while also highlighting their potential for broader application in language learning contexts.

CONCLUSION

Based on the findings of this research, it can be concluded that the mean score of students' listening comprehension significantly increased after using Spotify podcasts as a teaching tool. The comparison of pre-test and post-test scores reveals a substantial improvement in the average listening comprehension score from 61.14 to 81.46, indicating that Spotify podcasts effectively enhance students' ability to understand spoken English. This improvement demonstrates that incorporating digital media into the curriculum, specifically Spotify podcasts, can provide a valuable method for developing listening comprehension skills among high school students.

Furthermore, the study confirms that there is a significant positive effect of using Spotify podcasts on improving the listening skills of X IPA II students at SMAN 1 Samalanga. The results of the statistical analyses, including paired sample t-tests and N-Gain calculations, support the effectiveness of this method. The consistent increase in scores across the group suggests that the method benefits all students equally, fostering a more inclusive and stable learning environment. Therefore, integrating Spotify podcasts into teaching strategies offers a practical and efficient approach for educators to improve listening comprehension, while also highlighting the potential of digital media as a valuable educational tool in modern classrooms.

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THE EFFECTIVENESS OF ANIMATED VIDEO IN STUDENTS' SPEAKING SKILL (QUASI EXPERIMENTAL RESEARCH AT MTSN 6 ACEH UTARA)

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ABSTRACT

Speaking is a fundamental component of English language learning, yet many students in English as a Foreign Language (EFL) contexts continue to face difficulties developing this skill. These challenges often arise from limited exposure, low motivation, and the dominance of traditional, text-based teaching methods that fail to engage learners effectively. This study aims to examine the effectiveness of using animated videos as instructional media to enhance students' speaking skills. The research employed a quasi-experimental quantitative design involving 28 eighth-grade students at MTsN 6 Aceh Utara, divided equally into an experimental group taught through animated video media and a control group taught using conventional text-based learning. Data were collected through pre-tests and post-tests assessing five components of speaking performance: grammar, vocabulary, comprehension, fluency, and pronunciation. Statistical analysis using a t-test revealed a significant improvement in the experimental group's post-test mean score (73.71) compared to the control group (67.43), with a t-value of 2.07 exceeding the critical value of 1.70, indicating a significant difference between the two groups. The findings demonstrate that animated videos provide a dynamic, multisensory learning experience that enhances students' engagement, comprehension, and communicative competence. In conclusion, the integration of animated videos is an effective pedagogical strategy for improving EFL students' speaking performance and can be adopted to create more interactive and motivating English learning environments.

Keywords: *The effectiveness, Speaking Skill, Animated Video.*

INTRODUCTION

English serves as a global language that connects people across nations in education, technology, and international communication. In Indonesia, English proficiency has become a key competency for students to compete in the global era and to participate effectively in global academic and professional settings. Among the four essential language skills, namely listening, speaking, reading, and writing, speaking is considered the most complex because it requires learners to think quickly, manage vocabulary, and use grammar accurately in real-time interaction (Hongwilai and Kawboonsai, 2020). However, in the context of English as a Foreign Language (EFL), many students still experience difficulties in speaking fluently and confidently due to limited language exposure, low motivation, and monotonous teaching methods (Hakim in Puspitaningrum, 2021).

Several studies have emphasized that the use of engaging and contextual learning media can improve students' speaking performance. Audiovisual media, particularly animated videos, can enhance students' motivation and facilitate better comprehension by integrating sound, visuals, and non-verbal expressions (Karo Karo and Karo Rohani,

2022). Previous studies conducted by Ramadhani et al. (2020), Setianingrum et al. (2023), and Nasution (2022) demonstrated that animated videos have a positive impact on speaking fluency, pronunciation, and learners' enthusiasm. Nevertheless, most of these studies employed pre-experimental research designs with relatively small sample sizes, which may limit the generalizability of their findings.

The scientific novelty of this study lies in the application of a quasi-experimental design that compares the effects of animated video media and text-based learning on students' speaking skills. This approach allows for a more systematic comparison of instructional media in enhancing students' speaking performance. In addition, this research introduces standardized assessment criteria that encompass five speaking components, namely grammar, vocabulary, comprehension, fluency, and pronunciation, to produce more reliable and measurable outcomes.

Based on these considerations, this study addresses the following research problem, namely, whether the use of animated video media significantly improves students' speaking ability compared to text-based learning. This research question is formulated to examine the comparative effectiveness of different instructional media in supporting students' speaking development. Therefore, the objective of this research is to determine the effectiveness of animated videos in enhancing students' speaking performance at MTsN 6 Aceh Utara and to provide practical insights into the integration of multimedia-based instruction in EFL classrooms at the junior high school level.

METHOD

This study employed a quantitative approach using a quasi-experimental design to examine the effect of animated video media on students' speaking skills. The research was conducted at MTsN 6 Aceh Utara, involving 28 eighth-grade students divided into two groups: an experimental group (14 students) taught through animated video media, and a control group (14 students) taught using traditional text-based learning. Experimental research is divided into four types: (1) pre-experimental research, (2) Real Experimental research, (3) Quasi-Experimental research, (4) Factorial Experimental research, book citation (Sugiono, 2018). The quasi-experimental design was selected because it allows comparison between groups under natural classroom conditions without random assignment, providing practical validity for educational settings.

The instruments used in this study consisted of pre-tests and post-tests designed to measure students' speaking performance. The tests evaluated five components: grammar, vocabulary, comprehension, fluency, and pronunciation, book citation (Brown 2003). The experimental procedure consisted of three main stages: (1) administering the pre-test to both groups, (2) implementing the treatment, animated video-based instruction for the experimental group and text-based learning for the control group over three meetings, and (3) conducting the post-test to measure improvement after treatment.

Data collected from both groups were analyzed statistically to determine whether the use of animated videos produced a significant difference in speaking achievement. The data analysis involved testing for normality and homogeneity to ensure that the data met the assumptions for parametric testing. Subsequently, an independent samples t-test was applied to compare the mean scores of the two groups at a significance level of 0.05.

The statistical results were used to test the research hypothesis and determine the effectiveness of animated video media in improving students' speaking skills.

RESULT AND DISCUSSION

The pre-test results showed that the average scores of the experimental and control groups were relatively similar, with mean scores of 55.71 and 50.00, respectively, indicating that both groups had comparable speaking abilities before the treatment. Following the implementation of animated video media, the post-test results demonstrated a substantial difference between the two groups. The experimental group achieved a mean score of 73.71, while the control group obtained a mean score of 67.43. Furthermore, statistical analysis using an independent samples t-test revealed that the calculated t-value of 2.07 was higher than the t-table value of 1.70 at the 5 percent significance level, confirming that the difference in students' speaking achievement was statistically significant.

The findings suggest that animated video media positively influenced students' speaking performance. Animated videos provided both visual and auditory input that helped students understand pronunciation, vocabulary, and sentence structure within a meaningful context. The dynamic and colorful visuals captured students' attention, while contextual dialogues enabled students to practice speaking more naturally and confidently. This multimodal exposure supported better comprehension and language retention, which is consistent with multimedia learning theory, stating that learners process information more effectively when verbal and visual materials are combined (Mayer, 2009).

The improvement observed in the experimental group can also be attributed to increased motivation and reduced anxiety among students. Animated videos presented learning content in an engaging and enjoyable format, which encouraged active participation during speaking activities. This supportive learning atmosphere helped reduce students' fear of making mistakes and fostered greater confidence in expressing their ideas orally.

CONCLUSION

This study concludes that the use of animated video media is significantly effective in improving students' speaking skills. Students who were taught through animated videos achieved higher post-test scores and demonstrated better fluency, pronunciation, and vocabulary use compared to those taught using traditional text-based methods. Furthermore, animated videos provided both visual and auditory stimuli that helped learners understand language use within meaningful contexts, thereby making speaking practice more interactive and effective.

Animated video media also increased students' motivation and reduced anxiety by creating a fun and supportive learning atmosphere. The integration of animation encouraged active participation and enhanced learners' confidence in expressing ideas orally. Therefore, teachers are encouraged to incorporate animated videos into English instruction as an innovative and engaging medium to enhance speaking competence in EFL classrooms. Future research may explore the use of animated video media in

developing other language skills, such as listening comprehension or interactive speaking tasks, across different educational levels.

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THE EFFECT OF SHORT FILMS WITH BASIC DIALOGUE TO STUDENTS' LISTENING SKILLS

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ABSTRACT

This study investigates the effect of using short films with basic dialogue on the listening skills of seventh-grade students at SMPN 1 Tanah Pasir. The research employed a quantitative approach using a quasi-experimental design involving two groups: an experimental group taught through short films with basic dialogue and a control group taught through conventional methods. Data were collected through pre-tests and post-tests to measure students' listening comprehension performance. The results revealed that the experimental group achieved higher improvement scores compared to the control group, indicating a significant effect of short films with basic dialogue on students' listening skills. The use of visual and auditory media in short films provided contextual support that enhanced students' understanding, attention, and motivation during learning. These findings demonstrate that integrating short films with basic dialogue into English lessons can serve as an effective strategy to improve listening comprehension, particularly for junior high school students. Therefore, teachers are encouraged to apply this media in their teaching process to create more engaging and interactive learning experiences that promote better language acquisition.

Keywords: *Basic Dialogue; Experimental Research; Listening Skills; Short Films*

INTRODUCTION

Language is one of the most essential aspects of communication that enables people to express their ideas, feelings, and thoughts effectively. Among the international languages, English holds a significant role as a global language used in education, science, technology, and international relations. In Indonesia, English is taught as a compulsory subject from elementary to higher education levels to prepare students for global interaction and academic advancement. As a fundamental aspect of language learning, students are expected to master four basic skills—listening, speaking, reading, and writing—which are equally important for achieving communicative competence.

However, listening has been identified as one of the most challenging skills for students learning English as a foreign language. Observations at SMPN 1 Tanah Pasir revealed that many students experience difficulties in understanding spoken English due to limited exposure, a lack of vocabulary, and low motivation. Listening often receives less emphasis compared to other skills, such as reading and writing (Maulida, 2021). As stated by Gilakjani and Ahmadi (2011), listening is a vital component of communication and provides essential language input for learners to acquire the target language effectively. Therefore, improving listening comprehension is crucial for developing students' overall English proficiency.

To address these difficulties, teachers need to implement innovative and engaging learning media. One promising approach is the use of short films with basic dialogue.

According to Gagné and Briggs in Wilansari et al. (2023), learning media such as videos, films, and pictures can help students understand the material more easily by providing concrete examples and visual support. Short films combine both auditory and visual elements, allowing students to observe language in context and associate sounds with actions or expressions. Harmer (2001) explains that films and videos allow learners to see gestures and facial expressions that accompany spoken words, which enhances comprehension.

Recent research also supports the use of short films as visual media in language learning. Faramarzi and Abbasian (2020) found that using short English films containing simple dialogues significantly improved students' listening comprehension by providing contextual support and increasing attention and motivation. Similarly, Zohraeni and Iman (2017) reported that short English films effectively enhanced students' listening comprehension compared to conventional instruction.

Several studies have demonstrated the effectiveness of film-based learning in improving students' listening skills. Zohraeni and Iman (2017) found that short English films significantly improved students' listening comprehension compared to conventional teaching. Similarly, research by Pakpahan et al. (2023) showed that audiovisual media increased students' motivation and listening accuracy. However, most of these studies used general film materials without considering the linguistic level of junior high school learners. This study fills that gap by focusing on short films with basic dialogue that are simple, repetitive, and contextually relevant to beginner learners.

The novelty of this research lies in its emphasis on short films that contain basic English dialogue as a medium to improve listening comprehension among seventh-grade students. This approach integrates audio and visual elements that stimulate both cognitive and affective engagement, making the learning process more meaningful and enjoyable. By aligning the media content with students' language proficiency level, short films are expected to enhance their comprehension, focus, and retention.

Therefore, the purpose of this research is to determine the effect of short films with basic dialogue on students' listening skills at SMPN 1 Tanah Pasir. The findings are expected to contribute to the development of effective and innovative media-based strategies in teaching English, particularly in improving students' listening comprehension at the junior high school level.

METHOD

This research employed a quantitative approach with a quasi-experimental design to determine the effect of using short films with basic dialogue on students' listening skills. Two groups were involved: an experimental group taught through short films and a control group taught using conventional methods. Both groups received the same lesson duration and content, differing only in the instructional media used.

The population consisted of seventh-grade students of SMPN 1 Tanah Pasir in the academic year 2024/2025, totaling 125 students divided into four classes. Using simple random sampling, two classes were selected: one as the experimental class and one as the control class.

The instrument used was a listening comprehension test in multiple-choice form, administered as a pre-test and post-test. The pre-test measured students' initial listening ability, while the post-test evaluated their improvement after treatment. The treatment was conducted using two short animated films, *Days of the Week with Sid and His Sheep* and *The Very Hungry Caterpillar*, which contain simple, repetitive dialogues suitable for beginner learners. The learning process followed three stages—pre-teaching, whilst-teaching, and post-teaching—to guide students in understanding the content and developing comprehension skills.

The data obtained from both groups were analyzed quantitatively. Normality and homogeneity tests were applied to ensure data validity, followed by an independent sample t-test to determine whether there was a significant difference between the post-test scores of the experimental and control groups. The significance level was set at 0.05, meaning that if the p-value was lower than 0.05, the use of short films with basic dialogue had a significant positive effect on students' listening skills.

RESULT AND DISCUSSION

The results of this study show that the use of short films with basic dialogue had a significant effect on students' listening comprehension. The data were obtained from the pre-test and post-test administered to both the experimental and control groups. The experimental group, which was taught through short films, showed a remarkable improvement in their listening scores, with the mean score increasing from 61.33 to 85.00, while the control group increased only from 60.67 to 72.67. These findings indicate that short films with basic dialogue effectively enhanced students' ability to comprehend spoken English.

The results of the independent sample t-test revealed a t-value of 7.46 and a significance value (p) of 0.000, which is lower than 0.05. Therefore, the null hypothesis (H_0) was rejected, and the alternative hypothesis (H_1) was accepted, confirming that there was a significant difference between students taught through short films and those taught through conventional methods. This proves that using short films with simple and repetitive dialogue improves students' listening skills more effectively than traditional techniques.

The improvement occurred because the short films provided both visual and auditory inputs, allowing students to connect spoken words with contextual actions and facial expressions. This aligns with Harmer's (2001) statement that visual elements in films help learners interpret meaning more easily. Furthermore, the authentic dialogues presented in the films encouraged students to focus on pronunciation, intonation, and vocabulary in a meaningful context.

These findings are consistent with Zohraeni and Iman (2017) and Pakpahan et al. (2023), who found that the use of short films and audiovisual media increased students' motivation and comprehension. Faramarzi and Abbasian (2020) also emphasized that short English films containing simple dialogues can strengthen students' attention and retention by engaging both visual and auditory channels, leading to better comprehension outcomes. This is in line with Gagné's theory of instructional design, which highlights

that effective learning occurs when multiple sensory channels are activated simultaneously.

In addition, this study supports Rahmawati and Sari (2022), who demonstrated that the use of English short films effectively improved junior high school students' listening motivation and comprehension. Taken together, these findings confirm that integrating short films with basic dialogue into listening activities can make English learning more effective, engaging, and meaningful for students.

In summary, the results demonstrate that integrating short films with basic dialogue into English listening lessons significantly improves students' comprehension, motivation, and engagement. This approach not only supports students' linguistic development but also creates a more enjoyable and effective classroom environment.

CONCLUSION

Based on the findings of this study, it can be concluded that the use of short films with basic dialogue has a significant positive effect on students' listening skills. The results of the pre-test and post-test showed that students in the experimental group who were taught using short films achieved greater improvement compared to those in the control group taught through conventional methods. This indicates that short films, which combine visual and auditory elements, effectively enhance students' comprehension by providing meaningful context and stimulating motivation during the learning process. The simple and repetitive dialogues presented in the films help students understand pronunciation, vocabulary, and sentence structure more easily, leading to better listening performance. Therefore, integrating short films with basic dialogue into English language teaching can serve as an effective instructional strategy, particularly for junior high school students who are still developing their basic listening abilities. Future research may explore the use of different genres or longer films, as well as examine how students' learning styles and proficiency levels influence the effectiveness of audiovisual materials in English language learning.

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TEACHERS' APPROACHES IN TEACHING TO ENHANCE STUDENTS' ENGLISH-SPEAKING SKILL

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ABSTRACT

This study, entitled “Teachers’ Approaches in Teaching to Enhance Students’ English-Speaking Skill,” investigates how English teachers at SDIT Quantum School support students’ oral development in the 4th and 6th grades. Speaking is often considered the most challenging skill for Indonesian learners due to limited exposure, large class sizes, and a focus on grammatical accuracy. However, observations at SDIT Quantum School revealed that many students were confident in using English, leading to the question of what teaching approaches were applied. This descriptive qualitative study used classroom observation, field notes, and interviews to collect data. The findings show that teachers implemented Communicative Language Teaching (CLT), Total Physical Response (TPR), Task-Based Learning (TBL), Audio Lingual Approach, and Learner-Centered Approach. These approaches created engaging and interactive classrooms that encouraged students to speak English fluently and confidently, while peer collaboration and multimedia resources enhanced motivation and vocabulary growth. However, limited real-time feedback affected speaking accuracy. The study concludes that integrating communicative and learner-centered approaches effectively develops young learners’ speaking competence, and timely feedback should be emphasized for greater accuracy and long-term oral proficiency.

Keywords: *Communicative Approach; English Language Teaching; Speaking Skill; Teaching Approaches; Young Learners*

INTRODUCTION

English-speaking ability is an essential skill for learners in the global era. Despite years of instruction, many Indonesian students continue to experience difficulties in developing oral proficiency due to limited opportunities for practice and language exposure. Previous studies by Brown (2007) and Richards and Renandya (2002) emphasize that teaching approaches play a significant role in influencing students’ speaking performance. Approaches such as Communicative Language Teaching, Task-Based Learning, and Total Physical Response are considered effective in promoting meaningful communication and reducing learners’ anxiety in speaking classes.

Recent research by Chen (2024), Torres (2019), and Nguyen and Bui (2021) confirms that the integration of communicative and learner-centered strategies fosters fluency, interaction, and confidence among EFL learners. However, there remains a limited understanding of how these approaches are implemented in real Indonesian classroom contexts, particularly at the elementary school level. The novelty of this study lies in exploring how multiple teaching approaches, namely Communicative Language Teaching, Total Physical Response, Task-Based Learning, Audio Lingual Method, and learner-centered approaches, are integrated by teachers to enhance English speaking skills among young learners at SDIT Quantum School. This research seeks to answer the

question of what teaching approaches are used by English teachers in Grades 4 and 6 at SDIT Quantum School to improve students' English speaking skills. Therefore, this study aims to analyze the implementation of various instructional approaches in improving speaking proficiency among elementary school students.

METHOD

This study employed a descriptive qualitative research design, which is appropriate for exploring teaching practices within natural classroom settings. Data were collected through classroom observations, field notes, and semi-structured interviews conducted with two English teachers who taught Grades 4 and 6 at SDIT Quantum School in Banda Aceh. This approach enabled the researcher to obtain in-depth information regarding the instructional strategies and teaching approaches used to enhance students' English speaking skills.

Observation checklists and interview guides were used as research instruments to ensure consistency in data collection. Data analysis followed the model proposed by Miles and Huberman (2014), which consists of data condensation, data display, and conclusion drawing. Through this analytical process, the researcher identified recurring teaching approaches and examined how these approaches contributed to the improvement of students' English-speaking skills.

RESULT AND DISCUSSION

The findings revealed that both teachers implemented a combination of teaching approaches that were aligned with students' cognitive levels and language abilities. In Grade 4, the teacher used songs, repetition, and simple dialogues to develop students' vocabulary and pronunciation by applying Communicative Language Teaching, Audio Lingual Method, and Total Physical Response. In contrast, the Grade 6 teacher employed Communicative Language Teaching, Task-Based Learning, and learner-centered approaches by engaging students in group projects, presentations, and classroom discussions.

Communicative Language Teaching (CLT)

Both teachers applied Communicative Language Teaching through pair work, role plays, and class discussions to promote meaningful communication among students. These activities encouraged learners to actively use English in interactive classroom situations and to develop greater confidence in speaking. This finding is consistent with Chen (2024), who reported that Communicative Language Teaching is effective in increasing students' fluency and classroom participation.

Total Physical Response (TPR)

Commands such as "stand up" and "pick up the book," accompanied by physical gestures, helped young learners connect physical movement with language input. This approach enabled students to understand meaning through actions, which reduced anxiety and supported a more relaxed learning atmosphere. As a result, learners demonstrated

improved comprehension and greater willingness to participate in English-speaking activities.

Task-Based Learning (TBL)

Students were assigned real-life tasks such as narrating their daily routines and creating family trees to encourage the use of English in meaningful contexts. These activities promoted functional language use and supported contextual learning during speaking practice. This finding is consistent with Torres (2019), who reported that task-based instruction enhances learners' speaking fluency and motivation.

Audio Lingual and Learner-Centered Approaches

The Audio Lingual Approach helped reinforce students' pronunciation and grammatical accuracy through listening activities and structured repetition. This approach enabled learners to internalize correct language patterns before producing spoken language. In addition, the Learner-Centered Approach promoted learner autonomy by connecting learning topics with students' personal interests, such as sports and hobbies, which increased engagement during speaking activities.

Overall, the combination of these approaches fostered fluency, confidence, and collaboration among learners. However, a recurring limitation observed was the lack of immediate corrective feedback during speaking activities. Teachers tended to focus on fluency over accuracy, echoing Burns' (2019) assertion that feedback is essential for long-term language accuracy. Thus, while the implementation of multiple approaches successfully built students' confidence and fluency, future improvement should focus on integrating structured formative feedback to balance fluency with linguistic precision.

CONCLUSION

This study concludes that English teachers at SDIT Quantum School effectively employed multiple teaching approaches, namely Communicative Language Teaching, Total Physical Response, Task-Based Learning, Audio Lingual Approach, and learner-centered instruction, to enhance students' English speaking skills. These approaches fostered an engaging learning environment that promoted speaking fluency, learning motivation, and confidence among elementary school learners. However, the findings also revealed that limited corrective feedback during speaking activities remains a pedagogical challenge. Therefore, it is recommended that teachers incorporate more real-time feedback to improve students' grammatical accuracy and pronunciation. Overall, the integration of communicative and learner-centered methods is proven to be a powerful strategy for developing English speaking competence in elementary educational contexts.

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THE CORRELATION BETWEEN LEARNER AUTONOMY AND SPEAKING ACHIEVEMENT

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ABSTRACT

This study investigates the correlation between learner autonomy and speaking achievement among English Department students at UIN Sultanah Nahrasiyah Lhokseumawe. A quantitative correlational design was employed with 33 students from the 4th, 6th, and 8th semesters. Learner autonomy was measured using a 23-item questionnaire adapted from Oxford's framework, while speaking achievement was assessed from students' final exam scores in the Speaking in Academic Context course. Data were analyzed using Pearson Product-Moment correlation and a t-test with SPSS 27. The findings revealed a significant positive correlation ($r = 0.64$, $p < 0.05$) between learner autonomy and speaking achievement, with t-score (4.68) exceeding t-table (2.040). These results indicate that higher learner autonomy is associated with better speaking performance. The study highlights the importance of fostering autonomy in English language learning to improve speaking proficiency.

Keywords: *Learner autonomy; Speaking achievement; Correlational research*

INTRODUCTION

Speaking is one of the most important skills in learning a foreign language, as it reflects learners' ability to communicate and use the language effectively in real-life contexts. In Indonesia, speaking English remains a major challenge for many students, particularly at the university level, due to limited opportunities for practice, low confidence, and a high dependence on teacher guidance. As a result, many students struggle to reach their full potential in oral communication.

Learner autonomy, as introduced by Holec (1981), refers to learners' ability to take responsibility for their own learning process. This concept includes making decisions related to learning objectives, selecting appropriate learning strategies, monitoring learning progress, and evaluating learning outcomes. According to Oxford (1990), autonomous learners tend to be more motivated, strategic, and capable of achieving better results in language learning. In particular, learner autonomy has been shown to influence the development of speaking skills, as independent learners are more likely to seek additional practice opportunities and engage in meaningful communication.

Several studies have examined the relationship between learner autonomy and language proficiency. Minh and Ngoc (2020) found that Vietnamese EFL students with higher levels of learner autonomy achieved better speaking performance. Similarly, Nasution (2021) reported a significant correlation between learner autonomy and English proficiency among Indonesian students. In addition, Viola and Munir (2023) emphasized that learner autonomy fosters responsibility and confidence, which directly contribute to learners' language competence.

Despite these findings, research that specifically focuses on the correlation between learner autonomy and speaking achievement within the context of Indonesian Islamic universities remains limited. Most previous studies have tended to examine general language proficiency rather than speaking skills in particular. Therefore, this study aims to address this research gap by investigating the correlation between learner autonomy and speaking achievement among students of the English Department at UIN Sultanah Nahrasiyah Lhokseumawe.

METHOD

This research employed a quantitative correlational design to examine the relationship between learner autonomy and speaking achievement. The population consisted of 130 English Department students at UIN Sultanah Nahrasiyah Lhokseumawe, from which 33 students were randomly selected as the research sample. Learner autonomy was measured using a questionnaire that covered cognitive, metacognitive, affective, motivational, and confidence-related aspects. Speaking achievement data were obtained from students' final scores in the Speaking in Academic Context course, which were assessed based on fluency, accuracy, vocabulary, pronunciation, and comprehensibility. The data were analyzed using Pearson product-moment correlation and a t-test with the assistance of SPSS version 27.

RESULT AND DISCUSSION

The analysis showed that the students' average learner autonomy score was 91.15, while their average speaking achievement score was 91.39, both of which were categorized as very good. Pearson correlation analysis revealed a strong positive correlation between learner autonomy and speaking achievement, with a correlation coefficient of $r = 0.64$. The t-test further confirmed this result, as the obtained t value of 4.68 exceeded the t-table value of 2.040 at the 0.05 significance level. Therefore, the alternative hypothesis was accepted, indicating that learner autonomy is significantly correlated with students' speaking achievement. These findings are consistent with previous studies that reported significant positive relationships between learner autonomy and English language proficiency.

CONCLUSION

This study concludes that there is a significant, positive, and strong correlation between learner autonomy and speaking achievement among English Department students at UIN Sultanah Nahrasiyah Lhokseumawe. The findings indicate that higher levels of learner autonomy are associated with better speaking performance. Therefore, fostering learner autonomy can contribute to the improvement of students' speaking proficiency and should be considered an essential component in English language teaching and learning.

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STUDENTS' STRATEGIES IN REDUCING SPEAKING ENGLISH ANXIETY AT SENIOR HIGH SCHOOL (A DESCRIPTIVE QUALITATIVE AT ELEVENTH GRADE OF SMKN 8 LHOKSEUMAWE)

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ABSTRACT

Speaking anxiety is one of the common challenges faced by students in learning a foreign language, especially when required to speak English in front of others or in formal classroom situations. This condition often hinders their ability to communicate effectively and affects their overall learning performance. This study aims to identify the types of anxiety experienced by students when speaking English and to explore the strategies they use to reduce that anxiety. The research employed a descriptive qualitative method with observation and in-depth interviews as the main instruments. The participants were eleventh-grade students of SMKN 8 Lhokseumawe selected through purposive sampling based on the level and frequency of anxiety indicators. The observation of 29 students indicated that they mostly experienced state anxiety, which is temporary in nature. Furthermore, interviews with eight selected students revealed that four of them experienced trait anxiety and applied several strategies to overcome it, namely preparation, relaxation, positive thinking, seeking peers, ignoring the audience, dealing with social awkwardness, and sticking to something. The findings indicate that these strategies help students to manage their fear, build self-confidence, and enhance their English-speaking performance in the classroom.

Keywords: *Anxiety; Reducing; Strategies; Speaking.*

INTRODUCTION

Speaking is one of the most essential skills in learning English because it allows students to communicate and express their ideas effectively. However, many students face difficulties when speaking English due to fear of making mistakes, low self-confidence, and limited vocabulary. This condition often leads to speaking anxiety, which negatively affects students' participation and oral performance in class.

Previous studies, such as Horwitz et al. (1986), Liu and Jackson (2008), and Tien (2018), have shown that speaking anxiety is a common problem in foreign language learning, caused by factors such as fear of negative evaluation, communication apprehension, and test anxiety. Some researchers also emphasized that reducing anxiety can improve students' speaking ability and motivation. Nevertheless, each learner may apply different strategies to overcome their anxiety, depending on their experience and learning context.

The novelty of this research lies in identifying specific strategies used by Indonesian high school students to reduce their English-speaking anxiety within a vocational school context, which is still rarely explored. Unlike previous research focusing mainly on university students, this study highlights high school learners' coping mechanisms and their classroom experiences in managing speaking anxiety.

This research addresses two main questions concerning students' English-speaking anxiety. The first question focuses on identifying the types of speaking anxiety experienced by students, while the second question examines the strategies, they use to reduce their anxiety when speaking English. Therefore, this study aims to investigate the types of speaking anxiety faced by students at SMKN 8 Lhokseumawe and to describe the strategies they employ to manage and reduce their English-speaking anxiety.

METHOD

This study employed a qualitative descriptive research design to describe students' strategies in reducing English-speaking anxiety. The research was conducted at SMKN 8 Lhokseumawe with eleventh-grade students of the Office Automation and Governance program as participants. The main data source consisted of students' responses and opinions gathered through interviews, observations, and documentation.

The data collection process involved classroom observations to identify students' speaking anxiety behaviors during English learning activities. These observations were followed by semi-structured interviews conducted to explore the strategies students use to reduce their speaking anxiety. In addition, documentation in the form of photographs and field notes was used to support and strengthen the research findings.

The collected data were analyzed using Miles and Huberman's (1994) interactive model, which includes data reduction, data display, and conclusion drawing or verification. This approach enabled the researcher to organize the information systematically and draw conclusions about the types of anxiety experienced and the strategies used by students to manage it.

RESULT AND DISCUSSION

This research was conducted with 29 second-grade students at SMKN 8 Lhokseumawe to identify the types of anxiety experienced by students when speaking English. The study also aimed to explore the strategies used by students to reduce their speaking anxiety in classroom contexts. Data were collected through structured classroom observations and in-depth interviews to obtain comprehensive and reliable qualitative information.

Based on the observations, two main types of anxiety were identified among students: state anxiety and trait anxiety. State anxiety appeared dominantly, characterized by temporary nervousness, hesitation, or panic occurring during speaking activities and subsiding afterward. Trait anxiety, on the other hand, reflected a more consistent pattern of anxiety across various situations, not only during speaking in class.

Out of 29 students, most displayed state anxiety indicators such as fidgeting, trembling voices, avoiding eye contact, or hesitation before speaking. Meanwhile, four students (KZ, MA, FS, and CM) were identified as experiencing trait anxiety, showing prolonged worry, restlessness, and difficulty concentrating even beyond English classes.

Types of Students' Anxiety in Speaking English

The findings confirm that state anxiety is the most prevalent type among students learning English as a foreign language. This type of anxiety arises from situational

pressures, such as being suddenly asked to speak in front of the class or fear of negative evaluation. These symptoms — nervous gestures, trembling voices, or hesitation — are consistent with the theory proposed by Spielberger (1983), who defines state anxiety as a transient emotional condition triggered by specific situations perceived as threatening.

This situational anxiety can be explained through Horwitz et al. (1986), who argue that foreign language learners often experience “communication apprehension” and “fear of negative evaluation.” In this study, students’ nervousness and silence before speaking reflected exactly those fears. The dominant tendency for students to become calm after the speaking task (22 out of 29) reinforces the idea that their anxiety is temporary and context-dependent — it appears during performance but fades afterward.

In contrast, trait anxiety observed in four students represents a deeper psychological pattern. According to Endler & Kocovski (2001), trait anxiety is a stable personality characteristic causing individuals to respond with anxiety across different situations. The interviewed students reported persistent feelings of nervousness and overthinking even outside the classroom, which indicates internalized self-doubt rather than situational stress. This supports Ford & Courtois (2014), who emphasized that chronic anxiety can interfere with focus and academic performance if not managed properly.

Thus, the findings answer the first research question: students at SMKN 8 Lhokseumawe experience both state and trait anxiety, with state anxiety being more dominant. This aligns with previous studies by MacIntyre & Gardner (1991) and Rapee et al. (2005), who also found that language learners mostly suffer from temporary, context-based anxiety rather than long-term emotional tendencies.

Students’ Strategies to Reduce Speaking Anxiety

The study also found that students actively developed seven coping strategies to manage their anxiety when speaking English. These strategies reflect both cognitive and behavioral approaches.

1. Preparation

Students who practiced or reviewed materials beforehand showed reduced anxiety. This supports Kondo & Ying-Ling (2004), who classified preparation as one of the most effective coping strategies in language learning. By preparing, students build self-confidence and a sense of control over their performance.

2. Relaxation

Some students employed physical relaxation techniques, such as deep breathing, before engaging in speaking activities. These techniques helped students reduce tension and achieve a calmer state prior to oral performance. This finding is consistent with Benson and Klipper (1975), who demonstrated that relaxation responses can effectively lower physiological arousal during anxiety-provoking situations.

3. Positive Thinking

Students who reminded themselves that making mistakes is part of learning could minimize the fear of evaluation. This is supported by Beck’s (1976) Cognitive Theory, which emphasizes reframing negative thought patterns as a way to regulate emotional distress.

4. Seeking Peer Support

Students who discussed their speaking anxiety with friends reported experiencing emotional relief and increased comfort. Peer interaction provided opportunities for sharing experiences and gaining reassurance from others facing similar challenges. This finding reflects Bandura's Social Learning Theory (1977), which emphasizes the role of social support and modeling as important components of emotional regulation.

5. Ignoring the Audience and Social Awkwardness Strategy

Some students reported using mental strategies such as pretending that the audience did not exist or imagining them as inanimate objects during speaking activities. This technique allowed students to reduce self-focused attention and manage feelings of nervousness when speaking in front of others. This finding is consistent with Hofmann and Otto (2015), who describe this strategy as cognitive distancing, which helps individuals detach emotionally from perceived social threats and maintain composure during performance.

6. Holding Something

Holding an object, such as a piece of paper or a pen, provided students with a form of distraction and a sense of security during speaking activities. This physical engagement helped reduce nervousness and allowed students to focus more effectively on their speech. This finding is consistent with Horwitz et al. (1986), who reported that physical engagement can reduce tension during foreign language performance.

These strategies collectively demonstrate that students employ both traditional and innovative methods to manage their anxiety. Compared with Darmaida (2016) and Ani & Bambang (2023), who also found preparation, relaxation, and positive thinking as dominant coping strategies, this study adds three newly identified behaviors (ignoring the audience, social awkwardness, and holding objects), expanding the typology of coping methods among vocational high school students.

Scientific Implications

The findings indicate that speaking anxiety among EFL students is multifaceted, involving both temporary emotional states and long-term psychological tendencies. The study supports the hypothesis stated in the introduction — that students experience different types of anxiety and adopt diverse strategies to cope with them. Scientifically, it contributes to a deeper understanding that reducing foreign language anxiety requires both emotional regulation training (e.g., relaxation, positive thinking) and contextual classroom strategies (e.g., creating a supportive, non-judgmental environment).

CONCLUSION

This study aimed to identify the types of anxiety experienced by students when speaking English and to explore the strategies they applied to overcome that anxiety. The findings reveal that the majority of students at SMKN 8 Lhokseumawe experienced state anxiety, characterized by temporary nervousness and hesitation that appeared during speaking tasks and diminished afterward. A smaller number of students experienced trait anxiety, which persisted across learning situations and was associated with a deeper sense of self-doubt. These results indicate that English-speaking anxiety among vocational high

school students is primarily situational rather than dispositional, suggesting that contextual teaching practices play a central role in reducing anxiety.

Furthermore, the study discovered that students developed varied and creative coping strategies to manage their anxiety, including preparation, relaxation, positive thinking, seeking peer support, ignoring the audience, imagining the audience as inanimate objects, and holding an object to calm themselves. These strategies reflect both cognitive and behavioral efforts that enable students to regulate their emotions and perform better during speaking activities. The emergence of unique behaviors such as “ignoring the audience” and “holding an object” contributes a new perspective to the understanding of anxiety regulation in EFL contexts, particularly among vocational learners.

In conclusion, this research confirms the hypothesis that students experience different types of anxiety when speaking English and that they consciously employ multiple strategies to overcome it. The results underscore the importance of creating a supportive, low-anxiety classroom environment, where teachers provide psychological encouragement and emphasize that making mistakes is part of the learning process. Future studies may further investigate how instructional methods, teacher feedback, or digital media integration can enhance students’ self-confidence and reduce language anxiety more effectively.

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THE EFFECTIVENESS OF WORDWALL EDUCATIONAL GAME IN TEACHING THE SIMPLE PAST TENSE

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ABSTRACT

This paper was conducted to address the lack of student engagement and understanding in learning the simple past tense. The purpose of this study is to examine the effectiveness of Wordwall educational games in teaching the simple past tense. The approach used is quantitative with a quasi-experimental design, specifically the nonequivalent control group design. The sample consisted of 60 students from MTsS YMPI Tanjung Balai, divided into two groups: 30 students in the experimental group and 30 students in the control group. The research instruments were a pre-test and a post-test, each consisting of 25 multiple-choice questions. The results show that the experimental group, which used Wordwall games, achieved a higher average post-test score of 81.2, compared to the control group's score of 75.37 using traditional visual media (whiteboard). The t-test analysis showed a t value of 7.15, which is greater than the critical t-table value of 2.001. This indicates a statistically significant difference between the two groups, suggesting that Wordwall has a strong positive effect on students' understanding of the simple past tense.

Keywords: *Simple Past Tense, Wordwall Educational Game*

INTRODUCTION

Education plays a vital role in improving individual quality, as it stimulates the development of students' thinking, attitude, and action skills. In Indonesia, the educational process is conducted through formal, out-of-school, and family channels, with the formal pathway spanning levels from elementary school to university. One essential level within the formal educational structure is junior high school, where students begin to develop essential global skills, including English as an international language. To ensure that the learning process remains relevant, engaging, and focused on student needs, the Indonesian government has implemented a number of periodic curriculum changes, with the most recent update being the Independent Curriculum.

The Independent Curriculum is designed to address the challenges of globalization by preparing students for independence, creativity, and 21st-century competencies. In accordance with Ministerial Regulation No. 12 of 2024 concerning the structure of the junior high school curriculum, English is again on the agenda as an essential subject to enhance students' capabilities in global interactions. Despite the implementation of this policy, Indonesia still struggles to achieve adequate English language proficiency standards among students. The EF reported that ranks Indonesia 34th out of 44 countries in the Asian region, indicating that students' English proficiency remains a persistent issue that needs to be addressed (English Proficiency Index, 2024).

Understanding and applying grammar, especially tenses, is a crucial aspect of mastering English. Among the various tenses, the Simple Past Tense plays a vital role in communicating events or actions that took place and were completed in the past. It is used

to describe activities that began and ended at a specific point in time in the past (Azar, 1993). The use of the Simple Past Tense empowers learners to reconstruct past events and construct chronological narratives based on experiences (Aditya, 2014). Furthermore, the Simple Past Tense is also used to express unreal conditions and polite expressions, demonstrating its flexibility in various contexts (Greenbaum and Quirk, 1990). Structurally, tenses are classified into verbal and nominal forms. Regular verbs usually add -ed or -d (e.g., walk → walked, love → loved), while irregular verbs require memorization due to their unique changes (e.g., go → went, see → saw). Negative and interrogative tenses use the auxiliary did with the base form (e.g., "He doesn't go to school," "Does he go to school?"), while nominal tenses use was and were depending on the subject (e.g., "I am tired," "They are happy"). Mastering these tenses allows students to communicate accurately in recounts, narratives, and everyday conversations, which is essential for both oral and written proficiency.

However, difficulty using the Simple Past Tense remains a common problem among students. Observations conducted at MTsS YMPI Sei Tualang Raso on July 21, 2024, revealed that English teaching largely relies on conventional methods using blackboards, textbooks, and dictionaries. The absence of interactive and technology-based media results in repeated grammatical errors, especially when students attempt to construct sentences in the Simple Past Tense. Interviews with teachers also revealed a lack of pedagogical innovation and limited integration of digital tools, which hinders the effective implementation of the Independent Curriculum in practice.

To address these challenges, integrating technology-based media into the teaching process has become a promising solution. One such tool is Wordwall, an interactive digital platform that allows teachers to design engaging learning activities such as matching games, quizzes, group sorting, and random wheel games. This tool aligns with Piaget's constructivist theory, which promotes active learning, and Mayer's multimedia learning theory, which emphasizes the importance of combining verbal and visual elements (Mayer, 2009). By combining textual and interactive components, Wordwall increases motivation, encourages participation, and supports students in developing a deeper understanding of grammar.

Educational games like Wordwall are intentionally designed to encourage learning through play, enhancing students' problem-solving and critical thinking skills (Clark, 2022). These games provide an alternative to traditional classroom learning by creating a fun and immersive environment where students can actively engage with the learning content. Several important characteristics of educational games include clear learning objectives, structured rules, competition, challenges, fantasy elements, safety, and entertainment (Alessi and Trollip, 2001). These features stimulate curiosity, encourage collaboration, and make the learning process more meaningful.

In English language education, educational games have shown significant potential for improving vocabulary mastery, grammar comprehension, and learning motivation. Wordwall effectively increases students' vocabulary achievement, with an average score of 73.5% (Syamsidar et al., 2023). There was an increase in students' vocabulary scores from 56.43 to 74.67 after using Wordwall (Nurammida et al., 2024). Significant vocabulary gains through a true experimental design demonstrate the media's

ability to enhance engagement and learning outcomes (Shabrina et al., 2023). The use of game-based platforms like Baamboozle increased grammar scores from 72.8 to 91.9, demonstrating the effectiveness of digital gamification in improving language accuracy (Sakdiyah et al., 2024). Platforms like Kahoot! and Quizizz also contributed to improved grammar performance, supporting broader evidence that digital games can facilitate meaningful grammar learning (Nguyen Xuan Hong et al., 2024).

Wordwall itself offers several advantages over other platforms. It provides a variety of customizable templates, such as Match Up, Quiz, Open the Box, and Unjumble, which can be tailored to various topics and difficulty levels. Its accessibility via browsers and mobile devices makes it suitable for a variety of learning environments, including schools with limited resources. Furthermore, Wordwall automatically grades responses, allowing teachers and students to efficiently monitor progress (Gandasari & Prabudiani, 2021). Despite these advantages, Wordwall also has limitations, such as limited features for free accounts, dependence on a stable internet connection, and limited print customization options. Despite this, its overall impact on student engagement and motivation remains substantial, especially when compared to conventional teaching methods.

In the context of this research, Wordwall serves as a digital educational tool aimed at improving students' mastery of the Simple Past Tense. This research questions whether Wordwall has a significant impact compared to traditional visual media in teaching the Simple Past Tense to eighth-grade students at MTsS YMPI Tanjung Balai. Thus, this study not only aims to improve achievement scores but also to identify the extent to which Wordwall improves comprehension and retention of grammatical structures. These findings are expected to provide valuable insights for English teachers in selecting effective learning media and contribute to the development of literature on technology-based grammar learning in the context of the Indonesian Independent Curriculum.

METHOD

This research employed a quantitative approach based on scientific principles such as empiricism, objectivity, measurability, rationality, and systematics. This approach was chosen to produce quantitative data that could be processed and interpreted statistically. The research used a quasi-experimental design because the researchers wanted to compare the effectiveness of two different learning media without changing the existing classroom structure.

The research design employed a nonequivalent control group design, with two groups: an experimental group and a control group. Both groups were given pre-tests and post-tests, but received different treatments. The experimental group was taught using Wordwall media, while the control group used conventional visual media. The results of the post-tests were then compared to determine differences in learning outcomes between the two groups.

The population in this study was all 210 seventh-grade students at MTsS YMPI, spread across seven classes, from VII-1 to VII-7. From this population, a sample of 60 students was drawn using a random sampling technique. Class VII-2, consisting of 13

male and 17 female students, was designated as the experimental group, while class VII-1, also consisting of 13 male and 17 female students, served as the control group.

Research data was collected using a test instrument. The test consisted of a pre-test and a post-test, which served to measure students' abilities before and after learning. The test consisted of 25 multiple-choice questions, with 12 verbal questions and 13 nominal questions, covering positive, negative, and interrogative sentences. The format and difficulty level of the pre-test and post-test were kept the same, but the order of the questions and answer options was randomized to avoid memorization.

Ensuring the quality of the instrument, validity, and reliability tests were conducted. Content validity was determined through expert assessment, while construct validity was tested using the Pearson Product-Moment correlation in Microsoft Excel. A test item was declared valid if the r -count value was greater than the r -table. Meanwhile, instrument reliability was tested using the Kuder Richardson (KR-20) formula. The results of the reliability test were interpreted based on the following criteria: a KR20 value ≥ 0.70 is considered reliable, $0.50-0.69$ is considered moderately reliable, and <0.50 is considered unreliable.

This research procedure consisted of three main stages: preparation, implementation, and completion. In the preparation stage, the researcher observed school and student conditions, developed learning materials, and validated the instrument through a pilot test on eighth-grade students who had studied the same material. The implementation stage began with a pre-test for both groups, followed by the learning process according to the treatment determined by using a wordwall for the experimental group and visual media for the control group, and concluded with a post-test. The completion stage included data processing, analysis of the results, discussion, and drawing conclusions.

Data analysis was conducted using several statistical tests, namely the normality test, the homogeneity test, and the t-test. The normality test was conducted using the Lilliefors test to ensure normal data distribution, with the requirement that $L_{count} < L_{table}$. The homogeneity test was conducted using the F-test, where data were considered homogeneous if $F_h < F_t$. To test the hypothesis, an independent t-test was used, and the alternative hypothesis (H_a) was accepted if $t_{count} > t_{table}$, indicating a significant difference between the learning outcomes of the experimental and control groups.

Furthermore, the researchers calculated the effect size using Cohen's d to determine the extent of the treatment's influence on learning outcomes. Effect size values were interpreted into three categories: small ($E_s < 0.2$), medium ($0.2 \leq E \leq 0.8$), and large ($E_s > 0.8$). Through this systematic procedure and analysis, the study is expected to provide a clear picture of the effectiveness of using Wordwall media compared to conventional visual media in improving students' understanding of the simple past tense.

RESULT AND DISCUSSION

The results of this quasi-experimental research revealed a significant improvement in students' understanding of the simple past tense after being taught using different media. The experimental group, which was taught using Wordwall educational games, showed better progress than the control group, which was taught using visual

media. Before the treatment, the pre-test scores of both groups were relatively similar, with the experimental group scoring an average of 63.4 and the control group scoring 62.7. This indicates that both groups had nearly equal prior knowledge. However, after the treatment, the post-test results demonstrated a clear difference. The experimental group's mean score increased to 84.2, showing a gain of 20.8 points, while the control group only improved to 75.6, with a gain of 12.9 points. These results suggest that students in the experimental group had a better grasp of the material due to the use of interactive and engaging Wordwall activities.

An independent sample t-test was conducted to analyze whether this difference was statistically significant, and the result showed that the t-value was 3.85, which is greater than the t-table value of 2.00 at the 0.05 significance level, with a p-value of $0.000 < 0.05$. This means that the difference between the two groups is statistically significant, and the null hypothesis is rejected. These findings suggest that Wordwall educational games can significantly help students' grammar acquisition, particularly in learning the simple past tense. The interactive nature of the game promotes more active participation, encourages immediate feedback, and allows students to enjoy the learning process while reinforcing their understanding of sentence patterns and irregular verbs. Furthermore, the results of this study support previous research emphasizing the effectiveness of game-based learning in fostering better engagement and improving academic outcomes. Compared to visual media, which tends to be more passive, Wordwall's dynamic and learner-centered features create a more stimulating environment, making grammar instruction more meaningful and less monotonous. Therefore, the integration of educational games like Wordwall can be a valuable strategy for English teachers seeking innovative approaches to grammar instruction.

This study reveals that the use of Wordwall educational games has a significant effect on teaching grammar, particularly the simple past tense, compared to conventional visual media. This finding introduces a new perspective on the application of Wordwall, focusing not merely on vocabulary acquisition as in previous studies, but on the more complex domain of grammar, which students need for meaningful communication. The experimental results demonstrate that students who learned using Wordwall significantly outperformed those taught through visual aids, thus validating the effectiveness of this interactive platform in grammar instruction. While previous research, such as that by Syamsidar et al. (2023), which achieved 73.5% effectiveness in vocabulary aspects like pronunciation, spelling, and meaning, only emphasized vocabulary mastery, this study addresses a broader and deeper linguistic challenge.

Similarly, the works of Nurammida et al. (2024), Wordwall's success in this context can be attributed to its design, which enables immediate feedback and engaging activities that help students correct errors and internalize patterns in the use of the simple past tense, especially irregular verbs, which are often difficult for learners. Statistical analysis through a t-test showed a t-count of 7.15, which is far above the t-table value of 2.001, indicating a highly significant difference between the experimental and control groups. This confirms that learning grammar through educational games like Wordwall is more effective than traditional media such as whiteboards. Supporting this, Prensky's theory in *The Game Generation* highlights that today's learners are more engaged with

digital learning than with passive lecture methods. Krashen's Input Hypothesis further explains that language acquisition is best achieved when learners receive slightly challenging but comprehensible input ($i+1$), which Wordwall facilitates through its interactive tasks. According to Piaget's constructivist theory, learners understand complex information better when they are actively involved, and Wordwall provides that active learning environment by enabling students to engage, respond, and play, rather than passively memorize rules. Additional studies, such as that by Sakdiyah et al. (2024) with Baamboozle, and Nguyen with tools like Kahoot and Quizizz, also show that educational games can significantly support grammar mastery by providing competitive and enjoyable learning experiences.

Therefore, the findings in this study not only reinforce the role of digital games in language instruction but also extend the scope of Wordwall's use into more complex areas of language learning. The contribution of this study lies in its specific examination of grammar teaching, particularly the simple past tense using Wordwall, supported by a quasi-experimental nonequivalent control group design with objective pre- and post-test comparisons. This adds to the academic discourse on digital-based grammar instruction, especially in Indonesian MTs contexts with varying student backgrounds. Nonetheless, the study faces several limitations, including a small sample size, limited generalizability, the need for teacher training in digital tools, a focus only on written grammar skills, and the dependence on internet access and technological infrastructure. Despite these limitations, the study presents a valuable perspective on how Wordwall can transform grammar learning into a more effective and enjoyable process.

CONCLUSION

Based on the results of this study, there is a significant difference in learning outcomes between students who used Wordwall educational games and those who used conventional media such as whiteboards. The experimental group (VII-2) achieved an average post-test score of 81.2, while the control group (VII-1) scored 75.37. This difference is statistically significant, with a t-test value of 7.15 exceeding the t-table value of 2.001 at a 0.05 significance level, indicating that H_a is accepted and H_o is rejected. The use of Wordwall media in learning the simple past tense has a meaningful impact on students' grammar mastery compared to traditional methods. This suggests that Wordwall is effective in making learning more engaging and improving students' understanding. Based on these findings, it is recommended that teachers explore and implement interactive media like Wordwall, students be encouraged to use it independently, schools provide adequate infrastructure and training, and further research be conducted with broader samples or on different grammar topics using mixed methods approaches.

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EXPLORING THE ROLE OF DRILLING TECHNIQUE IN ENHANCING STUDENTS' TRANSLATION SKILLS (A QUALITATIVE DESCRIPTIVE STUDY)

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ABSTRACT

This study, entitled “Exploring the Role of Drilling Technique in Enhancing Students’ Translation Skills (A Qualitative Descriptive Study),” explored the use of the Drilling Technique to improve the translation skills of seventh-grade students at SMPN 6 Lhokseumawe. Translation skills were an important part of language learning, as they helped students understand foreign language texts and express meaning accurately between two languages. One structured technique that supported the development of translation was the Drilling Technique, which originated from the audio-lingual method and involved repeated practice of sentence patterns to help students internalize correct language forms. Many junior high school students faced difficulties in translating texts from English to Indonesian and from Indonesian to English, particularly in vocabulary, diction, and sentence structure. Therefore, this study aimed to explore how the Drilling Technique could enhance students’ translation skills. A qualitative descriptive method was employed, and data were collected through classroom observations and interviews with one English teacher and three purposively selected students. The findings showed that the teacher used several drills, including repetition, substitution, and question and answer techniques, which enhanced vocabulary, diction, sentence structure, and students’ confidence in translation. The study concluded that the Drilling Technique was an effective and practical method for improving translation accuracy, while feedback and consistent practice played a key role in developing fluency and confidence in translation.

Keywords: *Drilling Technique; Qualitative Descriptive Study; Translation Skills*

INTRODUCTION

Language learning is not only about mastering speaking and writing skills but also about understanding and conveying meaning between languages. Translation is an essential part of this process, as it allows learners to comprehend foreign texts and express ideas accurately in another language. According to Newmark (2005), translation involves transferring meaning from one language to another while maintaining the equivalence of message and style. In educational contexts, translation helps students develop deeper comprehension and linguistic awareness by connecting vocabulary, grammar, and context in real communication.

However, many junior high school students still face challenges in translating sentences or texts from English to Indonesian and vice versa. These challenges often include limited vocabulary, inaccurate diction, and incorrect sentence structure. Rahimi and Jabbari (2022) emphasized that lexical inaccuracy and weak grammatical understanding are among the most common problems in students’ translations. In line with this, Ediwarman and Pahamzah (2022) found that students tend to rely on literal word-for-word translation rather than focusing on meaning, which results in unnatural

and ambiguous expressions. These difficulties indicate the need for effective instructional techniques that provide structured and repetitive practice to strengthen translation skills.

One promising approach to address these issues is the Drilling Technique, a method that originates from the Audio-Lingual Method (Richards & Rodgers, 2014). Drilling emphasizes repeated practice of sentence patterns to help students internalize correct language forms automatically. Several studies have shown the effectiveness of drilling in improving linguistic accuracy and fluency (Mendrofa & Wijaya, 2022; Abrar & Widiati, 2024). Nugroho and Indriani (2020) also found that structured repetition through drilling significantly enhanced students' translation accuracy by reinforcing vocabulary recall and sentence formation. Nevertheless, most prior studies applied drilling mainly to speaking skills rather than translation. Therefore, the use of drilling to improve translation ability remains relatively underexplored, especially at the junior high school level.

The novelty of this research lies in its focus on examining how the Drilling Technique contributes to enhancing students' translation accuracy, particularly in vocabulary, diction, and sentence structure, through a qualitative descriptive approach. Unlike previous experimental studies, this research observes the authentic classroom implementation of drilling and explores both teacher and student experiences to provide deeper insights into how repetition can enhance translation skills in real contexts.

Based on these considerations, the main research problem of this study is formulated to examine how the Drilling Technique enhances the translation skills of seventh-grade students at SMPN 6 Lhokseumawe. This research question is intended to investigate the effectiveness of the Drilling Technique in supporting students' ability to translate texts accurately. Therefore, the purpose of this study is to explore the role of the Drilling Technique in improving students' translation skills, particularly in terms of vocabulary accuracy, appropriate diction, and correct sentence structure through qualitative classroom investigation.

METHOD

This study employed a qualitative descriptive method to explore how the Drilling Technique enhances students' translation skills. The qualitative descriptive approach was chosen because it allows the researcher to describe real phenomena as they naturally occur without manipulating any variables. This method was considered suitable for understanding how the Drilling Technique is implemented in the classroom and how it affects students' translation performance.

The research was conducted at SMPN 6 Lhokseumawe, located on Jalan Rancung Km 0.3, Sp. IV Batuphat Timur, Lhokseumawe, Aceh, Indonesia. The study took place over two days, on May 27 and 28, 2025. This school was selected because it has strong academic standards and actively implements English language instruction, yet students were still observed to struggle with translation accuracy, particularly in vocabulary, diction, and sentence structure. The data sources consisted of one English teacher who implemented the Drilling Technique and three seventh-grade students from Class VII-2 who were selected purposively. Purposive sampling was used because these participants were directly involved in the teaching and learning activities that applied the Drilling

Technique. The selected students represented different levels of classroom participation: one active, one moderate, and one less active. This variation was intended to provide a comprehensive picture of students' learning responses.

Data were collected through two main techniques: observation and interview. Observation was conducted to record how the teacher implemented the Drilling Technique in translation lessons and how students participated during classroom activities. The researcher used a structured observation checklist to ensure consistency in data collection. Semi-structured interviews were also conducted with the teacher and students to obtain deeper insights into their experiences, perceptions, and opinions about the effectiveness of the technique. This combination of observation and interview helped to triangulate data and increase validity.

To ensure the credibility of the data, several validation strategies were employed, including triangulation, member checking, and the use of supporting documentation such as photographs and interview transcripts. These strategies were applied to strengthen the trustworthiness and accuracy of the qualitative findings. In addition, reliability was maintained through inter-rater observation, in which a peer observer used the same observation checklist to confirm consistency in the data collection results.

Data analysis followed the framework of Miles, Huberman, and Saldaña (2019), which consists of three interactive stages: data reduction, data display, and conclusion drawing or verification. In data reduction, the researcher selected and simplified the raw data by identifying key themes related to vocabulary accuracy, diction, and sentence structure. The data display stage involved organizing the information into clear descriptions and thematic summaries supported by observation notes and interview quotations. Finally, conclusions were drawn and verified by rechecking the consistency between data sources to ensure that the interpretations were valid and based on evidence.

Through this method, the researcher was able to systematically describe the implementation of the Drilling Technique in the classroom. The analysis also focused on students' responses to the drilling activities during the learning process. Furthermore, the findings illustrated how this technique contributed to improving students' translation accuracy and increasing their confidence in translating texts.

RESULT AND DISCUSSION

The results of this study reveal that the implementation of the Drilling Technique significantly contributed to improving students' translation skills, particularly in the areas of vocabulary accuracy, diction, and sentence structure. The observation results showed that the teacher applied several types of drills, such as repetition, substitution, and question-and-answer exercises. During repetition drills, students repeatedly practiced English sentences after the teacher. This activity helped them internalize correct pronunciation, rhythm, and sentence patterns. Through substitution drills, students were encouraged to replace specific words in a sentence (such as verbs or subjects) with new ones while maintaining the grammatical structure. This practice improved their ability to identify consistent language patterns and use them flexibly in translation. In addition, question-and-answer drills encouraged students to respond actively in English, which

increased their confidence and helped them construct meaning accurately when translating.

The teacher's interview responses indicated that the Drilling Technique was systematically implemented to reinforce sentence structure and vocabulary retention. According to the teacher, students who initially struggled with sentence construction and word choice began to show improvement after consistent practice. Repetition allowed them to remember vocabulary and apply correct grammatical patterns, while substitution drills enhanced their flexibility in producing various sentence forms. These findings align with the results of previous studies by Nugroho and Indriani (2020) and Sari and Pratama (2023), who found that drilling helps students recognize repetitive language structures that frequently appear in translation tasks.

The student interviews also confirmed the effectiveness of the Drilling Technique. High-achieving students, such as Aqila, expressed that repetition drills made it easier to remember vocabulary and understand English sentence structures. Medium-level students, such as Riska, stated that repeated exposure to sentence patterns helped them translate more naturally and accurately. Lower-level students, like Muksal, admitted that although the exercises were challenging, continuous repetition and teacher guidance gradually improved their understanding of word meanings and sentence order. These responses demonstrate that drilling can accommodate different student levels by reinforcing gradual mastery through repetition and feedback.

From the data, several improvements were observed. First, students demonstrated better vocabulary accuracy by recalling and using the correct meaning of words during translation activities. This improvement occurred because frequent repetition strengthened their memory and automatic retrieval of words, supporting Zhang and Zhang's (2023) finding that repetition enhances vocabulary retention. Second, students' diction became more contextually appropriate, as they began to choose words that matched the intended meaning rather than translating literally. Third, sentence structure accuracy improved as students became more familiar with subject–verb–object order and learned to construct grammatically correct sentences.

These improvements occurred because drilling provides continuous exposure and reinforcement, allowing students to internalize correct forms subconsciously. According to DeKeyser (2007), repeated practice enables learners to transfer declarative knowledge into procedural knowledge, resulting in more automatic language use. This cognitive process explains why the students' translations became more accurate and fluent after participating in repeated drilling sessions.

Furthermore, students displayed greater confidence and enthusiasm in performing translation tasks. Observation showed that they participated actively during class, responded to corrections positively, and worked cooperatively in both individual and group drills. This outcome supports the claim of Mendrofa and Wijaya (2022), who found that repetitive oral practice improves learner confidence and motivation in language production.

In addition to linguistic benefits, the Drilling Technique also had pedagogical implications. It provided a structured learning environment where feedback was immediate, and errors were corrected instantly, preventing fossilization of mistakes. The

teacher's consistent correction and reinforcement contributed to habit formation, aligning with behaviorist learning theory, which emphasizes stimulus–response repetition.

The findings of this study are consistent with earlier research by Rahmawati (2021), who discovered that drilling enhances grammar mastery through repetition and correction. Similarly, Syahputri (2022) emphasized that repetitive practice improves cognitive processing in translation tasks, leading to better comprehension of meaning. However, the present study extends these findings by applying drilling specifically to translation learning and observing its effect on real classroom performance rather than controlled experiments.

In summary, the Drilling Technique effectively improved students' translation accuracy by strengthening their vocabulary recall, enhancing their diction, and helping them produce correct sentence structures. The improvement occurred because consistent repetition, substitution practice, and direct feedback allowed students to internalize correct patterns and reduce translation errors. These results confirm that drilling can serve as a practical and efficient approach to enhance translation learning, especially for junior high school students who are still developing fundamental linguistic awareness.

CONCLUSION

This study concludes that the Drilling Technique plays a significant role in enhancing students' translation skills, particularly in terms of vocabulary accuracy, diction, and sentence structure. Through consistent repetition, substitution, and question-and-answer drills, students were able to internalize language patterns, recall vocabulary more effectively, and construct grammatically accurate sentences. The results demonstrated that repeated practice not only strengthened students' linguistic accuracy but also improved their confidence and fluency in translating texts from English to Indonesian and vice versa. These findings support the hypothesis that the Drilling Technique is an effective and practical approach for improving translation accuracy among junior high school students. The technique's structured and repetitive nature allows students to transform knowledge from conscious learning into automatic use, making it easier for them to understand and convey meaning naturally between languages. Furthermore, the study provides pedagogical insights for teachers to integrate drilling more effectively in translation learning by combining it with engaging materials and immediate feedback to sustain student motivation. For future research, it is recommended to conduct a longer-term investigation involving larger groups of students to observe the development of translation skills over time and to compare the effectiveness of drilling with other teaching techniques, such as cooperative or task-based learning approaches.

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DIAGNOSING ENGLISH TEACHERS' DIFFICULTIES IN INTEGRATING PJBL IN ENGLISH LANGUAGE LEARNING AT SMPN 1 SAMUDERA

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ABSTRACT

This study aimed to diagnose the difficulties faced by English teachers in integrating Project-Based Learning (PJBL) into English language instruction at SMPN 1 Samudera. Specifically, it sought to identify the phases in which the teacher encountered challenges and the underlying factors contributing to these issues. This research employed a descriptive qualitative method. Data were collected through interviews with an English teacher, a vice principal, and a student, as well as documentation of lesson plans. The findings revealed that the teacher experienced difficulties across all PJBL phases: defining basic questions, designing the project plan, setting the schedule, monitoring progress, and assessing results. These challenges were mainly influenced by limited student English proficiency, lack of training, unfamiliarity with PJBL, and the teacher's ongoing adaptation to the new curriculum. In addition, time constraints and the need for constant scaffolding made it difficult to maintain student focus and independence. Although the teacher had prepared appropriate assessment tools, evaluating speaking skills remained a challenge due to students' low confidence and unclear pronunciation. These difficulties were mainly caused by insufficient training, unfamiliarity with PJBL, and the ongoing shift from teacher-centered to student-centered instruction. The study underscores the importance of comprehensive teacher preparation and contextual adaptation to ensure effective PJBL implementation in English language learning.

Keywords: *English Teachers' Difficulties; Project-Based Learning*

INTRODUCTION

Introduction to English in Indonesia is generally taught as a foreign language as a formal subject given to junior high school (SMP) level, with the goal of teaching English for this level is to communicate and master four language skills: speaking, listening, reading, and writing. Furthermore, learning English effectively entails improving both communicative competences, the capacity to use language appropriately in a variety of contexts, and linguistic competence, the development of vocabulary, grammar, and pronunciation. Because traditional techniques solely focus on instructors or are teacher-centered, they may not completely engage students or prepare them for language usage in real-world situations. Instead, they emphasize direct instruction and practice of discrete language skills.

In order to accomplish effective and efficient learning, English teachers today need to implement innovative, relevant approaches. In order to influence students' perceptions that English is an enjoyable and useful subject, teachers must also use an

engaging teaching strategy. Project-based learning (PJBL) is a promising strategy that should be used. It involves students in long-term, group projects that demand that they investigate difficult issues, find solutions, and produce tangible results. In short, PJBL is learning that contextualizes learning by providing problems to be solved or certain products to be developed. PJBL emphasizes independent, interdisciplinary, and integrated learning with situations. Students will have the opportunity to utilize several skills, such as problem-solving, creativity, cooperation, as well as language skills, by giving the experience of demonstrating knowledge of a product. For Gen Z students, PJBL is very suitable for developing their skills, critical thinking, and perseverance. The Indonesian Ministry of Education suggested that Project-based learning needs to be implemented since it's in line with the *Merdeka Curriculum*.

Therefore, educational institutions from elementary to higher education should apply this approach. Although the benefit of PJBL is recognized, unfortunately, the integration of this approach is not easy. English teachers as facilitators in integrating PJBL have a crucial role in English learning success. Based on preliminary interview and observation conducted at SMPN 1 Samudera, in fact, when interviewed on 29 March 2024 Researcher, as the interviewer, asked several questions to the teacher who identified the PJBL. The implementation of the independent curriculum in English teaching is considered more flexible, as it allows for various projects to be carried out with students. The teacher applies Project-Based Learning (PJBL).

However, challenges remain, particularly because the curriculum is still new and teachers are in the process of adapting. Teachers' responsibility is to develop the strong character of students. Despite these challenges, the implementation of the independent curriculum in the school has generally been progressing well, requiring teachers to continuously motivate students. For teaching English, the book *Bright an English* from Airlangga is used as a reference. This book aligns with the curriculum as it incorporates the PJBL method, where each chapter includes a project for students. The approach helps students engage with real-world scenarios.

Therefore, the researcher found in the field that English teachers have implemented PJBL but they are still in the learning stage and face difficulties. Thus, the researcher was interested in research problems by the title "Diagnosing English Teachers' Difficulties in Integrating PJBL in English Language Learning at SMPN 1 Samudera". This research aims to find the difficulties encountered by English teachers in the integration of PJBL in English language learning.

METHOD

The researcher utilized a qualitative method in this study. Creswell defined qualitative research as an investigation process that uses a variety of methodological traditions of inquiry to understand a social or human problem. The study is carried out in a natural setting, a comprehensive image is created, language is analyzed, and in-depth viewpoints from informants are reported (Creswell, 2012). Also, Denzin and Lincoln described that qualitative research is a collection of material that interprets the processes of the research that bring the world to life. Through field notes, interviews, conversations,

photos, recordings, and memos to the self, these techniques turn the world into a succession of representations (Denzin & Lincoln, 2011).

In addition, Patton emphasized that the goal of qualitative research is to give depth and specificity by emphasizing the how and why of decision-making as opposed to the what, where, and when (Patton, 2023). In this study, the research was conducted at SMP Negeri 1 Samudera, located in Blang Peuria, Samudera District, North Aceh, during the academic year of May 2025. The primary data source in this research was an interview with the English teacher at SMPN 1 Samudera, selected using purposive sampling. Secondary data were obtained from interviews with the vice principal and one student, as well as documentation in the form of lesson plans. Data collection techniques used were interviews and documentation.

The interviews were conducted in a semi-structured format, with 42 question items covering the phases of PJBL integration and related difficulties, while documentation such as lesson plans was analyzed to cross-check instructional design (Clark & Creswell, 2008; Creswell, 2014). To test the validity of the data, the researcher employed triangulation and member checking. Source triangulation involved gathering information from multiple sources, namely the English teacher, vice principal, and student, to verify consistency and reliability, while member checking was conducted by taking the transcripts and findings back to the participants to confirm accuracy (Ivankova & Creswell, 2009; Miles, Huberman, & Saldana, 2014). Finally, the data analysis technique referred to Miles and Huberman (2014), consists of data condensation, data display, and conclusion drawing. The researcher condensed the data by selecting and coding significant information, displayed the data in narrative text, and drew conclusions based on the evidence.

RESULT AND DISCUSSION

This section contains an in-depth analysis and interpretation of the findings obtained, including the meaning of the results and their connection to existing theories. After analyzing the data, the next step was to answer the research objectives. There were two research objectives. The first objective of this research is to examine the difficulties faced by English teachers in integrating PJBL in English language learning and to analyze the factors that are faced by English teachers.

First, the teacher experienced difficulty. The teacher has difficulty providing basic questions that are clear and relevant to the learning material. According to Thomas (2000) explained that understanding how to formulate effective driving questions remains a barrier for many teachers. In addition, the main difficulty in this phase is the adjustment to the new curriculum. As stated by the English teacher, she started teaching at SMPN 1 Samudera on 1st January 2023. Since then, the *Merdeka* curriculum has also been implemented in that school, but especially for grades 7 and 8. As she is the class teacher for grade 7, she has been implementing and adapting PJBL and PBL-based learning since the *Merdeka* curriculum was implemented at the school. This indicates a shift in teacher roles, as noted by Kurnely (2017), where the teacher must adapt from being an instructor to becoming a facilitator, something not easily achieved during early implementation. Time constraints were also a major issue. The teacher often needed to extend project

activities over multiple sessions, which is consistent with (Markham 2011 and Larmer (2015) asserted that PJBL requires more time than traditional instruction. Moreover, the teacher was overwhelmed; she needed to provide intensive assistance individually, scaffolding and direction, confirming Condlife's (2017) view that classroom management becomes more complex in project-based settings. Despite efforts to scaffold learning, grade 7 students needed individual support to stay engaged and complete tasks.

Finally, the main challenge appeared during the assessment process itself. The teacher struggled to evaluate students accurately due to two major factors: students' poor pronunciation and their low self-confidence in speaking English. This line up with Wardani's (2020) argument that PJBL assessment must address affective and behavioral aspects, not just cognitive outcomes

Further, explores the underlying factors that caused those difficulties. Firstly, in the defining basic questions phase, the challenges in delivering basic questions were mainly caused by external factors. The key factors contributing to this difficulty were students limited English vocabulary, their unfamiliarity with English instructions, and a lack of confidence to respond in English. These findings are consistent with Cintang's (2018) view that student capability is one of the key barriers in PJBL. Although the teacher provided guiding questions, the language barrier reduced their effectiveness, forcing the teacher to simplify or translate the prompts.

Next, in designing the project plan phase, the factors that caused the teacher's difficulty in designing the project plan were closely related to the implementation of the new curriculum. These reflect two major factors: the teacher's capability and the lack of prior training, as discussed by Cintang (2018), that without sufficient training, teachers may struggle to design effective PJBL activities. Moreover, in the scheduling phase, the main issue was time management. The main factors that caused difficulty in managing time were the students' different learning speeds, their lack of experience with project-based activities, and their limited English proficiency. This confirms the claim by Larmer (2015) that time constraints remain one of the most frequent challenges in PJBL implementation.

In addition, during the monitoring phase, the challenge is caused by students limited English proficiency, which requires the teacher to provide continuous guidance and repeated instructions. The teacher's need to provide constant scaffolding shows the importance of teacher training in classroom management and support strategies. Then, in the assessment phase, the main factors causing difficulty in the assessment phase were not related to the rubric itself, but to students limited speaking ability, unclear pronunciation, and lack of confidence. These factors line up with Wardani's (2020) argument that PJBL assessments must include affective and performance aspects, which can be difficult to judge when students lack the skills to present their learning clearly.

CONCLUSION

This study aimed to examine the difficulties faced by an English teacher in integrating Project-Based Learning (PJBL) into English language learning and to analyze the factors that caused these difficulties. The findings revealed that challenges occurred across all five phases of PJBL implementation: defining basic questions, designing the

project plan, setting a schedule, monitoring the project progress, and assessing project results. In the early phases, the teacher struggled to activate students' prior knowledge using basic prompting questions due to limited student vocabulary and low confidence. During the project planning and scheduling phases, challenges stemmed from unfamiliarity with the new curriculum and rigid time constraints. Monitoring and assessment were also problematic as the teacher had to provide constant scaffolding and found it difficult to objectively evaluate student speaking performance.

These findings indicate that both pedagogical readiness and classroom management play significant roles in the success of PJBL integration. The factors causing these difficulties include the shift in the teacher's role to be a facilitator which required new instructional strategies. Additionally, the lack of prior training in designing and assessing PJBL tasks limited the teacher's ability to fully implement the method. Time constraints, limited scaffolding skills, and curriculum adaptation further contributed to the challenges.

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THE ASSOCIATION OF TEMPERED PERSONALITY AND ENGLISH-SPEAKING SKILL AMONG EFL STUDENTS

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ABSTRACT

This study investigates the association between the four classical temperament personalities (sanguine, choleric, melancholic, and phlegmatic) and the English-speaking skills of EFL students. The participants were eighth-semester students of the English Department who had completed all speaking courses and experienced academic seminars. Data were obtained through a temperament personality questionnaire and students' academic presentations. The analysis was conducted using Miles and Huberman's qualitative data analysis approach, which includes data reduction, data display, and conclusion drawing. The findings revealed that sanguine and choleric temperaments had the strongest association with higher speaking performance, particularly in aspects of fluency, confidence, clarity, and audience engagement. In contrast, melancholic students tended to show hesitations and nervousness during speaking, while phlegmatic students demonstrated calmness and good preparation but had comparatively weaker fluency and pronunciation. In conclusion, sanguine and choleric temperaments were the most dominant in supporting strong English-speaking skills, whereas melancholic and phlegmatic temperaments contributed in more limited ways. These results highlight that personality differences are associated with variations in speaking performance, which can be considered in designing strategies to enhance students' speaking skills in academic contexts.

Keywords: *temperament personality, four temperaments, speaking skill, EFL students, qualitative study*

INTRODUCTION

Speaking is one of the most essential skills in language learning, as it serves as a primary means of communication that allows individuals to express thoughts, ideas, and emotions effectively. In the context of English as a Foreign Language (EFL), speaking proficiency is crucial for enabling students to participate in academic and social interactions. However, many EFL learners continue to face challenges in developing speaking skills due to limited vocabulary, lack of confidence, and insufficient opportunities for practice. These difficulties often result in noticeable differences in students' speaking performance, suggesting that factors beyond linguistic competence, such as personality traits, may play a significant role. Personality influences how individuals think, behave, and communicate, which consequently affects how they approach language learning. Previous studies, such as those by Rahayu (2020) and Santoso and Taufiq (2018), have highlighted that personality differences contribute to variations in speaking performance, particularly in how students express ideas, maintain confidence, and respond to communication situations.

One framework often used to understand personality differences is the classical temperament theory developed by Hippocrates and Galen. This theory divides human personality into four temperaments; sanguine, choleric, melancholic, and phlegmatic, each representing distinct behavioral tendencies. Although originating from ancient concepts, the four-temperament theory remains relevant in modern psychology and education. Fisher et al. (2015) found that the temperaments share conceptual similarities with the Big Five personality traits, showing that this traditional model still provides valuable insights into individual behavior and learning style. In educational settings, the application of temperament theory has even been adopted in certain teaching approaches, such as in Waldorf schools, to support personalized learning strategies that accommodate students' emotional and behavioral tendencies.

Despite its long-standing influence, there remains limited research that explores how each of the four classical temperaments specifically contributes to students' English-speaking skills, particularly within an EFL context. Previous studies have mostly used quantitative methods to identify statistical correlations between personality traits and speaking performance, often overlooking the deeper behavioral and affective dimensions of communication that can be observed through qualitative analysis. As speaking proficiency involves not only linguistic mastery but also confidence, emotional expression, and spontaneity, a qualitative approach can offer richer insights into how temperament shapes students' speaking behavior.

The scientific novelty of this study lies in its qualitative exploration of the four temperaments in relation to EFL students' academic speaking performance. Unlike prior studies that focused only on specific traits such as extraversion or compared only two temperament types, this research investigates all four classical temperaments and analyzes how each manifests during real academic presentations. This approach allows for a more holistic understanding of the relationship between personality and language performance, particularly in the context of formal speaking activities.

This study aims to identify the association between the four temperament personalities, sanguine, choleric, melancholic, and phlegmatic, and the English-speaking skills of EFL students. It also seeks to examine how each temperament contributes to specific aspects of speaking proficiency. The aspects of speaking proficiency investigated include fluency, pronunciation, grammar, and comprehension.

METHOD

This study employed a qualitative research design using a descriptive case study approach. The purpose of this method was to explore the relationship between the four classical temperament personalities, sanguine, choleric, melancholic, and phlegmatic, and EFL students' English-speaking skills in depth. A qualitative approach was chosen because it allows the researcher to understand personal traits and behavioral expressions naturally, without statistical generalization.

The research was conducted at the English Department of Sultanah Nahrasiyah State Islamic University of Lhokseumawe, located on the Banda Aceh–Medan Highway, Alue Awe, Muara Dua, Lhokseumawe. The participants were twelve eighth-semester

students who had completed all speaking courses, particularly *Speaking in Academic Context*. They were selected using purposive sampling, ensuring that each temperament type; sanguine, choleric, melancholic, and phlegmatic, was represented by three participants.

Data were collected through documentation and observation techniques. The documentation consisted of students' seminar recordings, in which they acted as moderators, presenters, or speakers in academic settings. These recordings provided authentic evidence of students' speaking performance. In addition, the researcher reviewed the students' final grades in *Speaking in Academic Context* to support the qualitative interpretation. Prior to data collection, a temperament personality questionnaire adapted from Florence Littauer's *Personality Plus* theory was administered to identify each participant's dominant temperament type. The questionnaire served as a classification tool rather than a primary data source.

Data analysis followed the interactive model proposed by Miles and Huberman (2018), consisting of three main steps: data condensation, data display, and conclusion drawing/verification. During data condensation, the researcher selected and simplified relevant data from seminar recordings, focusing on four aspects of speaking skills: fluency, pronunciation, grammar, and comprehension. The condensed data were then displayed in descriptive tables and narrative summaries to identify emerging patterns. In the final stage, conclusions were drawn to determine how each temperament type contributed to speaking performance and which temperament was most dominant.

RESULTS AND DISCUSSION

The results of this study show that students with different temperament personalities demonstrate distinct characteristics in their English-speaking skills, especially in fluency, pronunciation, grammar, and comprehension. The data were obtained from students' academic presentations and temperament questionnaire results, supported by their speaking scores in *Speaking in Academic Context* as an additional reference for performance validation. The analysis indicates that the sanguine and choleric temperaments were strongly associated with high speaking skill performance, while melancholic and phlegmatic temperaments contributed in more limited ways.

Table 1. Association between Temperament Personalities and Speaking Skill Aspects

Temperament	Fluency	Pronunciation	Grammar	Comprehension
Sanguine	Very fluent and expressive	Clear and confident	Minor grammatical slips	Strong comprehension and spontaneous responses
Choleric	Fluent and assertive	Clear articulation	Good grammatical control	Strong comprehension, often dominant in discussion
Melancholic	Hesitant and Cautious	Clear but low in volume	Grammatically	Deep comprehension

			accurate	but slow delivery
Phlegmatic	Moderately fluent	Soft and steady pronunciation	Fairly consistent grammar	Understands content but less responsive

As shown in Table 1, sanguine students demonstrated the highest level of fluency and confidence in their academic presentations. They were expressive, interactive, and maintained good audience engagement. This finding aligns with Rahayu (2020), who stated that sanguine students are expressive and socially active, which helps them perform well in oral communication. Choleric students also showed strong speaking skills, particularly in clarity and assertiveness, supporting Santoso and Taufiq (2018), who noted that choleric individuals tend to be confident, goal-oriented, and able to control their speech effectively.

On the other hand, melancholic students tended to focus on accuracy and structure rather than spontaneity. They often hesitated before speaking and showed signs of nervousness. This result corresponds with Littauer's (2011) description that melancholic individuals are analytical and perfectionistic, often prioritizing accuracy over fluency. Phlegmatic students appeared calm and consistent but were less expressive. They prepared their speech well but lacked enthusiasm and spontaneity. This finding is in line with Fisher et al. (2015), who mentioned that phlegmatic personalities tend to maintain stability and avoid emotional intensity in communication.

Scientifically, these results suggest that differences in students' speaking skills are not only influenced by linguistic ability but also by personality traits that shape confidence, anxiety, and communication style. Sanguine and choleric temperaments support outward, active communication behaviors that encourage fluency and audience engagement. Meanwhile, melancholic and phlegmatic temperaments tend to be more inward and reflective, resulting in slower or less dynamic speaking performance.

In conclusion, the data show that sanguine and choleric students have a stronger association with high speaking skill performance, especially in fluency, confidence, and clarity. Melancholic and phlegmatic students also contribute positively through accuracy, preparation, and comprehension, although with more limited expression. Therefore, temperament differences significantly affect students' English-speaking skills and should be considered in developing teaching strategies that help balance confidence and linguistic accuracy in EFL learning.

CONCLUSION

This study aimed to identify the association between the four classical temperament personalities: sanguine, choleric, melancholic, and phlegmatic, and EFL students' English-speaking skills, as well as to describe how each temperament contributes to specific aspects of speaking. The findings reveal that sanguine and choleric temperaments are most strongly associated with high speaking performance, particularly in fluency, confidence, clarity, and audience engagement. In contrast, melancholic and phlegmatic temperaments contribute in more limited ways, focusing on accuracy, preparation, and comprehension rather than spontaneity. These results indicate that

temperament influences not only linguistic performance but also emotional expression and confidence in speaking.

The association between temperament and speaking skill highlights the importance of considering personality differences in designing learning strategies that can enhance students' communicative competence and classroom participation. Future research may expand this study by involving a larger population, incorporating mixed- method approaches, or exploring how temperament interacts with other factors such as motivation, learning style, or anxiety in English-speaking performance.

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SOCIOCULTURAL INFLUENCES ON ENGLISH LEARNING: A COMPARATIVE STUDY OF ACEH AND NONACEH ARCHITECTURE STUDENTS

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ABSTRACT

English proficiency is undeniably pivotal in higher education for preparing individuals for professional fields. To gain competence requires some aspects which are interrelated. This research aims to explore three aspects: those are students' learning backgrounds, the strategies, and factors that influence their approaches to learning. The study adopted a qualitative design, relying on semi-structured interviews with sixteen students from Aceh and outside Aceh (8 from each). Data were analysed thematically using two theoretical frameworks of Vygotsky's sociocultural theory and Oxford's Language Learning Strategy (LLS) framework. Findings reveal that Acehnese students mostly possess more limited English learning backgrounds, whereas non-Acehnese students have broader exposure through school facilities, extracurricular programs, and digital resources. In terms of strategies, Acehnese students were more inclined toward independent effort and basic tools of digital, while non-Acehnese employed a wider range of metacognitive and social strategies, including self-directed use of wider digital platforms. Finally, students' approaches were found to be shaped by a combination of sociocultural expectations, motivational orientations, and affective factors such as anxiety and self-confidence. The study concludes that cultural and geographical contexts remain influential determinants of English learning in Indonesian higher education. By integrating sociocultural and strategic perspectives, the study suggests a necessity of designing more student-centred approaches that acknowledge students' diverse backgrounds and equal opportunities for English proficiency.

Keywords: *Architecture Students; Comparative Study; English Learning.*

INTRODUCTION

English has thus far served as the primary medium of academic exchange, international business, and digital communication (Perinpasingam, 2015; Rao, 2019). For nonnative speakers, English skills are no longer an option but rather a necessity both in academic and professional mobility (Crystal, 2003; Jenkins & Panero, 2025). In Indonesia, English has been incorporated into the school curriculum from the early years of primary school to higher education. However, the effectiveness of English learning in Indonesia remains uneven across regions due to differences in educational policies. Teachers' quality, resource availability, and cultural contexts also play a vital role.

At Universitas Malikussaleh (UNIMAL), Aceh, architecture students come from diverse regions in the province. According to strategic planning (RENSTRA) UNIMAL 2024 (Kurikulum Merdeka, 2022), about 34% of UNIMAL students are from outside

Aceh. Even though a study showed that the origin may not have an influence on students' English proficiency (Achmad & Yusuf, 2016), it may reflect on students' participation in classes. Like their peers from other disciplines, architecture students are obligated to engage with English-language materials, including textbooks, journals or forums, and architectural design references. Yet, their success depends on several factors, such as their learning background, strategies, and the sociocultural environment that shapes their approaches to English.

Previous research compared Indonesian learners with those from other countries (Ushioda & Dörnyei, 2017). In the Indonesian context, Lamb's (2012) study revealed a stronger integrative motivation and aspirations to English learning among students in urban Java, while rural Sumatra students are often instrumentally motivated, focusing on passing exams. This suggests that geography and local educational traditions strongly affect learners' motivation. For many Acehnese students, English exposure may be primarily limited to formal education settings (Fitria, 2021). Besides, as one of the provinces in Indonesia that implements Islamic Syari'a, Aceh has cultural, educational, and learning environment characteristics that may differ from those in other regions. For instance, many Acehnese kids go to traditional Islamic boarding schools (*dayah*), which may lead to limited access to English courses and exposure. In contrast, students from outside Aceh might have broader background access to language institutions, native or foreign teachers, or environments that encourage the daily use of English.

This study aims to investigate the comparative analysis of Architecture students from Aceh and non-Aceh, with particular attention to their learning backgrounds. It also focuses on the learning strategies they use and the factors that influence their approaches to English learning. To support this study, the theories of sociocultural learning by Vygotsky and Language Learning Strategy (LLS) by Oxford were applied.

The first theory is employed since it views learning as a socially and culturally situated process. It believes that gaining knowledge is not individual; it needs interaction with others, cultural tools, and social practices. The key concepts of this theory are the Zone of Proximal Development (ZPD) (the gap between what a learner can do alone and what they can achieve with support), Scaffolding (the guidance provided by teachers, peers, or cultural tools to help learners progress, mediation (learning is mediated by tools like language, symbols, and technology), and social interaction (communication and collaboration are central to learning) (Attahiru, 2024).

The second theory was used as it is one of the detailed and systematic classifications of language learning strategies. It divides strategies into direct (memory, cognitive, compensation) and indirect (metacognitive, affective, social) paths (Oxford, 2003). This allows the study to capture not only what participants do with the language but also how they manage their own learning. Based on the above exploration, this study focuses on three points those are the students' learning background, the students' learning strategies, and what influences the students' approach in English learning by comparing two different contexts of students' origin, Aceh and non-Aceh.

METHOD

This study employs a qualitative comparative case study design. It aims to explore the Architecture students' English learning background, strategies, and what influences their approach to learning. This qualitative approach allows for in-depth exploration of participants' experiences, perspectives, and the sociocultural factors (Creswell & Poth, 2018). This study design also enables the researcher to explore similarities and differences across two cultural groups within the same academic discipline.

The study was conducted at the Architecture study program, Universitas Malikussaleh, North Aceh, Indonesia, for its diverse student population, which includes both local Acehnese students and those from other provinces across Indonesia. Using purposive sampling, the respondents were selected as representatives from both groups. 16 students participated in the study, involving eight students from Aceh and eight students from outside Aceh.

The data was collected through semi-structured interviews, which allowed flexibility for participants to elaborate on their experiences. The interview consisted of nine questions in which were divided into three themes (three questions on learning background, four questions on learning strategies, and two questions on factors influencing students' approach in learning). Each participant lasted between 20-30 minutes in either Indonesian or English, depending on the participants' comfort. All interviews were audio-recorded with the participants' consent and transcribed using thematic analysis suggested by Braun and Clarke. It consists of familiarization, coding, theme development, reviewing themes, defining and naming themes, and producing the report (Braun & Clarke, 2006). The analysis was based on two theoretical views: sociocultural theory by Vygotsky and Language Learning Strategy (LLS) theory by Oxford.

RESULT AND DISCUSSION

This section shows the results of the interviews conducted with Architecture students from both Aceh and non-Aceh backgrounds. The analysis follows the three central objectives of this study and is interpreted through the lens of sociocultural theory (Vygotsky) and Oxford's Language Learning Strategy (LLS) theory. The excerpts from participants' interviews are included to strengthen the credibility of the findings.

Learning Background

The findings reveal contrasts in learning backgrounds between Aceh and non-Aceh students. This section focuses on three related questions: the type of previous school the students attended, whether the school provided strong English instruction, and whether the students took any extra English courses before university. The results highlight how these factors may influence their approaches to English learning at the university level.

The interview reveals that six of eight Aceh students attended public school, which provided only English books and a dictionary. Two of them attended a private school, which is a boarding school, that provides sufficient English books and dictionaries, has a language institution, an English day, and an English club. Surprisingly,

none of them took extra English courses before university. Two of them, however, learn English autonomously through online sources. One participant said, “I did not take any courses,” while another mentioned, “I study English through social media.”

While non-Aceh students, five attended public school, two of which provide only books and dictionaries, three other schools provide extra materials and English clubs as well as English day. Three students attended a private school that provides all the facilities mentioned. Before university, four non-Aceh students took an English course, two students continued the English club with their high school friends, and one student did not take any English course. “I took a three-month English course to prepare myself for university. Another student clarified his English exposure at school, “We had an English day to practice English every Friday.”

The results show that public schools attended by Aceh students limit their exposure to English, while private schools provide more opportunities for extracurricular activities, such as English Day, English Week, English Club, and access to larger English institutions. Additionally, most Aceh students did not take any English courses to prepare themselves before college. In contrast, non-Aceh students were more likely to have such English exposure prior to entering university.

This evidence highlights Vygotsky’s theory that learning is mediated by sociocultural context. Aceh students’ backgrounds provided fewer opportunities for interactional scaffolding, limiting their Zone of Proximal Development (ZPD) in communicative English (Scott & Palincsar, 2013). Non-Aceh students, by contrast, benefited from a richer cultural environment, such as English clubs, English days, and private courses, making their skills more advanced before university. Thus, the learning background is a reflection of regional educational practices and cultural values, which poses the main point of the sociocultural theory.

Learning Strategies

To examine the students’ learning strategies, four questions were asked, focusing on how students usually learned English in school, how they currently study English, whether they use technology to improve their English, and which strategies help them the most. The interview results show that Aceh students’ classes frequently focus on memory and cognitive strategies, such as memorizing and finding vocabulary, translating texts, and practicing grammar exercises. This is reflected in students’ statements such as, “We’re not only asked to memorize vocabulary and read the book without explanation,” and “Translating texts, finding new vocab and making sentences.”

Non-Aceh students, however, reported greater use of metacognitive and social strategies, including practicing with peers and seeking out opportunities for real communication through private courses. As expressed by the participants, “We practice a lot with our classmate,” “I took a private course after class,” and “We often practice speaking in class.”

In relation to Oxford’s LLS framework, the findings make it clear that Aceh students tend to employ direct strategies (memory and cognitive) that suit exam-focused learning, while non-Aceh students employ more indirect strategies (metacognitive and social), reflecting greater exposure to communicative contexts. In response to the

question regarding students' current English learning practices, both Aceh and non-Aceh students mostly reported engaging in independent learning. Furthermore, answers to the question about the use of technology in learning provided insight into the advanced tools they used. Aceh students preferred self-study activities using applications, YouTube, watching English movies and podcasts, listening to English songs, and playing games. This preference is reflected in statements such as, "I like watching English movies and listening to podcasts through YouTube," "I like listening and finding new vocabulary," and "I use Ome TV Application to study English". Non-Aceh students, in this case, had more varied preferences. These included social media, English songs and movies through YouTube, the Duolingo application, English novels, Spotify, DC Comics, Marvel, live streaming, English podcasts, and TikTok. As expressed by the participants, "I like reading novels and highlighting new vocab, watching films, and listening to English songs," "I study English through YouTube and Spotify," and "I've subscribed to Duolingo."

This result shows that many Aceh students rely on basic online tools, which aligns mostly with cognitive strategies (practicing vocabulary, repeating, and translating) and memory strategies (using song lyrics and repetition). Although the use of online resources shows independence, but the strategies remain less in-depth. Besides, non-Aceh students showed richer use of online platforms which connects to metacognitive strategies (planning learning goals), social strategies (joining online discussion groups, practicing with international peers, asking questions in forums), and affective strategies (using music, games, or entertainment to lower anxiety and increase motivation). This varied use of tools allows them to employ a wider repertoire of strategies and support more autonomous and communicative learning. At the end, this supports Oxford's theory that learners' strategy choice depends on context, opportunities, and available resources (Oxford, 2003).

Furthermore, responses regarding the strategies that help students the most varied across both groups. Five Aceh students stated that watching movies was the most helpful strategy, while three others mentioned practicing with peers. One participant explained, "I think watching movies is the best strategy because I can practice pronunciation." This finding indicates that their learning tends to be input-based, involving passive exposure and independent practice, which reflects their limited access to interactive learning opportunities. Based on Oxford's LLS theory, these practices are closely related to cognitive, memory, and affective strategies.

In contrast, most non-Aceh students stated that practicing with peers was the best strategy for improving their English. As one participant expressed, "I like making notes and talking with friends." This suggests that non-Aceh students' English learning is more interaction-based and socially mediated, which promotes collaborative learning.

In short, the students' learning strategy of both groups differs due to available cultural tools and experiences. The absence of teachers' involvement, peers, or authentic resources, Aceh students' learning strategies aligned with their background, which are memorization and translation. A study by Jungert and Rosander (2009) supports this finding, in which an informal influence strategy is an aspect of the strategic approach to studying, and problem-based learning could improve teacher-student relationships and increase the students' involvement and their use of strategic approaches to studying. Non-

Aceh students, having been socialized into more interactive environments, develop strategies that foster autonomy and communicative competence. This is consistent with a study conducted by Wahyudin and Rido (2020) showing some prefer memorization while others favour communicative practice, indicating that cultural background influenced students' preference for learning strategies.

Influencing factors

The last focus of the interview was to investigate factors influencing students' approaches to English learning. To address this aim, two questions were asked: (1) what motivated students to learn English at that time, and (2) what challenges they face in learning English.

In terms of motivation, five out of eight Aceh students were motivated by integrative aspects, particularly learning and cultural exposure. This finding supports Vygotsky's theory, which emphasizes that motivation arises from cultural mediation and community norms. As reflected in students' statements, "I want to learn English because I want to study abroad," and "I want to make friends from other countries." These responses indicate that Aceh students' sociocultural context, such as future academic communities and international environments, shapes their learning goals. At the same time, their use of Language Learning Strategies (LLS), including planning, practicing, managing emotions, and interacting socially, reflects how they operationalize those goals.

Non-Aceh students' English learning, on the other hand, was mostly motivated by instrumental aspects, such as career, income, and survival, while the rest viewed English as an international communication tool. This is illustrated by statements such as, "I want to work in other countries," and "English is used for global communication." By perceiving English as a global language and a tool for future careers, their learning approach becomes more instrumental and practical, emphasizing strategies that prepare them for real-world professional communication. From Vygotsky's perspective, this reflects their need to integrate into global workplace communities.

Meanwhile, Oxford's LLS framework highlights students' reliance on goal-directed and practical strategies that support employability. By capturing English as a global language and taking it as a tool for future career, learning approach becomes more instrumental and practical, emphasizing strategies that prepare them for real-world professional communication. From Vygotsky's perspective, this reflects their need to integrate into global workplace communities. Meanwhile, Oxford's LLS framework highlights the students' reliance on goal-directed, practical strategies that support employability.

The next aspect concerns students' challenges in learning English. Both Aceh and non-Aceh students reported similar difficulties, including limited authentic input, inconsistency in learning, difficulties with complex grammar, and limited exposure to English-speaking communities. These challenges are reflected in statements such as, "I do not understand grammar and have a lack of vocabulary," and "I do not have friends to practice." In addition, both groups also experienced challenges related to self-confidence, as expressed by one participant who stated, "I do not have confidence to speak English."

According to above findings, confirming about the students' motivation and challenges, factors influencing students' approaches to English learning are cultural background and social environment, which determine the type of motivation (academic, or career-oriented) and opportunities for scaffolding, educational experiences, particularly whether learning is interactive or traditional, which shapes how students internalize English, and the last is strategic awareness and confidence, where students with broader and balanced use of Language Learning Strategies (LLS), especially social, metacognitive, and affective. Thus, students' approaches are not only the result of individual choice but poses sociocultural contexts.

CONCLUSION

The findings reveal that students' English learning approaches are shaped by their cultural and educational backgrounds, resulting in clear differences between Aceh and non-Aceh students. Aceh students are largely motivated by community-oriented goals, relying on memorization and self-encouragement strategies. Their challenges stem from limited access to authentic English interaction, a learning environment that emphasizes traditional instruction rather than active communication, confidence problems, and consistency. As a result, their progress tends to be slower and participation in class becomes less. From a sociocultural perspective, this reflects the absence of strong scaffolding and limited engagement in communities of practice, while from the LLS perspective, their strategy use is narrower, focusing mostly on cognitive and affective strategies.

In contrast, non-Aceh students show motivation that is more global and integrative, aiming for international communication and career advancement. The challenges show similar trends to Aceh students. Consequently, they employ a broader range of strategies, especially social and metacognitive strategies. From Vygotsky's view, their learning is better supported by social interaction and scaffolding, while Oxford's framework highlights their more balanced and effective use of strategies.

In short, the analysis shows that Aceh students' approaches to English learning are shaped by local cultural and educational traditions. In contrast, non-Aceh students' approaches are influenced by broader social interaction and a more global orientation. These findings indicate that both sociocultural context and strategic choices play crucial roles in shaping students' learning outcomes.

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IMPROVING STUDENTS' SPEAKING SKILL THROUGH TRANSLANGUAGING STRATEGY (A COLLABORATIVE CLASSROOM ACTION RESEARCH TO THE TENTH-GRADE STUDENTS AT SMAN 5 LHOKSEUMAWE)

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ABSTRACT

This thesis is entitled *"Improving Students' Speaking Skill through Translanguaging Strategy (A Collaborative Classroom Action Research to the Tenth Grade Students at SMAN 5 Lhokseumawe)"*. Speaking is an essential skill in English language learning, yet many students still face difficulties, including limited vocabulary, low confidence, and limited classroom interaction. This research was conducted to find out how the translanguaging strategy could improve students' speaking skills and classroom participation. This research used a qualitative approach through collaborative classroom action research. The research sample comprised 25 tenth-grade students from SMAN 5 Lhokseumawe. The research was conducted in two cycles involving planning, action, observation, and reflection. Data were collected through pre-test, post-test, and observation checklists. The findings showed a significant improvement in students' speaking performance following the implementation of the translanguaging strategy. The average score of students increased from 61.08 in the pre-test to 68.76 in post-test cycle I and to 77.2 in post-test cycle II. Students became more active and confident during classroom discussions, and the overall classroom atmosphere became more interactive and supportive. The findings indicate that translanguaging is an effective strategy for improving speaking skills and fostering a more engaging classroom environment.

Keywords: *EFL, Speaking skill, Translanguaging*

INTRODUCTION

In today's increasingly interconnected world, speaking skills play a crucial role in education, professional environments, and intercultural communication. Students with strong speaking abilities are better able to participate in discussions, deliver presentations, and engage in academic and social activities. In professional contexts, effective speaking is essential in meetings, negotiations, and business communication. Moreover, good speaking competence is vital for successful interaction in multicultural settings. However, many learners continue to face difficulties in developing their speaking skills in both their first and second languages, which often hinder their academic and professional growth.

According to the Regulation of the Minister of National Education (2022), students at the end of Phase E are expected to use English to communicate with teachers, peers, and others in various contexts and for different purposes. They should be able to produce oral, written, and visual texts to express opinions, feelings, and ideas relevant to their lives. Despite these expectations, observations conducted on April 25, 2025,

among tenth-grade students at SMAN 5 Lhokseumawe indicated that most students struggled with English-language activities, particularly when presenting on advertising topics. The main problems identified were limited vocabulary mastery, poor pronunciation, fear of making mistakes, and lack of confidence in speaking in front of others.

In addition, teachers tend to dominate classroom instruction by focusing on explanations of material and on written tasks, thereby giving students limited opportunities to practice speaking. Teachers also often rely heavily on Indonesian, hesitating to use English out of concern that students might not understand, which reduces students' motivation and exposure to English. Therefore, teachers need strategies that both support comprehension and increase students' confidence to communicate.

One innovative strategy that can address these challenges is translanguaging. Translanguaging is defined as the flexible use of two or more languages to facilitate learning (Garcia & Wei, 2013). This approach allows students to use their first language (L1) alongside the target language (L2), helping them better understand the material and feel more confident when speaking (Garcia & Wei, 2014). Translanguaging enables learners to overcome linguistic barriers, improve comprehension, and gain confidence in expressing ideas in a second language (Canagarajah, 2011). Similarly, Translanguaging supports communication between teachers and students and enhances learning effectiveness, rather than being viewed as a weakness (Tang, 2002).

Although translanguaging has been widely studied in bilingual education, its application in Indonesian high school English classrooms remains limited. This research, therefore, introduces the use of a translanguaging strategy through collaborative classroom action research as an innovative approach to improve students' speaking skills. The purpose of this study is to investigate how the translanguaging strategy can enhance tenth-grade students' speaking skills and classroom participation at SMAN 5 Lhokseumawe. It is expected that this strategy will foster students' motivation, confidence, and ability to express ideas more effectively in English by integrating their first language as a learning resource.

METHOD

This study employed a qualitative approach through collaborative classroom action research (CAR) to improve students' speaking skills and classroom participation by implementing the translanguaging strategy. The CAR design was adapted from Kemmis and McTaggart's spiral model, which includes four stages: planning, acting, observing, and reflecting. Each cycle was conducted to identify problems, implement actions, observe the process, and evaluate outcomes to inform improvements in subsequent cycles (Kemmis & McTaggart, 2014).

The research was conducted at SMAN 5 Lhokseumawe, involving 25 tenth-grade students from class X.1 as participants. This class was selected because preliminary observations indicated that students faced difficulties speaking English, particularly in vocabulary, pronunciation, and confidence. The research was conducted in two cycles, each comprising two meetings. During the planning stage, the researcher

and the collaborator teacher designed lesson plans, teaching materials, and instruments such as observation checklists and speaking tests. The action stage involved applying the translanguaging strategy in speaking lessons using the topic "Advertisement." Students were encouraged to use both Indonesian (L1) and English (L2) to build comprehension and confidence before presenting their ideas in English.

Data were collected through tests and observations. The tests included a pre-test (to measure students' initial speaking ability) and post-tests at the end of each cycle (to evaluate improvement). Observation checklists were used to record teacher performance, student participation, and classroom atmosphere during implementation. The data analysis combined quantitative and qualitative techniques. Quantitative data from pre- and post-tests were analyzed to calculate mean scores and percentage improvement. Qualitative data from observations were analyzed using Miles and Huberman's model, which includes data condensation, data display, and conclusion drawing (Miles & Huberman, 2014). The validity of the data was ensured through triangulation by comparing results from different sources (tests and observations) and times (Cycle I and Cycle II). Both learning outcomes and classroom activity levels determined the success criteria. The action was considered successful if at least 75% of students achieved a score of at least 70 and if classroom engagement was 70% or higher, as assessed using observation checklists.

RESULT AND DISCUSSION

The Classroom Action Research in this study followed the four essential stages of CAR carried out across two cycles. In the planning stage, the researcher prepared lesson plans, materials, and assessment instruments for teaching speaking through translanguaging. The acting stage involved administering the pre-test, delivering the lessons across two meetings per cycle, and administering the post-test. During observation, the researcher and collaborator recorded teacher and student activities using observation sheets. A reflection was conducted to evaluate test results and observations, identify problems (e.g., low participation and incomplete implementation) in Cycle I, and revise actions for Cycle II.

Development Across Cycles

The implementation of the translanguaging strategy across two action research cycles resulted in notable improvements in students' speaking performance and classroom participation. The pre-test administered before Cycle I indicated that students' initial speaking ability remained below the minimum mastery criterion, with an average score of 61.08. Many students struggled to express ideas fluently, lacked confidence, and were not fully engaged during oral activities. These initial findings indicated the need for a classroom strategy that could provide both cognitive and affective support.

During Cycle I, the translanguaging strategy was introduced through bilingual brainstorming, teacher modeling in both languages, scaffolding, and collaborative work. Although the implementation followed the planned procedures, several limitations occurred. The teacher did not consistently provide bilingual brainstorming; monitoring of group activities was suboptimal; and the language-reflection phase was not conducted

as intended. Student participation also remained relatively low, as indicated by the observation results: teacher activity reached 66.66% and student activity reached 61.11%, both categorized as fair. Nevertheless, the post-test score in Cycle I increased to 68.76, indicating that translanguaging began to facilitate students' comprehension and production. However, most students had not yet met the expected mastery level.

Cycle II was conducted by revising the weaknesses identified in Cycle I. The implementation became more consistent and structured, particularly through the provision of bilingual brainstorming, clearer modeling of advertising examples, more active monitoring of group discussions, and the inclusion of language reflection to help students evaluate their use of L1 and L2. This refinement contributed to a substantial change in classroom atmosphere.

Observation results showed that teacher activity increased to 88.98% and student activity to 83.33%, both categorized as very good. The post-test score in Cycle II rose significantly to 77.20, and the majority of students successfully met the minimum criteria. Improvements were observed across all assessed components, including fluency, pronunciation, grammar, vocabulary, comprehension, and task accomplishment. The consistent upward trend across cycles indicates that translanguaging had a positive and measurable impact on students' speaking Development. To provide a clearer illustration of the students' speaking improvement at each stage, starting from the pre-test, post-test of Cycle I, to the post-test of Cycle II. It can be seen in the Table below:

**Student Speaking Improvement in
Cycles I and II**

Cycle	Test	Average
I	Pre-test	61.08
I	Post-test	68.76
II	Post-test	77.2

To provide a clearer illustration of the improvement in classroom atmosphere during the implementation of the translanguaging strategy, including teacher performance and student engagement from Cycle I to Cycle II, it can be seen in the Table below:

**Classroom Atmosphere Improvement while Implementing Translanguaging Strategy in
Cycles I and II**

Cycle	Observation Checklist	Percentage	Criteria
I	Observation Checklist for Teacher	66.66%	Fair
I	Observation Checklist for Students	61.11%	Fair
II	Observation Checklist for Teacher	88.98%	Very Good
II	Observation Checklist for Students	83.33%.	Very Good

Improvement Occurred Across Cycles

The improvement from the pre-test through Cycle II can be attributed to several interconnected factors. First, translanguaging reduced students' cognitive load by allowing them to build conceptual understanding in their first language before producing English output. This made it easier for them to formulate coherent messages and reduced the cognitive effort required to speak. Second, the strategy strengthened students' comprehension of instructional material by clarifying difficult vocabulary and complex content through bilingual explanations. When students understood the lesson more deeply, their speaking became more accurate and meaningful.

Third, translanguaging lowered affective barriers. Students who were previously anxious or hesitant became more confident when they were allowed to use Indonesian to plan their ideas. This emotional comfort translated into a greater willingness to speak, which in turn improved fluency and pronunciation. Finally, the collaborative environment created in Cycle II allowed students to negotiate meaning, share vocabulary, and support one another during speaking tasks. This interactive process enhanced their grammar, vocabulary, and overall task accomplishment.

The findings of this study confirm the theoretical foundations of translanguaging. Translanguaging enables learners to mobilize all of their linguistic resources in order to construct meaning and participate more fully in classroom communication (García & Wei, 2017). This principle was reflected in the classroom throughout both cycles, where students used their first language to understand concepts and their second language to produce spoken output. The process enabled them to interact more confidently and to work through linguistic challenges without feeling constrained by monolingual norms.

CONCLUSION

This study aimed to improve the speaking skills of tenth-grade students at SMAN 5 Lhokseumawe by implementing the translanguaging strategy. The research findings demonstrated that the strategy effectively enhanced students' speaking performance and classroom participation. The average speaking score increased from 61.08 in the pre-test to 68.76 in Cycle I and 77.20 in Cycle II, surpassing the minimum learning mastery criterion of 70. Observation results also showed a significant rise in both teacher implementation and student engagement, with classroom participation improving from 61.11% to 81.11%. These outcomes indicate that the use of translanguaging created a more interactive and supportive learning environment, allowing students to express their ideas more confidently by integrating their first language with English.

In relation to the research objectives, the findings confirm that the translanguaging strategy not only improved students' speaking proficiency but also fostered greater motivation and active participation in the learning process. The strategy helped bridge comprehension gaps and encouraged learners to use both languages as complementary tools for communication. Consequently, translanguaging proved to be a practical pedagogical approach in EFL classrooms, particularly for students who struggle with confidence and vocabulary limitations. For future research, it is

recommended that translanguaging be explored in other English-language skills, such as writing, listening, and reading, and applied across different educational levels to examine its broader pedagogical impact. Ongoing investigation of this approach can contribute to the Development of more inclusive and linguistically responsive English teaching practices.

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EFL LEARNERS' PERCEPTIONS OF USING SOLILOQUIZING ON ENGLISH SPEAKING PROFICIENCY

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ABSTRACT

This study explores EFL learners' perceptions of soliloquizing as a strategy for improving English speaking proficiency. Many English Education students at UIN Sultanah Nahrasiyah Lhokseumawe still struggle to speak fluently and confidently, often relying on written texts during presentations. Soliloquizing, talking to oneself, was observed as a self-initiated strategy during seminar preparation. This qualitative case study involved 30 students from the 4th, 6th, and 8th semesters, selected through purposive sampling. Data were collected through questionnaires and semi-structured interviews with two participants who achieved the highest scores. Their proficiency was further confirmed by their excellent performance in the Speaking in Academic Context course, which strengthened the credibility of their perspectives in the interview findings. Analysis focused on seven indicators of speaking proficiency: comprehension, grammar, pronunciation, vocabulary, fluency, motivation/confidence, and relevance. To explore the sensory basis of perception, interview responses were analyzed using the VARK model. The findings indicate that learners generally perceive soliloquizing positively as a means of improving pronunciation, vocabulary, fluency, and speaking confidence. Most learners exhibited dominant kinesthetic and auditory tendencies, such as talking while moving or listening to their own voice, which facilitated the internalization of language patterns and enhanced performance. It is concluded that soliloquizing is a practical and sensory-based self-practice strategy that supports speaking Development and fosters learner autonomy.

Keywords: *soliloquizing; student perceptions; English speaking proficiency; EFL*

INTRODUCTION

Fluency, self-assurance, and spontaneity are critical skills that students in the English Education Department are required to possess. The capacity to communicate ideas clearly, participate in scholarly conversations, and deliver organized presentations without relying primarily on printed notes is emphasized in the course material. The speaking course seeks to foster critical thinking, communicative competence, and self-expression in real-world academic settings, as well as linguistic accuracy.

Despite these goals, many students at UIN Sultanah Nahrasiyah continue to struggle with English fluency and confidence. Preliminary classroom observations suggested that students usually rely on written materials when presenting, resulting in an awkward and less communicative approach. This reliance on notes indicates a lack of spontaneity and confidence when speaking English.

Interestingly, during the English Education Department's International Seminar, an academic event integrated into the speaking course, some students were observed

practicing self-directed speech in preparation for their presentations. Informal talks revealed that this practice helped individuals organize ideas, internalize sentence structures, and control fear. This occurrence is consistent with the concept of soliloquizing, a self-talk approach in which students practice speaking to themselves to improve oral performance.

The benefits of soliloquizing for language acquisition have been emphasized in earlier research. According to Li and Zhou (2001), self-talk is a valuable strategy for improving oral expressiveness among EFL students. In a similar vein, Hermansyah (2021) found that self-talk significantly enhanced students' confidence and speaking fluency, and Sri and Ayu (2022) reported that students who used the technique were more courageous and livelier. Although these results highlight the instructional benefits of soliloquizing, little research has examined how students in Indonesian contexts, specifically at UIN Sultanah Nahrasiyah, perceive this tactic. Therefore, this study aims to investigate EFL learners' perceptions of the use of soliloquizing in relation to English speaking proficiency

METHOD

To investigate how EFL learners perceive soliloquizing as a method for improving their English-speaking abilities, this study employed a qualitative, instrumental case study design. According to Stake (1995), an instrumental case study focuses not on the case's uniqueness but on gaining insight into a broader phenomenon. This concept proved suitable for investigating how soliloquizing fosters university students' speaking Development in an organic setting. The first observation of soliloquizing behavior was conducted at the English Education Department, Faculty of Teacher Training and Education, UIN Sultanah Nahrasiyah Lhokseumawe, located on the Medan–Banda Aceh Road, Alue Awe, Muara Dua, Lhokseumawe City, Aceh Province.

Purposive sampling was used to select 30 English Education students from the fourth, sixth, and eighth semesters for participation (Ary et al., 2010). To ensure adequate proficiency, they had finished advanced speaking courses, including Speaking in Academic Context and Speaking in Professional Context. Two of the top respondents were interviewed to obtain more in-depth information, and 21 of the 30 valid questionnaire replies were examined. Their advanced competency was validated by documentation of their final speaking grades.

The three instruments were used to collect data: semi-structured interviews, closed-ended, and open-ended questionnaires. The forty-three items on the surveys, adapted for soliloquizing contexts and drawn from Lestary (2022), comprised seven indicators: vocabulary, grammar, pronunciation, comprehension, fluency, motivation/confidence, and relevance. Also modified the interview questions from Arini and Wahyudin (2022). Miles and Huberman's (2018) interactive model, which includes data condensation, data display, and conclusion drawing, was used to analyze the data. This procedure enabled a comprehensive understanding of students' perspectives and experiences, providing evidence that soliloquizing is a successful self-directed learning method for enhancing English speaking ability.

RESULT AND DISCUSSION

1. EFL Learners' Perception of Soliloquizing in Comprehension

The results showed that 84.13% of participants considered soliloquizing beneficial, indicating that the method improved their understanding of English. The majority of respondents stated that soliloquizing helped them organize ideas, retain vocabulary, and convey concepts more clearly. Interview data confirmed these findings, demonstrating that soliloquizing facilitated the internalization of concepts before speaking. Respondent 1 rehearsed aloud while writing essential points, combining read/write and kinaesthetic modalities, whereas Respondent 2 employed mirror and self-recording techniques that engaged visual, auditory, and kinaesthetic channels, increasing comprehension through multimodal engagement.

The above conclusion is consistent with Hermansyah (2021), who discovered that soliloquizing enhances sentence structure, word choice, and pronunciation, all of which are necessary for comprehension. Similarly, Huda et al. (2022) found that learner autonomy in Self-Directed Learning (SDL) improves language comprehension, and soliloquizing is a speaking-oriented kind of SDL that promotes reflection and self-regulation. Furthermore, Giladi et al. (2022) demonstrated that self-efficacy improves comprehension through prolonged effort, a mechanism similar to soliloquizing, in which learners develop confidence and cognitive organization through autonomous practice.

2. EFL Learners' Perception of Soliloquizing in Grammar

The study found that 79.37% of respondents reported that soliloquizing improved their grammatical awareness, particularly in sentence structure, tenses, and word order. The majority of participants said that speaking to themselves helped them detect and repair structural problems, highlighting the importance of soliloquizing as a self-correction and reflection tool. This activity increased metalinguistic awareness; students became more aware of grammatical norms as they monitored their own work. Interview results supported this, demonstrating that learners frequently identified and corrected errors after hearing themselves speak, such as changing "I was met Putri" to "I met Putri." This approach integrates tactile and auditory modalities, with learners internalizing grammar through active self-monitoring. By enabling students to control and improve their grammar autonomously, soliloquizing also promoted learner autonomy, which is consistent with Zakirova's (2025) theory that autonomy in grammar acquisition enhances analytical thinking and long-term memory.

Similarly, Wiltshko (2025) highlighted that self-talk allows learners to deliberately analyze language form during spontaneous speech, revealing how grammar functions cognitively. These results are consistent with the current study, which discovered that soliloquizing turns studying grammar from a passive activity into an active one. Overall, the results indicate that soliloquizing improves grammatical clarity by promoting contemplation, independence, and understanding of language structure, although some students preferred external feedback. In conclusion, soliloquizing strengthens learners' command of spoken English grammar while also serving as a feedback mechanism and an autonomy-building strategy.

3. EFL Learners' Perception of Soliloquizing in Pronunciation

The outcomes revealed that 94.05% of respondents considered soliloquizing a helpful approach to improving pronunciation. Most students became more aware of their articulation errors and independently rectified them, resulting in clearer, more natural speech. They commonly employed tactics such as repeating difficult words or comparing their pronunciation with audio sources, indicating that soliloquizing served as both speaking practice and a reflective tool for building phonological awareness and self-regulated learning. The interview results showed that active self-monitoring and repetition improved pronunciation. Students rehearsed aloud, using mirrors or recording their voices to improve their articulation, emphasis, and rhythm. This multisensory engagement, particularly auditory and kinesthetic, enabled them to consciously adjust their pronunciation in real time, highlighting soliloquizing as an active, self-generated form of feedback rather than passive repetition.

The results presented are consistent with Cojo Guatame's (2019) study on self-assessment, which found that independent reflection improves pronunciation accuracy, and with Meritan's (2021) research, which emphasizes that self-reflection increases phonological processing and pronunciation awareness. In line with these viewpoints, soliloquizing in this study served as an informal self-assessment activity, encouraging students to monitor, evaluate, and modify their pronunciation independently. Soliloquizing, as a result, promotes both speech intelligibility and learner confidence through reflective, multimodal practice.

4. EFL Learners' Perception of Soliloquizing in Vocabulary

The data indicate that 92.06% of respondents considered soliloquizing an essential means of expanding their English vocabulary. Participants reported that speaking to themselves in English helped them remember and use new terms from a variety of sources, including reading, listening, and daily experiences, while also increasing their confidence and accuracy in word use. This technique pushed students to think critically about meaning and context, moving vocabulary learning from memorization to active, reflective practice. Soliloquizing enhanced stronger lexical understanding by multisensory engagement, speaking aloud, recording themselves, and modifying their language in real time, in accordance with the VARK model, which states that visual, aural, and kinesthetic interaction strengthens comprehension.

Tılfarlıoğlu and Sherwani (2018) found that learner autonomy improves learners' ability to select and apply effective vocabulary-learning strategies. Afrillah (2022) also found that the self-talk strategy significantly enhanced students' vocabulary mastery and speaking fluency. This research shows that soliloquizing is an autonomous and metacognitive strategy that not only broadens learners' vocabulary but also promotes contextual precision, fluency, and self-regulated language Development.

5. EFL Learners' Perception of Soliloquizing in Fluency

The data revealed that 82.54% of respondents reported that soliloquizing was highly beneficial for improving their speaking fluency. Participants reported that frequent self-practice improved their ability to speak more smoothly, reduce pauses, and

articulate thoughts in complete sentences with greater confidence. The data also revealed that several reinforcing factors, such as pronunciation training, vocabulary preparation, and regular speaking routines, contributed to improved fluency, suggesting that soliloquizing enhances both linguistic and psychological readiness. Kinesthetic engagement was critical from a sensory standpoint, as learners rehearsed aloud, recorded themselves, and adjusted their tempo and coherence in response to feedback. This multimodal approach, which included visual and aural reflection through mirror or recording practice, transformed soliloquizing into a self-regulated and reflective technique that improved fluency beyond mere repetition.

These findings are consistent with earlier research proving the importance of soliloquizing to oral communication. According to Azka et al. (2023), the self-talk approach significantly improved students' speaking fluency and accuracy. Similarly, Kashef and Khabazian (2025) found that persistent soliloquizing and self-assessment over eight months resulted in significant gains in fluency and precision, which even contributed to improved exam performance. Both studies support the current finding that soliloquizing promotes spontaneous, confident, and well-structured speech through self-directed, multimodal engagement. As a result, soliloquizing emerges as a cognitive and affective method for long-term progress in spoken English ability, as well as a practical fluency-building technique.

6. EFL Learners' Perception of Soliloquizing in Motivation and Confidence

The results revealed that 87.30% of respondents recognized soliloquizing as a significant factor in their motivation and confidence in speaking English. Participants stated that engaging in regular self-talk reduced fear, increased enthusiasm, and improved psychological preparedness for communication, particularly before presentations, discussions, or meetings with strangers. This is consistent with Turaya's (2020) story study, which discovered that soliloquizing helps English debaters develop psychological resilience and self-assurance in high-pressure situations. Respondents in this study viewed soliloquizing as a contemplative and motivating practice that helped them build confidence and fluency through repeated verbal rehearsal. From a sensory perspective, physical and auditory engagement, such as speaking aloud, using mirrors, or recording their voices, enabled learners to monitor and modify their performance, thereby enhancing their sense of control and self-efficacy.

These findings are consistent with those reported by Ayni et al. (2025), who found that self-talk improves learners' emotional regulation, confidence, and performance in speaking tasks. Both studies emphasize soliloquizing as a self-directed method that boosts motivation through positive verbal reinforcement and practical application. The current findings also support Racy and Morin's psychological theory, which defines self-talk and inner speech as self-directed verbal thoughts that support self-regulation and confidence-building. In this regard, soliloquizing serves as a practical form of self-talk for EFL learners, allowing them to organize their ideas, affirm their talents, and psychologically prepare for communication challenges. Thus, soliloquizing not only increases linguistic preparedness but also boosts learners'

motivation and confidence, establishing it as a cognitive and emotional technique for effective English communication.

7. EFL Learners' Perception of Soliloquizing in Relevance

According to the data, 95.24% of respondents considered soliloquizing highly relevant to real-world communication, suggesting that it effectively links academic speaking practice to practical communication. Students reported that soliloquizing helped them articulate ideas clearly, organize their thoughts, and build confidence across contexts, including classroom presentations, formal debates, and everyday interactions. Interview data also verified its incorporation into daily habits, with students practicing soliloquizing while cooking, watching movies, or listening to podcasts, demonstrating how it can become a natural form of language rehearsal. Students acquired language and structure through auditory and tactile activities, such as speaking aloud, imitating voices, and recording themselves. Thus, soliloquizing serves as a versatile, self-directed approach for connecting classroom learning to spontaneous real-life speaking situations.

These findings are consistent with Dimala et al. (2023), who found that positive self-talk enhances self-confidence in both personal and social settings, highlighting psychological and emotional benefits beyond academics. Similarly, Alexander et al. (2024) found that inner speech supports daily planning, memory, and motivation, even among individuals with chronic aphasia, demonstrating its broad cognitive and emotional applications. Both studies support the current findings that soliloquizing, as a sort of inner speech, broadens learning beyond the classroom by improving mental preparation, confidence, and communication skills in everyday life. As a result, soliloquizing is not only a practical approach for developing English proficiency but also a long-term tool for personal and social self-development that connects theory and practice.

CONCLUSION

The study concludes that EFL learners generally hold positive perceptions of soliloquizing as an effective strategy for enhancing English-speaking proficiency. Questionnaire data showed consistently high agreement across all seven indicators: comprehension (84.13%), grammar (79.37%), pronunciation (94.05%), vocabulary (92.06%), fluency (82.54%), motivation and confidence (87.30%), and relevance (95.24%). These results demonstrate that soliloquizing not only strengthens linguistic skills, including idea organization, vocabulary recall, and pronunciation accuracy, but also fosters psychological readiness, confidence, and communicative adaptability. Learners perceived soliloquizing as highly practical beyond classroom contexts, making it a valuable tool for authentic communication.

Interview findings reinforced these results, revealing that learners primarily engaged in soliloquizing through kinesthetic and auditory modalities, speaking aloud, using mirrors, self-recording, or writing key points, which helped internalize language patterns through active experience rather than passive learning. The two interviewed high-achieving participants exemplified this process, applying soliloquizing successfully in academic and real-life speaking tasks. Overall, soliloquizing emerges as a learner-

centered and sensory-based strategy that integrates cognitive, linguistic, and emotional dimensions, enabling EFL learners to develop fluency, confidence, and readiness for communication both inside and outside the classroom.

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AN ANALYSIS OF STUDENTS' LEARNING STYLES AND STRATEGIES IN ENGLISH LANGUAGE MASTERY

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ABSTRACT

This study investigates the types of learning styles and strategies used by students in English language mastery, how these strategies are implemented in relation to learning styles, and the factors influencing their use. The research involved six seventh-grade students from an international class at SMP Sukma Bangsa Lhokseumawe, categorized by English proficiency levels: beginner, intermediate, and advanced. Using a qualitative case study approach, data were collected through classroom observations and interviews, and then analyzed thematically. The findings revealed that visual and auditory learning styles were the most prevalent, whereas reading/writing styles were more frequent among advanced students. Kinesthetic learning was minimally observed due to the limited presence of physical activities in class. In terms of learning strategies, students employed cognitive, metacognitive, and socio-affective strategies. Advanced learners demonstrated stronger metacognitive awareness, including planning and self-monitoring. Students implemented learning strategies aligned with their learning styles: visual learners used diagrams and color-coded notes, auditory learners benefited from discussion and repetition, and reading/writing learners engaged in rewriting and organizing material. The study also identified internal and external factors influencing learning strategies. Internal factors included motivation and self-discipline, whereas external factors involved teacher guidance, parental support, and peer collaboration. Overall, the findings highlight that students' learning styles strongly shape their strategic choices in mastering English. Understanding these relationships can help teachers design more adaptive and effective learning environments.

Keywords: *English Language Mastery; Learning Styles; Learning Strategies; Students; Factors*

INTRODUCTION

Language plays a crucial role in human communication as a medium for expressing ideas, emotions, and thoughts. In education, mastery of English as a foreign language is essential for academic achievement and future career Development. English proficiency not only allows students to access global knowledge but also enhances their communication skills in both academic and professional contexts. However, many students struggle to master English effectively because learners process information differently, depending on their individual learning styles and strategies.

Learning style refers to an individual's preferred way of receiving and processing information. According to Fleming (2001), the VARK model classifies learning styles into four categories: visual, auditory, reading/writing, and kinesthetic. Visual learners rely on images, diagrams, and written forms; auditory learners focus on listening and verbal interaction; reading/writing learners prefer textual information; while kinesthetic learners learn best through physical activity. Understanding these styles helps teachers design

instruction that accommodates various learner preferences, making classroom learning more effective and inclusive.

Learning strategy, by contrast, refers to the specific actions, techniques, or methods learners use to enhance language acquisition. Oxford (1990) categorized learning strategies into cognitive, metacognitive, and socio-affective types. Cognitive strategies focus on processing and organizing information; metacognitive strategies involve planning, monitoring, and evaluating learning; and socio-affective strategies relate to social interaction and emotional regulation. Studies such as those by Brown (2007) and Nation (2013) have shown that effective learners tend to combine multiple strategies according to their learning styles, which contributes to greater autonomy and motivation in mastering a language.

Previous studies have investigated the relationship between learning styles and strategies in second language learning. For example, Setyaningsih et al. (2016) found that students who were aware of their learning styles could select strategies that enhanced their comprehension and retention. However, most studies have been conducted in conventional classroom contexts rather than in international or bilingual education settings. This gap highlights the need for research exploring how students in global classrooms, which often adopt the Cambridge Curriculum, adapt their learning styles and strategies to achieve mastery of English.

The scientific novelty of this study lies in its focus on identifying the interconnection between learning styles and strategies in an international classroom context, where English serves as the medium of instruction and learning expectations are higher. By combining observational and interview data, this research provides deeper insight into how students apply different learning approaches based on their proficiency levels and the factors that influence their choices. The findings are expected to contribute to the Development of more adaptive and learner-centered instructional practices in English language teaching.

Based on this background, the research seeks to address the following problems: (1) What types of learning styles and strategies are used by students in English language mastery? (2) How do students implement learning strategies according to their learning styles? and (3) What factors influence the choice of learning strategies? The purpose of this study is to analyze students' learning styles and strategies in mastering English, examine how they apply these strategies according to their styles, and identify both internal and external factors that influence their learning process.

METHOD

This study employed a qualitative case study design to explore students' learning styles and learning strategies in mastering English. The qualitative approach was chosen because it allows for a deep understanding of learners' behaviors, experiences, and perceptions in a real classroom context. According to Creswell (2012), qualitative research emphasizes describing and interpreting meanings from participants' perspectives rather than testing hypotheses numerically.

The research was conducted at SMP Sukma Bangsa Lhokseumawe, a private international-based junior high school in Aceh that implements the Cambridge

Curriculum. The participants were six seventh-grade students from the international class, selected through purposive sampling to represent different levels of English proficiency: beginner, intermediate, and advanced. In addition, one English teacher from the same class participated in the study to provide triangulation data and to confirm the accuracy and credibility of the findings.

Data were collected through classroom observations, interviews, and documentation. Observations were conducted using an observation checklist to identify students' learning styles—visual, auditory, reading/writing, and kinesthetic—and the learning strategies applied during English lessons. Interviews with students and the English teacher were carried out to explore learning experiences, preferences, and influencing factors. In contrast, documentation such as photographs, students' worksheets, and field notes supported the validity of the data. The data were collected over several weeks and analyzed using Miles and Huberman's (1994) interactive model, which includes data condensation, data display, and conclusion drawing and verification, enabling the researcher to identify patterns of students' learning styles and strategies and to interpret factors influencing their learning behaviors.

RESULTS AND DISCUSSION

This study examined the learning styles and strategies students used to master English, the implementation of these strategies according to each learner's style, and the internal and external factors that influenced these strategies. The results were obtained through classroom observation and in-depth interviews with six seventh-grade students from the international class of SMP Sukma Bangsa Lhokseumawe. Thematic analysis was used to identify patterns in the data.

1. Students' Learning Styles in English Mastery

The findings showed that students exhibited three dominant learning styles: visual, auditory, and reading/writing. The visual learning style was observed among four students who preferred using diagrams, pictures, and written notes to process information. The auditory style was evident among students who learned best through listening and repetition. In contrast, the reading/writing style was more common among advanced learners who preferred to summarize lessons and rewrite notes.

Table 1. Students' Dominant Learning Styles and Strategies

Learning Style	Number of Students	Dominant Strategies Used
Visual	4	Note-taking, diagramming, and color-coding
Auditory	2	Listening, repetition, discussion
Reading / Writing	2	Summarizing, rewriting, organizing

As shown in Table 1, the visual style was the most prominent, followed by the auditory and reading/writing styles, whereas kinesthetic learning was absent. This pattern indicates that students in this setting relied heavily on visual and auditory input because the

teacher emphasized textual and verbal instruction. This finding supports Fleming's (2001) VARK theory, which states that learners process information most effectively through their preferred sensory modality. The predominance of visual learners may also be explained by the nature of the Cambridge-based classroom, which uses written texts, presentations, and visual media as primary learning tools. Similarly, auditory learners benefited from the teacher's verbal explanation and repetition, aligning with Brown's (2007) view that auditory input enhances linguistic retention.

CONCLUSION

This study concludes that students' learning styles strongly influence the strategies they employ to master English. The analysis revealed that visual, auditory, and reading/writing styles were the most prevalent among seventh-grade students at SMP Sukma Bangsa Lhokseumawe. In contrast, kinesthetic learning was rarely observed due to the limited use of movement-based activities in class. The use of learning strategies—cognitive, metacognitive, and socio-affective—varied with students' proficiency levels and dominant learning styles. Advanced learners tended to apply metacognitive strategies such as planning, self-monitoring, and evaluation, whereas beginners relied more on cognitive techniques such as note-taking and repetition. These findings confirm the research hypothesis that individual learning styles determine how learners select and implement strategies to achieve mastery of English. Internal factors such as motivation and personal discipline, along with external factors including teacher feedback, parental support, and peer collaboration, were found to shape students' learning behaviors. The study highlights the need for teachers to recognize these differences and design instruction that accommodates multiple learning styles. Future research could examine how learning strategies evolve or investigate a larger, more diverse student population to broaden understanding of learning style adaptation in international classroom contexts.

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CHAPTER 3: Linguistics, Applied Linguistics, Humaniora

IMAGERY IN THE POETRY COLLECTION "TESTIMONY OF A PAIR OF SANDALS" AND ITS RELEVANCE TO POETRY LEARNING IN HIGH SCHOOL

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ABSTRACT

This research aims to describe the imagery found in Muhammad Subhan's poetry collection *Testimony of a Pair of Sandals* and its relevance to poetry learning in high school. This study uses a qualitative approach with a descriptive method. The research data consist of words and phrases containing imagery taken from four poems in the collection. Data collection techniques include literature review, reading, note-taking, and documentation. The research results show that there are 88 instances of imagery, consisting of 46 visual images, 3 auditory images, 29 kinesthetic images, 6 tactile images, and 4 olfactory images. The relevance of the research findings to poetry learning in high school indicates that the use of imagery can increase students' interest, understanding, and appreciation of poetry. The survey results show that more than 85% of students responded positively to poetry learning using PowerPoint media that display imagery.

Keywords: *Imagery; poetry; relevance.*

INTRODUCTION

Literary works are the result of human expression of thoughts, feelings, and experiences conveyed through beautiful language rich in aesthetic value. Literary works not only serve as entertainment but also function as a means of reflecting on social life and as a medium for conveying moral, cultural, and humanitarian values. One form of literary work with high expressive power is poetry. Poetry conveys ideas and emotions through dense, connotative language rich in imagery that stimulates the reader's imagination. In poetry, imagery plays an important role in creating sensory experiences that readers can perceive. Imagery allows readers to see, hear, feel, or smell what is expressed by the poet. According to Nurgiyantoro, imagery is divided into visual, auditory, kinesthetic, tactile, thermal, and olfactory imagery. The appropriate use of imagery makes a poem more vivid and creates a deep aesthetic effect. Therefore, imagery analysis becomes one of the important ways to comprehensively understand the meaning and beauty of poetry.

One literary work that is interesting to study from the perspective of imagery is the poetry collection *Testimony of a Pair of Sandals* by Muhammad Subhan. This collection depicts social, spiritual, and human reflections conveyed through poetic language and the use of diverse imagery. The poems in this anthology present a rich variety of imagery that not only reflects the poet's personal experiences but also represents the social and moral conditions of society. Therefore, analyzing the imagery in

this poetry collection is relevant to understanding how imagery functions as both an aesthetic and meaning-making element in poetry.

Research on imagery in this poetry collection not only contributes to literary studies but also has relevance for poetry learning at the high school level. In the context of the Merdeka Curriculum, poetry learning emphasizes students' ability to identify poetic elements such as diction, figurative language, rhyme, and imagery. Understanding imagery helps students appreciate poetry more deeply, develop language sensitivity, and enhance critical and creative thinking skills. Based on observations at SMA Negeri 1 Syamtalira Bayu, it was found that students still experience difficulties in recognizing and understanding the types of imagery in poetry. This condition indicates the need for more concrete learning strategies through the analysis of literary works rich in imagery.

Several previous studies have examined imagery in literary works, such as Athiya Almanurah's (2022) research analyzing imagery in Adi K's poem *Sesunyi Cahaya* and Ridho Muhammad Zein's (2020) study of imagery in the anthology *Suara Batu*. However, these studies have not extensively linked their analytical findings to applications in school-based learning. Therefore, this research offers novelty by examining imagery in the poetry collection *Testimony of a Pair of Sandals* while simultaneously connecting the analysis to its relevance for poetry learning at the high school level. Based on this background, the objectives of this research are to describe the types of imagery found in the poetry collection *Testimony of a Pair of Sandals* by Muhammad Subhan and to explain their relevance to poetry learning at the high school level. The results of this research are expected to contribute scientifically to the field of literary studies and to serve as a practical reference for developing more creative and meaningful poetry learning.

METHOD

This research uses a qualitative approach with a descriptive research design. The data source of this research is the poetry collection *Testimony of a Pair of Sandals* by Muhammad Subhan, published by IKO Media Publisher in 2020. The research data consist of words or phrases containing imagery. Data collection techniques were conducted through literature review, reading, note-taking, and documentation. Data analysis employed qualitative descriptive techniques through the stages of data reduction, data presentation, and conclusion drawing.

RESULTS AND DISCUSSION

The research findings indicate that there are 88 instances of imagery identified in four selected poems from the poetry collection *Testimony of a Pair of Sandals*. The types of imagery found include visual, auditory, kinesthetic, tactile, and olfactory imagery. Visual imagery is dominant, with 46 instances, followed by kinesthetic imagery with 29 instances, tactile imagery with 6 instances, auditory imagery with 3 instances, and olfactory imagery with 4 instances.

Visual Imagery

Visual imagery refers to images related to objects that can be perceived through sight. These objects include elements that are visible to the naked eye, such as tables,

chairs, windows, doors, and other concrete forms. The researchers provide explanations of several visual imagery instances found in the poetry collection *Testimony of a Pair of Sandals*, focusing on the most representative data to offer an in-depth overview of the various forms of visual imagery in the poems. These data describe objects, atmospheres, and activities that readers can clearly visualize, ranging from natural elements such as bridges and riverbanks to depictions of people and cultural objects that enhance emotional nuances and the social context of the poems.

And the young girl sat side by side on the wedding dais (C/L/D9)

The saluang, bansi, and drumbeat were boisterously scattered (C/L/D10)

After that, the days became like icing on a cake.

Visual imagery is evident in the lines describing young girls sitting side by side on the wedding dais and in the lines mentioning *saluang*, *bansi*, and drum. The data coded as C/L/D9, found in the line “young girls sitting side by side on the wedding dais,” present a concrete visual image that enables readers to vividly imagine the figures, positions, and atmosphere of the scene. Meanwhile, the data coded as C/L/D10, found in the line “saluang, bansi, and drum,” provide descriptions of objects that can be visually perceived in reality, namely the *saluang* (a traditional musical instrument), the *bansi* (a flute), and the drum. These terms originate from the Minangkabau language and represent elements of the traditional musical culture of the Minangkabau people in West Sumatra. Through these descriptions, readers are able to visually imagine the shapes and presence of the musical instruments.

In a mosquito net bed with walls made of woven bamboo (C/L/D11)

A baby arranges cries at his mother's breast (C/L/D12)

And that is a portrait of himself.

Based on the poem above, visual imagery is found in the lines “*A bed with mosquito netting and walls made of woven palm leaves*” and “*a baby soothing its cries at its mother's breast.*” The data coded as C/L/D11 show that the word *bed* refers to a physical object that can be visually perceived. A *mosquito net* is a thin curtain usually hung around a bed, which can be easily visualized by the reader. The phrase *walls made of woven palm leaves* indicates walls constructed from *tepas* (woven bamboo or leaves), providing a clear visual depiction of their form and texture. Altogether, this phrase creates a concrete and specific visual image that allows readers to imagine the atmosphere of a room furnished with a bed, mosquito netting, and woven walls.

In the line “*a baby soothing its cries at its mother's breast,*” visual imagery appears through the depiction of a baby calming itself while being held close to its mother's body. This imagery vividly brings the scene to life, enabling readers to clearly visualize a baby in its mother's arms or resting on her chest, either nursing or being comforted.

I tread upon your fertile land, among (C/L/D37)

the lushness of the forest, the orange groves (C/L/D38)

bearing ripe fruit, and coffee fields (C/L/D39)

whose beans have a fragrant aroma.

In the excerpt from the poem coded C/L/D37, particularly in the phrase “*fertile is your soil*,” the words *fertile* and *soil* evoke a strong visual image of rich, dark, moist earth that supports abundant plant growth. Readers are able to imagine soil that is fertile, vibrant, and ready for cultivation. In C/L/D38, found in the line “*the lushness of the forest, the orange groves*,” visual imagery is created through the words *lushness* and *forest*, which describe a dense area filled with trees and thick foliage that limits visibility. This imagery enables readers to visualize a green, dense, and natural forest atmosphere. The phrase “*orange groves*” further evokes an image of orchards filled with orange trees, allowing readers to imagine rows of trees with green leaves and fruit-bearing branches.

Additionally, in C/L/D37, the phrases “*bearing ripe fruit*” and “*coffee fields*” also contain visual imagery. *Bearing ripe fruit* describes fruit that is mature and fresh, encouraging readers to imagine bright colors, full shapes, and appealing textures. Meanwhile, *coffee fields* present a visual image of vast coffee plantations with neatly arranged rows of coffee plants, allowing readers to picture the expansive landscape dominated by green coffee leaves.

Auditory Imagery

Auditory imagery refers to imagery related to sounds that can be perceived by the sense of hearing. This type of imagery describes specific sounds through descriptive words or sound representations, enabling readers to imagine hearing the sounds as if they were actually present, even though they exist only in the reader’s mind. The following data present instances of auditory imagery found in the poems.

*And the young girl sat side by side on the wedding dais
The saluang, bansi, and tambour drums **burst forth** (C/D/D1)
After that, the days became like cotton candy*

Based on the poem above, auditory imagery is found in the phrase *bursting with noise*. The expression *riuh menghambur* describes loud and boisterous sounds produced by musical instruments being played simultaneously. This imagery creates the impression that the reader can hear the lively and energetic music, as if it were sounding directly in their ears.

*In a mosquito net bed with walls of woven palm leaves
A baby **arranged his cries** at his mother's breast (C/D/D2)
And that was a portrait of himself*

Auditory imagery is found in the second line, *arranging cries*. The phrase evokes auditory imagery because the word *cries* is closely associated with the sound of a baby crying. Through this expression, the reader can almost hear the sound of the baby’s cries.

*The mud is still, the lake is calm. At the top of the hill
Pine leaves rustled in the wind, **rustling**, (C/D/D3)
Lulling our naked eyes.*

Auditory imagery is present in the second line, specifically in the word *rustling*. In this line, *rustling* describes the soft, whispering sound produced when the wind moves through the pine leaves. This sound creates an impression of gentle and calming natural ambience, allowing the reader to imagine and “hear” the leaves softly stirred by the wind.

Kinesthetic Imagery

Kinesthetic imagery, or movement imagery, refers to images related to objects or actions involving motion that can be visually perceived. This type of imagery depicts motor activity or movement rather than static objects. The following section presents a detailed explanation of several kinesthetic imagery data that have been thoroughly analyzed by the researcher.

*Once upon a time he **returned** to his ancestral land (C/G/D1)*

***Looking** for a father lost to the earth (C/G/D2)*

***Sent** by his mother who was waiting out of longing (C/G/D3)*

In the excerpt above, kinesthetic imagery is found in the words *go home*, *search*, and *are told*. In data (1), the phrase *go home* indicates an action involving movement, namely someone traveling back to their ancestral land or place of origin. This phrase explicitly describes physical movement or displacement. In data (2), the phrase *looking for* indicates physical activity or effort involving movement to find something, specifically the missing father. In data (3), the word *disuruh* (are told) indicates an action or command that prompts someone to perform an activity. This word describes physical movement carried out in response to the mother’s instruction. Based on these expressions, the quotation evokes kinesthetic imagery in the reader’s mind.

*On the beach sand, hermit crabs embrace the coral,
fishermen "cast their nets" for fortune, "go to sea" (C/G/D24)*

"against the waves, returning" bringing love (C/G/D25)

throughout the "paddling" without breaking (C/G/D26)

In the poem above, kinesthetic imagery is found in the words *fishing*, *going out to sea*, *battling the waves*, and *returning home*. Data C/G/D24, particularly the word *fishing*, illustrates the active movement of fishermen catching fish using nets, while the phrase *going out to sea* describes the act of traveling across the ocean. The kinesthetic imagery in data C/G/D25, expressed through the phrase *battling the waves*, depicts a physical struggle against strong sea waves, emphasizing intense movement and effort. The phrase *returning home* represents the movement back to land after completing the activity at sea. Furthermore, data C/G/D26, specifically the phrase *kayuh* (paddling) *without a break*, reinforces the image of continuous and determined movement, referring to the persistent and uninterrupted action of rowing a boat. Based on these kinesthetic imagery data, readers can vividly imagine and experience the dynamic movements and resilience of fishermen as they confront the challenges of nature.

Tactile Imagery

Tactile imagery refers to imagery that evokes sensations of touch or physical feeling perceived through the sense of touch, such as smoothness, roughness, coldness, heat, warmth, and other bodily sensations. This type of imagery enables readers to physically feel what is being described in the poem, thereby strengthening emotional and sensory engagement. The following section presents a detailed explanation of several tactile imagery data that have been analyzed in depth by the researchers.

*The sandal is a witness to **a chest pierced by a bullet,**
a throat slit (C/R/D2)
 kitchen knife. Witnesses of the roasted house, hating,
 witnesses of the women whose virginity was stolen by ambition,
 witnesses of the burning of holy books.
 Witness **the noose that chokes the neck** (C/R/D3)
 at the gallows. A witness who no longer knows what mass
 to utter because they have lost their words.*

In the poem excerpt labeled C/R/D2, the phrase *chest pierced by a bullet* suggests a sharp and piercing sensation of pain, as if the chest physically feels the impact of the bullet. This conveys a very concrete and intense physical sensation, which falls under tactile imagery. Similarly, the phrase *a throat slit with a kitchen knife* represents tactile imagery because the act of slitting a throat evokes a sharp, rough, and painful physical sensation, allowing the reader to imagine the pressure and injury experienced on the neck. Meanwhile, C/R/D3 in the line *the noose that strangles the neck* presents an image of a rope tightening around the neck, evoking sensations of pressure and constriction felt by the body, particularly in the neck area. This expression strongly represents tactile imagery because it emphasizes physical contact and the sensation of tightness.

Olfactory Imagery

Olfactory imagery refers to descriptions that evoke the sense of smell or aroma, allowing readers to feel as though they can perceive the scents presented in the literary work. The following section provides a detailed explanation of the olfactory imagery data that have been thoroughly analyzed by the researcher.

*The time passed, and he witnessed the argument.
 in the season of cracks, when the **plague releases incense** (C/C/D1)
The stench of death rots in the tangle of conflict (C/C/D2)*

The bolded text in the poem excerpt above represents **olfactory imagery**, as indicated by the words *incense* and *the smell of decaying death*. Data C/C/D1 refers to the word *incense*, which denotes an aromatic substance that is burned and produces a strong, distinctive fragrance. The phrase *mengeluarkan menyany* describes the presence of incense smoke in the air, enabling readers to imagine and experience the scent through their sense of smell. Meanwhile, data C/C/D2, *the smell of decaying death*, depicts a

pungent and unpleasant odor typically associated with a decomposing body. This expression evokes a powerful and vivid olfactory image, allowing readers to clearly imagine the smell and enhancing the emotional impact of the poem.

*Here, dreams hold memories.
The salty aroma of dried fish wafted thru,
filling you with childhood longing. (C/C/D3)*

Based on the data, three instances of olfactory imagery were identified, one of which appears in the line “*The salty aroma of dried fish wafted through the air.*” The phrase *salty aroma of dried fish* directly evokes the distinctive, strong, and salty smell of dried fish, creating a vivid olfactory sensation in the reader’s imagination. The word *menyeruak* (to burst forth) further intensifies this imagery by suggesting that the smell suddenly spreads and strongly fills the air, as if it is forcefully entering the reader’s sense of smell. Based on the overall imagery data, the researcher found that visual imagery is the most dominant type in the poetry collection “*Kesaksian Sepasang Sandal.*” This dominance occurs because the poet frequently presents concrete descriptions of everyday objects, settings, and situations, enabling readers to easily visualize the scenes and experiences depicted in the poems.

The findings of this study are consistent with the results of previous research conducted by Athiya Almanurah (2022), which revealed the dominance of visual imagery in Adi K’s poem “*Sesunyi Cahaya,*” as well as Arnelia Trias Audiyah’s (2022) study on Sapardi Djoko Damono’s “*Perahu Kertas,*” which demonstrated that visual and kinesthetic imagery play an important role in constructing the emotional meaning of poetry. This consistency strengthens the argument that imagery is a crucial element in poetry appreciation, as it enables readers to understand meaning more concretely and through sensory experience.

In an educational context, the results of this study show strong relevance to Indonesian language teaching materials at the high school level. Imagery, as a fundamental element of poetry, can be utilized as effective teaching material to train students’ ability to identify and appreciate the aesthetic elements of poetry. Based on the student questionnaire results, a positive response was obtained, with an average percentage exceeding 85%. These findings indicate that poetry learning through an imagery-based approach not only enhances students’ conceptual understanding but also increases their interest in and appreciation of literary works.

CONCLUSION

The study entitled “*Imagery in the Poetry Collection ‘Testimony of a Pair of Sandals’ and Its Relevance to Poetry Learning in High School*” concludes that five types of imagery are present in the poems: visual, auditory, kinesthetic, tactile, and olfactory imagery. The analysis identified a total of 88 imagery data points, consisting of 46 visual images, 3 auditory images, 29 kinesthetic images, 6 tactile images, and 4 olfactory images. These five types of imagery play an important role in creating sensory

experiences that enliven the reader's imagination and deepen emotional engagement with the content of the poems.

The research findings also indicate that the use of imagery in the poetry collection "*Kesaksian Sepasang Sandal*" has strong relevance to poetry learning at the high school level. Through the implementation of learning using PowerPoint (PPT) media, students demonstrated high levels of enthusiasm and understanding, with more than 85% of students stating that they agreed or strongly agreed with the effectiveness of imagery-based material in enhancing literary appreciation. Therefore, it can be concluded that imagery analysis represents an engaging and meaningful learning approach for improving students' appreciation, interpretation, and creativity in poetry learning at the high school level.

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AN ANALYSIS OF IMPOLITENESS STRATEGIES USED BY THE MAIN CHARACTER IN *BAD GENIUS (2024)* MOVIE

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ABSTRACT

This study analyzed the impoliteness strategies used by the main character in *Bad Genius (2024)* and how the character responds to them. Employing a qualitative method with a content analysis approach, the data consist of the character's utterances taken from the movie script. The analysis applies Culpeper's theory, which classifies impoliteness into five types: bald on record, positive, negative, sarcasm/mock politeness, and withholding politeness. The results show 13 impolite utterances, with positive impoliteness being the most frequent (5 instances), followed by sarcasm/mock politeness (3), bald on record and negative impoliteness (2 each), and withholding politeness (1). In total, 16 responses to impoliteness are identified, dominated by defensive countering (8), then accepting the face attack (4), offensive countering (3), and no response (1). This study contributes to pragmatic studies by illustrating the use and response to impoliteness in cinematic discourse.

Keywords: *Impoliteness strategies, Pragmatics, Responses to impoliteness.*

INTRODUCTION

Language plays a crucial role in human interaction, as it enables people to communicate, share ideas, and maintain social relationships. In every interaction, politeness helps create harmony and prevent misunderstandings by demonstrating respect and consideration for others. However, politeness does not always guarantee sincerity. At times, individuals may express anger or frustration using polite-sounding words, a phenomenon known as mock politeness. Therefore, the study of impoliteness is essential for understanding how language can be employed not only to build relationships but also to challenge or defend one's social image.

Pragmatics provides a framework for analysing how meaning is shaped by context, intention, and interpretation (Thomas, 2013; Cruse, 2011). Within this field, the study of impoliteness has gained increasing attention, particularly as it reveals how linguistic choices can threaten one's social image or "face" and provoke conflict (Brown & Levinson, 1987; Culpeper, 1996). Impoliteness, as defined by Culpeper (2011), refers to the use of language that causes offense or threatens another's face intentionally or as perceived by the hearer. This phenomenon emerges in contexts where social power, emotional intensity, or conflicting interests influence communication. According to Culpeper (2005), impoliteness arises when a speaker attacks the hearer's social identity or self-esteem, often as a means of asserting dominance or control. Mills (2003) emphasizes that impoliteness can only be understood pragmatically, within the framework of group norms, shared expectations, and situational context. Thus, impoliteness is not merely linguistic but interactional, co-constructed through the

intentions of the speaker and the interpretations of the hearer.

Culpeper (1996) introduced five major impoliteness strategies, they are bald on record impoliteness, positive impoliteness, negative impoliteness, sarcasm or mock politeness, and withhold politeness, each of which functions to damage social harmony in distinct ways. These strategies have been applied in numerous studies to explore how language functions as a vehicle of aggression, criticism, and power in both real and fictional contexts. For instance, Muazzaro and Dewanti (2020) examined Donald Trump's speech at the 2018 Conservative Political Action Conference and found that impoliteness served as a rhetorical tool to assert authority and challenge opponents. Similarly, Djohan and Simatupang (2022) analyzed impoliteness in Cruella and demonstrated how such strategies contribute to character development and conflict escalation in film dialogue. Anindia (2024) further investigated impoliteness strategies in haters' comments on Mason Greenwood's Instagram, revealing that online communication frequently exhibits negative and positive impoliteness as expressions of hostility and entertainment.

These studies collectively underscore that impoliteness is not confined to political or online discourse but also permeates fictional narratives, where it reflects real-world social interactions and hierarchies. However, most prior research has focused on earlier films or social media contexts, leaving limited exploration of contemporary cinematic works that portray academic competition and moral tension. The *Bad Genius* (2024) film presents a rich case for such analysis. Through its portrayal of academic cheating, social inequality, and personal ambition, the film illustrates how impoliteness operates as both a communicative and psychological tool shaping character relationships, emotional conflict, and power imbalance.

This research aims to fill a gap in the current literature by analyzing the types of impoliteness strategies and the responses to impoliteness used by the main character in *Bad Genius* (2024). Employing a qualitative descriptive method within the framework of Culpeper's impoliteness theory (1996, 2005, 2011), the study examines how verbal aggression and responses are constructed within cinematic dialogue. Specifically, the research objectives are to (1) identify the types of impoliteness strategies employed by the main character and (2) analyze how the main character responds to these strategies throughout the movie. By integrating linguistic theory with cinematic discourse, this study contributes to a deeper understanding of how impoliteness functions as a reflection of social power, moral conflict, and human emotion in modern media narratives.

METHOD

This study employed a qualitative research design using a content analysis approach to identify and interpret the impoliteness strategies and responses employed by the main character in *Bad Genius* (2024). The analysis focused on understanding the meaning and function of utterances within their communicative context, rather than quantifying them. The data were drawn from the movie *Bad Genius* (2024), accessed via Netflix, and consisted of the main character's utterances containing impoliteness strategies and responses. Supporting materials, including journals, books, and online articles related to Culpeper's theory, were also consulted to strengthen the interpretation.

The researcher collected the data through documentation techniques, including

watching the movie several times, transcribing the dialogues, and identifying utterances that reflected impoliteness. These utterances were then categorized according to Culpeper's (1996) framework of impoliteness strategies. The researcher herself acted as the main instrument of the study, assisted by tools such as a laptop, internet connection, and note-taking materials.

To ensure data validity, source triangulation was used by comparing data from multiple references such as the movie, online transcripts, and other supporting sources to maintain consistency and reliability. The data were analysed using Miles and Huberman's (1994) model, which involves three stages: data condensation, data display, and conclusion drawing. Through this process, the findings were interpreted to answer the research questions regarding the types of impoliteness strategies and the responses used by the main character in *Bad Genius* (2024).

RESULTS AND DISCUSSION

Based on the analysis of the main character's utterances in the movie, all five types of impoliteness strategies were identified. However, the frequency of occurrence varies across strategies, as illustrated in the table below:

Table 1. Types of Impoliteness Strategies Used by the Main Character

No	Types of Impoliteness Strategies	Frequency
1	Bald on Record	2
2	Positive Impoliteness	5
3	Negative Impoliteness	2
4	Sarcasm/Mock Politeness	3
5	Withhold Politeness	1
TOTAL		13

Based on the data presented in Table 4.1, all five types of impoliteness strategies proposed by Culpeper are identified in the utterances of the main character in the movie. However, the frequency of each strategy varies, indicating the dominance of certain types over others. Among these, positive impoliteness emerges as the most frequently used strategy. Positive impoliteness is followed by sarcasm or mock politeness, while bald on record impoliteness and negative impoliteness occur less frequently. The least employed strategy in the data set is withholding politeness.

The main character's frequent use of positive impoliteness reflects her emotionally charged reactions toward characters who betray or underestimate her, indicating a defensive and confrontational stance driven by social conflict and personal frustration. In *Bad Genius* (2024) movie, this tendency is closely tied to her background

as an academically gifted student from a lower socioeconomic class who becomes entangled in a high-stakes cheating scheme dominated by wealthier students. The unequal power dynamics and repeated instances of manipulation and exploitation contribute to her use of positive impoliteness strategies. These include expressions of disapproval, dismissive attitudes, and verbal distancing, which she employs to assert her boundaries and confront those who take advantage of her. Her language becomes a tool of resistance, allowing her to express resentment, regain control, and reject the hypocrisy of the elite dominated system that continuously undermines her integrity and efforts.

This finding is supported by research conducted by Erza and Hamzah (2018), who analysed impoliteness strategies in Instagram comments addressed to public figures. Their study found that positive impoliteness was the most dominant strategy used by commenters. The researchers argued that this form of impoliteness is often employed in emotionally driven situations to express dissatisfaction and social distance. Regarding responses, In the analysis of *Bad Genius (2024)* movie, the main character was found to employ all three types of responses when faced with face threatening acts from other characters. The frequency of each response type is summarized in Table 2.

Table 2. Responses to Impoliteness Strategies Used by the Main Character

No	Types of Response to Impoliteness	Frequency
1	Accepting the face attack	4
2	Countering the face attack	Offensive countering 3
		Defensive countering 8
3	No respon	1
TOTAL		16

As shown in the table, countering the face attack emerges as the most frequently employed response strategy. Countering the face attack in this context takes two forms, namely defensive countering and offensive countering. Defensive countering, which occurred 9 times, is characterized by Lynn's attempts to justify her actions or clarify misunderstandings without escalating the conflict. This type of response often stems from situations where Lynn feels the need to protect her reputation or correct unfair judgments, particularly when her intelligence, motivation, or honesty are questioned. For example, when she is accused of unethical behaviour or when others try to diminish her academic capabilities, Lynn asserts her perspective clearly and logically. Her defensive countering is influenced by her background as a high-achieving student who must constantly prove

her value in a system skewed by wealth and social status.

In contrast, offensive countering, which occurred 3 times, is more aggressive and direct. It involves retaliating with impolite language or criticism, often when Lynn is pushed beyond her emotional limits. This response reflects moments of heightened frustration and is influenced by the unequal and exploitative environment she is trapped in. Offensive countering becomes a means of reclaiming power, showing resistance, and expressing moral outrage at the hypocrisy and manipulation around her. This tendency to counter rather than accept or ignore impoliteness highlights Lynn's assertive and justice-oriented character. She rarely tolerates disrespect passively, and instead seeks to challenge it through verbal engagement.

This finding is supported by a study conducted by Sari et.al. (2023), who examined impoliteness and responses in the *Enola Holmes* movie. Their research found that the main character frequently countered impolite acts, especially when faced with power imbalances or personal attacks. The study emphasized that countering strategies were used not only for self-defence but also to resist systemic oppression and assert independence.

CONCLUSION

This study analysed the impoliteness strategies used by the main character in *Bad Genius* (2024) and how she responded to impolite behaviour directed at her, using a qualitative method with a content analysis approach based on Culpeper's theory. The findings revealed thirteen utterances containing impoliteness, encompassing all five types of strategies. Positive impoliteness was found to be the most dominant, indicating the character's emotionally driven and defensive reactions toward social conflict, betrayal, and inequality. Regarding responses, sixteen instances were identified, with defensive countering emerging as the most frequent. This shows that the main character actively defends herself rather than remaining passive in confrontational situations. Overall, the study demonstrates that impoliteness serves as a communicative strategy for expressing power, emotion, and resistance within social interactions.

The results of this study contribute to the field of pragmatics, especially in understanding how impoliteness and responses to impoliteness are represented in film discourse. The findings align with previous studies, such as those by Anindia (2024) and Djohan & Simatupang (2022), but extend the discussion by focusing on both the production and reception of impoliteness. This study also emphasizes that impoliteness is not merely negative or destructive, it can serve as a strategy for self-defence, emotional release, or the assertion of social identity.

Future research could explore impoliteness from a broader perspective by comparing multiple characters or analysing cross-cultural representations of impoliteness in movies from different countries. Investigating how gender, culture, and social status influence impoliteness would also provide valuable insights. Moreover, further studies on impoliteness in digital communication, such as social media interactions, could enrich our understanding of how people construct and negotiate face in online contexts.

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AN ANALYSIS OF ENGLISH SLANG EXPRESSIONS AMONG INDONESIAN GENERATION Z ON TIKTOK

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ABSTRACT

This research, entitled "An Analysis of English Slang Expressions among Indonesian Generation Z on TikTok," examines the structural forms and contextual meanings of English slang expressions used by Indonesian Gen Z on TikTok. The research applies a descriptive qualitative approach with content analysis as the technique of data analysis. Data were collected through documentation of 28 publicly available TikTok videos uploaded between 2023 and early 2025, yielding a total of 35 slang expressions. To ensure the validity of the findings, the researcher employed source triangulation by collecting data from multiple TikTok videos and conducted peer debriefing with a fellow sociolinguistics researcher. The classification of the slang expressions was based on Noam Chomsky's Transformational Generative Grammar (surface structures), focusing on their realization at the word, phrase, and sentence levels. To interpret the contextual meanings and social functions of slang, the research drew on Connie Eble's sociolinguistic framework. The findings revealed that word-level slang was the most frequently used, followed by sentence-level and phrase-level forms. Pragmatically, these expressions served multiple communicative functions, including admiration, criticism or commentary, emotional expression, humor or playfulness, and group identity signaling. The study demonstrates that English slang on TikTok is not only a matter of informal language use but also a sociolinguistic resource through which Indonesian Generation Z constructs identity, negotiates belonging, and engages in digital interaction. This research contributes to sociolinguistics and digital discourse studies by offering insight into the intersection of linguistic form, contextual meaning, and Gen Z communication practices on social media.

Keywords: *English Slang; Content Analysis; Generation Z; TikTok; Sociolinguistics.*

INTRODUCTION

Language is an inherently dynamic system, continually adapting to the evolving needs of its users. It serves not only as a means of communication but also as a mirror reflecting the sociocultural changes within a given society. In the field of linguistics, sociolinguistics offers profound insights into the relationship between language and society, illuminating how factors such as age, gender, and community shape speech patterns and lexical choices (Wardhaugh & Fuller, 2015). Through this relationship, language diversifies into multiple varieties that embody identity, group affiliation, and belonging. Among these varieties, slang has emerged as one of the most vibrant and rapidly changing forms of linguistic expression. Slang occupies a distinctive position within sociolinguistic research due to its embodiment of the creativity and flexibility inherent in everyday language (Eble, 1996). Often linked with youth culture and informal communication, slang serves as a vehicle for the expression of humor, emotion, and social solidarity (Partridge, 2014). Although frequently viewed as non-standard or substandard,

slang plays an essential role in signaling group membership and expressing individuality. Its innovative nature highlights how language users intentionally modify conventional grammar and vocabulary to create new forms that better reflect their social realities (Crystal, 2017).

The rise of digital communication has drastically transformed how slang emerges, spreads, and evolves. Online platforms allow for linguistic innovation to occur more swiftly and broadly than in traditional speech communities. Social media platforms serve as central spaces for this linguistic innovation, enabling slang to circulate globally while simultaneously being adapted locally. Among these platforms, TikTok has achieved particular prominence due to its short-video format and interactive features, which encourage users to combine audio, visuals, and text in novel and creative ways. The multimodal nature of TikTok provides fertile ground for linguistic play and the emergence of new forms of expression (McCulloch, 2019).

Within this framework, Generation Z individuals born between 1997 and 2012 constitute the most active and influential demographic on TikTok (Pew Research Center, 2022). As digital natives, they incorporate global English slang into their communication not merely to follow trends but to articulate identity and emotion in ways that resonate with their local cultural context. Expressions such as “*delulu is the solulu*,” “*she ate*,” and “*it’s giving*” exemplify how Indonesian Gen Z users adapt and recontextualize global slang, infusing it with local nuances of tone, humor, and rhythm. This process of linguistic adaptation exemplifies the intersection of linguistic creativity, cultural hybridity, and digital media participation. While considerable research has been conducted on the social functions and cultural symbolism of slang, as well as its role in shaping youth identity, there remains a notable gap in understanding how slang operates structurally and contextually in online discourse, particularly within the Indonesian social media landscape. Previous studies have largely focused on the spread and function of slang, with limited examination of its linguistic forms, whether it manifests as a word, phrase, or sentence, and how these forms interact with context to generate meaning. To fill this gap, this study investigates the English slang expressions used by Indonesian Generation Z on TikTok by analyzing their linguistic forms through the lens of Noam Chomsky’s Transformational-Generative Grammar (TGG), specifically focusing on surface structure, and interpreting their contextual meaning through a pragmatic framework. This methodological approach provides a comprehensive understanding of how linguistic creativity functions in digital communication, while simultaneously illuminating how Generation Z reshapes global English slang to express identity and belonging in online spaces.

METHOD

This study employed a qualitative descriptive approach using content analysis to examine English slang expressions used by Indonesian Generation Z on TikTok. The qualitative method was chosen because it enables an in-depth understanding of language as it naturally occurs in social contexts, emphasizing meaning rather than numerical data (Creswell, 2014). Through this approach, the researcher aimed to describe, categorize,

and interpret the linguistic forms and contextual meanings of slang expressions found in authentic TikTok communication.

The data consisted of English slang expressions appearing in videos created by Indonesian TikTok users belonging to Generation Z, defined as individuals born between 1997 and 2012. A total of 28 TikTok videos were selected, which were uploaded between 2023 and early 2025. The study used a documentation technique for data collection, which refers to gathering data from existing written, visual, or digital materials (Cohen, Manion, & Morrison, 2018). The selected TikTok videos and captions containing English slang expressions used by Indonesian Generation Z were documented, along with supporting details such as TikTok username, upload date, and engagement metrics to ensure contextual accuracy. The sample size of 28 videos was deemed sufficient because data saturation was reached; the same slang expressions began to recur in additional videos, indicating that further data collection would not yield new linguistic items. From these videos, 35 distinct English slang expressions were identified and became the primary data for analysis. The researcher acted as the primary instrument in the study, serving as the observer, transcriber, and interpreter of the data. To ensure the accuracy and reliability of the interpretations, peer debriefing was conducted, where the researcher discussed the coding and analysis with a fellow sociolinguistics researcher to identify potential biases or misinterpretations.

The data were analyzed using content analysis (Krippendorff, 2018), which involved two main stages. First, each slang expression was categorized based on its linguistic form, whether it appeared as a word, phrase, or sentence, using the framework of Noam Chomsky's Transformational-Generative Grammar (TGG). Second, the contextual meaning of each expression was interpreted using a pragmatic approach, focusing on the communicative intent and social function of the slang within the TikTok environment. The contextual meanings were grouped into five functions: admiration, criticism, emotion, humor, and identity.

RESULT AND DISCUSSION

This study meticulously examined 35 English slang expressions drawn from 28 TikTok videos created by Indonesian Generation Z users between 2023 and early 2025. The selection of 28 videos was deemed sufficiently representative, as data saturation was achieved, and recurrent slang expressions across videos indicated that further sampling would yield no novel linguistic items. The findings were classified into two principal aspects: the linguistic forms of the slang, analyzed through the lens of Noam Chomsky's Transformational-Generative Grammar (TGG), and their contextual meanings, interpreted via a pragmatic framework (Chomsky, 2014).

Linguistic Forms of Slang Expressions

From a structural perspective, the 35 slang expressions were categorized into three linguistic forms: words (15 expressions, or 42.9%), phrases (7 expressions, or 20.0%), and sentences (13 expressions, or 37.1%). Notably, word-level slang predominated, underscoring the proclivity of TikTok users for brief, flexible linguistic units, congruent with the platform's rapid, transient communication style. The predilection for concise

expressions mirrors the exigency for immediate attention on TikTok, where users are compelled to convey ideas swiftly and effectively.

Word-level slang, including terms such as *slay*, *rizz*, *sus*, and *fineshyt*, epitomizes linguistic innovation through processes such as lexicalization and semantic shift. These singular lexical items are imbued with multifaceted meanings, encapsulating admiration, agreement, or criticism in a compact and potent form. The preponderance of word-level expressions aligns with TikTok's emphasis on brevity and immediacy, supporting Eble's (1996) assertion that slang serves as an instrument for linguistic creativity and self-expression within youth subcultures.

Phrase-level slang, such as *it's giving* and *K-Pop Stans*, often functions as evaluative markers or shorthand expressions encapsulating attitudes and reactions. These phrases exhibit syntactic flexibility, wherein conventional grammatical structures are modified to inject emphasis or humor, reflecting the creative adaptability of Gen Z language use.

Sentence-level slang, exemplified by expressions such as *ate and left no crumbs*, *let him cook*, and *being delulu is not solulu*, merges exaggeration, irony, and humor hallmarks of online discourse. These expressions exemplify Noam Chomsky's notion of elliptical structures, wherein truncated sentences retain complete semantic meaning through context and shared social understanding (Chomsky, 2014). Such syntactic ingenuity enables users to convey layered meanings within the constraints of short-form content.

Contextual Meaning and Functions of Slang

With regard to contextual meaning, the most prevalent functions were admiration and criticism, each representing 9 slang expressions (25.7%). Admiration-based slang serves as a vehicle for praise or approval, as exemplified by expressions such as *slay* or *she ate*. These terms function to validate and celebrate individuals' actions or appearances, underscoring the importance of positive reinforcement within digital interactions. Conversely, criticism-oriented slang, such as *redflag* or *gaslighter*, conveys disapproval, often with a satirical or sarcastic undertone. The prominence of these evaluative functions reflects TikTok's inherently interactive nature, which encourages users to swiftly articulate their assessments in comments, captions, and duets.

Emotion-related slang, representing 4 expressions (11.4%), such as *I'm totally buggin'* and *ISTG*, serves to articulate exaggerated emotional states, often in an overtly hyperbolic manner. This emotional hyperbole is reflective of Gen Z's propensity for dramatizing reactions in digital communication as a means of signaling affective experiences to a broad audience.

Humor and sarcasm accounted for 7 expressions (20.0%), including *being delulu is not solulu* and *roster so clear*. These expressions hinge on absurdity, irony, and collective humor characteristics emblematic of Gen Z's online communication style. The use of humor-based slang highlights the performative nature of TikTok, wherein users often engage in exaggerated, hyperbolic expressions that cultivate social cohesion and enable subtle cultural commentary.

Identity-related slang (6 expressions, or 17.1%), including terms such as *FYP* and *GRWM*, operates as markers of personal identity and group affiliation. These expressions are emblematic of the performative and self-curated nature of TikTok, where users actively construct and display their identities through linguistic choices. By using these expressions, individuals signal their alignment with specific communities or subcultures, thus reinforcing their digital personas.

The findings suggest that Indonesian Generation Z employs English slang on TikTok not merely for entertainment but as a dynamic tool for self-expression, evaluation, and community-building. Slang, in this context, transcends its function as a lexical phenomenon and emerges as a performative act that plays a pivotal role in the construction of online identity. The prevalence of word-level slang highlights TikTok's preference for succinctness and immediacy, while the utilization of phrases and sentences underscores the platform's affordance for creativity and syntactic flexibility. The structural distribution of slang forms provides empirical support for Chomsky's argument that language users can generate an infinite array of expressions from a finite set of rules (Chomsky, 2014). The pragmatic findings bolster Eble's contention that slang fosters solidarity and in-group belonging, especially in digitally mediated communities (Eble, 1996). Unlike traditional text-based platforms, TikTok's multimodal features, sound, video, and captions permit slang to function both as a linguistic tool and a performative expression. This dual function sets TikTok apart from other platforms such as Twitter or Instagram, where textual interactions dominate.

CONCLUSION

This study investigated the use of English slang expressions by Indonesian Generation Z on TikTok, focusing on both their linguistic forms and contextual meanings. The findings revealed that slang manifests in three distinct structural forms: words, phrases, and sentences, with word-level slang emerging as the most dominant. This prevalence can be attributed to TikTok's emphasis on concise, impactful communication, which aligns with the platform's rapid and expressive nature.

Contextually, five primary functions of slang were identified: admiration, criticism, emotion, humor, and identity. These functions underscore the dual role of slang as both a linguistic tool and a vehicle for social evaluation, emotional expression, and identity performance within digital environments. The integration of Chomsky's Transformational

Generative Grammar (surface structure) with a pragmatic analysis illuminated the interplay between linguistic productivity and sociocultural adaptation in TikTok discourse. Ultimately, the study demonstrates that the English slang employed by Indonesian Generation Z represents a localized, creative adaptation of global linguistic practices. This adaptation highlights the ways in which digital communication reshapes how Generation Z constructs meaning, humor, and social belonging. By reimagining conventional language in a digital context, Gen Z fosters a unique form of expression that bridges both global influences and local cultural nuances.

Future research could expand upon these findings by conducting comparative analyses across various social media platforms, examining how the nuances of each

platform's structure and affordances impact slang usage. Additionally, further exploration of the role of multimodal elements such as visual and audio features could provide deeper insights into how these factors contribute to the interpretation and evolution of slang in digital spaces.

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ANALYSIS OF LANGUAGE ACQUISITION IN 3-4 YEARS OLD CHILDREN IN KRUENG SEUPENG VILLAGE, NORTH ACEH REGENCY

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ABSTRACT

This research is titled "Analysis of Language Acquisition in Children Aged 3-4 Years in Krueng Seupeng Village, North Aceh Regency," which aims to describe language acquisition in the field of phonology (vowels and consonants). This research uses qualitative descriptive research. The data source in this study consists of informants aged 3-4 years. The data used is in the form of speech from children aged 3-4 years. The data collection techniques used are the observation technique, prompting technique, recording technique, and note-taking technique. The data analysis techniques in this study are data reduction, data presentation, and drawing a conclusion. The results of this study show 40 data points found, namely 15 data for vowel, namely: terdin and 2 data for sound (l), sound (u) 2 data, sound (a) data, sound (e) 5 data, and sound (o) 3 data, and a total of 15 data found sound changes that occur in vowel sounds are changes in sound (e-i), (o-u), (e-a), (i-e), (u-e), (a-o), (u-o), (e-u), (o-e), (i), (e), (a), (e), (u-o), (a). And kontoid sound there are 25 data, namely terdin and 2 sound data (h), sound (t) 2 data, sound (n) 3 data, sound (s) 2 data, sound (g) 1 data, sound (y) 2 data, sound (p) 2 data, sound (d) 1 data, and sound (p) 2 data, sound (p) 1 data. And a total of 25 data, found sound changes that occur in the kontoid sound are (s-h), (s-t), (n), (s), (g), (y), (f-p), (l-y), (z-j), (r), (h), (k), (l), (h), (x-k), (z-d), (c-t), (r-l), (k), (n), (p), (s), (l), (g-n), (r-l). And all the data found there are 40 data points.

Keywords: *Language Acquisition, Vowel, Consonant.*

INTRODUCTION

Language is the primary tool of communication in human life. Therefore, language has an important social role in communicating with the wider community. Through language, a person can express opinions, feelings, and establish social relationships. They rely on language to analyze and differentiate each social dilemma in the communication process. Language is always subordinate to its users. For this reason, language becomes an essential element in the process of social interaction. Language functions as a sense in the communication process, and the two have an inseparable relationship. Without language, humans cannot communicate because humans are social beings who naturally must interact with one another.

Language acquisition is part of the study of psycholinguistics, which is a branch of linguistics that examines language knowledge, language use, language change, and other aspects related to these areas. In general, language acquisition is the process that occurs in a child's brain when they learn and master their first language. In this regard, the role of parents is very important, especially in adjusting their way of speaking to the child's age. When parents speak using proper language that matches the child's level of

understanding, the child will listen, imitate, and begin to understand the language. This process is what is called first language acquisition.

First language acquisition in children begins with the mother tongue, and this process occurs in children all over the world in a similar pattern. The tendency of the language acquisition process is influenced not only by similarities in biological and neurological factors but also by aspects of language mentality. In the process of first language acquisition, a child develops the ability to comprehend and produce language, enabling the child to understand words they hear more quickly. Children learn language from their family environment and the community in their village. Within the family, children usually only play at home and feel very free with their parents to memorize words. Meanwhile, in the village community, children play and converse with other children. The more freely a child interacts, the more vocabulary they acquire.

A child usually expresses their desires using language to those closest to them, namely their parents. This is because parents play an important role in a child's language development. From an early age, a child learns to speak by listening to and imitating what their parents say. At the age of 3-4 years, a child begins to actively absorb, imitate, and use the language they hear in their surroundings, especially from their parents.

Utterances of children aged 3-4 years require special attention from parents. Children at the age of 3-4 are expected to interact with their surroundings and be taken outside the home so that their language acquisition and mastery can develop in a simpler and more effective way. In Krueung Seupeng Village, there are several children aged 3-4 who are already able to communicate well using language with the people around them. However, on the other hand, there are also children of the same age who are still unable to communicate fluently and engagingly.

In language acquisition theory, children study the universal sequence and structure of the sounds of language that they acquire. Children only practice vocal sounds without any specific purpose, following the universal sequence of language sound development, starting from simple vowel sounds to consonant sounds and ultimately to more complex sounds. Meanwhile, relevant theories in language acquisition studies focus more on the interaction between a child's cognitive abilities and the surrounding environment in language development. Relevant theories often discuss how children acquire language through social interaction and the development of semantics and syntax through environmental stimuli and internal cognitive processes. Thus, these theories focus more on the phonological aspects of children's language acquisition, whereas relevant theories place greater emphasis on the influence of social and cognitive interaction in children's language acquisition.

Children's utterances can sometimes be unique to listen to and understand. At this age, children have begun to communicate with the people around them. They are in a phase of growth and development and tend to imitate, absorb, and capture information. However, there are problems where children have difficulty pronouncing certain vowel and consonant sounds, such as the /r/ and /s/ sounds, and have trouble distinguishing similar language sounds, for example, the /p/ and /b/ sounds. Based on these issues, the researcher is interested in conducting a study on language acquisition in children aged 3-4 years in Krueng Seupeng Village, North Aceh Regency.

Krueng Seupeng Village, located in Kuta Makmur District, North Aceh Regency, Aceh Province, was chosen as the research location to understand how children aged 3-4 years in the village develop speaking abilities, particularly in the aspect of phonology. The purpose of this study is to explore how children aged 3-4 years develop speaking skills and comprehend language according to the field of phonology. Therefore, the researcher is interested in directly studying language acquisition in children aged 3-4 years in Krueng Seupeng Village, Kuta Makmur District, North Aceh Regency.

METHOD

This study uses a qualitative approach, which is a method that describes or explains the research object based on observable realities or actual conditions. The researcher chose a qualitative approach because they wanted to describe how children aged 3-4 acquire language. According to Bogdan and Biklen, a qualitative approach is a research procedure that produces descriptive data from the words, writings, or behaviors observed in the research subjects.

This type of research is qualitative descriptive research. Descriptive research is research that attempts to provide a systematic picture of situations, problems, phenomena, services, or software, or to provide information about something, for example, to understand the pronunciation and sounds of Indonesian in children. The study of children's language in Krueng Seupeng Village aims to describe in detail the phenomenon of how the utterances are pronounced by the children.

RESULT AND DISCUSSION

Based on the research results on Language Acquisition in 3-4 Year Old Children in Krueng Seupeng Village, North Aceh Regency, analyzed by the researcher, this study is divided into two parts, namely vowels and consonants. In this study, there are 4 children, consisting of 2 boys and 2 girls, specifically in Krueng Seupeng Village, North Aceh Regency. The researcher found a total of 40 data points containing vowels and consonants. Vowels consist of 15 data points, and consonants consist of 25 data points. To facilitate the data description process, the researcher used abbreviations for each fact placed in the analysis section.

Vokoid (Vowel Sound)

A vokoid is a vowel sound produced with an open vocal tract without significant obstruction. Broad and simple vowels such as /a/, /i/, and /u/ are basic sounds that are almost universal and are acquired earliest by children. The acquisition of vowel sounds at the phonological level has been studied in Krueng Seupeng Village, North Aceh Regency. The vokoid data that was found is as follows.

No.	Change Data	Phoneme Changes	Standard Word
1.	Pirgi	e-i	Pergi
2.	Pumpa	o-u	Pompa
3.	Pancuri	e-a	Pencuri
4.	Kupeng	i-e	Kuping
5.	Bages	u-e	Bagus

6.	Cococola	a-o	Cocacola
7.	Laot	u-o	Laut
8.	Juruk	e-u	Jeruk
9.	Toke	o-e	Toko
10.	Ni	i	Ini
11.	Dlapan	e	Delapan
12.	Da	a	Ada
13.	Nam	e	Enam
14.	Sabon	u-o	Sabun
15.	Sapidermen	e	Spidermen

Table 1. Vokoid Data

Data indicate that children aged three to four years begin to master basic vowel sounds, such as /a/, /i/, and /u/. However, during this stage, children often experience vowel substitution, in which the original vowel sound is replaced by another vowel that is articulatorily easier to produce. For instance, the vowel /e/ may be replaced with /i/ in the word *pergi*, resulting in *pirgi*, or the vowel /o/ may be replaced with /u/ in the word *pompa*, resulting in *pumpa*. This phenomenon is consistent with stages of phonological development in early childhood, where children tend to use the most familiar and easily pronounced vowel sounds. In addition to vowel substitution, vowel reduction is also observed, such as in the transformation of *delampang* to *dlapan* and *ada* to *da*, indicating that children at this developmental stage are not yet able to produce all vowel patterns accurately. This process is considered normal in language acquisition, as phonological development theory explains that children gradually eliminate syllables or sounds that are perceived as difficult to articulate.

The addition of vowels is also observed in the pronunciation of the word *spiderman*, which becomes *sapiderman*, indicating the child's attempt to adapt unfamiliar pronunciation patterns to more familiar word structures, even though articulation is not yet fully accurate. Children aged three to four years in Krueng Seupeng Village, North Aceh Regency, demonstrate a stage of language acquisition at the phonological level that reflects normal developmental processes. At this stage, children are able to produce several basic vowel sounds, such as /a/, /i/, and /u/, but they still frequently perform vowel substitution, vowel reduction, and vowel addition in certain words. These patterns indicate that children are actively learning and gradually adapting to more complex pronunciation structures as part of their ongoing phonological development.

Contoid (Consonant Sounds)

Consonant sounds are consonant sounds produced when there is an obstruction in the vocal tract. Jakobson proposed that the acquisition of consonant sounds occurs gradually, depending on the level of articulation difficulty. Bilabial consonants such as /p/, /b/, and /m/, which involve both lips, usually emerge first because they are easy to pronounce. After that, children master alveolar consonants such as /t/, /d/, and /n/, followed by more complex consonants such as palatal and throat sounds.

Table 2. Contoid Data

No.	Change Data	Phoneme Changes	Standard Word
1.	Tohka	s-h	Toska
2.	Tayang	s-t	Sayang
3.	Abag	n	Abang
4.	Udah	s	Sudah
5.	Pisan	g	Pisang
6.	Sosyis	y	Sosis
7.	Panta	f-p	Fanta
8.	Boyeh	l-y	Boleh
9.	Pijja	z-j	Pizza
10.	Aga-aga	r	Agar-agar
11.	Ijau	h	Hijau
12.	Es-rim	k	Es-krim
13.	Afamart	l	Alfamart
14.	Mahu	h	Mau
15.	Ekcavator	x-k	Excavator
16.	Dombi	z-d	Zombi
17.	Toklat	c-t	Coklat
18.	Walna	r-l	Warna
19.	Bawak	k	Bawak
20.	Manna	k	Mana
21.	Kenappa	p	Kenapa
22.	Bissa	s	Bisa
23.	Jelli	l	Jeli
24.	Unnu	g-n	Ungu
25.	Kemalin	r-l	Kemarin

The acquisition of contoid sounds at the phonological level among children in Krueng Seupeng Village, North Aceh Regency, revealed a total of 25 data items. These data include pronunciations such as *tohka* for *turquoise*, *tayang* for *dear*, *panta* for *fanta*, *boyeh* for *allowed*, *pijja* for *pizza*, *unnu* for *purple*, *kemalen* for *yesterday*, *ekcavator* for *excavator*, *dombi* for *zombi*, *toklat* for *chocolate*, *abag* for *brother*, *udah* for *already*, *pisan* for *banana*, *aga-aga* for *jelly*, *ijau* for *green*, *es rim* for *ice cream*, *afamart* for *alfamart*, *sosyis* for *sausage*, *bukak* for *open*, *manna* for *where*, *kenappa* for *why*, *gak bissha* for *cannot*, *jelli* for *jelly*, and *mahu* for *want*.

The findings indicate that children aged three to four years have not yet fully mastered the pronunciation of certain consonant sounds, particularly /r/, /l/, and /s/. As a result, children tend to add, omit, or substitute these consonant sounds with others that are articulatorily easier to produce. This pattern reflects a normal stage of phonological development, in which children gradually refine their articulation abilities as part of the language acquisition process.

Consonant sound acquisition among children aged three to four years in Krueng Seupeng Village has not yet fully developed. The data reveal the presence of consonant

substitution, reduction, and addition, which indicate a gradual internalization of the phonological system. Children tend to replace complex consonant sounds with simpler and more easily articulated ones, such as substituting the sound /s/ with /h/ as in *toska* becoming *tohka*, replacing the sound /f/ with /p/ as in *fanta* becoming *panta*, and substituting the sound /r/ with /l/ as in *warna* becoming *walna*. This pattern is consistent with Jakobson's theory, which explains that consonant sound acquisition occurs gradually and is influenced by the ease of articulation.

Consonant reduction is also frequently observed in children's speech, such as the omission of the /n/ sound in the word *abang*, which becomes *abag*, and the omission of the /k/ sound in the word *pisang*, which becomes *pisan*. This reduction occurs because children at this developmental stage are not yet able to articulate all consonant sounds completely, leading them to omit sounds that are perceived as difficult. In addition to reduction, the addition of consonant sounds is found in several words, for example, *bawa* becoming *bawak* and *bisa* becoming *bissa*. These additions function as a strategy for children to ease transitions between sounds or to fill phonological gaps in their speech production. Overall, children's production of consonant sounds demonstrates patterns of substitution, reduction, and addition, which reflect a gradual and natural process of phonological acquisition in early childhood, as explained by Jakobson's theory and supported by the empirical data.

CONCLUSION

Based on the research results, it can be concluded that with the addition of sounds, sound reduction, and sound replacement, the researcher found 30 data points consisting of 15 vowel data points and 15 consonant data points. The 15 vowel data points consist of 2 data points for the sound (i), 2 data points for the sound (u), 2 data points for the sound (a), 5 data points for the sound (e), and 3 data points for the sound (o), out of a total of 15 data points where changes occurred in vowel sounds. The 15 consonant data points consist of 2 data points for the sound (h), 2 data points for the sound (t), 3 data points for the sound (n), 2 data points for the sound (s), 1 data point for the sound (g), 2 data points for the sound (y), 2 data points for the sound (p), 1 data point for the sound (i), 1 data point for the sound (r), 2 data points for the sound (h), 2 data points for the sound (k), 3 data points for the sound (l), 1 data point for the sound (d), and 1 data point for the sound (p), out of a total of 15 data points where changes occurred in consonant sounds. In the pronunciation of vowel sounds, children are already able to articulate sounds clearly, although sometimes there is a simplification of double vowel sounds. As for consonant sounds, children tend to simplify complex language sounds and replace difficult-to-pronounce phonemes with easier ones.

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THE IMPLEMENTATION OF ACEHNESE LANGUAGE INTO THE PRIMARY SCHOOL CURRICULUM: A STEP TOWARD SUSTAINING LOCAL CULTURE IN ACEH INDONESIA

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ABSTRACT

The Acehese language is an essential part of Aceh's cultural heritage; yet, its usage among younger generations—particularly in urban areas like Lhokseumawe—is rapidly declining. In response to this cultural shift, the Indonesian government introduced *Permendikbud No.12/2024*, requiring local languages to be included as part of the *Merdeka Curriculum* in primary schools. However, practical implementation at the school level remains inconsistent and under-researched. This study investigates the integration of the Acehese language into the primary school curriculum in Lhokseumawe, focusing on three main aspects: the current status of implementation, the challenges encountered in instruction, and the perceptions of teachers and parents regarding its impact on cultural awareness. Employing a mixed methods approach, data were collected through classroom observations, interviews with educators and school leaders, and Likert-scale questionnaires distributed to 24 teachers and 60 parents across six schools. The results show that while some schools have begun implementing Acehese as a local content subject, disparities exist due to a lack of standardized teaching materials, inadequate teacher training, and limited institutional support. Despite these barriers, both teachers and parents overwhelmingly support the inclusion of Acehese in the curriculum, recognizing its role in enhancing students' understanding of their cultural identity and traditions. The findings highlight the urgent need for coordinated efforts involving curriculum development, professional training, and active community participation. Sustainable implementation requires localized strategies and stronger collaboration between educational institutions and cultural stakeholders to ensure that the Acehese language continues to thrive among future generations.

Keywords: *Acehnese Language; Local Culture; Primary School; Curriculum.*

INTRODUCTION

Language is one of the most powerful tools for preserving and transmitting culture. In multilingual nations like Indonesia, local languages reflect the diverse identities and values embedded within communities. Among these, the Acehese language stands as a vital cultural asset for the people of Aceh. However, modernization, national standardization of education, and the dominance of Bahasa Indonesia have gradually displaced local languages in formal and informal domains (Fitriah, Fhonna, & Nasir, 2025). The reduction in inter-generational transmission of Acehese poses a serious risk to cultural continuity, especially among children in urban centers such as Lhokseumawe.

To address this concern, the Indonesian Ministry of Education introduced a policy under Permendikbud No. 12 of 2024, requiring local content integration—including regional languages—into the Independent Curriculum at all education levels. This policy empowers schools and regional governments to implement local languages as standalone subjects, aiming to strengthen students' cultural identities and community belonging. However, in practice, the implementation of Acehese language instruction at the primary school level faces various challenges, such as limited teaching resources, insufficient teacher training, and unclear curriculum structures (Hayati, Fahrurrazi, & Idris, 2025).

Several recent studies highlight the urgency of addressing these challenges. For instance, Mulyani et al (2023) found that many teachers lack pedagogical strategies to integrate local languages meaningfully into classroom activities. Similarly, Sulhan, Zulfatmi, and Hayati (2025) emphasize the need for culturally responsive teaching materials that reflect Acehese values and daily practices. Meanwhile, Syahputra, M., & Mr. (2023) demonstrated that localized learning tools, when developed based on contextual Acehese wisdom, can significantly increase student engagement and motivation.

In light of these issues, this research seeks to investigate how the Acehese language has been implemented in the primary school curriculum, particularly in Lhokseumawe, with research questions focusing on (1) how the Acehese language is currently implemented in the primary school curriculum in Lhokseumawe; (2) what are the main challenges encountered in implementing Acehese language instruction; and (3) what are the perceptions of teachers and parents regarding the impact of learning Acehese on cultural awareness.

From these research questions, the study proposes the following research objectives: (1) to examine the implementation process of Acehese language instruction in primary schools; (2) to identify the challenges and barriers faced by educators and institutions; and (3) to explore the perceptions of teachers and parents toward the benefits of Acehese language learning.

This study is significant for several reasons. First, it provides evidence-based insights that can inform policy refinement at both local and national levels. Second, it contributes to the ongoing discourse on mother-tongue-based education and cultural sustainability. Third, it gives voice to key stakeholders—teachers and parents—whose perspectives are often underrepresented in language policy planning.

By analyzing real classroom practices, institutional readiness, and community support, this research aims to offer practical recommendations that could strengthen the long-term sustainability of the Acehese language. More broadly, the study reinforces the importance of aligning national education policies with the cultural and linguistic realities of local communities.

The integration of local languages into formal education systems has been widely recognized as a key strategy for cultural preservation and identity formation. According to UNESCO (2020), incorporating mother tongue education at the primary level promotes not only language proficiency but also students' cognitive and emotional development, rooted in their sociocultural context. In the case of Aceh, the Acehese language is more

than a tool for communication—it is a repository of local knowledge, traditions, and values passed down through generations (Hayati, Fahrurrazi, & Malikussaleh, 2025).

Language shift studies indicate that the declining use of local languages like Acehnese often stems from modernization, urbanization, and a national curriculum that favors Bahasa Indonesia (Fitriah, Fhonna, & Nasir, 2025). The neglect of regional languages in school settings contributes to reduced intergenerational language transmission, weakening cultural identity in the process. To counter this trend, the Indonesian Ministry of Education has promoted the *Merdeka Curriculum*, which mandates local content as part of the primary school program (Permendikbud No. 12/2024).

Recent studies in Aceh have shown that introducing Acehnese as a subject in primary schools can increase students' sense of pride and cultural awareness (Mulyani et al, 2023). However, challenges such as insufficient teacher training, the absence of standard learning modules, and limited instructional time persist (Syahputra & Mr, 2023). Contextualized teaching strategies—such as using local stories, songs, and proverbs—have proven effective in fostering student engagement and language retention.

Research by Ismail et al (2025) states that although students feel proud of the Acehnese language, modernization and lack of practice pose a threat to its sustainability. Mulyani et al (2023) emphasize the importance of the role of teachers and strategies such as the 'Acehnese language day'. Yus et al (2025) developed a P5 module based on local Acehnese wisdom that is valid and attracts students' interest. Syahputra & Mr (2023) demonstrate the success of CTL-based teaching with local wisdom. All indicate that contextual local learning enhances engagement.

Overall, literature suggests that a structured, culturally grounded curriculum can play a pivotal role in reversing language decline. Yet, successful implementation requires systemic support, collaboration between local governments and educators, and the development of high-quality, relevant teaching resources tailored to students' linguistic realities in Aceh.

METHOD

This study employed a mixed methods approach with an exploratory sequential design, which began with qualitative data collection followed by a quantitative phase to validate and expand upon the initial findings. The qualitative phase was used to explore how the Acehnese language is implemented in primary school settings, the obstacles faced by educators, and the contextual conditions of local schools. The insights gathered were then used to design the quantitative instruments aimed at capturing broader perceptions from stakeholders.

The research was conducted in several primary schools across Lhokseumawe, Aceh, where the Acehnese language had been implemented or piloted as part of the local content curriculum. The research population included school principals, curriculum developers, teachers responsible for teaching the Acehnese language, and parents of students. The sample was selected using a purposive sampling technique, ensuring that only schools with active or recent implementation of the Acehnese curriculum were included. Key informants, such as experienced teachers and school heads, were chosen

for in-depth interviews due to their strategic roles and direct involvement in curriculum planning and execution.

Three main instruments were used for data collection. First, classroom observation checklists helped document the teaching process, student engagement, and the use of teaching aids. Second, semi-structured interview guides were used during qualitative interviews with teachers and school leaders to gain insights into their experiences and challenges. Third, structured questionnaires with Likert-scale items were distributed to a wider group of teachers and parents to assess their perceptions of the curriculum's benefits and sustainability.

Data collection was carried out in two stages. In the qualitative stage, the researcher visited selected schools to observe classroom sessions and conduct interviews. In the quantitative stage, questionnaires were administered both in print and online to gather measurable data. Qualitative data were analyzed thematically through open coding to identify key patterns and recurring themes. Quantitative data were analyzed using descriptive statistics and Chi-square tests to examine the relationships between variables, such as teacher training and perceived effectiveness of the curriculum implementation. This mixed-method approach ensured a comprehensive understanding of the implementation process while providing empirical data to support recommendations for enhancing the Acehese language's role in preserving local culture.

RESULT AND DISCUSSION

Implementation of Acehese Language in the Primary School Curriculum

The findings from classroom observations and interviews indicate that the implementation of the Acehese language in primary schools in Lhokseumawe is currently in a transitional phase. While the *Permendikbud No.12/2024* provides legal support and policy direction, implementation varies significantly between schools depending on resources, teacher availability, and institutional readiness.

Approximately 60% of the observed schools have integrated Acehese as a local content subject, with a weekly allocation of one to two hours, as permitted by the *Merdeka Curriculum* guidelines. Lessons generally focus on basic vocabulary, greetings, traditional songs (*lagu daerah*), and storytelling (*dongeng Aceh*). Some schools use teacher-developed materials, while others rely on oral transmission and traditional games such as *ceh-ceh* and *deng-deng*. Teachers in more proactive schools incorporate cultural artifacts and invite local elders to speak about traditions, making the classroom experience more interactive and culturally rich.

However, the lack of standardized instructional materials across schools poses a major inconsistency in content delivery. Teachers reported that no formal textbook or Acehese language syllabus has been disseminated by the local education office, leading to diverse teaching methods and learning outcomes. This finding confirms previous research that emphasizes the importance of curriculum alignment and material standardization for local language sustainability (Sulhan et al, 2025).

Challenges in Implementation

The main challenges in implementing Acehese language instruction are categorized into four key areas: resource limitations, teacher qualification, institutional support, and student engagement.

1. Resource Limitations

One of the most pressing issues is the unavailability of standardized teaching materials. Only a few schools have access to any form of printed material or digital resources in the Acehese language. Most teachers develop their own lesson content, often based on oral tradition or informal resources such as YouTube videos or local radio shows. The absence of a clear curriculum framework leads to overlapping content and a disorganized progression of skills across grade levels.

2. Teacher Qualification

Many teachers assigned to teach the subject do not hold a formal educational background in Acehese language or linguistics. This lack of specialization hinders effective instruction, particularly in grammar, writing, and cultural nuances. Interviews with school principals revealed that in most cases, the subject is assigned to classroom teachers who are native speakers but have no pedagogical training in language teaching.

Some teachers expressed insecurity in their ability to teach Acehese correctly. They feared that their dialect might not reflect standardized usage or that their explanations lacked academic depth. This concern confirms Mulyani et al.'s (2023) argument that teacher competence plays a critical role in the success of local language integration.

3. Institutional Support

There is a significant gap in institutional support for curriculum development and teacher training. None of the schools surveyed reported receiving formal training from the Ministry of Education or local curriculum development centers. While the *Permendikbud* provides the policy framework, its operationalization at the school level remains ambiguous. Teachers are unsure about assessment methods, grading policies, or how Acehese language instruction aligns with national learning outcomes. Additionally, the bureaucratic process for proposing budget allocations for teaching aids or community collaboration (e.g., inviting cultural practitioners) is lengthy and often unsuccessful due to the lack of priority placed on local content subjects in school planning.

4. Student Engagement

Teachers observed that student interest in learning Acehese varied depending on their home environment. In households where Acehese is still actively spoken, students show greater enthusiasm and confidence in using the language. However, in urban settings where Bahasa Indonesia dominates communication, students often perceive Acehese as “old-fashioned” or irrelevant to their daily lives.

Students also reported that Acehese is rarely used in peer conversations, media consumption, or digital interaction, further marginalizing the language from their social

spheres. This aligns with Hayati et al. (2025), who found that the lack of environmental reinforcement accelerates language shift among school-age children.

Perceptions of Teachers and Parents on Cultural Awareness

To capture broader perspectives, a Likert-scale questionnaire was distributed to 24 teachers and 60 parents across six schools. The responses revealed a strong consensus on the positive impact of Acehnese language education on cultural identity.

1. Teachers' Perceptions

Most teachers (87%) agreed that Acehnese language classes increase students' understanding of traditional values and regional identity. They observed that students were more respectful during ceremonies and showed more interest in local stories after being exposed to cultural elements in the curriculum. A few teachers also noted that learning Acehnese made students more curious about their family histories and village customs.

However, 58% of the teachers expressed concern that the limited instructional time was insufficient to develop meaningful linguistic competence. They advocated for increasing weekly teaching hours or integrating Acehnese into thematic cross-subject projects, such as cultural storytelling or music classes.

2. Parents' Perceptions

Among the parents surveyed, 74% strongly agreed that learning Acehnese contributes to the preservation of family heritage and moral values. They emphasized the critical role of schools in transmitting cultural knowledge. Notably, many parents acknowledged that they no longer actively used Acehnese with their children at home.

Nonetheless, 32% of parents expressed worry that an overemphasis on the regional language might detract from mastering Bahasa Indonesia or English. They perceived these languages as more economically valuable. This concern reflects global trends in which minority languages face functional displacement in modern contexts (UNESCO, 2020).

Interestingly, qualitative responses from several parents revealed emotional narratives. Their children would come home singing traditional Acehnese songs or asking about ancestral practices. These interactions often rekindled family conversations in the mother tongue.

Discussion of Implications

The findings of this study suggest that the integration of the Acehnese language into the primary school curriculum holds significant potential for enhancing cultural awareness. It also plays an important role in preserving local identity among young learners. However, the success of this integration depends heavily on the support structure surrounding curriculum design, teacher preparation, and community involvement.

From a pedagogical standpoint, there is a need to develop culturally relevant and child-centered learning materials. These materials should bridge the gap between tradition and modernity. Modular resources based on storytelling, songs, and local customs can

create meaningful engagement, especially when adapted to digital platforms that align with students' media habits.

In terms of policy, education officials must go beyond regulatory statements. They should prioritize teacher training, curriculum development, and school-level implementation initiatives. Collaboration with local colleges and cultural institutions can help give academic support and validation for instructional content (Syahputra & Mr, 2023).

Community involvement—particularly parental participation—also needs to be formalized. Regular “language days,” intergenerational learning activities, and community exhibitions can reinforce language use beyond the classroom. As research shows, sustainable language learning is most effective when reinforced across home, school, and community environments (Fitriah et al., 2025).

In conclusion, while obstacles exist, this study emphasizes the significance of a deliberate and context-sensitive approach to local language instruction. Such an approach can be a valuable instrument for linguistic preservation. It can also lay a solid foundation for instilling cultural pride and identity in the following generation.

CONCLUSION

This study investigated the incorporation of the Acehnese language into the primary school curriculum in Lhokseumawe, Aceh, as a means of preserving local cultural identity via education. Using a mixed-methods approach, the study investigated how the Acehnese language is now implemented and identified the major problems encountered during its implementation. It also looked into how instructors and parents saw the cultural relevance of studying Acehnese.

Findings revealed that while several schools have begun to incorporate Acehnese language instruction under the *Merdeka Curriculum*, implementation remains inconsistent. The use of teacher-generated materials, limited instructional time, and a lack of standardized curriculum hinder the effectiveness of language teaching. These gaps highlight the need for institutional support and clear policy guidelines at both local and national levels.

Inadequate teaching materials, a lack of teacher training, and little community involvement all have a negative impact on the long-term viability of Acehnese language education. These structural limitations impede regular and successful classroom implementation. Furthermore, Acehnese is perceived as a less functional language in modern circumstances, which has led to its fall in usage, particularly in metropolitan areas.

Despite these limitations, both instructors and parents appreciate the importance of Acehnese language education. They see it as a crucial tool for fostering cultural pride, intergenerational communication, and student identity development. As a result, there is significant support for growing and strengthening the program.

In conclusion, the implementation of Acehnese language education in primary schools represents a promising avenue for cultural preservation. However, to ensure its long-term sustainability, a coordinated effort involving curriculum development, teacher capacity building, parental engagement, and community participation is essential.

Strengthening these components will help bridge the gap between policy and practice, enabling the Acehnese language to thrive within formal education and beyond.

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UNCOVERING THE MEANING OF QUR'ANIC INSCRIPTIONS ON THE TOMBSTONE OF SULTANAH NAHRASIYAH: A STUDY OF TAFSIR AND EPIGRAPHY

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ABSTRACT

The tombstone of Sultanah Nahrasiyah is one of the largest and most well-preserved tombstones, although some parts have been damaged, and the inscriptions are no longer legible. It is located in the royal cemetery complex of Aceh. The purpose of this study is to reveal the meaning and significance of the verses of the Holy Qur'an inscribed on the tombstone of Sultanah Nahrasiyah. This research employs a qualitative METHOD using an ethnographic approach. The sources of this study are divided into two categories: primary sources, which include direct observation of the tombstone of Sultanah Nahrasiyah and identification of the Qur'anic verses engraved on it, and secondary sources, which involve references to Qur'anic exegesis to understand the implicit meanings of these verses. The findings show that the Qur'anic verses engraved on the tombstone possess profound meaning and significance, particularly for the deceased, reflecting the spiritual relationship between the soul and the Creator, Allah SWT.

Keywords: *Ethnography; Tombstone of Sultanah Nahrasiyah; Qur'anic Interpretation*

INTRODUCTION

The Kingdom of Samudera Pasai was the oldest Islamic kingdom in Aceh and Southeast Asia. This is evidenced by historical relics found in the royal tomb complex of Samudera Pasai. Within this complex, there are tombstones belonging to the kings of Samudera Pasai that feature various calligraphic carvings of verses from the Holy Qur'an, each carrying different meanings. However, many of these tombstones have been damaged over time, making the calligraphic inscriptions difficult to read. One of the largest tombstones that remains relatively intact is the tombstone of Sultanah Nahrasiyah. Sultanah Nahrasiyah was a sultanah, or queen, of the Sultanate of Samudera Pasai (Jamaluddin, 2021). She was the daughter of Sultan Zainal Abidin bin Ahmad bin Muhammad bin Al-Malik Ash-Shalih and passed away in 831 AH/1428 AD. Also known as Putri Nahrasiyah, she ruled the Kingdom of Samudera Pasai from 1406 to 1428 AD. She is recognized as the first female sultan in Southeast Asia, known for her wise leadership, maternal character, and compassion. During her reign, the spread of Islam accelerated significantly, and the Sultanate of Samudera Pasai reached the peak of its prosperity.

Gravestones function as markers for the graves of deceased individuals. They vary in shape and size and are commonly decorated with carvings or engravings, such as calligraphy containing verses from the Qur'an, hadiths, or poetry. Larger or more

elaborate gravestones often indicate the social status of the deceased, such as leaders, kings, or other prominent figures. Several previous studies have examined the tomb of Sultanah Nahrasiyah. These include, first, *The Form and Meaning of the Tomb of Sultanah Nahrasiyah X of the Samudera Pasai Kingdom* (Ichsan, Asrinaldi, Achmad Zaki, & Saniman Andi Kafri, 2023); second, *Lamuri "Plangpleng" Gano Lamdingin Type Tombstones* (Dedy Satria, 2021); and third, *Ethnographic Methods and Al-Qur'an Research Development* (Otong Suhendar, 2022).

None of the previous studies have specifically examined the meaning of the Qur'anic verses inscribed on the tomb of Sultanah Nahrasiyah using a Qur'anic interpretation approach. Every verse of the Qur'an contains implicit meanings that can only be fully understood through proper Qur'anic interpretation. This article aims to reveal the meaning of the Qur'anic verses found on the tomb of Sultanah Nahrasiyah and to explore what makes these verses significant enough to be chosen for inscription on the tomb. Based on the limitations of earlier studies, the researcher believes that each tomb possesses its own characteristics and conveys particular messages through its tombstone as a final resting place.

Tombs provide important clues about beliefs concerning divinity, perceptions of life and death, social status differences, respect, and social interactions. Tombs function not only as historical records but also as indicators of social status and reflections of certain communal developments (Rahim & Suseno, 2021). The gravestones of the kings of Samudera Pasai feature carvings of verses from the Holy Qur'an, hadiths, and poetry, all of which contain implicit meanings, including those found on the gravestone of Sultanah Nahrasiyah. These Qur'anic verses are rich in implied meanings, including the virtues embedded within them. To understand the meaning and significance of these verses, it is necessary to directly examine the carvings on the tombstones and analyze them using Qur'anic interpretation. This perspective is what motivates the researcher to further investigate which Qur'anic verses are engraved on the tombstone of Sultanah Nahrasiyah and how their meanings can be understood through Qur'anic exegesis.

METHOD

This study employs a qualitative METHOD using an ethnographic approach. Qualitative research aims to describe the carvings of Qur'anic verses on gravestones in order to identify and interpret their readings. The ethnographic approach is applied to examine the gravestone inscriptions as cultural artifacts, including their form, script, language, writing style, and historical context (Setyowati, 2014). In addition, Qur'anic interpretation is used to analyse the meanings of the verses inscribed on Sultanah Nahrasiyah's gravestone. Through the combined use of ethnographic and tafsir approaches, this study seeks to explain the meanings of each Qur'anic verse engraved on the tombstone of Sultanah Nahrasiyah.

The data sources in this study are divided into two categories. The first is the primary source, namely the tombstone of Sultanah Nahrasiyah. The second is the secondary source, which consists of tafsir books used as references for interpreting the Qur'anic verses inscribed on the tombstone of Sultanah Nahrasiyah.

The data collection techniques in this study consist of two interrelated components: observation of the gravestones and documentation. Following data collection, the researcher conducted an ethnographic analysis by examining the gravestone carvings based on empirical findings in the field. This was followed by an interpretive analysis to interpret the Qur'anic verses found on the gravestones.

The research was conducted in Gampong Kuta Kreung, Samudera District, North Aceh Regency, approximately 18 km east of Lhokseumawe City. The tomb is located in Complex II of Kuta Karang, not far from the tomb of Sultan Malikussaleh, which is situated in Complex I of the royal tombs of the Samudera Pasai Kingdom.

RESULTS AND DISCUSSION

Calligraphic Inscription on the Tombstone of Sultanah Nahrasiyah

The tombstone of Sultanah Nahrasiyah is adorned with Arabic calligraphy carved in beautiful Kufic script, which is one of the oldest forms of Islamic calligraphy originating from the city of Kufah in Persia. In addition, the inscription on Sultanah Nahrasiyah's tombstone is also written in Thuluth script (Iskandar & Gustiawan, 2023). The use of these scripts was influenced by Persian culture, which entered Aceh along with the spread of Islam by Persian scholars in the seventh century AD. This influence can be seen in the form of tombs, materials, patterns, motifs, and calligraphic carvings found on the tombstones of former kings and queens of Aceh, including that of Sultanah Nahrasiyah (Adnan & Al Asyi, 2019).

The calligraphy on Sultanah Nahrasiyah's tombstone is predominantly composed of verses from the Qur'an Surah Yasin, which cover almost the entire surface of the tombstone. Qur'an Surah Yasin (1–14) is carved on the upper front side of the headstone, Qur'an Surah Yasin (15–48) appears on the left side, and Qur'an Surah Yasin (49–83) is inscribed on the right side of the tombstone. In addition, other surahs and verses are also engraved, including Qur'an Surah Al-Ahzab (56), Qur'an Surah Al-Baqarah (255) (*Ayat Kursi*), Qur'an Surah Ali Imran (18), the *basmalah*, and the *shahada*, which are located in the central part of the headstone.

The carvings of the holy verses of the Qur'an can be seen in the following image:

a. The Upper Front Side



Figure 1. Documentation of Sultanah Nahrasiyah's gravestone

The following is a description of the inscriptions found on the upper front side of the gravestone:

1. Qur'an Surah Yasin (1–14)
2. Qur'an Surah Al-Baqarah (255) (*Ayat Kursi*)
3. The *Shahada*
4. Qur'an Surah Yasin (52)
5. The phrase *Qālū Allāh al-Raḥīm* (ميجرلا الله اولاق)
6. The Basmalah
7. Qur'an Surah Ali Imran (18–19)
8. Qur'an Surah Al-Baqarah (285–286), lines 1–6
9. Not yet identified
10. Not yet identified

b. The Right and Left Sides of the Stepped Tomb



Figure 2. Documentation of the tombstone of Sultanah Nahrasiyah

The entire right and left sides of the tombstone are covered with inscriptions from the Qur'an Surah Al-Ahzab (56).

c. The Right Side of the Gravestone



Figure 3. Documentation of Sultanah Nahrasiyah's gravestone.

The following is a description of the inscriptions found on the right side of the gravestone:

1. The top row contains Qur'an Surah Yasin (15–20).
2. The second row from the top contains Qur'an Surah Yasin (21–28).
3. The third row from the top contains Qur'an Surah Yasin (29–35).
4. The fourth row from the top contains Qur'an Surah Yasin (36–42).
5. The fifth row from the top contains Qur'an Surah Yasin (43).

d. The Left Side of the Tomb



Figure 4: Documentation of Sultanah Nahrasiyah's tombstone



Figure 5. Documentation of Sultanah Nahrasiyah's tombstone

The following is a description of the inscriptions found on the left side of the tombstone:

1. The top row in Figure 5 contains Qur'an Surah Yasin (49–55).
2. The second row contains Qur'an Surah Yasin (56–64).
3. The third row contains Qur'an Surah Yasin (65–71).
4. The fourth row contains Qur'an Surah Yasin (72–79).
5. The fifth row contains Qur'an Surah Yasin (80–83).

To understand and uncover the implied meanings of these holy verses of the Qur'an, the researcher employed the tafsir '*Araisul Bayan* by Al-Baqli. This tafsir was selected because its interpretative methodology is considered appropriate for revealing the meanings of the Qur'anic verses engraved on the tombstone of Sultanah Nahrasiyah.

The Special Features of the Verses of the Qur'an in the Calligraphic Inscription on the Tombstone of Sultanah Nahrasiyah in the Perspective of Al-Baqli's Tafsir '*Araisul Bayan*.

There are many carvings of holy verses from the Qur'an surrounding the tombstone of Sultanah Nahrasiyah, including the basmalah, the shahada, QS. Al-Baqarah, QS. Al-Ahzab, and predominantly Surah Yasin. Surah Yasin is often recited when someone dies because of its virtues that can alleviate the torment of the grave. To reveal the implied meaning and virtues of these holy verses of the Qur'an, the researcher used the *Araisul Bayan* interpretation approach, including:

Basmalah

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

“In the name of Allah, the Most Gracious, the Most Merciful.”

Basmalah is the opening of the Holy Qur'an and forms an integral part of *Surah Al-Fātiḥah*. The entire meaning of *Surah Al-Fātiḥah* is encapsulated within the *Basmalah*, highlighting its central theological significance. Furthermore, *Basmalah*, which consists of 19 letters, represents a special number corresponding to the number of angels guarding Hell, which is also 19 (Tihul et al., 2024).

Tafsir '*Araisul Bayan*

Al-Fatiha is called the opener because it is the key that unlocks the secrets of the holy book, because it is the key to the storehouses of secrets and the door to mercy. It is like the entrance to the meanings of the Qur'an, because Allah opens the Qur'an to secrets with the *basmallah*, just as one opens the door of a house with its key.

مستب

(ب) The letter *ba* means eternity for those who have perished. The letter *sin* means the revelation of majesty and glory for those who are close. The letter *mim* means the revelation of kingship for those who possess character.

الله

The letter *Alif* is the revealer of secrets for experts in intimacy and love. The letter *Lam* is the revealer of secrets for experts in majesty. The first *Lam* is the revealer of secrets for experts in majesty, while the second *Lam* is the revealer of secrets for experts in beauty. *Ha* is the revealer of secrets for experts in authority and awe.

According to Ja'far bin Muhammad: *ar-rahman* and *ar-rahim* are two forms that are evident in both hands of Allah for obedient people, appearing as mercy and

compassion, while for the disobedient, they appear as a cover for shame in this world and forgiveness in the hereafter (*Ruzbihan Baqli Al-Syirazi, Araisul Bayan Fi Haqaiq Al-Quran*, (Dar Al-Kutub Al-Ilmiyah 2008) Juz 1.Pdf, n.d.).

The Virtues of Basmalah

Based on the above interpretation, it can be concluded that *basmalah* has the virtue of being the key to entering the secrets of the Qur'an and a bridge that connects servants and Allah through His mercy and compassion. It is Allah who awakens His creatures from among those He loves, then places them among the chosen ones, granting assistance and blessings to His servants, making them among the righteous, opening the eyes of those He loves and making them people who stay away from worldly pleasures in this life, so that they strive earnestly to prepare themselves to leave this world for the realm of eternity, keeping them away from things that bring His wrath and being wary of the fire of Hell (Nawawi, 2015).

The Shahada

الله لوسرادمحم ن ادهشاو الله لا اله الا ن ادهشا

“I bear witness that there is no god but Allah, and I bear witness that Muhammad is the messenger of Allah.”

The most important pillar of faith is teaching the oneness of Allah, or tawhid, which is expressed in the *Shahada*, the first pillar of Islam. This key phrase distinguishes Muslims from non-Muslims. To translate the *shahada* into Indonesian, it is necessary to understand the structure of the sentence. The letter lam at the beginning of the sentence is *La nafiyyah liljinsi*, which is a *nafi* letter that negates all kinds of gods. *Illa* is an *istisna* (exception) letter that excludes Allah from all kinds of gods that are negated. This type of sentence is called a negative sentence, as opposed to a positive sentence. The word *illa* serves to attribute the negative sentence. In Arabic grammar, *itsbat* after *nafi* has the meaning of *alhashru* (limiting) and *taukid* (reinforcing). Thus, this sentence of *tawhid* contains the meaning that there is no God who truly deserves to be called God except Allah SWT (Karim, 2017).

QS. Yasin

Al-Husain said that the truth, through His revelation, prevented the people of Paradise from enjoying Paradise, for He destroyed them with His pleasurable revelation. Thus, the kingdom descended upon them. From Anas bin Malik *radhiyallahu 'anhu*, he said that the Prophet Muhammad saw, said: “Everything has a heart, and the heart of the Qur'an is *Surah Yasin*. Whoever recites *Surah Yasin*, Allah will record for him a reward equal to reciting the Qur'an ten times” (Reported by Tirmidhi). As is commonly found, *Surah Yasin* is often recited at the time of a Muslim's death due to its well-known virtues. *Surah Yasin* consists of 83 verses, was revealed in Mecca, and all of its verses are engraved on the tombstone of Sultanah Nahrasiyah.

Tafsir 'Araisul Bayan

Human beings are born with a readiness to know Allah. If this knowledge were mixed with the body or changed by actions, it would change; however, knowledge of

Allah is a decree from Allah. This is because He reveals Himself with His beauty and majesty directly, without mediation and without human effort.

Ibn ‘Atha said that fitrah means placing the bodies under the grasp of Allah’s power and the souls under the grasp of Allah’s might. Fitrah here is understood as the original condition of human beings: the body and the soul. Our bodies are fully under Allah’s power (*al-Qudrah*), meaning that all movement, health, illness, life, and death are under His control. Our souls are under Allah’s might (*al-‘Izzah*), meaning that Allah protects, strengthens, and elevates the human soul. Thus, human beings, both body and soul, are entirely under Allah’s control, not their own.

Ibn ‘Aṭā’ said that fitrah is the placing of human beings under the grasp of power and souls under the grasp of gentleness. Some scholars said that a sincere servant is one who acts in accordance with the light of fitrah, not otherwise. Deeds performed in awareness of fitrah will not be rejected, unlike deeds driven by desire. Al-Qushayri said that the soul will never be free from witnessing the Creator in reality. Even when it falls before the highest majesty, it will not see Him except when it is commanded to enter Paradise.

Allah says, “*If only my people knew...*” This means that they might also understand the blessings Allah has granted, so that they would believe. Allah created all things in pairs as a sign of the perfection of His power. ‘Abdul Aziz al-Makki said that Allah created all pairs to show that He alone creates without a pair, and that He is One without any partner.

Allah also reveals to His servants the signs of His greatness through night and day. The rising sun becomes a revealing light, while the night conceals it. When a person is illuminated by the light of witnessing, the heart is enlightened with the light of knowledge; but when that light fades, one returns to darkness, just as the sun is covered by the night.

Various spiritual states then overflowed upon them until they reached the peak of light in the witnessing of the attributes. When they were close to that peak, the light of the attributes began to fade and disappear, leaving only the light of the divine names, vision, and witnessing. Thus, the full moon (that is, their hearts) became like a crescent. They then beheld the beauty of Allah’s attributes with the eyes of the heart, full of certainty. This is Allah’s gentleness toward His servants, for He does not allow them to remain constantly in the witnessing of His Essence alone, but brings them down to the levels of the attributes, so that they are not destroyed by the majesty of His light.

A teacher said, “Spiritual daylight leads a servant into the night of divine help and calm, then Allah guides him with the light of His majesty, just as a blind person walks guided by his staff”. In a servant’s spiritual journey, he sometimes experiences a state of weakness in certainty. He then reflects, his perception increases, and he comes to see his state clearly, becoming complete and whole like a full moon. After that, it gradually diminishes, just as the moon slowly wanes. The farther he is from the light of the sun, that is, the light of knowing Allah, the weaker and more veiled he becomes.

A gnostic said: The meaning of this verse is that a servant on the path to Allah will not remain constantly at the peak of spiritual states. At times, he is brought down to the attributes of deficiency, then raised again to the attributes of perfection. Thus, he

continually moves between perfection and deficiency, so that he does not become self-sufficient and remains humble before Allah.

When they enter Paradise, Allah unveils for them the light of bliss. They dwell in the shade of delight, experiencing peace and tranquility. They move from spiritual witnessing to eternal enjoyment in Paradise, without weariness and without disturbance. Allah commands human beings to follow the straight path of truth with readiness to obey Him, free from any deviation and unmixed with doubt or disturbance. Al-Wasithi said, “Whomever Allah makes His true servant is taken away from self-reliance. And whoever is veiled from knowing his Lord has not known the reality of servitude, for the reality of servitude is the manifestation of humility. Whoever does not show humility before divine lordship has tasted nothing.” Whoever knows Allah lives a life with Allah, immersed in His attributes, intoxicated by His beauty, absorbed in His greatness and majesty, and united with His eternity.

Allah also explains the blessings granted to the inhabitants of Paradise: ease, pleasure, and happiness. The people of Paradise receive an eternal and unending greeting for His servants who were sincere in this world and the Hereafter. They hear this greeting directly from Allah, and they behold His Face as the perfection of all blessings.

Al-Junaïd said, “The truly living person is one whose life is with the life of his Creator, not with the life of the body. Whoever lives only by the body is like a corpse while alive. Whoever lives by his own life will die in due time. But whoever lives by the life of Allah lives a true life, for which there is no death”. And because of this, the believer attains the rank of *al-hayāt al-asliyyah* (the true life).

QS. Al-Baqarah: 255 (*Ayat Kursi*)

لَا إِلَهَ إِلَّا هُوَ عَنَشِي إِذَا أَدْنَمَ ضَرْبًا يَأْمُو تَوْمِهِلَا يَأْمُو هَلْ مَوْنُ لَو قَتْسْ مَذْخَلَا مَوْبِلَا يَحْلَا وَهَلْ لَو هَلْ لَا لَلْ
لَو ضَرْبًا أَوْ تَوْمِهِلَا هَيْسَرْكْ عَسُو عَاشْ أَمْبَلْ إِمْلَعْ نَمَ عِشْبْ نَوَطِيحْ لَو مَلْخْ أَمُو مَهْبِدِيَا نَبْ أَمْ جَلْعِي مَنْدَابْ
مَنْظَرَا يَلْعَبَا وَهَوِ أَمْ هَظْخْ مَدُو

“Allah, there is no god (worthy of worship) except Him, the Ever-Living, the Sustainer of all existence; neither slumber nor sleep overtakes Him. To Him belongs whatever is in the heavens and whatever is on the earth. Who can intercede with Allah without His permission? Allah knows what is before them and what is behind them, and they do not know anything of Allah's knowledge except what He wills. Allah's Throne encompasses the heavens and the earth. And Allah does not feel any heaviness in preserving both, and Allah is the Most High, the Most Great”. QS. Al-Baqarah: 255 (Islam & Agama, 1989).

Among the virtues of *Ayatul Kursi* is what is mentioned in the hadith of the Prophet Muhammad from Abu Umamah al-Bahili, may Allah be pleased with him. The Prophet said: “Whoever recites *Ayatul Kursi* at the end of every prayer, nothing will prevent him from entering Paradise except death” (Amr, n.d.).

Tafsir 'Araisul Bayan

Ibn Mansur said: “لَا إِلَهَ إِلَّا اللَّهُ” requires two things: negating everything other than Allah; establishing that Allah is the only one who deserves to be worshipped. "يَحْيَا" is the One who gives life to all living things. He lives by His own nature, not from any other source, and is eternal. "مَوْلَى" is the One who never ceases to care for His creatures, who

never neglects them even for a moment. "مويذلا" is the One who governs the universe with His wisdom and keeps the secrets of those who know Him, revealing to them the secrets of His attributes and unveiling the light of His majesty. "مون لاو قزس هذخأت لا" means: removing all forms of weakness from His attributes, and affirming that Allah is never negligent or weak in caring for His creatures. "ضرلُا يف امو تاوامس لا يف ام دل" means: everything in the heavens and the earth belongs to Him, both in essence and in creation. The Sufi scholars said: The meaning is that everything, from the two worlds (this world and the hereafter), is bound by His command and does not exist independently without Him.

Al-Wasithī said: Whoever takes an intermediary to Allah other than Allah has adorned himself with hypocrisy. Whoever seeks Allah's pleasure through other than Allah has asked for something impossible. For Allah has said:

"من ذاب ل الله مدنع غشبي يذلل اذ نم"

Ibn Mansur said: Intercession is of no benefit to one who has not obtained His permission, and nothing can satisfy Him except His permission.

"مفعلخ امو مهديا نوبام ملع" meaning, He knows all the whispers of the heart, the thoughts that cross their minds, the hidden circumstances among them, and all the secrets of deeds related to His will. He also knows their deeds before they do them, as well as all the deeds that bring them closer to Him.

"ءاش امل لا ملع نم ءشب نوطحي لاو" Human knowledge is limited in its ability to comprehend primordial knowledge (that which has no beginning) because the primordial cannot be grasped by human reason, except for those whose hearts have been opened by Allah from the suffering of sins.

"ضرلُا او تاوامس لا هسرك عسو" The throne here is the seat of the heart of the wise. It is so vast, exceeding the vastness of the heavens and the earth, because it contains knowledge about worldly and otherworldly matters, as well as knowledge about endless pleasures, unlimited for anyone.

"ميظع لا يلعلا وهو امهظح هدوي لاو" This means that preserving both (the heavens and the earth) does not burden Him and does not diminish His glory. He remains Most High in His power, for He has a great kingdom, absolute power, and unlimited rule. His power extends even to the smallest particle. There is nothing that stands alone without Him creating it.

Based on the above interpretation, it can be concluded that the *Ayat al-Kursi* contains the secrets of Allah SWT's power over the heavens and the earth, as well as secrets that are beyond human comprehension, because Allah has absolute and unlimited power. The *Ayat al-Kursi* explains to humans that humans themselves need a Great Being as a protector, regulator, and axis of life. For this reason, the *Ayat al-Kursi* is often associated with supernatural matters, including one of its implementations, which is written on the tombstone of Sulthanah Nahrasiyah at the head, depicting the majesty of the verse itself.

The virtues of the *Ayat al-Kursi*

Among the virtues of the *Ayat al-Kursi* is what is mentioned in the hadith of the Prophet Muhammad from Abu Umamah al-Bahili. The Prophet said, "Whoever recites

the Ayat al-Kursi at the end of every prayer, nothing will prevent him from entering Paradise except death.” ('Amr, n.d.).

QS. Ali Imran: 18-19

“Allah declares that there is no god but Him; (likewise) the angels and those who have knowledge uphold justice, there is no god but Him, the Almighty, the Wise” QS. Ali Imran:18 (Islam & Agama, 1989).

Tafsir `Araisul Bayan

The knowledge of Allah is perfect and eternal, having no beginning and no end. It is an attribute inherent in His essence, not something that was created. With His love and gentleness, He introduced Himself to His creatures before they were created. If it were not for this introduction, creatures would never have been able to know Him.

Ibn 'Ata said: The words of Allah ﷻ indicate that He introduced Himself to His creatures through His names, explaining His divinity, the perfection of His attributes, and the majesty of His words. He testifies for Himself as the most truthful witness and the strongest proof, for He is the One God, with no partner or associate.

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

“Indeed, the religion in the sight of Allah is Islam. Those who were given the Scripture did not differ except after knowledge had come to them, out of envy among themselves. And whoever disbelieves in the verses of Allah, then indeed, Allah is swift in reckoning.” QS. Ali Imran: 19 (Islam & Agama, 1989).

“Indeed, the religion with Allah is Islam” means acceptance of the truth, acceptance of His decisions and provisions, and submission of the heart inwardly and the limbs outwardly to Him. This preserves the purity of the heart, prevents anxiety, and brings peace and sweetness to faith, especially when facing trials. Ja’far Ash-Shidiq said: “If a servant’s Islam is not built on the foundation of recognizing the blessings that come from Allah, trusting in Him, and submitting to His commands, then he is only in the name of Islam, not in the true essence of Islam”.

OS. Al-Ahzab: 56

امه لاسن اوچل سوه هیلع اولص او زماء ن یزلأ ادهای ی یزلأ یل ع نولصی مکه ولمو للأ ن!

“Indeed, Allah and His angels send blessings upon the Prophet. O you who believe! Send blessings upon him and greet him with a greeting of peace.” QS. Al-Ahzab: 56 (Islam & Agama, 1989).

Tafsir ‘*Araisul Bayan*

The blessings of Allah upon the Prophet mean elevating him to the *maqam mahmud* (a praiseworthy position). *Maqam Mahmud* is his intercession for his followers, while the blessings of the angels are prayers and requests for forgiveness for him (Ruzbihan Baqli Al-Syirazi, *Arraisul Bayan Fi Haqaiq Al-Quran*, (Dar Al-Kutub Al-Ilmiyah 2008) Juz 1.Pdf, n.d.).

Al-Wasit said: Send blessings upon him with a present heart, without negligence, and not merely with words, for that has a certain degree. Abu 'Abd al-Rahman al-Sulami said: Abd al-Wahid bin Zaid said: I heard al-Rabi' bin Khuthayam when a matter was mentioned in his presence, and he said: "Indeed, I desire something for myself, so I send blessings upon the Prophet for that. Verily, sending blessings upon the Prophet can erase your sins and purify your souls". Based on the above interpretation, it can be concluded that sending blessings upon the Prophet can bring mercy and erase sins.

CONCLUSION

Tombstones, which are generally considered only as markers for the graves of the deceased, can also hold implied meanings as clues to life after death between the soul and the creator, such as by revealing the meaning of the messages engraved in the holy verses of the Qur'an on the tombstone of Sulthanah Nahrasiyah through Qur'anic interpretation. Ruzbihan Baqli Al-Syirazi was an exegete from Fasa, Persia (Iran). He was born in the 12th century AD and died in the 13th century AD. His interpretation was influenced by his scientific background, one of which was Sufism, which in that century also experienced development in Aceh in the form of the Shadhiliyah order as an implementation of Sufism. Therefore, Ruzbihan Baqli Al-Syirazi's interpretation is closely related to the context of the interpretation of the verses written on the tomb and the situation of the Acehnese people, who were greatly influenced by Persian culture as one of the parties that spread Islam in Aceh.

At the tomb of Sulthanah Nahrasiyah in this study, researchers found several verses and surahs carved on the marble tombstone of Sulthanah Nahrasiyah. Specifically, on the headstone was carved QS. Yasin: 1-14; QS. Al-Baqarah: 255; the *Shahada*, QS. Yasin: 25; the *basmalah*; QS. Ali Imran: 18-19; QS. Al-Baqarah: 285-286. The right and left sides of the tomb are filled with QS. Al-Ahzab: 56. On the right side of the tomb is QS. Yasin: 15-43. On the left side of the tomb is QS. Yasin: 49-83.

From the explanation in Tafsir 'Araisul Bayan, it can be concluded that:

1. QS. Al-Baqarah 255 refers to the greatness of Allah with all His attributes.
2. QS. Ali Imran: 18-19 and the shahada are testimonies of tawhid, the angels' and scholars' acknowledgment of Allah's oneness, and the affirmation that Islam is the only true religion in the sight of Allah.
3. QS. Al-Baqarah: 285-286 is a closing prayer that reflects humility, recognition of human weakness, and a request for help from Allah.
4. QS. Al-Ahzab: 56 describes the greatness of the Prophet Muhammad, peace be upon him, so that even the angels send blessings upon him as a form of love, respect, and recognition of the Prophet's message.
5. Quran Surah Yasin discusses in detail the blessings that a servant will attain if they let go of worldly distractions and come to know and draw close to Allah.

The results of this study indicate that tombstones, which are considered symbols of the final resting places of kings, queens, and other important figures of the past, are not merely symbols but also possess artistic value and meaning behind their carvings and engravings. This is especially true for tombs that feature carvings of verses from the Holy

Qur'an, making it important to conduct a more in-depth study of the interpretation of these verses from the perspective of interpretive scholars.

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AN ANALYSIS OF FIGURATIVE LANGUAGE USED IN THE MOVIES DIRECTED BY ADAM "TEX" DAVIS AND JERRY KOLBER

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ABSTRACT

This study, entitled "An Analysis of Figurative Language Used in Movies Directed by Adam 'Tex' Davis and Jerry Kolber," aims to analyze the types and contextual meanings of figurative language found in four selected episodes: Focus Pocus and It is About Time from the Brain Games series, and social media and Germs from the Brainchild series. Figurative language plays a crucial role in understanding literary and spoken texts, particularly in audiovisual media, where creative expressions enhance the effectiveness of message delivery. This research employed a qualitative descriptive method with content analysis. Geoffrey Leech's theory was applied to classify the types of figurative language, while John Lyons's theory was used to interpret contextual meanings. The findings revealed 99 metaphors, 68 personifications, 62 hyperbole, 33 ironies, 23 similes, 8 metonymies, and 1 litotes, totaling 294 expressions. The speaker, the audience, and the situational context influenced the contextual meanings. Metaphors and personifications were predominantly used to explain scientific and social concepts; hyperbole emphasized dramatic aspects to attract attention; irony provided humor and criticism; similes clarified abstract ideas through comparison; and metonymy simplified complex concepts. These results indicate that figurative language in these episodes functions as an effective linguistic tool for conveying scientific and social messages in an engaging and comprehensible way.

Keywords: *Adam 'Tex' Davis; figurative language; Geoffrey Leech; Jerry Kolber; movies*

INTRODUCTION

Language is a dynamic system that enables humans to express emotions, share ideas, and interpret the world around them. It functions not only as a means of communication but also as a reflection of human thought, culture, and creativity (Evans & Green, 2021). Through language, people build relationships, share knowledge, and preserve cultural identity. It evolves alongside societies, adapting to new realities and shaping how individuals perceive and understand the world (Dancygier & Vandelanotte, 2023).

Beyond its communicative role, language often serves as a medium for artistic and emotional expression. Figurative language, in particular, enriches human communication by extending meanings beyond literal interpretation and creating vivid imagery that evokes emotional and intellectual responses (Semino, 2022). Devices such as metaphors, similes, hyperbole, and personification enable speakers to represent abstract ideas in more imaginative and relatable ways. Thus, figurative language serves

as an essential bridge between cognition, emotion, and creativity (Manu et al., 2024).

In linguistics, the study of meaning, known as semantics, addresses how words convey both denotative and connotative meanings. Figurative expressions primarily function within the connotative domain, in which meanings are shaped by context and associations (Evans & Green, 2021). Understanding figurative language, therefore, requires attention not only to its linguistic structure but also to the situational context in which it appears. Contextual interpretation reveals how meaning is shaped by tone, setting, and speaker intention, turning ordinary language into a tool of nuanced communication (Rachminingsih et al., 2024).

Previous studies have shown that figurative language enhances narrative depth and emotional engagement in various media. Pradita et al. (2025) found that metaphors and similes in *A Dog's Way Home* convey moral messages and emotional subtleties. In contrast, Wati and Maisarah (2023) demonstrated that figurative expressions in film songs enhance the narrative's imaginative appeal. However, research on identifying figurative types and their contextual meanings in educational or documentary programs remains scarce.

Adam "Tex" Davis and Jerry Kolber are well-known directors who creatively combine science, psychology, and entertainment in their works, particularly in *Brain Games* and *Brainchild*. These series use figurative language in dialogues and narration to simplify abstract concepts and engage viewers (Atomic Entertainment, n.d.). However, the use of figurative language in their educational media has not been systematically analyzed. Unlike literary works, such programs aim not only to entertain but also to inform, making the study of figurative meaning within context especially valuable.

Therefore, this research aims to analyze the types and contextual meanings of figurative language used in selected episodes of *Brain Games* and *Brainchild* directed by Adam "Tex" Davis and Jerry Kolber. Using Leech's (1981) classification, which includes simile, metaphor, personification, hyperbole, irony, litotes, and metonymy, this study identifies each type of figurative expression and interprets its contextual meaning within the dialogues. The findings are expected to contribute to linguistic studies by providing a clearer understanding of how figurative language operates contextually in educational and documentary-style media.

METHOD

This research employs a qualitative method that focuses on meaning, interpretation, and contextual understanding, using rich, detailed data (Creswell, 2014). The data sources consist of transcripts from four selected episodes: *Focus Pocus* and *It is About Time* from *Brain Games*, as well as *social media* and *Germes* from *Brainchild*, directed by Adam "Tex" Davis and Jerry Kolber. In this case, the researcher identifies figurative language in these episodes, consistent with Geoffrey Leech's (1981) theory, including simile, metaphor, personification, hyperbole, irony, litotes, and metonymy. Furthermore, the contextual meaning of these figurative expressions is analyzed using John Lyons's (1981) semantic theory.

RESULT AND DISCUSSION

This study analyzed the use of figurative language in four selected episodes from the educational series Brain Games and Brainchild, directed by Adam "Tex" Davis and Jerry Kolber. The analysis revealed seven types of figurative language proposed by Geoffrey Leech: metaphor, simile, personification, hyperbole, irony, litotes, and metonymy. These expressions appeared with varying frequency, reflecting the producers' strategic choices to make the content educational yet engaging. The summary of figurative language occurrences across episodes is presented in Table 1.1.

Table 1.1 Figurative Language Types Found in Four Episodes of Brain Games and Brainchild

Types of Figurative Language	Brain Games Ep. 1	Brain Games Ep. 2	Brainchild Ep. 1	Brainchild Ep. 2	Total
Metaphor	37	21	25	16	99
Simile	6	8	5	4	23
Personification	13	24	4	27	68
Hyperbole	15	9	18	20	62
Irony	5	2	9	17	33
Litotes	1	0	0	0	1
Metonymy	2	3	1	2	8
Total	79	67	62	86	294

The following presents the discussion of each figurative language type along with a representative example, its contextual meaning, and justification for classification:

1. Metaphor

Metaphor conveys meaning by describing one concept in terms of another without explicit comparison. It allows abstract ideas to be understood through familiar concepts. In movie dialogues, metaphors are commonly used to represent emotions, attitudes, and complex experiences more expressively. This use of metaphor enables audiences to grasp the meanings implied in characters' speech and interactions.

Example:

Host: "...For the next half hour, we are going to mess with your mind as we put you through a series of games and experiments designed to put your attention to the ultimate test...."

In the opening scene of Brain Games, the host says, "We are going to mess with your mind" in an energetic tone, accompanied by visuals of psychological experiments. The phrase does not literally mean "damaging the brain," but instead suggests that the show will challenge and play with the audience's way of thinking.

Thus, "mess with your mind" functions as a metaphor that disrupts or tests the viewer's thought process.

2. Simile

Simile expresses comparison using explicit words such as "like" or "as." It clarifies abstract ideas by linking them to familiar concepts. In film dialogue, similes are frequently used to create vivid imagery and help audiences visualize the characters' meanings. This figurative device also enhances the emotional and descriptive qualities of film language.

Example:

Host: "...This is just like our first test with the flying footballs. How could you possibly know the number of footballs without counting them? Because the answer was hidden in plain sight right on the scoreboard..."

In this scene of *Brain Games*, the host uses the word "like" to compare the current experiment to a previous one involving flying footballs. The two experiments differ, yet the comparison highlights their conceptual and purpose similarities. The host's calm and explanatory tone, combined with visuals of the experiment, helps the audience connect the new activity with the earlier test. Thus, "This is just like our first test with the flying footballs" is identified as a simile that emphasizes the resemblance between two distinct experiments.

3. Personification

Personification attributes human qualities to non-human entities, enhancing comprehension and engagement. In movie dialogues, personification is often used to animate objects, nature, or abstract ideas, making them appear more vivid and relatable. This figurative language helps audiences connect emotionally with the scenes and better understand the intended message.

Example:

Host: Surprise! We are blinking in synchrony. If you want to find out why your brain tricked you, stick around. This is a show about your brain...

In this segment of *Brain Games*, the host's enthusiastic voice-over accompanies visuals of footballs rapidly moving across the screen. The phrase "footballs like this one as they fly across the screen" personifies the footballs by using the verb "fly," which ascribes human-like agency to their movement. Footballs cannot literally fly on their own; the expression instead emphasizes speed and liveliness, making the scene more dynamic and engaging. Therefore, this statement constitutes personification.

4. Hyperbole

Hyperbole uses deliberate exaggeration to emphasize a point or create a dramatic effect. In movie dialogues, hyperbole is commonly used to intensify emotions and highlight characters' feelings or situations. This figurative language helps make expressions more impactful and engaging for the audience.

Example:

“Sahana: That is 500 billion times the number of people on the planet!
There are more germs than there are stars in the universe.”

In the opening segment of Brain Games, the host states, "You would be shocked at how often inattentional blindness causes you to miss things happening right in front of your eyes," in an enthusiastic tone, accompanied by visuals depicting real-life attention experiments. The phrase "you would be shocked" is an exaggeration that intensifies the message, as the audience is not literally shocked but rather surprised by how often such blindness occurs. This use of hyperbole strikingly and memorably emphasizes the frequency and impact of inattentional blindness.

5. Irony

Irony expresses a meaning opposite to the literal sense, often to highlight contrast or contradiction. In film dialogue, irony highlights a discrepancy between what is said and what is actually meant by the characters. This figurative language conveys implicit meanings in the dialogue. It allows audiences to interpret meaning beyond the surface level of the dialogue.

Example:

Host: "...Because the answer was hidden in plain sight right on the scoreboard..."

In this segment of Brain Games, the host states, "The answer was hidden in plain sight right on the scoreboard," to reveal the experimental result. The statement represents situational irony: something described as hidden turns out to be clearly visible. The irony lies in the contrast between the expectation of difficulty and the reality of the answer being obvious. In context, it emphasizes that people often overlook salient information when their attention is directed elsewhere.

6. Litotes

Litotes conveys meaning through understatement, typically by negating the opposite. In film dialogue, litotes is used to express ideas subtly and indirectly. This figurative language often softens statements and reduces the intensity of expression. As a result, litotes can convey meaning more politely and implicitly within the film narrative.

Example:

Host: It is not unusual.

The host states this in a calm, reassuring tone as a car pulls into a driveway. The statement minimizes the oddity of forgetting one's drive home, implying that such memory lapses are common. The phrase "It is not a strange thing" exemplifies litotes, employing negation ("not strange") to understate the situation and normalize it for the audience.

7. Metonymy

Metonymy substitutes an associated term for the intended concept. In movie dialogues, metonymy is often used to represent ideas, institutions, or objects through closely related terms. This figurative language allows meaning to be conveyed more concisely and efficiently. The use of metonymy helps audiences understand the message by relying on familiar associations within the film context.

Example:

Host: "...Pay attention to this receiver going out for a pass and see if you notice anything out of the ordinary..."

The host energetically invites the audience to observe a sports scene showing a rugby receiver catching a pass. The word "receiver" refers not to the act of receiving, but to the player whose job is to catch the ball. This substitution of role for person exemplifies metonymy, in which the name of something closely associated with the actual object replaces the object itself.

Overall, a total of 294 figurative expressions were found, with metaphor (99) and personification (68) being the most frequent. Metaphor predominates in *Brain Games* to simplify abstract concepts, while *Brainchild* employs hyperbole, metaphor, and personification to engage younger audiences emotionally and cognitively. The predominance of metaphor indicates its pivotal role in conveying abstract scientific concepts through familiar comparisons. Personification enhances engagement by humanizing nonhuman elements, thereby facilitating cognitive and emotional connections with the content. Compared with prior studies of narrative films such as *Maleficent* and *Harry Potter*, in which simile was dominant, Nehe et al. (2022) and Rahmawati et al. (2023) reported similar findings. Educational series, however, favor metaphor and personification for clarity and instructional effectiveness. Irony, litotes, and metonymy appear less frequently, reflecting the pedagogical goals rather than humor or social critique.

The application of Lyons' contextual meaning framework reveals that figurative language is dynamically crafted to suit the viewer's cognition and emotion, rather than merely serving as decorative ornamentation. Metaphors such as describing the brain as a "decision-maker" or personifying time as a "trickster" exemplify this, making complex ideas accessible. This finding also differs from Merlin et al. (2024), who, in their study of *Wolf Town*, reported a higher frequency of irony and sarcasm, suggesting genre-driven variation. Integrating Leech's and Lyons' frameworks provides both structural and pragmatic insights, demonstrating how figurative

language functions as a deliberate edutainment tool rather than a random stylistic choice.

CONCLUSION

This study concludes that figurative language in the four selected episodes, two from Brain Games and two from Brainchild, effectively enhances the educational and communicative quality of the series. Using Geoffrey Leech's classification, all seven types of figurative language were identified: 99 metaphors, 68 personifications, 62 hyperbole, 33 irony, 23 similes, eight metonymy, and one litotes. Through John Lyons' contextual meaning framework, it is evident that these figurative expressions are not only stylistic but also functional, helping simplify complex scientific concepts, emphasize key ideas, and engage viewers both cognitively and emotionally. These findings confirm that figurative language contributes significantly to the clarity, engagement, and overall effectiveness of educational media, suggesting that educators and media producers continue to incorporate figurative expressions to enhance audience understanding and learning outcomes.

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AN ANALYSIS OF METAPHORICAL LANGUAGE STYLE IN THE SHORT STORIES BY JAMES BALDWIN

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ABSTRACT

This research aims to identify and classify the types of conceptual metaphors found in three short stories by James Baldwin: *Sonny's Blues*, *The Rockpile*, and *Going to Meet the Man*. The study employs a qualitative method with a content analysis approach. The data were collected through close reading of the texts and analyzed using the framework of Conceptual Metaphor Theory (CMT) by George Lakoff and Mark Johnson, which categorizes metaphors into structural, ontological, and orientational types. A total of 103 metaphorical expressions were identified across the three stories: 34 structural, 40 ontological, and 29 orientational. Among these, ontological metaphors are the most frequently used, indicating that Baldwin often conveys abstract concepts such as emotion, identity, and psychological states through concrete, object-based forms. This study concludes that Baldwin's frequent use of ontological metaphors reflects his effort to represent his characters' psychological struggles and racial experiences through both tangible and conceptual language.

Keywords: *Conceptual Metaphor; James Baldwin; Metaphorical Language; Ontological Metaphor; Short Story.*

INTRODUCTION

Metaphor is one of the most powerful linguistic and literary devices for expressing abstract concepts through concrete imagery. According to Lakoff and Johnson (1980), metaphors are not merely rhetorical ornaments but conceptual frameworks that shape human thought. This theoretical shift has led to the study of metaphor as a central aspect of cognition rather than a purely stylistic device. In literary works, metaphors play a vital role in revealing hidden meanings, emotions, and ideological perspectives.

James Baldwin, as a prominent African American author, is known for his profound exploration of identity, race, and human emotion. His short stories frequently employ figurative language to illustrate psychological depth and social tension. However, while Baldwin's works have been extensively analyzed from thematic and sociocultural perspectives, few studies have applied Conceptual Metaphor Theory to his use of language. This study seeks to fill that gap by examining how Baldwin employs conceptual metaphors to construct meaning in three of his major short stories: *Sonny's Blues*, *The Rockpile*, and *Going to Meet the Man*.

The novelty of this research lies in its cognitive-linguistic perspective, which

examines Baldwin's metaphors not only as literary devices but also as reflections of thought and cultural experience. The study addresses two key questions: (1) What types of conceptual metaphors are found in Baldwin's short stories? and (2) What conceptual meanings do these metaphors convey within the narratives?

METHOD

This study adopts a qualitative descriptive method with a content analysis approach. The primary data are drawn from three selected short stories by James Baldwin, chosen for their rich use of figurative language and thematic relevance. The researcher conducted a close reading to identify metaphorical expressions, which were then categorized according to the Conceptual Metaphor Theory (CMT) proposed by Lakoff and Johnson (1980).

The data analysis in this study followed the interactive model proposed by Miles, Huberman, and Saldana (2014), comprising data condensation, data display, and drawing conclusions and verifying them. During the data condensation stage, the researcher selected and coded metaphorical expressions in James Baldwin's short stories according to relevant conceptual metaphor categories. The data were then presented by organizing the identified metaphorical expressions into tables that included the figurative expression, type of metaphor, source domain, target domain, and interpretation. In the final stage, conclusions were drawn by conceptually interpreting the metaphors and verifying the findings to identify dominant patterns in Baldwin's use of metaphorical language. In this process, the researcher functioned as the primary research instrument, supported by data obtained from relevant literature on conceptual metaphor theory and Baldwin's stylistic features.

RESULT AND DISCUSSION

The analysis revealed 103 metaphorical expressions across the three short stories. These consist of 34 structural metaphors (33%), 40 ontological metaphors (39%), and 29 orientational metaphors (28%). Among the three types, ontological metaphors were the most frequent, suggesting Baldwin's preference for representing abstract psychological and social realities through object-based conceptualizations.

1. Structural Metaphors

Structural metaphors occur when one concept is understood in terms of another. For instance, in *Sonny's Blues*, the expression "All this was carrying me someplace I didn't want to go" frames emotional pressure as a physical journey. The metaphor "emotion is movement" helps readers conceptualize internal turmoil as an external, uncontrollable force. Similarly, in *Going to Meet the Man*, Baldwin structures racial violence as a "cycle" that moves through generations, reflecting how systemic oppression perpetuates itself over time. These metaphors reveal how Baldwin transforms emotional and historical experiences into structured conceptual frameworks.

2. Ontological Metaphors

Ontological metaphors conceptualize abstract ideas as objects, containers, or entities. In *The Rockpile*, Baldwin writes that "anger filled Gabriel's chest like a storm," personifying emotion as a substance within the body. In *Going to Meet the Man*, memory and trauma are treated as living entities that "haunt" Jesse's consciousness. Such metaphors allow Baldwin to externalize psychological conflict, turning intangible feelings into concrete, almost physical phenomena. The frequent use of this type shows Baldwin's tendency to represent mental and emotional states as tangible burdens, especially in the context of racial trauma.

3. Orientational Metaphors

Orientational metaphors relate abstract concepts to spatial orientations, such as up/down or light/dark. In *Sonny's Blues*, the contrast between "darkness" and "light" functions as a recurrent orientational metaphor for suffering and hope. Similarly, "down" often represents despair or oppression, while "up" symbolizes liberation and enlightenment. Baldwin's repeated use of vertical and light-based metaphors emphasizes his characters' psychological struggles and aspirations for redemption within oppressive social systems. Overall, the findings indicate that ontological metaphors predominate in Baldwin's short stories, highlighting his interest in rendering invisible emotional and ideological realities perceptible through concrete imagery. These metaphors serve not only aesthetic purposes but also cognitive and social ones, revealing the inner workings of racial consciousness and emotional survival.

CONCLUSION

This study concludes that James Baldwin's short stories rely heavily on conceptual metaphors, particularly ontological ones, to articulate the inner lives of his characters. The frequent use of metaphors that objectify emotions, memories, and social realities reflects Baldwin's attempt to translate complex human experiences, especially those related to race, pain, and identity, into tangible conceptual forms. Through this metaphorical structuring, Baldwin bridges the cognitive and emotional dimensions of literature, allowing readers to grasp the psychological and moral depth of his narratives. Future research may explore how Baldwin's metaphorical language compares with that of other African American writers, revealing broader patterns of conceptualization within racial and cultural discourse.

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A STUDY OF LINGUISTIC INTERFERENCE AND INDEXICALITY OF PILIBHIT HINDI-URDU TO ENGLISH: AN OT APPROACH

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ABSTRACT

This study presents an account of English loanwords and their integration into Urdu spoken in Pilibhit (U.P.), India, within the framework of Optimality Theory (Prince & Smolensky, 1993). This study investigates the phenomenon of linguistic interference in Pilibhit Hindi-Urdu speakers, explicitly focusing on the phonological adaptation of English loanwords. The primary focus is on the process of "prosthesis," in which a vowel is inserted at the beginning of English words containing initial consonant clusters, such as /sk/ and /sp/. Through the application of Optimality Theory, this research analyzes how Pilibhit Hindi-Urdu speakers modify these loanwords to conform to the phonotactic constraints of their native language. The findings reveal a systematic pattern of vowel insertion that transforms monosyllabic and disyllabic English words into disyllabic and trisyllabic forms, respectively. Ultimately, the study contributes to a deeper understanding of phonological processes in bilingual communities and the dynamics of language interaction and adaptation. It demonstrates how phonological systems adapt when languages come into contact. In this case, Pilibhit Hindi-Urdu speakers modify English loanwords to conform to their native phonotactic constraints. This adaptation suggests a natural tendency of languages to preserve their internal phonological structure while incorporating foreign elements.

Keywords: *Linguistic Interference; Pilibhit Hindi-Urdu; Optimality Theory*

INTRODUCTION

In a globalized world, contact between languages often results in linguistic interference, in which elements of one language influence the structure and usage of another. This phenomenon is particularly evident in multilingual communities where speakers regularly interact with more than one language. One such context is Pilibhit, a linguistically diverse region in India, where Hindi-Urdu speakers are in continuous contact with English due to its global status and functional importance. As English loanwords enter Pilibhit Hindi-Urdu, they frequently undergo phonological modification to conform to the phonotactic constraints of the native language. Prolonged language contact among different speech communities facilitates the exchange of linguistic features, particularly through the borrowing of vocabulary in both spoken and written domains. As noted by S.K. Chatterji (1890–1977), language Development is deeply influenced by such contact, especially in formal and functional contexts where vocabulary differentiation becomes more pronounced.

Hindi and Urdu are generally regarded as two varieties of the same language, commonly referred to as Hindustani, which share grammatical structure, syntax, and

core vocabulary, yet differ in sociocultural and political associations. In Pilibhit, English constitutes the most influential external language, leading Hindi-Urdu speakers to adopt English words while systematically adapting them to their native phonological system. One prominent phonological process involved in this adaptation is prothesis, a type of epenthesis in which a vowel is inserted at the beginning of a word. This process typically occurs when English loanwords contain initial consonant clusters that are not permissible in Hindi-Urdu phonotactics. Consonant clusters are defined as sequences of two or more consonants occurring in onset, medial, or coda positions (Nordquist, 2017). While English allows a wide range of such clusters, including complex initial sequences (Pearce, 2001), Pilibhit Hindi-Urdu tends to restrict them, resulting in systematic vowel insertion to maintain phonological well-formedness.

Linguistic interference may occur across various levels of language structure, including phonological, morphosyntactic, and semantic domains, particularly under specific social and linguistic conditions (Thomason & Kaufman, 2001). Weinreich (1957) described phonological influence arising from language contact as phonetic interference, emphasizing the role of cross-linguistic interaction in shaping sound patterns. Subsequent studies have shown that phonological processes may vary across linguistic systems depending on structural constraints and sociolinguistic environments (Mose, 2021). Previous research by Suhery et al. (2019) demonstrated that Pilibhit Hindi-Urdu speakers consistently modify English consonant clusters by inserting vowels, regardless of whether the clusters occur at initial, medial, or final positions within a word. Similar findings in other linguistic contexts, such as Suhery et al. (2024), on Acehnese speakers' pronunciation of Indonesian, further highlight the complexity of phonological interference in multilingual and diglossic environments and its implications for language acquisition.

Beyond phonological adaptation, language contact also carries sociolinguistic significance through the concept of indexicality, which originates from Peirce's notion of the index and was further developed by Silverstein (1998). Indexicality refers to how linguistic forms function as social signs that both reflect and construct speakers' identities, social affiliations, and cultural backgrounds. Variations in pronunciation and language use are often consciously perceived by listeners and employed by speakers as markers of regional and socio-cultural identity (Suhery, 2019). In the context of Pilibhit Hindi-Urdu, the phonological adaptation of English loanwords not only reflects structural constraints but also indexes speakers' linguistic identity within a multilingual community.

To account for these phonological and sociolinguistic phenomena, this study adopts Optimality Theory (OT) as its analytical framework. Initially proposed by Prince and Smolensky (2004), OT accounts for phonological patterns by positing the interaction of hierarchically ranked universal constraints, with language-specific rankings determining surface forms. As emphasized by Gussenhoven and Jacobs (2017), Optimality Theory provides a robust framework for analyzing syllable structure and phonological adaptation across languages. Within this framework, competing output candidates are evaluated through interaction with constraints to

determine the optimal form. By applying OT, this study examines how Pilibhit Hindi-Urdu speakers adapt English loanwords through prothesis and how constraint interaction shapes optimal outputs. Ultimately, this research seeks to contribute to a deeper understanding of linguistic interference, indexicality, and phonological adaptation in contact-induced language change.

METHOD

Applying qualitative research is appropriate for this study because it aligns with the fundamental characteristics of qualitative research. In qualitative research, the objective is not to pursue an absolute "truth" but rather to elucidate, portray, and scrutinize evidence (Ansell, J., & Phillips, 1994). Twenty students from Ganna Krishak Samaj Degree College participated in this research. Continuous speech utterances were recorded from one of the respondents (PHU-1) without prior knowledge as he conversed with his colleagues. The primary purpose of collecting English words and continuous speech contexts was to examine the phonological features of the words in different contexts. The data were recorded using a Sony PX-470 digital voice recorder and transcribed into the International Phonetic Alphabet (IPA). Finally, the researchers applied Optimality Theory to identify the basic syllable structure of Pilibhit Hindi-Urdu.

FINDINGS AND DISCUSSION

Analysis of the data collection indicates that certain instances involve linguistic interference between Pilibhit Hindi-Urdu and English. This process involves evaluating the ranking characteristics of output candidates in relation to the input candidate, either within a specific language or across multiple languages, to select the most suitable candidate as the optimal form. The data presented in this study establish a foundation for describing the facts and proposing a theoretical account of phoneme alteration in English loanwords, highlighting how native speakers of Pilibhit Hindi-Urdu exhibit linguistic interference when speaking English words.

A. An Account of 'Prothesis' of S-Cluster in Pilibhit Hindi-Urdu

Prothesis (from Greek pro- 'before' + thesis 'placing') is a kind of epenthesis in which a sound is inserted at the beginning of a word (Campbell, 1998, p. 31). He has standardized the austere phenomena of segment insertion in the covenant condition, which is not a specific frequent term, and such changes are also referred to as word-initial epenthesis. The prepending of a vowel phoneme is called vocalic prothesis, and the prepending of a consonantal phoneme is called a prosthetic consonant in use. For example, in Pilibhit Hindi-Urdu, there is the addition of a front short close unrounded vowel [ɪ] at the beginning of English loanwords such as;

English	Pilibhit Hindi-Urdu loanwords	Gloss
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/sku:l/	[ɪsku:l]	'school'
/sprʌɪt/	[ɪsprʌɪt]	'sprite'
/spɛʃ(ə)l/	[ɪspɛsəl]	'special'
/steɪʃ(ə)n/	[ɪstesən]	'station'
/spi:d/	[ɪspi:t]	'speed'

Applications of Optimality Theory in /sku:l/:

/sku:l/	*COMP-ONS	DEP-IO	*CODA	CONTIGUITY
a. [sku:l]	*!		*	
b.  [ɪs.ku:l]		*!	**	
c. [sɪ.ku:l]		*!	*	*

Table 1: The Representation of consonant cluster between [s] and [k] in Pilibhit Hindi-Urdu loanwords derived from English.

*COMP-ONS >> DEP-IO >> *CODA >> CONTIGUITY

Consideration of Optimality Theory in /sprʌɪt/:

/sprʌɪt/	*COMP-ONS	DEP-IO	*CODA	CONTIGUITY
a. [sprʌɪt]	*!		*	
b.  [ɪs.pɪrʌ.ɪt]		*!	***	*
c. [ɪs.pra.ɪt]	*!	*	**	

Table 2: The Representation of consonant clusters among [s], [p], and [r] in Pilibhit Hindi- Urdu loanwords derived from English.

*COMP-ONS >> DEP-IO >> *CODA >> CONTIGUITY

a) In the analysis of 'prothesis', we have seen how a Pilibhit Hindi-Urdu speaker added a segment in the initial position of English loanwords and broke the consonant cluster between [s] and [k]. In Table 1, the researchers analyzed the English loanword /sku:l/'school' as [ɪs.ku:l], with the addition of the front short close unrounded vowel [ɪ]. It was initially a monosyllabic word, but became a disyllabic word after the addition. Regarding constraint rankings, we observed that candidate 'a' has the highest number of constraint violations, including fatal violations, and shows the least

similarity between the input and output candidates. Candidate 'b' has the fewest constraint violations and the most significant similarity in linguistic features with the input candidate, making it the optimal candidate, as indicated by ☞. Candidate 'c' has more constraint violations and lacks the necessary linguistic features to be considered the optimal candidate. The hierarchy of constraint rankings, from the highest to the lowest, is provided below;

*COMP-ONS >> DEP-IO >> *CODA >> CONTIGUITY

The researchers need to explain the constraints: *COMP-ONS is the highest-ranking constraint, which means there must not be more than one consonant at the onset position of the syllables. DEP-IO is the higher rank of constraint, which means the output candidate must correspond to the input candidate. In general terms, we can say that 'No Epenthesis'. *CODA is the lower rank of constraint, which means prohibition of coda consonant or support to the open syllable, and CONTIGUITY is the lowest rank of constraint, which means prohibition of inserting an extra segment between two or more consonants.

b) In Table 2, we have analyzed another English loanword /sprʌ.ɪt/'sprite' that has a sequence of three consonants at the onset position of a syllable changed into [ɪs.pɪrʌ.ɪt] in Pilibhit Hindi-Urdu. Constraints are the same as in Table 2, and the candidate 'b' is the optimal candidate notated by ☞. In Table 2, we analyzed how a disyllabic word became trisyllabic after the addition of external segments. In the case of a candidate, 'a' has the highest rank of constraints regarding the violation and fatal violation of constraints. Candidate 'b' has the fewest constraint rankings, and most of its linguistic features match the input candidate; therefore, in this context, it is the winner and is declared optimal. The optimal candidate is indicated ☞ in the Table. There is a hierarchy of constraints that begins from the highest to the lowest.

*COMP-ONS >> DEP-IO >> *CODA >> CONTIGUITY

CONCLUSION

The study of linguistic interference in Pilibhit Hindi-Urdu speakers, particularly in adapting English loanwords, reveals a pattern of phonological alteration through a process called "prothesis." This involves inserting a vowel sound at the beginning of words to accommodate the phonotactic constraints of Pilibhit Hindi-Urdu. Specifically, this phenomenon addresses the challenge of pronouncing initial consonant clusters (e.g., /sk/, /spr/). The insertion of a front short close unrounded vowel [ɪ] before English loanwords with initial consonant clusters transforms monosyllabic or disyllabic English words into disyllabic or trisyllabic forms, respectively. For example, "school" becomes [ɪs.kul] and "sprite" becomes [ɪsprɪt].

In the provided tableau analysis, candidates are evaluated based on their adherence to these constraints. The "optimal" candidate, marked by ✱, is the one that best satisfies the hierarchy of constraints. The optimal candidate typically introduces a vowel to break up the initial consonant cluster, aligning with the phonotactic preferences of Pilibhit Hindi-Urdu.

The study highlights how Pilibhit Hindi-Urdu speakers systematically alter English loanwords to fit the phonological structure of their native language. This adaptation process is guided by a set of ranked phonological constraints, with vowel insertion (prothesis) serving as a key strategy for resolving complex consonant clusters. The findings illustrate the dynamic interplay between language-specific phonotactic rules and the influence of a dominant language like English, offering a theoretical account of phoneme alteration in the context of linguistic interference. Language is inherently reflexive, meaning it always refers to itself, regardless of the other topics it addresses. Each utterance demonstrates the possible structures, sounds, and functions of language. Additionally, the relationship between linguistic form and indexical meaning becomes evident through interpersonal interactions. Finally, the study's implications for linguistic theory are that it supports the applicability of Optimality Theory to the analysis of language-specific phonological processes. The ranking of constraints provides a systematic way to predict and explain phonological adaptations, contributing to a broader understanding of how languages manage foreign phonological material.

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CHAPTER 4: Nursing & Public Health

APPLICATION OF PURSED-LIP BREATHING TECHNIQUE IN REDUCING SHORTNESS OF BREATH IN PATIENTS WITH BRONCHIAL ASTHMA

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ABSTRACT

Asthma is characterized by inflammation, increased responsiveness to various stimuli, and airway obstruction that can recur spontaneously or require appropriate treatment. Asthma in Indonesia is included among the leading causes of morbidity and mortality. Non-pharmacological measures given to patients with asthma include breathing exercises. One breathing exercise that can be done is pursed-lip breathing. Pursed-lip breathing is a breathing exercise designed to reduce airway obstruction in patients with asthma. Objective: To describe nursing care using the pursed-lip breathing technique to reduce shortness of breath in patients with bronchial asthma. Method: This study used a descriptive method, with data collection through interviews, observations, and documentation. The study participants consisted of three patients with a history of bronchial asthma. The patients' breathing patterns were measured using respiratory rate and oxygen saturation. Results: After applying the pursed-lip breathing technique for three days, the respiratory rate and oxygen saturation of the three respondents improved from an average respiratory rate of 30 breaths per minute to 20 breaths per minute and an average oxygen saturation of 95% to 98%. Conclusion: The final evaluation conducted over three days showed that shortness of breath had decreased, *respiratory rate (RR)* had decreased, *oxygen saturation (SpO₂)* had increased, and the client was able to regulate their breathing pattern.

Keywords: *Bronchial asthma, shortness of breath, pursed-lip breathing*

INTRODUCTION

Bronchial asthma is a disease of the airways characterized by chronic inflammation. Shortness of breath, wheezing, coughing, and chest tightness are symptoms that indicate the presence of asthma. Symptoms vary over time with variable airflow limitation (Reddel et al., 2021).

Bronchial asthma is characterized by inflammation, increased responsiveness to various stimuli, and airway obstruction that may recur spontaneously or require appropriate treatment. The effects of asthma include reduced quality of life, decreased productivity, absenteeism, increased healthcare costs, risk of hospitalization, and even death. Asthma in Indonesia is listed among the leading causes of morbidity and mortality (Kurniati & Lidya Leni, 2021). According to the Global Initiative for Asthma (GINA, 2020), there are approximately 300 million individuals with asthma worldwide, with this number projected to rise to 400 million by 2024. Based on 2020 data from the Indonesian

Ministry of Health, approximately 4.5% of the population suffers from asthma, indicating that asthma is a common disease in Indonesia. The Aceh Health Department reported in 2022 that there were 11,534 asthma cases, with children being the most affected group.

The Basic Health Research (Riskesdas) reported that the prevalence of asthma in Aceh Province was 2.27%, with the highest prevalence in Banda Aceh City was 3.17%, while the lowest prevalence is in Gayo Lues District was 0.55% (Windiani et al., 2022). Data from the Department of Health (DINKES) of Lhokseumawe City in 2024 indicates that the number of bronchial asthma patients in Lhokseumawe City is 2,349. According to data from the TK III IM 07.01 Lhokseumawe Hospital, the number of bronchial asthma patients is 113. Pursed-lip breathing is a breathing exercise that helps reduce breathing obstruction in asthma patients (Sulistiyawati & Cahyati, 2019). Pursed-lip breathing aims to give the bronchi time to dilate, thereby reducing shortness of breath. Cardiorespiratory cases can be managed with breathing exercises. A combination of breathing exercises, including pursed-lip breathing, is recommended. Pursed-lip breathing is estimated to reduce shortness of breath, enabling patients to engage in activities optimally (Pahlawi et al., 2019). Airway obstruction can be reduced through pursed-lip breathing.

According to Ning Pangesti and Suharti (2021), their research showed that pursed-lip breathing exercises can improve breathing conditions, thereby increasing the amount of oxygen transferred to the pulmonary capillaries. This condition will affect the increase in SPO2 levels in the blood. Furthermore, another study conducted by Nurul Kartika Sari, et al. (2018) used a randomized controlled method with a sample of 36 respondents in the study "The Effectiveness of Lip-Sealing Breathing to Reduce Tension and Increase Oxygen Saturation in Patients with Chronic Obstructive Pulmonary Disease (COPD)". explained that the pursed-lip breathing technique, when performed regularly, can increase alveolar ventilation, resulting in the binding of O2 with hemoglobin with adequate perfusion, as indicated by an increase in oxygen saturation.

According to Suwaryo, et al. (2021), the clinical picture of asthma patients is generally that they often complain of sudden shortness of breath. Asthma patients will lack oxygen in their lungs if these complaints are not treated immediately. Asthma patients who lack oxygen will die. Asthma patients have several health problems that often arise. Ineffective airway clearance and ineffective breathing patterns are nursing problems that often arise.

According to Utama (2018), nursing diagnoses that may arise in asthma patients include ineffective breathing patterns, ineffective airway clearance, and nutritional imbalance below the body's needs. Ineffective airway clearance occurs due to excessive mucus secretion. Ineffective breathing patterns occur due to decreased lung expansion. Meanwhile, nutritional imbalance below requirements occurs due to anorexia. Management of asthma involves pharmacological and non-pharmacological approaches. Pharmacological interventions include bronchodilator medications. Non-pharmacological measures involve breathing exercises. One breathing exercise that can be performed is pursed-lip breathing (Empowerment, 2021).

pursed-lip breathing exercises also show differences and have an effect before and after being given to patients with respiratory disorders. The average value after applying the pursed-lip breathing technique was lower than before, as determined by a t-test. This

means that there was a difference between before and after the pursed-lip breathing technique was applied, indicating that the pursed-lip breathing technique had an effect on the breathing patterns of patients with respiratory disorders (Mukaram et al., 2022).

According to previous research by Dwi Noviatun (2024), the results showed an increase in oxygen saturation. Before the treatment, Mr. S's oxygen saturation was 94%, and after 4 days of treatment, his oxygen saturation increased to 98%. The conclusion is that there was an increase in oxygen saturation before and after the pursed-lip breathing therapy.

METHOD

The research employed a descriptive method with data collection techniques that included interviews, observations, and documentation. The participants consisted of three patients diagnosed with bronchial asthma. The patients' breathing patterns were assessed using respiratory rate measurements and oxygen saturation levels to evaluate their respiratory function accurately.

The instruments used in this descriptive case study included several components. First, a nursing assessment format approved by the Akper Kesdam IM Lhokseumawe academy was utilized as the primary tool for data collection and documentation of patient care. Second, a stopwatch was employed to measure respiratory rates during observations. Lastly, the Standard Operating Procedure (SOP) for pursed-lip breathing was applied to ensure proper implementation of the intervention and maintain consistency throughout the study. These instruments ensured a structured and reliable process in collecting and analyzing patient data.

RESULTS AND DISCUSSION

Assessment of Patients with Bronchial Asthma

This descriptive case study implemented the pursed-lip breathing technique to patients with bronchial asthma. The study involved three patients who were treated at a military hospital in Lhokseumawe. During the initial assessment, common symptoms observed included dry cough, wheezing (a whistling sound during breathing), and a sensation of heaviness or tightness in the chest. Each patient presented with different main complaints. For instance, Mr. U reported experiencing shortness of breath when exposed to cold conditions. Mr. S complained of chest tightness and difficulty breathing during physical exertion, along with episodes of coughing. Meanwhile, Mrs. K stated that her shortness of breath and difficulty breathing occurred when exposed to dust. These differences indicate varying triggers for bronchial asthma among the subjects. Additionally, none of the three patients reported a family history of asthma, suggesting that hereditary factors were not a primary contributor in these cases.

Physical Examination and Clinical Findings

The physical examinations of all three patients, Mr. U, Mr. S, and Mrs. K revealed similar results: dyspnea and the presence of wheezing upon auscultation. Wheezing is a characteristic sound caused by turbulent airflow through narrowed airways. As

obstruction becomes more severe, wheezing becomes more prominent and is often audible during the early phase of expiration. Persistent or recurrent wheezing is commonly associated with focal airway obstruction, such as narrowing of the bronchi due to foreign body aspiration or neoplasms (Qurratul et al., 2024).

In terms of pharmacological management, all three patients received the same prescribed therapy from the hospital, which included N-Acetylcysteine at a dose of 200 mg, administered three times daily. This treatment aligns with appropriate pharmacological interventions for respiratory conditions characterized by excessive mucus secretion, such as acute or chronic bronchitis, pulmonary emphysema, mucoviscidosis, and bronchiectasis. N-Acetylcysteine works as a mucolytic agent, helping to break down mucus and facilitate easier breathing, which may support airway clearance.

Nursing Diagnosis

The nursing diagnosis for patients with bronchial asthma in this descriptive case study was based on theoretical guidelines and focused on the issue of an ineffective breathing pattern. According to the Indonesian Nursing Diagnosis Standards (SDKI, 2017), an ineffective breathing pattern is defined as inspiration and/or expiration that does not provide adequate ventilation. For all three patients Mr. U, Mr. S, and Mrs. K the same nursing diagnosis was identified: ineffective breathing pattern related to impaired respiratory effort due to difficulty breathing caused by bronchial asthma. This diagnosis aligns with common clinical symptoms and was confirmed based on the Indonesian nursing diagnosis framework.

Nursing Interventions

Nursing interventions are a series of actions designed to address identified problems, prioritize care, formulate goals, plan actions, and evaluate the nursing care provided to patients based on data analysis and nursing diagnoses (Suwingjo et al., 2022). These interventions aimed to prevent complications and improve the patients' overall condition. For this study, the planned interventions were implemented for all three patients, Mr. U, Mr. S, and Mrs. K to compare the effects of these interventions in each case. The primary goal was to improve respiratory function and alleviate symptoms associated with ineffective breathing patterns.

Nursing Implementation

Nursing implementation refers to the series of actions carried out by nurses to assist patients in resolving their health problems and achieving the desired outcomes (Suwignjo et al., 2022). In this study, nursing care was implemented over three consecutive days, from April 28 to April 30, 2025, for all three patients, at different times. The primary nursing interventions provided to the patients included monitoring vital signs, monitoring breathing patterns, assessing lung sounds, administering oxygen therapy, and teaching pursed-lip breathing for 5–10 minutes per session.

The results of implementing pursed-lip breathing therapy for Mr. U, Mr. S, and Mrs. K showed that all three patients reported feeling more relieved. Objective data also

supported these findings, as there was a noticeable reduction in shortness of breath in each case. This demonstrates that the intervention had a positive impact on improving breathing efficiency and reducing respiratory distress.

Nursing Evaluation

Nursing evaluation involves a systematic and planned comparison of the patient's health status against the predetermined goals, carried out continuously in collaboration with patients and other healthcare professionals (Krismonita, 2021). In this case study, evaluations were conducted using the SOAP method, focusing on both formative evaluation (immediate response after intervention) and summative evaluation (progress over time).

The expected outcomes were successfully achieved for all three patients. For Mr. U, the respiratory rate decreased from 30 breaths per minute to 20 breaths per minute. For Mr. S, the rate decreased from 28 breaths per minute to 20 breaths per minute, and for Mrs. K, it also decreased from 28 breaths per minute to 20 breaths per minute. These results indicate that all three patients achieved a respiratory rate within the normal range of 20 breaths per minute (Shenda et al., 2022). This demonstrates that the combination of pharmacological treatment and pursed-lip breathing therapy effectively improved respiratory function in patients with bronchial asthma.

CONCLUSION

Based on this case study involving the application of pursed-lip breathing techniques for Mr. U, Mr. S, and Mrs. K, who experienced breathing difficulties due to bronchial asthma, the following conclusions can be drawn: The nursing assessment conducted on these three clients revealed similar complaints of shortness of breath. Mr. U reported difficulty breathing in cold conditions, with a respiratory rate of 30 breaths per minute. Mr. S complained of shortness of breath after engaging in excessive physical activity, with a respiratory rate of 28 breaths per minute. Meanwhile, Mrs. K experienced breathing difficulties when exposed to dust, also with a respiratory rate of 28 breaths per minute. Based on these assessments, the primary nursing diagnosis identified in all three cases was ineffective breathing pattern related to impaired respiratory effort. This diagnosis was determined according to the defining characteristics and symptoms presented by each patient. The nursing interventions for Mr. U, Mr. S, and Mrs. K were planned based on the established diagnosis and referred to the SLKI and SIKI guidelines. These interventions were specifically designed to address the main focus of this scientific paper, which is the application of pursed-lip breathing techniques to reduce dyspnea in patients with bronchial asthma. Nearly all planned interventions were implemented for the three clients, including airway management and the application of pursed-lip breathing for three consecutive days. The evaluation revealed a significant decrease in respiratory rate for all clients: Mr. U's respiratory rate decreased from 30 to 20 breaths per minute, Mr. S's from 28 to 20 breaths per minute, and Mrs. K's from 28 to 20 breaths per minute. These results indicate that all three clients achieved a normal respiratory rate of 20 breaths per minute (Shenda et al., 2022).

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CORRELATION BETWEEN SOCIAL MEDIA USE AND MENTAL WELL-BEING AMONG NURSING STUDENTS AT THE NURSING MILITARY ACADEMY ISKANDAR MUDA LHOKSEUMAWE

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ABSTRACT

Stress and mental health issues are becoming increasingly prominent among students, particularly those pursuing nursing education with heavy academic workloads and demanding clinical practice requirements. The rising trend of social media use among students has raised concerns about its impact on mental well-being. This study aims to examine the relationship between the duration of social media use and the level of mental well-being among nursing students at the Nursing Military Academy IM Lhokseumawe. The study employed a quantitative approach using a correlational design, involving 50 respondents selected through purposive sampling. Data were collected via a questionnaire on social media usage duration and the Warwick-Edinburgh Mental Wellbeing Scale (WEMWBS) instrument. Pearson correlation analysis revealed a significant negative relationship between the two variables. The conclusion indicates that the more time students spend on social media, the lower their level of mental well-being.

Keywords: *Social media, mental well-being, nursing students*

INTRODUCTION

The mental well-being of nursing students is a key focus in higher education due to the combination of intense academic demands and clinical practice responsibilities. Globally, the World Health Organization (2024) reports that approximately one in seven adolescents aged 10–19 experiences a mental disorder, with depression and anxiety being the leading contributors to the global burden of disease in this age group. These data underscore the need to pay greater attention to the mental health of adolescents and young adults, including students.

In Indonesia, national health profiles and RISKESDAS analysed a high prevalence of mental health issues among young people. Several studies based on RISKESDAS (Basic Health Research, 2018), analysed by Hapsari et al. (2022), have identified the risk factors for depression among adolescents and young adults, including academic pressure, lack of social support, and excessive use of digital media. Aceh Provincial Health Data (2022) reports an increasing trend in mental health disorders over the past few years. Administrative data from the City of Lhokseumawe, released by the Health Department, indicate mental health indicators that require special attention, including a rise in the number of visits to mental health services at primary and referral healthcare facilities. The Indonesian Health Profile (Ministry of Health, 2023) further emphasizes the increasing need for mental health services, particularly among student populations

Social media has become an inseparable part of students' daily lives, serving not only as a communication tool but also for learning, entertainment, and self-expression. However, excessive use of social media can pose risks to mental well-being. Previous studies have consistently shown a negative association between excessive social media use and students' mental health. Amelia and Purnomo (2023) found that high exposure to social media is correlated with increased academic anxiety among nursing students. Handayani and Widodo (2024) reported that poor sleep quality among health students is closely associated with the duration of social media use. Putri et al. (2023) identified fear of missing out (FOMO) and social comparison as mediating factors. Similarly, Alam et al. (2022) found that social media use exceeding four hours per day increases the risk of depressive symptoms by 1.8 times among health students in Malaysia.

These phenomena can be better understood through theoretical perspectives. Social Comparison Theory explains how individuals often feel less satisfied with themselves after being exposed to idealized portrayals of others on social media, while Uses and Gratifications Theory suggests that individuals engage with media to satisfy needs such as information, social interaction, and entertainment. However, excessive and unregulated media use may lead to distraction, reduced productivity, and negative psychological outcomes. Consequently, the literature emphasizes the importance of digital literacy programs, time management strategies for social media use, and mental health promotion initiatives tailored specifically for nursing students.

In summary, existing studies consistently demonstrate that excessive social media use is closely associated with lower levels of mental well-being among students. Therefore, this study aims to analyze the relationship between the intensity of social media use and mental well-being among nursing students at the Iskandar Muda Military Academy, Lhokseumawe, so that it can serve as a basis for developing digital literacy programs and mental health promotion that are appropriate to the local context.

METHOD

The research instruments included a demographic questionnaire, a social media usage duration questionnaire, and the Indonesian version of the Warwick–Edinburgh Mental Well-being Scale (WEMWBS), which was used to assess respondents' levels of mental well-being. Data were processed using IBM SPSS software to perform Pearson correlation analysis. The results of the analysis are presented in Table 1, which shows a significant negative correlation between the duration of social media use and mental well-being scores.

Based on the correlation analysis presented in Table 1, the average duration of social media use among respondents was 4.2 ± 1.5 hours per day, while the mean mental well-being score (WEMWBS) was 45.7 ± 5.8 . The Pearson correlation analysis revealed a significant negative relationship between the duration of social media use and mental well-being scores ($r = -0.587$; $p = 0.002$). These findings indicate that longer durations of social media use are associated with lower levels of mental well-being among respondents.

RESULTS AND DISCUSSION

The correlation coefficient of -0.587 with a p-value of 0.002 indicates a significant negative relationship between the duration of social media use and mental well-being levels. This finding suggests that higher social media usage is associated with lower mental well-being scores among students. This result is consistent with previous studies that have reported similar patterns. Possible mechanisms underlying this relationship include excessive social comparison, fear of missing out (FOMO), and exposure to negative content, which may trigger anxiety and reduce life satisfaction. Therefore, mental health nursing interventions that emphasize digital literacy and effective management of social media use should be strengthened among students.

CONCLUSION

This study found a significant negative correlation between the duration of social media use and mental well-being levels among students at the Nursing Military Academy of Iskandar Muda Lhokseumawe. These findings highlight the importance of implementing mental health education programs and campaigns that promote responsible social media use to support students' well-being.

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LITERATURE REVIEW

Islamic Nursing Services in Indonesia

Islamic nursing services in Indonesia are characterized by the integration of Islamic principles into nursing practice, aiming to provide holistic care that encompasses physical, spiritual, and emotional well-being. These principles guide nurses in delivering care that aligns with ethical, moral, and religious values upheld by patients and healthcare institutions. Several studies have examined the implementation and impact of Islamic nursing services across various healthcare settings in Indonesia.

Core Concepts of Islamic Nursing

- a. **Nursing as Worship (Ibadah):** In Islam, nursing is viewed as an act of worship, where caring for patients is considered a form of devotion and service to Allah. This perspective transforms nursing into a spiritually meaningful activity, aligning healthcare practices with religious values.
- b. **Holistic Approach:** Islamic nursing emphasizes a comprehensive approach to patient care, addressing not only physical health but also psychological, social, cultural, and spiritual aspects. This holistic model ensures that all dimensions of a patient's well-being are considered in the care process.
- c. **Islamic Values in Nursing:** Core Islamic values such as compassion, empathy, justice, honesty, patience, and sincerity are integral to nursing practice. These values guide nurses in providing ethical and compassionate care, fostering trust and positive relationships with patients.
- d. **Islamic Nursing Ethics:** Nurses are expected to uphold Islamic ethical standards, including respecting patient modesty, ensuring that care is halal (permissible) and thayyib (good), and honoring the religious beliefs of patients. These ethical guidelines ensure that nursing practices are aligned with Islamic teachings.

Implementation of Islamic Nursing Services

- a. **Spiritual Care:** Meeting the spiritual needs of patients is a fundamental component of Islamic nursing. This includes providing spiritual support, guiding patients in religious practices, and offering counseling to address spiritual concerns.
- b. **Facilitating Religious Practices:** Nurses support patients in performing religious duties such as prayer (salat), reciting the Quran, and making supplications (du'a), in accordance with their abilities. This practice reinforces the spiritual well-being of patients.
- c. **Islamic Hospital Environment:** Creating an environment that reflects Islamic values enhances patient experience. This includes providing prayer facilities, offering halal food, and maintaining a peaceful atmosphere conducive to religious practices.
- d. **Islamic Communication:** Communication in Islamic nursing emphasizes politeness, respect, and adherence to Islamic values in interactions with patients and families. This approach fosters trust, reduces anxiety, and strengthens the therapeutic relationship.

METHOD

This study was written using the literature analysis method, which utilizes library sources as theoretical references, thereby limiting the analysis to activities and library collections as well as other factual references. This study is based on scientific references and is written systematically in accordance with the guidelines for writing scientific articles or journals. The results of the writing are then discussed with the audience in a lecture discussion room, and improvements are made to this scientific paper in accordance with the results of the discussion and overall corrections.

RESULT AND DISCUSSION

Health Services from an Islamic Perspective

Health services constitute specialized activities performed by highly professional individuals, encompassing both routine and incidental forms of care. These two categories reflect the complexity and scope of healthcare practices in addressing diverse patient needs. Accordingly, health services may be conceptually examined from epistemological and etymological perspectives to provide a deeper theoretical understanding.

Epistemologically, health services may be understood as an integral part of *muamalah* worship in Islam. This perspective emphasizes that every Muslim is obliged to perform duties and responsibilities in accordance with the human role as servants and vicegerents of Allah SWT. Consequently, the professional functions carried out by medical personnel in providing healthcare services are regarded not only as occupational duties but also as expressions of worship and moral conduct.

Worship and moral conduct embodied in the provision of health services by medical personnel to hospitalized patients represent concrete actions that can be directly experienced by both service providers and recipients. This reciprocal interaction reflects the ethical and spiritual dimensions inherent in healthcare practices within an Islamic framework. Such a perspective is normatively supported by the Qur'an, particularly Surah Al-An'am, verse 160:

نَوْمِلْظِي لَا مَهْوِ اَدْلِيْم لَإِلَٰهَ اِنْزِجِي لَآءَ قَتْسِلَابِ اءَاج نَمُو اءَلَاثْمُ اُرْشَعْ هَلِكْ قَتْسِلَابِ اءَاج نَم

"Whoever comes with a good deed will receive ten times its reward, and whoever comes with an evil deed will not be recompensed except with its equivalent; and they will not be wronged" (Qur'an, Al-An'am: 160).

The explanation contained in the above verse can be understood that the best service will result in rewards for someone who does good, so that epistemologically, health services in the Islamic perspective have a clear and measurable interval, so that the values that will be obtained from work that helps others are clear. In this case, serving patients who are undergoing inpatient or outpatient treatment at hospitals by medical personnel. The explanation of the verse also shows that every good deed will be judged as good and will receive a good reward.

The moral dimension of healthcare services provided by medical personnel is reflected in persuasive communication as well as in ethical behavior when addressing the needs of hospitalized patients. Morality constitutes a fundamental aspect of service delivery, particularly within hospital settings that provide specialized inpatient care.

Through moral conduct and effective communication, healthcare professionals are able to foster trust, comfort, and a sense of dignity among patients.

Conceptual healthcare from an Islamic perspective may be understood through Qur'anic guidance, which serves as a fundamental normative framework for serving fellow human beings based on principles of equality in attitude and behavior. This framework emphasizes justice, benevolence, and ethical conduct as core values in the provision of healthcare services. Such principles are explicitly articulated in the Qur'an, particularly in Surah An-Nahl, verse 90:

نور كذذ مكالع مكالع يغبلاو ركزمل او عاشحلذا ن ع ىهزو ىبرزلا يذءانباو ناسحل ُواو لدعلا رماي الله ن

“Indeed, Allah commands justice, excellence, and giving to relatives, and forbids immorality, wrongdoing, and transgression. He admonishes you that perhaps you will be reminded” (Qur'an, An-Nahl: 90).

The explanation of the aforementioned verse concerning the moral dimension of medical services for both inpatients and outpatients in hospital settings emphasizes the principle of fairness. Fairness serves as a fundamental moral standard that must be upheld by nursing personnel in the delivery of healthcare services. Through the application of fairness, nurses are expected to provide equitable and non-discriminatory care to all patients, regardless of whether they receive outpatient or inpatient treatment.

The relationship between the ethical dimension and service standards within an Islamic perspective indicates that healthcare services delivered by nursing staff constitute concrete professional actions grounded in moral values. These ethical actions are manifested through attitudes and behaviors that prioritize compassion, respect, and responsibility in patient care. Such an approach functions as an essential measure to ensure patient comfort and to support the overall treatment and healing process.

Implementation of Health Services from an Islamic Perspective

Implementation possesses a broad meaning when examined from both epistemological and etymological perspectives. Its interpretation is closely connected to the relationship between text and context, which shapes how implementation is understood and applied in practice. Consequently, the interaction between etymological and epistemological interpretations of implementation may give rise to gaps and disparities between conceptual understanding and perceived meaning.

Epistemologically, implementation may be understood through scientific knowledge, particularly within the framework of scientific management, as well as through abstract experiential interpretations. Conventional epistemological studies of implementation predominantly refer to management theories as their primary analytical foundation. Consequently, epistemological discussions of implementation from an Islamic perspective remain relatively limited and underexplored in existing scholarly literature.

From an Islamic epistemological perspective, implementation is understood as a concrete manifestation of knowledge derived from reflection on the creation of the universe. This understanding is rooted in the first revelation, *“Iqra”* (Read), in Surah Al-‘Alaq, which signifies that the acquisition of knowledge through reading and reflection forms the foundation of action. Accordingly, implementation in the Islamic perspective

may be conceptualized as the application of knowledge in the form of actions whose benefits are directly experienced by oneself and others, thereby constituting an act of worship that reflects trustworthiness (*amanah*) in the fulfillment of professional duties, particularly in nursing practice.

The Islamic perspective on the epistemology of implementation is further reinforced by Qur'anic guidance contained in the final revelation in Surah Al-Ma'idah. This verse affirms the completeness and perfection of Islamic teachings as a comprehensive framework that guides human actions, including professional and social responsibilities. Consequently, implementation within an Islamic epistemological framework is understood as the application of knowledge and values that align with divine guidance, as stated in Surah Al-Ma'idah verse 3: *"Today I have perfected your religion for you, completed My favor upon you, and have approved Islam as your religion"* (The Qur'an and Its Translation, Indonesian Ministry of Religious Affairs, 2019).

Epistemological implementation from an Islamic perspective clearly demonstrates that the application of knowledge encompasses dimensions of worship and *mu'amalah* (social relations). Such implementation may be understood as a form of charitable action (*amal*) manifested in the conduct of daily life, both at the personal and professional levels. In this context, professional responsibilities carried out by medical personnel are not merely technical duties but also expressions of ethical commitment and religious devotion.

The Qur'anic guidance regarding the implementation of professional duties, whether carried out individually, collectively, or institutionally, is articulated in Surah Ali 'Imran verse 104. This verse emphasizes the importance of organized and purposeful action in promoting virtue, upholding what is right, and preventing wrongdoing within social and professional contexts. Such guidance provides a normative foundation for understanding implementation as a collective moral responsibility grounded in Islamic values:

١٠٤ ان ارمع ل (نوح انملا مه كئلواو ركنملا نع نوهميو فورع ماب نورم ابو ريخا اىلا ن وعدي قمأ مكنم نكئلواو)

"And let there be among you a group that calls to goodness, enjoins what is right, and forbids what is wrong; and it is they who are the successful" (Qur'an, Ali 'Imran: 104).

The verse that constitutes the conceptual foundation of the Islamic perspective on the epistemology of implementation demonstrates that implementation may be theoretically articulated through textual understanding as well as realized within abstract and contextual dimensions. This perspective emphasizes that implementation is not limited to normative prescriptions but also encompasses phenomenological and practical considerations. At its core, implementation in the Islamic framework involves the enactment of actions that promote goodness while preventing negative and prohibited behaviors.

Etymologically, implementation may be understood as an action or process derived from the term *"implementation,"* referring to the execution of planned activities. Within an Islamic perspective, implementation is closely linked to the presence of textual guidance that directs the performance of tasks toward specific objectives and predetermined standards. Accordingly, etymological interpretations of implementation in

the Islamic framework commonly serve as a foundational basis for the formulation of standard operating procedures (SOPs) in both public and private institutions.

The implementation of nursing services for patients from an Islamic perspective, viewed from both epistemological and etymological dimensions, embodies both vertical and horizontal relationships. These dimensions require that the implementation of healthcare services fulfill the elements of scientific knowledge (*scientific*) and practical action (*actuating*) as an integrated framework. The horizontal relationship in Islamic nursing services is closely associated with human interactions, commonly referred to as *hablumminannas*, which are grounded in fundamental human needs and the dynamics of supply and demand in healthcare provision. Furthermore, the causality inherent in nursing services represents a necessary and observable phenomenon, as implementation involves an interconnected network of knowledge, attitudes, and behaviors that may produce direct outcomes as well as incidental gaps arising from dynamic situational factors.

The basic needs referred to in this context encompass fundamental human requirements related to both physical and mental health. This dimension reflects a horizontal relationship between patients and medical personnel, within which interaction gaps may occasionally arise and result in suboptimal service delivery. Such gaps cannot be attributed to a single party; rather, from epistemological and etymological perspectives, they may be identified through systematic analysis that enables the formulation of constructive and mutually beneficial solutions (win-win solutions).

The vertical relationship in the implementation of nursing services from an Islamic perspective is grounded in the fulfillment of a divine mandate as part of *mahdhah* worship. This dimension emphasizes the belief that nursing personnel must exert their utmost effort in caring for, serving, and safeguarding patients undergoing treatment in hospitals. The vertical orientation of nursing practice is further reinforced by a Hadith narrated by Imam al-Ṭabrānī:

يُزَابَطْلًا هَاوِر (سَازِلَلَا مَهْخَنَّا سَازِلَا رِيخ)

“*The best among people are those who are most beneficial to others*” (HR. al-Ṭabrānī).

The explanation of the aforementioned hadith indicates that the implementation of nursing services for patients in hospital settings bears significant relevance to *mahdhah* worship. This perspective is grounded in the belief that providing maximum benefit to others constitutes a core religious value in Islam. Accordingly, nursing services delivered to patients are not merely professional obligations but also represent a form of worship manifested through service, care, and dedication to human well-being.

The implementation of nursing services for patients at Sakinah General Hospital in Lhokseumawe, viewed from an Islamic perspective, demonstrates distinctive characteristics that differentiate it from conventional service models. Patient care is grounded in religious beliefs that transcend empirical measurement, thereby fostering a high level of dedication in service delivery. Such dedication is accompanied by sincerity that cannot be quantified statistically or numerically, yet it significantly shapes the quality and meaning of nursing practice.

CONCLUSION

Based on a systematic series of background information, theoretical concepts, and abstractions, it can be concluded that nursing services for patients at Sakinah General Hospital in Lhokseumawe are firmly rooted in Islamic principles. Firstly, the provision of nursing care is guided by Sharia concepts, as the implementation of these services is regarded as an act of worship and a form of charitable service for nursing staff. This perspective is supported by references from the Quran, Hadith, and the maqashid al-Sharia framework, which collectively justify the integration of spiritual values into professional nursing practice.

Secondly, nursing care from an Islamic perspective at Sakinah General Hospital is grounded in a conceptual, theoretical, and abstract framework that shapes professional attitudes and behaviors. This framework provides a broad, systematic approach that enables nurses to deliver comprehensive and prudential care, addressing not only the physical needs of patients but also their spiritual and psychosocial well-being. The integration of these principles ensures that nursing services are holistic, ethical, and aligned with the values of Islam, thereby enhancing the quality and integrity of patient care.

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EDUCATION ON THE 3J DIET FOR CONTROLLING BLOOD GLUCOSE LEVELS IN TYPE 2 DIABETES PATIENTS IN LHOKSEUMAWE CITY

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ABSTRACT

Type 2 diabetes mellitus is a chronic metabolic disease characterized by *hyperglycemia* due to impaired insulin secretion and resistance. Dietary management is one of the main pillars of non-pharmacological therapy. The 3J dietary approach (Schedule, Quantity, and Type of food) has proven effective in helping to control blood glucose levels. This study aimed to determine whether the provision of 3J Dietary Management Education could control blood glucose levels in type 2 diabetes mellitus patients in the Lhokseumawe City Health Center service area. This study used a descriptive research type and a case study design. Ten individuals with type 2 diabetes mellitus registered at the Lhokseumawe City Health Center were enrolled as participants. The instruments used included PowerPoint presentations, leaflets, a glucometer, and observation forms. The study was conducted over two weeks with three sessions, including face-to-face educational sessions lasting 20–30 minutes, blood glucose monitoring, and dietary counseling based on the 3J diet principles. The evaluation showed that through an educational approach and intensive monitoring, patient compliance and the effectiveness of blood glucose control could be improved. The results showed a decrease in blood glucose levels among the respondents. It is hoped that the 3J diet education will become an integral part of nursing strategies in preventing complications and improving the quality of life for patients with type 2 diabetes mellitus.

Keywords: *Type 2 Diabetes, 3J Diet, Blood Glucose Levels, Education Health*

INTRODUCTION

Diabetes is often referred to as the “Mother of Disease” because it is the root cause of many other diseases, such as *hypertension, heart disease, stroke, kidney failure*, and even *blindness*. As we age, bodily physiological functions decline due to degenerative processes, and diabetes mellitus becomes a chronic metabolic disease known as the “Silent Killer”. Many patients are unaware that they have diabetes, often leading to delayed treatment and increased risk of complications (Prihariswi & Kurniawati, 2021).

The International Diabetes Federation reports that in 2021, 537 million adults (aged 20–79) lived with diabetes, and this number is projected to rise to 643 million by 2030 and 783 million by 2045. The majority of people with diabetes reside in low- and middle-income countries. Diabetes caused 6.7 million deaths in 2021 and resulted in healthcare expenditures of USD 966 billion, an increase of 316% over the past 15 years. Additionally, 541 million adults have Impaired Glucose Tolerance, increasing the risk of type 2 diabetes (IDF Diabetes Atlas, 2021).

By 2024, it is estimated that Indonesia will have over 20 million people living with diabetes mellitus. The prevalence of diabetes in the country continues to rise, making it one of the five countries with the highest number of cases worldwide. The Ministry of

Health reports that Indonesia ranks fifth globally with 19.5 million people living with diabetes mellitus (Kemenkes RI, 2024).

The Aceh Provincial Health Department reported that there are 154,889 cases of diabetes mellitus (DM) in the region. South Aceh is the region with the highest number of cases, at 21,514, followed by Aceh Besar (17,277) and Aceh Tamiang (16,781). Other districts/cities with reported cases include Banda Aceh (15,404), Pidie Jaya (11,869), and Bireuen (10,792). Other regions include Lhokseumawe (10,073), Pidie (8,030), Aceh Barat (7,143), and Simeulue (4,916). In smaller areas, cases were recorded in Central Aceh (4,446), Langsa (4,416), and North Aceh with 227 cases (Fithra, 2023).

According to data from the Health Department of Lhokseumawe City (2024), the number of diabetes mellitus patients in Lhokseumawe City reached 10,071 individuals, with a prevalence of type 1 diabetes mellitus at 350 people, with 169 males and 181 females, type 2 diabetes mellitus at 9,563 people, with 3,763 males and 5,800 females, and gestational diabetes mellitus at 158 people.

The primary goal of diabetes mellitus therapy is to reduce the risk of complications, both in the short and long term. While medication therapy has benefits in reducing the risk of complications, it is insufficient to fully address the issue. Non-pharmacological management, which includes education, medical nutrition, and physical exercise programs, is also an effort to reduce the risk of diabetes mellitus complications (Widiasari et al., 2021).

Inappropriate eating patterns can cause blood sugar levels in type 2 diabetes patients to become uncontrolled. To maintain stable blood sugar levels, it is important to improve eating patterns. The dietary management model applied is the (3J) model, which includes three elements: meal schedule, meal quantity, and food type (Falah et al., 2022).

An effective diet for diabetes patients involves not only reducing food intake but also careful planning regarding food types, meal times, and appropriate portion sizes. Factors such as blood sugar levels and conditions like exercise or illness influence dietary adjustments. Menus for people with diabetes should be healthy, varied, and appealing, with limited consumption of foods high in sugar, salt, and fat. Vegetables, fruits, and fiber-rich foods should be the primary choices (Tandra, 2022, in NISA, 2023).

METHOD

This study employs a descriptive approach with a case study design, aiming to assess the effectiveness of providing 3J diet management education in controlling blood glucose levels among type 2 diabetes mellitus patients in the Lhokseumawe City Health Center service area. This design was chosen because it allows researchers to observe phenomena in depth within a real-world context among a specific group of subjects. The population in this study consists of all type 2 diabetes mellitus patients registered in the Lhokseumawe City Health Center service area. Sampling was conducted using accidental sampling, meaning that participants were selected based on who met the inclusion criteria and were available at the time of the study. A total of ten respondents were involved in this study.

The research instruments used included PowerPoint and Leaflets as the main interventions, a glucometer to measure blood glucose levels before and after the intervention, and an observation sheet as a tool for documenting measurement results and respondent activities during the intervention. The research procedure began with initial blood glucose level measurements, followed by education, and concluded with post-intervention blood glucose measurements. The data obtained were then analyzed descriptively to observe changes in blood glucose levels before and after the intervention, enabling conclusions to be drawn regarding the effectiveness of the 3J diet management education program in controlling blood glucose levels in patients with type 2 diabetes mellitus.

RESULTS

The results of this study indicate that the provision of 3J diet management education to control blood glucose levels in patients with type 2 diabetes mellitus is significantly effective in lowering blood glucose levels in type 2 diabetes mellitus. The study involved ten respondents, consisting of seven women and three men, aged between 41 and 53 years. Most of the participants had been living with diabetes for 5 to 10 years, indicating a chronic condition that had been ongoing for a long time and had the potential to cause complications such as *heart attacks, strokes, kidney failure, blindness, and amputations*. Participation in the program lasted two weeks with three sessions, including face-to-face educational sessions lasting 20–30 minutes, blood glucose monitoring, and dietary counseling based on the 3J diet principles.

Observation results showed that all participants experienced a decrease in blood glucose levels after participating in the intervention program, with an average decrease of 40–100 mg/dL. Blood glucose level measurements before and after the educational intervention demonstrated a significant decrease in blood glucose levels in most participants. Out of the ten participants involved, eight participants experienced a decrease in blood glucose levels following the intervention. For example, Mrs. N experienced a decrease from 260 mg/dL to 160 mg/dL, and Mr. R from 280 mmHg to 180 mg/dL. Similarly, Mrs. D, Mrs. Sr, Mrs. S, Mrs. W, Mrs. N, Mrs. L, Mr. D, and Mr. Y showed a decrease in blood glucose levels of 40 to 80 mg/dL.

DISCUSSION

The discussion of this study highlights the significant role of providing 3J diet management education in controlling blood glucose levels as a non-pharmacological intervention in lowering blood glucose levels in patients with type 2 diabetes mellitus. The intervention, conducted three times over 14 days with each session lasting 20–30 minutes, was proven to significantly lower blood glucose levels.

This study is consistent with the findings of a previous study by Fentiana et al. (2024) titled “3J Diet Management Education to Control Blood Glucose Levels in Diabetes Mellitus Patients”. The activity was divided into three stages: pre-education, education, and evaluation. The pre-education phase involved measuring blood glucose levels and assessing patients’ knowledge of diabetes diet management.

The education phase provided counseling on the principles of diabetes diet management using a focus group discussion approach involving the patient's family. The evaluation phase reassessed patients' knowledge through a questionnaire and measured blood glucose levels again. After patients with diabetes mellitus were educated on the application of the 3J principles, their knowledge and that of their families improved, leading to greater discipline and adherence to dietary management, thereby controlling their blood glucose levels. Interview results indicated that patients understood diabetes dietary guidelines, and observations during home visits confirmed that family meal menus aligned with the teachings.

CONCLUSION

Based on the results of the research and discussion conducted, it can be concluded that providing education on the 3J diet management to control blood glucose levels is an effective non-pharmacological intervention in lowering blood glucose levels in patients with type 2 diabetes mellitus in the Lhokseumawe City Health Center working area. The implementation of the 3J diet management education program for controlling blood glucose levels on a routine basis, despite its short duration and limited frequency, has proven effective in reducing blood glucose levels among all participants. This effectiveness is supported by previous theories and research findings that emphasize that discipline and adherence to diet management can help control blood glucose levels.

The results of this study can be utilized, particularly in the service area of the Lhokseumawe City Health Center. It is hoped that 3J diet education for controlling blood glucose levels will become an integral part of nursing strategies in preventing complications and improving the quality of life for patients with type 2 diabetes mellitus.

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LEADERSHIP SUPPORT FOR PATIENT SAFETY IN PATIENT SAFETY INCIDENTS AT EAST ACEH HOSPITAL

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ABSTRACT

Patient safety incidents are unintentional events when providing care to patients. However, Leadership Support is needed in an effort to minimize the occurrence of patient safety incidents. This study aims to analyze the relationship between the need for Leadership Support and Patient Safety Incidents at East Aceh Hospital. The METHOD uses quantitative research with a descriptive analytical design using a cross-sectional approach. The population in this study were all health workers at East Aceh Hospital, totaling 520 health workers, with a sample of 226 respondents obtained by a simple random sampling technique. Data collection using questionnaires and analyzed with the Chi-Square statistical test at a 95% confidence level. The results showed that there was no relationship between the Relationship of Leadership Support for Patient Safety and Patient Safety Incidents at East Aceh Hospital, with a value (OR = 1.08 95% CI = 0.44- 2.63 P = 0.852). Conclusion: The results of this study can be used as input and a source of information for the community and health workers regarding Leadership Support for Patient Safety and Patient Safety Incidents in Hospitals.

Keywords: *Support, Leadership, Patient Safety, Hospital*

INTRODUCTION

A patient safety incident is an unintentional event during patient care. A patient safety incident can, but is not always, the result of a flaw in a system or process design, system failure, equipment failure, or human error. If patient safety is not implemented properly, it can have a negative impact, reducing patient trust in healthcare services and leading to a decline in hospital quality (Hermawan, 2022).

In reality, patient safety incidents in hospitals still occur even though a good safety system has been implemented. However, the facts show that if the existing competencies and values or culture do not support it, it cannot run optimally. Therefore, it is necessary for hospital management to create a patient safety culture in an effort to minimize incidents related to patient safety aspects. The World Health Organization (WHO) recorded 134 million adverse events occurring annually and 2.6 million deaths per year due to unsafe services. Patient safety incidents involving medical errors occurred in the United States, the United Kingdom, Denmark, and Australia, with a range of 3.2% - 16.6%. Meanwhile, in Brazilian hospitals, surgical adverse events accounted for 21.8% of 60 cases, 90% of which were preventable incidents, and more than two-thirds resulted

in mild or moderate damage (WHO, 2020). This indicates that the complexity of healthcare delivery in hospitals poses a high risk to patient safety (Daud, 2020).

Research Sinurat & Lusya (2018) states that the first thing to consider when implementing a patient safety culture is the leader's commitment to safety. Creating a strong patient safety culture and reducing adverse events requires effective leadership to instill a culture that clearly supports employee efforts and is non-punitive. A ward head who is able to know himself well is one of the characteristics of a good leader. Strong self-awareness can build empathy, which will foster a sense of closeness and trust among subordinates, facilitating collaboration in achieving goals (Adhyatma, 2019).

Patient safety incidents consist of adverse events (AEs), near misses (NMS), non-injury events (NMS), potential injury events (PIDs), and sentinels (Wanda, 2020). Every patient safety incident in a hospital should be recorded and reported to the hospital's patient safety team. However, in practice, reporting of patient safety incidents in hospitals is not optimal, so that officers tend to report incidents that cause injuries; officers are afraid to report patient safety incidents (Wanda, 2020). Reporting of patient safety incidents is directly submitted to the PMKP Committee or the head of the PMKP Committee, which is actually the report submitted to the hospital patient safety coordinator (Tristantia, 2018).

The results of a preliminary survey conducted by researchers at East Aceh Hospital on June 11, 2024, regarding two cases of patient safety incidents that threatened patient safety. This is not considered good because reporting patient safety incidents should be used as lessons learned to prevent similar incidents from recurring. To improve quality and patient safety, East Aceh Hospital has a quality committee and a patient safety officer within it. According to a study conducted by the World Health Organization, unsafe care causes 64 million patients to become disabled and 2.6 million to die each year. This indicates that an unsafe culture regarding patient safety leads to accidents and deaths (WHO, 2020).

Based on research data by Anggriyanti et al. (2018), in Indonesia, errors in the service process are caused by various factors, including 85% by healthcare workers and 15% by equipment. Meanwhile, another study by Tampubolon et al. (2022) stated that 28.3% of patient safety violations were committed by nurses. Nurses must be aware of their role in patient safety in hospitals and must actively participate in ensuring its success. Personal characteristics (employees) can describe the characteristics inherent in a person in life and life, especially in acting and behaving. Hardari (2002), says that the characteristic Robbins (2001) states that Organizational commitment is to the degree to which an employee sides with a particular organization and its objectives, and intends to maintain membership in a particular organization. Greenberg and Baron (2003) state that organizational commitment is a psychological status that marks employee relations with the organization, and has implications in determining the decision to continue membership in the organization.

METHOD

This study employed an analytical survey method with a cross-sectional approach. The population comprised 520 healthcare workers at East Aceh Hospital. The sample comprised all healthcare workers at East Aceh Hospital, totaling 226 respondents. Random sampling was used as the sampling technique, which involves randomly selecting members of the population without regard to strata within the population (Syapitri et al., 2021).

This research was conducted at East Aceh Hospital on July 20-24, 2024. This research has been ethically tested at Prima Indonesia University, number: 070/KEPK/UNPRI/VII/2024. The instrument in this study was a questionnaire from the Hospital Survey on Patient Safety Culture (HSOPSC) developed by AHRQ, (2004). The validity test of the questionnaire was a patient safety incident instrument on July 18, 2024, at Bireuen Hospital. The results of the validity test on 20 respondents showed that the pairwise correlation value for each question was between 0.6809-0.8885, so the question was declared valid. The results of the reliability test showed that the Cronbach Alpha value in 20 respondents was 0.956, which means that the question was reliable. This study used the chi-square statistical test.

RESULT AND DISCUSSION

Table 1 indicates that the average age of healthcare workers is 20-29 years, i.e., 45.58%. Females are more prevalent at 81.42%. More workers have 6-10 years of service, i.e., 45.13%. The highest level of education is nursing graduates, i.e., 71.24%. The highest level of education is a bachelor's degree, i.e., 49.12%. The highest employment status of workers is non-civil servants, i.e., 67.70%. Most workers work in the ward, i.e., 61.50%.

Table 1. Respondent Characteristics

No	Variable	f	%
Age			
1	20-29 year	103	45,58
2	30-39 year	98	43,36
3	40-49 year	18	7,96
4	50-59 year	7	3,10
Gender			
1	Laki-Laki	42	18,58
2	Perempuan	184	81,42
Years of service			
1	<1 year	11	4,87
2	1-5 year	86	38,05
3	6-10 year	102	45,13
4	11-15 year	27	11,95
E d u c a t i o n			

No	Variable	f	%
1	Nurse	161	71,24
2	Doctor	16	7,08
3	Midwife	10	4,42
4	Pharmacist	15	6,64
5	Nutritionist	4	1,77
6	Public Health	8	3,54
7	Laboratory Analyst	6	2,65
8	X-ray Technician	6	2,65
Educational level			
1	Diploma	109	48,23
2	Bachelor's Degree	111	49,12
3	Master's Degree	6	2,65
Employee Status			
1	Civil Servants	73	32,30
2	Non-civil servants (Service and Honorary)	153	67,70
Work unit			
1	Polyclinic	22	9,73
2	Treatment Ward	139	61,50
3	Laboratory	6	2,65
4	Psychiatric Ward	9	3,98
5	Radiology	6	2,65
6	Pharmacy	14	6,19
7	Emergency Room	17	7,52
8	Nutrition	4	1,77
9	Maternity Ward	9	3,98

Table 2 indicates that leadership support for patient safety was the highest at 52.65%. The highest number of patient safety incidents was never, at 90.27%.

Table 2. Frequency Distribution of Leadership Support and Patient Safety Incidents

No	Variable	f	%
1	Leadership Support		
	Good	119	52,65
	Not good	107	47,35
2	Patient safety incidents		
	Once	22	9,73
	Never	204	90,27

Table 3 indicates that 90.65% of healthcare workers with good leadership support for patient safety and never experienced a patient safety incident. Meanwhile, 10.08% of healthcare workers with poor leadership support and never experienced a patient safety incident, with an OR of 1.08, meaning that healthcare workers with leadership support for patient safety were able to prevent a patient safety incident once. The statistical results obtained a p-value of 0.852, indicating that there is no significant relationship between leadership support for patient safety and patient safety incidents.

Table 3. Leadership Support for Patient Safety with Patient Safety Incidents

Table 3: Leadership Support for Patient Safety with Patient Safety Incidents						
Variabel	Patient safety incidents				OR	p- val
	Once		Never			
	n	%	n	%		
Leadership Support						
Good	10	9,35	97	90,65	1,08	0,85
Not good	12	10,08	107	89,92		

The Relationship of Leadership Support for Patient Safety to Patient Safety Incidents. The analysis of the relationship between leadership support for patient safety and patient safety incidents showed that good leadership support and never experiencing a patient safety incident were 90.65%. Meanwhile, poor leadership support and never experiencing a patient safety incident was 10.08%, with an OR value of 1.08, meaning that staff with management support for patient safety were able to prevent a patient safety incident once. The statistical results obtained a p-value of 0.852, meaning there was no significant relationship between management support for patient safety and patient safety incidents.

According to Antonakos et al. (2021), patient safety culture was assessed using the MOSPSC dimensions. The dimensions with the lowest scores were work pressure and pace (46%), followed by leadership support for patient safety (62%). This conclusion aligns with Nabila et al. (2023), who found a regression coefficient of 5.633 and a p-value of 0.038, showing that management support for a patient safety culture has a significant impact on patient safety events.

Teamwork among healthcare workers requires a leader to direct and coordinate every service provided to patients to avoid misperceptions or misinformation (Wahyuda et al., 2024). Strong leadership will encourage improved implementation of a patient safety culture by the team providing care to patients. One aspect of leadership in implementing a patient safety culture is management support. Managers, as leaders, coordinate the implementation of a patient safety culture and are able to influence strategies and efforts to mobilize healthcare workers within their jurisdiction to jointly implement a patient safety culture. Effective leaders are those who are able to read and sort out conditions that will ultimately impact a patient safety culture (Pasaribu, 2020).

The results of a study conducted by Wahyuda et al. (2024) stated that there is a significant relationship between leadership and patient safety culture, where out of 68 nurses, most nurses had effective leadership support, namely 37 nurses (54.4%). This study's results are supported by research conducted by Suryatin (2018), which also stated

that there is an influence of leadership style on patient safety culture. According to Tevžič et al. (2021), patient safety culture uses the MOSPSC. Dimensions that have low scores in this measurement are leadership support for patient safety (39.2%) and work pressure and speed (11.5%).

Further research by Antonakos et al. (2021) showed that the leadership dimension of patient safety had a low score and needed improvement. The leadership dimension is defined as leaders actively supporting patient quality and safety, placing a high priority on improving patient care processes, not ignoring errors, and making decisions based on what's best for patients. The low score for the leadership dimension of patient safety was due to a lack of leadership support in communication between management and the workforce, which impacted the implementation of a patient safety culture.

According to the author's assumption, leadership support for patient safety towards patient safety incidents has a significant influence on reducing hospital patient safety incidents. In creating a culture of patient safety, leaders are needed who are able to establish and communicate a clear safety vision, appreciate and empower staff to achieve the vision, and are able to be good role models. Building a culture of patient safety in hospitals is the obligation and responsibility of all staff working in hospitals, especially medical personnel who are in direct contact with patients, such as doctors and the role and other medical personnel who are responsible for patients.

CONCLUSION

This study used primary data, namely data collected directly from the field by distributing questionnaires to respondents. There was no relationship between the Relationship of Leadership Support for Patient Safety and Patient Safety Incidents at East Aceh Hospital, with a value (OR = 1.08 95% CI = 0.44- 2.63 P = 0.852). The results of this study can be used as input and a source of information for the community and health workers regarding Leadership Support for patient safety towards patient safety incidents in hospitals.

Leadership support for patient safety in patient safety incidents has a significant impact on reducing hospital patient safety incidents. In creating a patient safety culture, leaders are needed who are able to establish and communicate a clear safety vision, appreciate and empower staff to achieve the vision, and are able to be good role models.

Leadership is crucial in implementing patient safety. Leaders must have a strong and clear commitment to supporting staff in implementing patient safety. Therefore, management and the workforce need to improve communication to optimally implement a culture of patient safety.

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NUTRITION-CONSCIOUS FAMILY BEHAVIOR (KADARZI) TOWARDS STUNTING INCIDENCE IN TODDLERS

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ABSTRACT

Stunting is not only a problem of physical growth impairment, but also causes toddlers to become ill easily. In addition, there are also impairments in brain and intelligence development as well as growth and development, making stunting a major threat to the quality of life and human resources. KADARZI behavior is family behavior that is nutrition-conscious, meaning families that consistently practice nutritional behaviors as measured by five indicators: regularly weighing body weight, providing exclusive breastfeeding, consuming a varied diet, using iodized salt, and taking nutritional supplements as recommended. This study is a quantitative study with a cross-sectional design. The results of the Chi-Square test in the table show that 52 respondents (32.9%) exhibited Kardashian behavior with normal stunting rates, and 55 (34.8%) respondents did not exhibit Kardashian behavior and a Chi-Square. This test obtained a significant value, namely a p-value of $0.032 < 0.05$, so it can be concluded that H_0 is rejected and H_a is accepted, which means that there is a relationship between KADARZI behavior and stunting in toddlers. Family behavior that is aware of good nutrition is the key to preventing stunting in toddlers. The application of five indicators of nutrition-conscious families, including regular weight measurement, exclusive breastfeeding, consumption of a varied diet, use of iodized salt, and recommended nutritional supplements, can reduce growth barriers in infants and overcome stunting in toddlers.

Keywords: *Stunting, Kadarzi, Toddlers*

INTRODUCTION

Indicators Human development indicators set out in the 2020-2024 National Medium-Term Development Plan (RPJMN) relate to reducing the prevalence of stunting to 14% in 2024 from 27% in 2019. This target is also in line with the Indonesian government's commitment to achieving Sustainable Development Goal (SDG) 2.2.1, namely the prevalence of stunting (short and very short) in children under five years of age. Achieving this target still faces various challenges and persistent health issues, such as nutritional status. Based on data from the 2021 Nutritional Status Study (SSGI), the stunting and wasting rates in Indonesia are 24.4% and 7.1%, respectively. These figures exceed the World Health Organization (WHO) standard, which requires that a region be classified as having good nutritional status if the prevalence of short children is less than 20% and the prevalence of thin children is less than 5%. Based on these criteria, Indonesia is classified as having a chronic-acute problem, with a prevalence of short children of more than 20% and a prevalence of thin children of 5% or more (Minarto, 2009).

Efforts to combat and prevent stunting can be made by increasing health promotion about KADARZI Family Awareness Behavior, which is nutrition-aware family behavior, in other words, families who always practice nutritional behavior as seen in 5 indicators,

namely: weighing body weight regularly, providing exclusive breastfeeding, consuming a variety of foods, using iodized salt, and consuming nutritional supplements as recommended. Based on the Decree of the Minister of Health of the Republic of Indonesia (Kepmenkes RI) Number: 747/Menkes/SK/2007, KADARZI is defined as a family that is able to recognize, prevent, and overcome nutritional problems in each of its members. The general objective of the KADARZI Program is for all families to be nutrition-aware. The specific objective of KADARZI is to increase the ease with which families and communities can obtain nutrition information and quality nutrition services. A nutrition-aware family is a family that is able to apply balanced nutritional behavior in their daily lives, is able to understand, recognize, and overcome nutritional problems that occur in each of its members. The failure to achieve the KADARZI indicators will certainly result in nutritional problems (Simon MED, 2018).

The results of Hamid et al.'s (2022) research indicate that there is a relationship between KADARZI behavior and stunting in toddlers by applying five indicators that families must implement in determining their children's nutrition to avoid stunting. The KADARZI program is a community health center program aimed at improving the quality of toddler growth and development. The indicators of success in fostering toddler growth and development include not only improvements in the health and nutritional status of toddlers, but also optimal development in terms of mental, emotional, social, and independence skills (Sanansari, 2020).

Nutrition-Aware Families (KADARZI) is an approach developed to raise public awareness, especially within families, about the importance of balanced nutrition and healthy eating habits. This program aims to improve families' knowledge, attitudes, and behaviors in implementing nutritious eating habits to create a healthy and intelligent generation. According to Sumarni (2019), KADARZI plays a crucial role in improving the quality of life for communities, especially in preventing malnutrition and diseases associated with unhealthy eating habits.

One of the main focuses of the KADARZI theory is empowering families to understand and manage the appropriate nutritional needs of each family member. KADARZI emphasizes the importance of involving housewives in educating family members to choose nutritionally balanced foods, as well as providing knowledge about the benefits of consuming healthy and affordable local foods. Suryani (2020) explains that nutrition-aware families will be better able to utilize the resources available around them, such as vegetables, fruits, and other local products, to meet their nutritional needs without having to rely on instant foods or junk food that are less nutritious.

In addition, the KADARZI program also focuses on a community-based approach that involves cooperation between the community, health workers, and the government. Wijayanti (2021) states that the success of the KADARZI program is highly dependent on support from various parties, such as continuous education about nutrition and the provision of adequate facilities to support healthy eating habits at the family level. Through proper education and learning about the importance of balanced nutrition, KADARZI can reduce stunting, malnutrition, and non-communicable diseases caused by poor eating habits.

Stunting is a condition of growth disorder in toddlers characterized by height below the standard for their age, caused by chronic malnutrition during the first 1000 days of life, from pregnancy to the age of two. According to the WHO (2019), stunting occurs due to insufficient nutritional intake, repeated infections, and inadequate health care during this critical period. Stunting not only affects children's physical growth but also impacts their cognitive development and learning abilities, which can affect their future productivity. Therefore, preventing stunting in toddlers is crucial to ensure optimal growth and development.

The causes of stunting in toddlers are complex and involve nutritional, health, sanitation, and social environmental factors. Indriani (2020) states that factors contributing to stunting include poor nutrition, frequent infections due to poor sanitation, and inadequate parenting practices, such as non-exclusive breastfeeding or late introduction of complementary foods. Limited access to health services and low parental knowledge about good nutrition also play a major role in exacerbating this condition. In addition, socioeconomic factors also have an influence, as families with low economic status are more vulnerable to malnutrition and stunting.

Prevention of stunting can be achieved through various approaches, such as improving the nutrition of pregnant women, exclusive breastfeeding, and providing nutritionally balanced complementary foods. Haryanto (2021) explains that efforts to prevent stunting must begin during pregnancy by ensuring that mothers receive adequate nutrition, including iron and folic acid supplements. In addition, it is important to promote nutritious eating patterns in toddlers, including providing foods with nutritional variety appropriate for the child's age. These appropriate and holistic interventions are expected to reduce the prevalence of stunting, which in turn will improve the quality of human resources in the future.

METHOD

This study is a quantitative study with a cross-sectional research design, namely a descriptive study with research subjects who were observed, measured, and asked to answer questions only once, and aimed to analyze the relationship between family behavior and nutritional status (KADARZI) with the incidence of stunting. The location of this study was in the working area of the Banda Sakti Community Health Center in Lhokseumawe City, which consists of 26 villages. The population in this study was families with toddlers in the working area of the Banda Sakti Health Center in Lhokseumawe City. Sampling was done using stratified random sampling, which gives every element (member) of the population an equal chance of being selected as a sample member proportionally, with a sample size of 158 respondents. The data in this study were obtained through the distribution of a Nutrition-Aware Family (KADARZI) behavior questionnaire consisting of five indicators that must be met in accordance with nutrition-aware families in alert villages (Indonesian Ministry of Health No. 747/Menkes/SK/VI/2007). Stunting was measured using TB/U using a microtoise. The data analysis stage involved univariate analysis (frequency distribution) and bivariate analysis, which sought relationships based on the Chi-Square test if the p-value was ≤ 0.05 .

RESULTS AND DISCUSSION

Based on the results of the Chi-Square test, a p-value of $0.032 < 0.05$ was obtained, which means that Kadarzi behavior is closely related to the incidence of stunting in toddlers, through the Kadarzi indicator that must be implemented to prevent stunting. Kadarzi is a simplification of the PUGS (General Guidelines for Balanced Nutrition) program from the Ministry of Health of the Republic of Indonesia, which is considered valid and reliable. In addition, this program is easy to apply in the community to address toddler nutrition issues, especially stunting. The Kadarzi indicators are exclusive breastfeeding, consumption of a variety of foods, consumption of iodized salt, and consumption of nutritional supplements as recommended. Kadarzi is a family where all members understand and practice balanced nutrition behaviors, enabling them to address nutrition and health issues. This program was initiated by the government with the hope that families can identify, prevent, and address nutrition issues among their members through their own efforts (Agustina, 2020).

A family is considered to have good nutrition awareness if it regularly weighs its children, practices exclusive breastfeeding, consumes a variety of foods, consumes iodized salt, and takes nutritional supplements as needed. This program is carried out by improving families' nutritional knowledge and behaviors that do not support the handling of nutritional problems, such as stunting. The standard percentage of achievement is 80%, so according to research by Rodiah et al. (2018), the achievement of this program can be categorized as not yet good and already good. The category that is not yet good is exclusive breastfeeding (61.8%), while the other indicators are already classified as good because the achievement percentage is $\geq 80\%$, with the use of iodized salt having the highest achievement percentage. The nutritional status of very short toddlers is 5.8% and short toddlers is 16.5%. The results of the study by Agustina et al. were slightly different, with the percentage of achievement of the Kadarzi indicator ranging from 33.7% to 98.9%. The indicator with the lowest achievement in this study was exclusive breastfeeding, with a value of only 33.7%.

In addition, consumption of a variety of foods and weight measurement achievement rates are still below 80%, with values of 57.6% and 72.8% respectively, and a stunting rate of 41.3%. The low achievement of the exclusive breastfeeding indicator is also shown in Apriani's study, with an achievement rate of only 57.1%. Wijayanti and Nindya's study showed different results. Although the percentage of exclusive breastfeeding has not reached 80%, the value has reached 77.8%. Each indicator is significantly related to the nutritional status of toddlers (TB/U) with a p-value < 0.05 . Family awareness of providing a variety of foods for toddlers has exceeded the standard achievement level. A varied diet is one that contains balanced nutrition with nutrients that are sources of energy (staple foods containing carbohydrates), nutrients that build and maintain body tissue (side dishes from animal and plant-based foods containing protein), reserve nutrients (fats), and regulatory nutrients (vegetables and fruits containing vitamins and minerals). Toddlers with a diverse diet tend to have good nutritional status. A positive relationship has been shown between the type and frequency of meals and the nutritional status of toddlers. Humans need 40 different nutrients that must be obtained

from a diverse diet in the right amounts, including breast milk and consumption of plant and animal foods as well as fortified foods (Wijayanti, 2017).

The application of this behavior within the family is certainly influenced by factors such as education level, employment status, family income, maternal nutrition knowledge, and program cadre participation. Maternal education contributes significantly to improving nutritional adequacy. The prevalence of stunting is lower in children whose mothers have a higher level of education. The incidence of stunting is lower or decreases by 16.5% in children whose mothers have basic education. This decline is seen from the second year of life. Deformities are diseases caused by nutritional imbalances during the golden period, not growth hormone disorders or the result of certain diseases. Toddlers still experience growth spurts, which are usually accompanied by periods of increased energy and appetite (Qonitun, 2015).

During slower growth, children's contributions and appetites also decline. Parents must understand the difference between appetite and food intake in school-age children in order to respond appropriately to any illness in their children. Toddlers need complete nutrition for their growth and brain and body development. Childhood is a critical period for children, so their nutritional needs must be balanced in terms of both quantity and nutritional content. Children will be healthy, disease-free, and intelligent if they achieve a balanced diet. A balanced diet is achieved through foods that meet nutritional needs related to age and activity levels in order to achieve a normal body weight. Nutrients for young children include carbohydrates, protein, fat, vitamins, water, and minerals (Hariyadi, 2011).

Toddlers need 75-90% carbohydrates, 10-20% protein, and 15-20% fat. Young children need building materials, energy, and regulatory substances. In line with Riana's (2021) research, the most influential KADARZI behavioral indicators are exclusive breastfeeding and consumption of a variety of foods. Consuming a variety of foods can reduce the risk of stunting in toddlers by 13.175 times compared to consuming a less varied diet. Family awareness in providing a diverse diet for toddlers is crucial. Toddlers with a diverse diet tend to have good nutritional status. Research by Yosika et al. shows that the majority of mothers' knowledge about KADARZI is categorized as insufficient (48 people, 84.2%), and KADARZI behavior is categorized as insufficient (31 people, 54.3%). Therefore, it is concluded that to address nutritional problems in toddlers, parents should pay more attention to their children's dietary intake by adopting a Kadarzi lifestyle to reduce the risk of nutritional problems, particularly stunting in toddlers (Yosika, 2023).

CONCLUSION

The results of the study can be concluded that good nutrition awareness among families is key to preventing stunting in toddlers. The application of five indicators of nutrition awareness among families, including regular weighing, exclusive breastfeeding, consumption of a varied diet, use of iodized salt, and consumption of recommended nutritional supplements, can reduce growth impediments in infants. There is a need for socialization and development of more effective and easily understood messages about balanced nutrition for the community. Socialization can be carried out by simplifying messages about balanced nutrition, especially for households. The main objective for mothers of toddlers is the overall impact of the program to address the issues of balanced

nutrition messages and KADARZI behavior, as well as to address the issue of stunting in toddlers.

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DEVELOPMENT OF DIGITAL EDUCATION SERVICES BY NURSES IN MODERN WOUND CARE

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ABSTRACT

This study aims to develop and evaluate the effectiveness of digital education services led by nurses in enhancing modern wound care management. Using a pre-experimental one-group pretest-posttest design, the study involved 35 patients with chronic and acute wounds who participated in a three-month digital education program. The intervention comprised educational videos, infographics, and interactive consultations delivered via mobile-friendly platforms to improve patients' knowledge and self-care practices. Data were collected through validated questionnaires assessing knowledge and confidence before and after the intervention, alongside monitoring changes in clinic visit frequency. The results demonstrated a significant increase in patient knowledge scores from 62.5% to 80.1% ($p < 0.001$) and a substantial rise in self-care confidence reported by 85% of participants. Additionally, unnecessary clinic visits decreased by 40%, indicating enhanced patient autonomy and optimization of healthcare resources. The digital platform also served as a collaborative medium for nurses' continuous professional development. This study contributes original insights into the integration of nurse-managed digital education in wound care, highlighting its potential to improve patient outcomes and healthcare efficiency. The findings suggest that digital health education can overcome barriers such as geographic limitations and time constraints, providing accessible and effective wound care guidance. Future research is recommended to explore the long-term clinical effects and scalability of this intervention across diverse healthcare settings. The study underscores the value of digital innovation in nursing education and its role in advancing patient-centered care practices.

Keywords: *Digital Education, Nurse-led Intervention, Wound Care.*

INTRODUCTION

Wound care is one of the essential aspects of healthcare services that requires a comprehensive and evidence-based approach. Advances in medical technology have driven the adoption of modern wound care methods, which emphasize optimal moisture balance, appropriate dressing selection, and infection prevention to accelerate healing and minimize complications. However, in many regions, the level of public knowledge regarding proper wound care practices remains low, which may increase the risk of wound deterioration and lead to higher healthcare costs.

In the era of digital transformation, nurses have a significant opportunity to develop health education services through online platforms. The use of digital media, such as social media, instant messaging applications, and educational videos, can expand the reach of information while providing targeted and effective education. This strategy

aligns with the nurse's role as a health educator, who not only delivers direct care but also educates communities to enhance self-care independence.

The development of digital education services in modern wound care is a strategic step to address limitations of time, distance, and resources. Through this approach, communities can access accurate, easy-to-understand, and applicable knowledge that can be practiced independently at home. Furthermore, this innovation supports the achievement of the Sustainable Development Goals (SDGs), particularly Goal 3, which aims to ensure healthy lives and promote well-being for all at all ages.

The role of nurses in developing health education services has evolved alongside advances in digital technology. Beyond providing direct care, nurses are now expected to utilize digital media as a means of counseling, health promotion, and continuous patient support. Digital-based education enables patients to access accurate, reliable, and easy-to-understand information anytime and anywhere. Thus, the presence of digital educational media not only improves patient knowledge but also fosters more consistent healthy behaviors.

The rapid development of digital technology has significantly transformed healthcare services, including wound care. Digital education services provided by nurses offer accessible, flexible, and personalized learning opportunities for patients with chronic or acute wounds. According to recent studies, digital media—such as mobile applications, educational videos, and teleconsultations—enable patients to gain a better understanding of wound care procedures, enhance compliance with treatment plans, and improve overall wound healing outcomes (Ousey & Milne, 2020; Posnett & Franks, 2008).

Nurses play a critical role in developing and implementing digital education interventions. These interventions not only focus on providing accurate and evidence-based information but also aim to foster patient engagement and empowerment. Research has shown that digital education can complement face-to-face care, particularly in remote areas or for patients with mobility limitations (Green et al., 2018). In addition, digital tools can help monitor patient progress, provide timely feedback, and reduce the frequency of unnecessary clinic visits, thus improving both cost-effectiveness and patient satisfaction (Santamaria et al., 2015).

Furthermore, the integration of digital education into modern wound care aligns with the broader healthcare trend toward patient-centered care. The use of interactive content, visual demonstrations, and two-way communication platforms facilitates better understanding and retention of information. This approach not only improves the quality of nursing care but also supports the professional development of nurses by equipping them with digital competencies necessary for contemporary practice (World Health Organization, 2021).

METHOD

This study employed a pre-experimental design with a one-group pretest–posttest approach to evaluate the effectiveness of digital-based health education on modern wound care. The research was conducted at the independent nursing clinic *Perawatan Luka Ns. Liza Phonna, S.Kep., CWCCA* in 2024. The study population consisted of all patients with

chronic wounds who visited the clinic during the research period. A purposive sampling technique was applied to recruit participants based on predetermined inclusion and exclusion criteria, resulting in a sample of 20 respondents.

Data on patients' knowledge and attitudes toward modern wound care were collected using a structured questionnaire validated by wound care experts. The questionnaire comprised sections on demographic data, knowledge assessment, and attitude measurement using a Likert scale. Digital-based health education was delivered via a short, animated educational video combined with verbal explanation, focusing on principles of wound hygiene, moisture balance, infection prevention, and the importance of regular wound assessment.

Paired t-tests were used to compare pretest and posttest scores. The tests' significance level was set to $p < 0.05$. The relevant institutional review board granted ethical approval, and all subjects provided informed consent prior to data collection.

RESULT

Nurses designed and tested a digital education program for current wound care over a three-month period. The program involved a total of 35 patients. The data were processed and evaluated to determine the intervention's impact on patient knowledge and participation.

Table 1. Patient Knowledge and Behavioral Outcomes Pre- and Post-Digital Education Intervention

Indicator	Pre-Intervention Mean $\pm$ SD	Post-Intervention Mean $\pm$ SD	p-value
Knowledge Score (%)	62.5 $\pm$ 8.4	80.1 $\pm$ 6.7	< 0.001
Confidence in Self-Care (%)	45	85	—
Unnecessary Clinic Visits (%)	30	18	—

Table 1 summarizes the pre- and post-intervention knowledge scores and related indicators. The mean knowledge score significantly increased from 62.5% (± 8.4) before the intervention to 80.1% (± 6.7) after the digital education program, indicating an improvement of 17.6% ($p < 0.05$). Additionally, 85% of patients reported feeling more confident in self-managing their wound care after participating in the program. The frequency of unnecessary in-person visits decreased by 40%, demonstrating improved patient autonomy and reduced burden on healthcare resources.

DISCUSSION

The significant improvement in patient knowledge after the digital education intervention supports the effectiveness of nurse-led digital platforms in modern wound care education. This aligns with the theory of adult learning, which emphasizes self-directed learning and the use of multimedia to enhance comprehension (Knowles, 1984).

The use of videos, infographics, and interactive modules caters to different learning styles and improves patient engagement.

The reduction in unnecessary clinic visits indicates that patients became more autonomous in managing their wounds. This increased autonomy reflects greater confidence and is consistent with the Chronic Care Model, which emphasizes patient self-management and empowerment (Wagner et al., 2001). This finding also aligns with research by Gilmartin et al. (2021), which reported that telehealth interventions improve patient self-efficacy in wound care.

However, some patients still preferred face-to-face consultations for complex cases. This preference reflects the ongoing need for blended care approaches that combine digital and in-person services. In addition, the digital platform facilitated professional knowledge sharing among nurses, supporting continuous professional development as suggested by Li et al. (2009).

Theoretically, this study reinforces the integration of digital health education into nursing practice to enhance patient outcomes. Practically, it suggests that digital education services can optimize resource use, improve patient independence, and potentially reduce healthcare costs. Future research should explore long-term clinical outcomes and scalability across diverse healthcare settings.

CONCLUSION

The development and implementation of nurse-led digital education services in modern wound care significantly improved patient knowledge and confidence in self-care. The digital platform effectively reduced unnecessary clinic visits, enhancing patient autonomy and optimizing healthcare resources. These findings support the integration of digital health education into standard nursing practice as a promising approach to improve wound care outcomes. Future studies are recommended to evaluate the long-term clinical impact and feasibility of scaling this intervention in various healthcare settings.

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RELATIONSHIP BETWEEN BREASTFEEDING EFFICACY AND EXCLUSIVE BREASTFEEDING AMONG PRIMIPARA MOTHERS IN THE WORKING AREA OF THE BABAH BULOH COMMUNITY HEALTH CENTER, NORTH ACEH

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ABSTRACT

Breast milk is the most important source of nutrition for infants aged 0-6 months and is excellent for their growth and development. The nutrients contained in breast milk are essential for growth, health, and the immune system of children. Exclusive breastfeeding is a condition in which infants are fed only breast milk without any additional food during the first six months. Exclusive breastfeeding plays a crucial role in maintaining the health, sustainability of life, and immune system of infants. Several factors influence exclusive breastfeeding, one of which is breastfeeding self-efficacy. The aim of this study is to investigate the relationship between breastfeeding self-efficacy and exclusive breastfeeding among primiparous mothers in the service area of the Babah Buloh Health Center in North Aceh. The study employs a cross-sectional approach. The sample consisted of breastfeeding mothers with infants aged 0–6 months in the Babah Buloh Health Center area in North Aceh. The correlation test used in this study was the Spearman rank test. The results of the study using the Spearman correlation test yielded a p-value of 0.00 (<0.05), indicating a relationship between breastfeeding self-efficacy and exclusive breastfeeding in the Babah Buloh Health Center service area, with a correlation coefficient of 0.482, indicating a moderate relationship with a positive correlation direction. There is a relationship between breastfeeding self-efficacy and exclusive breastfeeding among primiparous mothers in the service area of the Babah Buloh Health Center.

Keywords: *Breastfeeding self-efficacy, Exclusive breastfeeding, Primipara*

INTRODUCTION

Breast milk is the most important source of nutrition for infants aged 0-6 months and is very good for their growth and development. The nutrients contained in breast milk are essential for growth and development, health, and the immune system in children (Octaviyani & Budiono, 2020). The global community has responded to this by establishing strategies to achieve exclusive breastfeeding rates, including in Indonesia. One of the key focuses in the field of nutrition is the provision of exclusive breastfeeding for newborns up to six months of age. To date, exclusive breastfeeding remains an indicator of achievement in the Indonesian Healthy Program (PIS), which the Indonesian government uses to improve public health.

Exclusive breastfeeding is a condition where the baby is only given breast milk without any additional supplements in the first six months (Dewi, 2022). Exclusive breastfeeding plays an important role in maintaining the health, sustainability of life, and immunity of the baby. Inadequate breast milk can lead to nutritional imbalances, which may affect the baby's growth and development (Syafriani, 2022).

According to the World Health Organization (WHO) in 2020, approximately 44% of infants aged 0-6 months worldwide receive exclusive breastfeeding, while the target set is 50%. Based on data from the Ministry of Health, in 2021, the percentage of exclusive breastfeeding among infants aged 0-6 months reached 71.58%, an increase of 69.62% from the previous year. According to the 2021 Riskesdas data, exclusive breastfeeding reached 2.3 million infants in Indonesia, accounting for 52.5%, a decrease of 12% from 64.5% in 2019 (Octaviyani & Budiona, 2020).

In Aceh itself, the exclusive breastfeeding coverage rate in 2017 was 56% (Aceh Health Office, 2018). Furthermore, the exclusive breastfeeding coverage rate in 2018 was 57.6% (Banda Aceh Health Office, 2019). Meanwhile, in North Aceh, the exclusive breastfeeding coverage rate remained low in 2022 at 30% of the 50% target (North Aceh Health Office, 2022).

The factors that can influence the success of exclusive breastfeeding are closely related to barriers in interventions provided to mothers, health workers, and other parties involved directly or indirectly. Regarding self-efficacy, a mother's belief in her ability to breastfeed or provide breast milk to her baby is called Breastfeeding Self-efficacy (BSE). There are several interventions that can be implemented for postpartum mothers to enhance BSE, including Breastfeeding Self-Care (BSC).

Breast milk is crucial for infants because it provides a natural immune system that formula milk cannot fully replicate. The underdeveloped digestive system of infants may lead to risks such as *allergies, diarrhea, digestive disorders, constipation, and coughing*. Breastfeeding also promotes more mature growth and development compared to formula feeding (Wahyuni, 2021).

The formula contains fat, protein, whey, casein, carbohydrates, energy, minerals, sodium, potassium, calcium, phosphorus, chloride, magnesium, and iron. The protein content in breast milk is higher than in formula milk. The higher casein content in formula milk forms relatively hard clumps in the baby's stomach, making it difficult to digest when the baby is given formula milk and potentially causing allergies in the baby (Audihani, 2022).

Exclusive breastfeeding is influenced by various factors, including age, education, occupation, economic status, and previous breastfeeding experience. A study conducted on 164 mothers after childbirth showed that mothers' confidence in breastfeeding is influenced by their previous breastfeeding experience (Thi, Ngo, Chou, Gau, & Liu, 2019). The sustainability of breastfeeding behavior is influenced by three important factors: knowledge, self-efficacy, and support (Vincent, 2015).

Various studies on breastfeeding behavior (BSE) have shown that interventions such as support, education, information, and counseling can effectively increase BSE among pregnant women and breastfeeding mothers. Higher BSE levels in mothers have

a significant impact on the breastfeeding process. This influence is both direct and meaningful, as reported by Brockway, Benzies, and Hayden (2017).

In line with Mudah Arimbi's (2021) research, which states that low maternal self-efficacy/confidence in breastfeeding is associated with failure in exclusive breastfeeding. The study results showed that all informants (6 individuals) had low breastfeeding self-efficacy and none successfully provided exclusive breastfeeding; they only provided breast milk for less than one month. The researcher stated that mothers' breastfeeding self-efficacy can influence the success of exclusive breastfeeding.

METHOD

The design of this study used a cross-sectional method. Data on self-efficacy in breastfeeding and breastfeeding practices were collected randomly. This study used purposive sampling techniques. The population in this study included 100 mothers with infants aged 0-6 months in the working area of the Babah Buloh Community Health Center in North Aceh.

RESULT AND DISCUSSION

A. Characteristics of Respondents

The respondents in this study were 100 breastfeeding mothers in the villages of Gp Teungoh, Gle Dagang, Punteut, Teupin Reusep Babah Buloh, North Aceh. The characteristics of the respondents include the mother's age, the baby's age, parity, the mother's occupation, the mother's highest level of education, the family's economic status, the baby's weight, the baby's length, exclusive breastfeeding, and breastfeeding self-efficacy, which can be analyzed from the following table:

Table 4.1 Frequency Distribution of Breastfeeding Based on Respondent Characteristics in the Babah Buloh Aceh Utara Work Area

Characteristics	Frequency	Percentage (%)
Mother's Age		
Low Risk (20-35 Years old)	82	82%
High Risk (<20 and > 35 Years Old)	18	18%
Baby's Age		
Exclusive Breast Milk Minimum (0-6 Months)	70	70%
Exclusive Breast Milk (7-24 Months)	30	30%
Paristas		
Primigravida	58	58%
Multigravida	42	42%
Education		
Low Education (SD,SMP)	20	29%
High Education (SMA, D3, S1)	71	71%
Work		

Job	53	53%
Jobless	47	47%
Economics Status		
Lower Middle Class	80	80%
Upper Middle Class	20	20%
Breastfeeding Self-Efficacy		
Low	16	16%
Medium	34	34%
High	50	50%
Exclusive Breast Milk		
Exclusive Breast Milk	66	66%
Not Exclusive Breast Milk	34	34%
Total	100	100%

Table 4.1 shows that most respondents in the Babah Buloh Community Health Center working area in 2025 are in the low-risk age group of 20-35 years old, totaling 82 or (82%). The average age of infants is 0-6 months old, totaling 70 or (70%) infants. The average parity or number of children born to mothers in the Babah Buloh Health Center service area is 58 (58%). Breastfeeding mothers in the Babah Buloh Health Center service area have a high level of education, with 71 (71%) respondents, and the majority of mothers are not employed, with 47 (47%) respondents.

The average economic status of respondents was lower-middle class, with 80 (80%) respondents. The level of confidence among mothers in breastfeeding was quite high, with 34% or 34 (34%) breastfeeding mothers reporting high confidence. Meanwhile, the majority of mothers provided exclusive breastfeeding to their infants. The percentage of mothers practicing this reached 66% or 66 babies out of the total sample.

B. Relationship Between Breastfeeding Self-Efficacy and Exclusive Breastfeeding
C.

Table 4.2 Relationship Between Breastfeeding Self-Efficacy and Exclusive Breastfeeding in the Work Area of the Babah Buloh Health Center, North Aceh, in 2025 (n=100)

Giving Exclusive Breastfeeding							
		Eksklusif		Not Eksklusif			P
		N	%	N	%	N	100,0
Breastfeeding Self-Efficacy	Low	5	31	11	69	16	100,0
	Medium	17	50	17	50	34	100,0
	High	44	88	6	12	50	100,0
Total		66	66	34	34	100	100,0

Table 4.2 shows that most respondents had a fairly high level of confidence in providing exclusive breastfeeding, with 66 (66%) respondents providing exclusive

breastfeeding and 34 (34%) providing non-exclusive breastfeeding. This indicates that there is a relationship between breastfeeding self-efficacy and exclusive breastfeeding. The correlation coefficient (r) obtained was 0.48, indicating a moderate positive correlation between breastfeeding self-efficacy and exclusive breastfeeding. Here, correlation means that the higher the mother's level of confidence in breastfeeding, the higher the rate of exclusive breastfeeding.

1. Characteristics of Respondents Based on Breastfeeding Self-Efficacy Levels

Based on the results of a study conducted on breastfeeding mothers in the Babah Buloh Community Health Center working area, the level of confidence of mothers when breastfeeding was sufficient, with most mothers at a sufficient level of 34 (34%) and a high level of 50 (50%). These results indicate that breastfeeding mothers in the Babah Buloh Health Center service area have fairly good confidence in breastfeeding their babies.

Breastfeeding Self-Efficacy (BSE) is useful for assessing and identifying mothers who are at high risk of stopping breastfeeding prematurely. When mothers believe they can provide sufficient breast milk for their babies, the body responds positively by producing more breast milk through hormones that increase milk production (Nur, 2019). Providing suggestions to boost mothers' confidence is one strategy to address breastfeeding issues using positive affirmations (Susanti, 2022).

2. Characteristics of Respondents Based on Exclusive Breastfeeding

Based on the results of research conducted in the Babah Buloh Community Health Center working area, most mothers exclusively breastfed their babies, with 66 (66%) respondents providing exclusive breastfeeding. This shows that breastfeeding in this area is quite successful. Breast milk contains components such as immune factors, digestive enzymes, leukocytes, proteins, and hormones that are essential for meeting the nutritional needs of infants from birth until six months of age (Syafriani, 2022). Programs aimed at improving breastfeeding practices, particularly exclusive breastfeeding, have a significant impact on the nutritional health of both mothers and infants. However, in Indonesia, exclusive breastfeeding is still not a common practice. Key issues hindering the implementation of exclusive breastfeeding include socio-cultural factors, low awareness of the importance of breastfeeding, and insufficient support from health services and medical personnel (Mukaromah, 2023).

3. Relationship Between Breastfeeding Self-Efficacy and Exclusive Breastfeeding

Based on the results of the study using Spearman's correlation test, the p -value was 0.00 (<0.05), indicating a relationship between breastfeeding self-efficacy and exclusive breastfeeding in the Babah Buloh Health Center area, with a coefficient of 0.482, showing a moderate correlation with a positive direction. Mothers who breastfeed and have high self-efficacy regarding their ability to provide breast milk for their babies will feel calm, comfortable, and relaxed while breastfeeding, resulting in increased breast milk production.

Zahra's (2020) study showed that there is a relationship between maternal self-efficacy and exclusive breastfeeding. Based on the results of the study, high self-confidence in breastfeeding mothers tends to result in successful exclusive breastfeeding, whereas mothers with low self-confidence in breastfeeding tend to choose not to breastfeed their babies, with the possibility of switching to formula milk. Research conducted by Yudhie (2022) shows a strong relationship between social support, knowledge, attitude, and self-efficacy with breastfeeding behavior. Mothers with low breastfeeding confidence are more likely to use alternative techniques when facing breastfeeding challenges. Alternative techniques involve the use of formula milk.

CONCLUSION

Breastfeeding self-efficacy and exclusive breastfeeding had a correlation coefficient of 0.482, indicating a somewhat good link. This research implies that mothers who are more confident in their ability to breastfeed are more likely to use exclusive breastfeeding. In other words, as a mother's self-efficacy grows, so does the prevalence of exclusive breastfeeding.

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CHAPTER 5: Social Sciences & Environmental Science

NAVIGATING THE CHANGING LANDSCAPE OF EDUCATION AND JOBS: PREPARING FOR A CHANGING WORLD

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ABSTRACT

Education in the twenty-first century is undergoing profound change as technological innovation, globalisation, and shifting labour market demands reshape how societies learn and work. This paper examines the intersection of education and employment by analysing the integration of digital tools, adaptive and personalised learning systems, project-based learning, and the introduction of coding and robotics education. It examines the challenges that students face, including mental health concerns, financial barriers, and time-management issues, as well as the obstacles educators encounter, such as managing workloads, adapting to technological change, and maintaining classroom engagement. Using a theoretical and conceptual approach, the study reviews existing scholarship, identifies emerging patterns, and reflects on practical implications for preparing learners for future work. The analysis reveals that education must go beyond traditional knowledge transfer and emphasise interdisciplinary learning, creativity, and lifelong adaptability. At the same time, inclusivity, affordability, and accessibility must remain central to ensure equitable participation in an evolving system. The findings show that the future of education and the future of work are inseparable: success in one domain depends on progress in the other. By embracing innovation while addressing systemic challenges, education can empower learners with the resilience and competencies necessary for navigating a rapidly changing world.

Keywords: *Education, Future of work, adaptive learning, Lifelong learning, Technology in education*

INTRODUCTION

Education has long been regarded as the foundation of social and economic progress. As Nelson Mandela remarked, “Education is the most powerful weapon which you can use to change the world.” In the twenty-first century, this statement resonates more strongly than ever, as education is increasingly shaped by technology, globalisation, and evolving career requirements. The central challenge lies not only in imparting knowledge but also in preparing students for the uncertainties of the future job market.

The significance of this study lies in its exploration of the evolving relationship between education and work in the context of rapid global change. Ongoing digital transformation demands essential skills such as critical thinking, adaptability, and innovation, which traditional education systems often fail to sufficiently emphasise. At the same time, persistent inequalities in access to educational resources continue to create barriers for many learners. Therefore, this study addresses the research problem of how education systems can effectively adapt to global changes to prepare learners for the demands of a rapidly evolving job market. To achieve this aim, the study examines

current trends reshaping education, analyses the challenges faced by both students and educators, evaluates the role of adaptive and personalised learning, highlights the importance of interdisciplinary skills for future employment, and proposes strategies for aligning education with workforce requirements.

Scholars have recognised that education is deeply connected to economic and social transformation. The *World Economic Forum's Future of Jobs Report 2023* highlights the increasing demand for skills such as analytical thinking, problem-solving, and digital literacy. UNESCO emphasises the need for inclusivity and lifelong learning, framing education as a public good essential for sustainable development (*Reimagining Our Futures Together*). The rise of adaptive learning systems, powered by artificial intelligence, is transforming pedagogy by personalising learning experiences (Holmes, Bialik, and Fadel). Similarly, project-based learning has gained attention as a method for enhancing creativity and real-world problem solving skills (Thomas). Researchers also point to the growing importance of interdisciplinary education, in which fields such as Science, Technology, Engineering, and Mathematics (STEM), the humanities, and the social sciences intersect to produce holistic thinkers.

Despite these innovations, challenges remain. Studies reveal that students often face mounting mental health pressures due to academic expectations (Huang and Zhou), while teachers struggle with technological adaptation and workload stress (*Teachers and Teaching in the Digital Age*). This literature suggests both opportunities and gaps: while technology provides tools for innovation, systemic inequalities and institutional inertia continue to hinder progress.

METHOD

This paper adopts a theoretical and conceptual approach, drawing insights from secondary sources including reports by international organisations (UNESCO, WEF, OECD) and scholarly articles on education and workforce trends. The analysis synthesises key themes rather than presenting primary empirical data. By comparing emerging patterns across studies, the paper identifies challenges and strategies that are globally relevant.

RESULTS AND DISCUSSION

The findings reveal two broad themes: opportunities driven by innovation and barriers arising from systemic challenges. Technological integration has expanded access to learning through online platforms and AI-powered tools, enabling more personalised and flexible learning experiences. Adaptive and personalised learning allows students to progress at their own pace, thereby increasing engagement and retention (Holmes, Bialik, and Fadel). In addition, project-based learning promotes collaboration, creativity, and critical problem-solving skills (Thomas). Coding and robotics education equip students with essential technical competencies for a technology-driven economy, while interdisciplinary approaches foster holistic understanding and innovation across fields (Beane).

However, significant challenges remain. Students face considerable pressures, including high performance expectations, financial inequality, and mental health

concerns (Huang and Zhou). Educators, in turn, contend with classroom management demands, workload stress, and the ongoing need to update their digital competencies (Teachers and Teaching in the Digital Age). Beyond these individual challenges, systemic barriers particularly unequal access to digital resources risk widening the global educational divide. Overall, the analysis suggests that education systems must strike a balance between innovation and inclusivity. Preparing learners for future employment requires rethinking assessment models, fostering lifelong learning, and ensuring affordable and equitable access to education for all.

CONCLUSION

Education and work futures are deeply interconnected. As economies continue to evolve, education systems must move beyond rote learning to cultivate adaptability, critical thinking, and creativity. By integrating advanced technologies, supporting educators, and addressing systemic barriers, schools and universities can better prepare students to thrive in uncertain futures. To achieve this, education systems should take several strategic steps. First, schools and higher education institutions should ensure that technology is affordable and accessible so that all learners can benefit from digital tools. Second, teachers should receive adequate training to use new technologies with confidence and effectiveness. Third, education systems must prioritise students' mental health by providing appropriate support and guidance to help them manage stress. Fourth, learning should not end with formal schooling; opportunities for lifelong learning and reskilling must be made available to learners of all age groups. Finally, teaching approaches should place greater emphasis on interdisciplinary learning and project-based activities, which foster creativity, collaboration, and the ability to solve real-world problems.

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GAYO TRADITIONAL WEDDING PROCESSION IN BENER MERIAH REGENCY

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ABSTRACT

This study aims to analyze the Gayo traditional wedding procession in Bener Meriah Regency, explore the meanings and cultural values embedded in Gayo customary marriage, as well as the obstacles faced by the community during the implementation of wedding ceremonies. This research employs a qualitative approach with an ethnographic design. The findings indicate that the Gayo traditional wedding consists of several stages, namely bersiengonen (initial proposal), munginte (formal proposal), amal nome nipi jege (ritual sleep), betelah (negotiation of marriage requirements), mujele emas (delivery of gold), beguru (traditional advice), jule bai (escorting the groom), akad nikah (Islamic marriage contract), and jule beru (escorting the bride). Each stage carries specific symbolic meanings and cultural values. However, the practice faces obstacles, including economic constraints, the influence of modernization and external cultures, and the lack of understanding among the younger generation regarding the meanings and values embedded in the Gayo traditional wedding procession.

Keywords: *Cultural Values, Gayo Customary, Wedding Procession*

INTRODUCTION

Indonesia is a country characterized by diverse ethnic groups and cultures, where marriage is governed by various customs and traditions that differ across regions. This richness of wedding traditions is a cultural asset that must be preserved, particularly amid globalization, which threatens the sustainability of national identity (Arsya & Badrun, 2023). One community with unique and meaningful wedding traditions is the Gayo people, who are spread across Central Aceh, Bener Meriah, and Gayo Lues. Although they belong to Aceh Province, the Gayo ethnic group has distinctive customs, language, and cultural practices that differ from those of surrounding ethnic groups (Mahara & Misgiya, 2020; Islami, 2018).

Marriage in Gayo tradition is not merely a bond between two individuals but also represents values of solidarity, social harmony, and religiosity. The wedding procession, from *ngekik* (matchmaking) and *berih* (expression of intent) to the post-wedding ceremony, contains deep symbolic meanings and involves active participation from families and the wider community. Each stage is designed to strengthen social ties and ensure cultural continuity (Nasution, 2019; Bakti & Anismar, 2022).

However, these traditions face serious challenges due to modernization. Changes in the mindset of younger generations and the influence of external cultures potentially reduce interest in customary practices, which are often perceived as impractical or outdated. If not addressed, Gayo traditional wedding practices may gradually decline and weaken the cultural identity of the Gayo community. Therefore, further research is necessary to document, analyze, and revitalize these traditions so that they remain relevant and sustainable.

METHOD

This study employs a qualitative approach with an ethnographic design to obtain an in-depth understanding of the phenomenon. The research focuses on understanding and interpreting the meanings of events, interactions, and behaviors of the subjects from the researcher's perspective rather than on statistical analysis. The study was conducted in Bale Atu and Paya Gajah villages, Bukit District, Bener Meriah Regency, Aceh Province, starting on January 2, 2025. The informants included key local community and traditional leaders (*tokoh adat*). Data were collected holistically through interviews, observations, and documentation. Data analysis followed a taxonomic and iterative approach, in which the collected data were summarized, categorized, and presented in narrative form. Preliminary conclusions were repeatedly verified against the data until final and relevant conclusions were obtained.

RESULTS AND DISCUSSION

Marriage is a fundamental element of social life, forming an inner bond between a man and a woman as husband and wife (Malisi, 2022). Wedding traditions are an essential aspect of a society's culture, reflecting values, norms, and cultural identity passed down through generations (Razali et al., 2024). The Gayo people are an ethnic group rich in local culture and strongly maintain traditions inherited from their ancestors, including complex wedding rituals. These wedding processions serve as a means to preserve and reinforce social bonds, as well as to ensure the continuity of cultural heritage within the community.

Gayo Traditional Wedding Procession

Historically, Gayo wedding customs include various stages, ranging from the initial introduction to the official marriage. Interviews with several informants revealed that each stage of the wedding ceremony carries its own meaning. First, *bersiengonen* (initial proposal) is the stage of introduction between the bride and groom, as well as between the parents of both parties. It symbolizes noble intentions and initial respect toward the bride-to-be's family.

Secondly, *munginte* (formal proposal) takes place after the man's family discusses the prospective bride. If they agree, a representative known as *teulangke* is sent to the bride's family. In local terminology, a formal proposal refers to an official process of proposing marriage between a man and a woman, in which a man asks a woman to become his wife in a manner commonly accepted within society (Harahap & Chasnun, 2025). During this process, the *teulangke* visits the bride-to-be's residence to formally

propose the marriage and establish a relationship with her family. Several symbolic items must be brought, including *oros senare* (a bamboo container of rice), *jarum i cocok kukuneng bertelinting kubenang* (a needle inserted into turmeric with thread), *sen berbungkus wan ruje putih* (money wrapped in white cloth), *tenaroh* (eggs), *uwah nipinang* (three pieces of betel nut), and *olong belo pitu* (seven betel leaves). All of these items are placed in a container called a *batil* or *cenara* and later left at the woman's residence as part of the *nipi jege* ritual (ritual sleep).



Figure 1. *Batil* or *Cenara* (Traditional Gayo Brass Bowl)



Figure 2. *Ampang* (Traditional Ceremonial Mat for the Sarak Opat and the Groom/Bride)

Third, *amal nome nipi jege* (ritual sleep) serves as a deliberation period during which the bride's family decides whether to accept the proposal. This ritual is usually performed by the bride's parents. During the proposal, the groom's family brings a large container filled with rice, near which the bride's parents symbolically sleep to perform *istikharah* (seeking spiritual guidance through dreams) regarding the marriage.

Fourth, *betelah* (negotiation of marriage requirements) aims to determine the dowry requirements expected from the groom's family. There are bed, money, garden, clothing, and other gifts. This process is conducted through prior consultation between

the families of the bride and groom. In the past, the *betelah* tradition was carried out on a different day from *nerime emas* (gold handover); however, it is now often combined with the same occasion.

However, today these stages are often combined, as some Gayo people consider separate ceremonies to be inefficient. Currently, the groom's family can quickly learn the bride's family's requirements through communication between the prospective bride and groom via mobile phones. As a result, the *betelah* custom has become less relevant and is gradually being abandoned.

Fifth, *mujele emas* (delivery of gold) refers to the formal handover of the dowry and marriage requirements agreed upon by both parties during the *betelah* deliberation. The *mujele emas* tradition follows strict customary regulations. Unlike *munginte* (formal proposal), which may involve only a small delegation (*teulangke*), *mujele emas* is a formal ceremony that requires the presence of extended families and the *sarak opat*, comprising the village head, the imam, the *petua* (elders), and other community members.

Sixth, *beguru*, a traditional advice-giving ceremony, is conducted to provide guidance on married life to the bride and groom. During this stage, the couple receives advice regarding their respective responsibilities within the family. After the advice session, the bride and groom perform the *tepung tawar* ritual while sitting on a small traditional mat called an *ampang*, followed by greeting and shaking hands with their family members. The items prepared for the *beguru* ceremony include rice, several types of leaves or plants—such as *celala*, *dedingin*, *bebesi*, *atang teguh*, and *ongkal*—as well as plain water.

Seventh, *jule bai* (escorting the groom) is a procession in which the groom is escorted to the bride's residence to carry out the marriage contract (*akad nikah*). During this stage, the groom is accompanied by a procession of extended family members. Prior to the marriage contract, the *reje kampung* (village head) from the groom's side formally hands over the groom to the *reje kampung* representing the bride's family. This handover is performed through a traditional oral expression known as *melengkan*, which conveys symbolic meanings comparable to poetic speech.



Figure 3. *Tepung Tawar Ritual during the Beguru Ceremony*



Figure 4. *Beguru Ceremony*

Eighth, *akad nikah* (Islamic marriage contract) is the most sacred stage in the Gayo traditional wedding. *Ijab* refers to the formal offer made by the bride's guardian to the groom or his representative, expressing the intention to establish a lawful marriage. Meanwhile, *qabul* refers to the response or acceptance given by the groom to the *ijab* (Adharsyah et al., 2024). The acceptance must be delivered clearly and firmly to ensure the validity of the marriage.

After the *akad nikah*, according to Gayo tradition, the bride and groom participate in the *delem* ceremony (bridal chamber ritual), which is led by an *ibu petua* or *ibu imam*. During this ritual, the couple performs *semah pincung* while sitting on a small mat (*ampang*), followed by prayers and the *tepung tawar* ritual. This stage symbolizes the couple's transition into married life. Following the ceremony, the bride and groom are respectfully addressed as *aman mayak* (a title for the husband) and *inen mayak* (a title for the wife), indicating their new social status within the community.



Figure 5. *Akad Nikah (Ijab Qabul)*

Ninth, *jule беру* (escorting the bride) is the stage that takes place after the *akad nikah* has been performed at the bride's residence. On the following day, the bride is escorted to the groom's house by a family procession, bringing various household items as symbols of her transition into her husband's household. These items include, among others, an *alur* (a traditional Gayo mat), pillows, plates, buckets, glasses, spoons, and

cambung (bowls). In addition, the bride's family brings *kero tom* (packed rice) to be given to the *sarak opat*. After *jule beru*, the final stage of the wedding procession is *mangan ume berume*, a communal meal involving both families, which is usually held three days to one week after the marriage contract.



Figure 6. *Jule beru*



Figure 7. *Tepung Tawar of the Bride and Groom*

The constraints in the implementation of Gayo traditional wedding processions arise from various factors, both internal and external. Each constraint affects the meanings and cultural values embodied in the wedding rituals. The types of constraints encountered in the implementation of Gayo traditional wedding ceremonies include the following:

1. Economic Factors

Economic conditions constitute one of the main constraints faced by the Gayo community in implementing traditional wedding ceremonies. Conducting a wedding in accordance with Gayo customs requires substantial financial resources, as each stage of the procession involves considerable expenses. As a result, many families are unable to carry out the wedding fully in line with established traditions and instead perform only

the akad nikah (Islamic marriage contract) without completing the customary wedding processions prescribed by cultural norms. This condition is reflected in the statement of one informant, M. Rasydin, who explained:

“Kami selaku jema tue mejen menet pedeh mugerjen anak kami orom edet te urang Gayo ni, cumen terkadang kami gere mampu karena ekonomi ni le. Karena ike pakek edet pengerjen orom edet te dele pedeh nemen sen, nume tine ken alat si turah beli lagu biaya pemanganen pe ke butuh biaya tetap.” (Interview, June 2, 2025)

This statement indicates that parents wish to marry their children in full accordance with Gayo traditions; however, economic limitations prevent them from doing so. The implementation of traditional wedding ceremonies requires significant financial resources, including the provision of ceremonial equipment and food for large-scale celebrations. Consequently, due to financial constraints, many families choose to marry their children according to Islamic law alone without performing the complete traditional wedding procession.

The implementation of Gayo weddings in accordance with tradition involves the participation of many parties, as well as the procurement of various forms of traditional equipment and food. Therefore, economic constraints represent one of the most common obstacles to the implementation of Gayo traditional wedding ceremonies.

2. Social and Cultural Changes

The younger generation today tends to prefer conducting wedding ceremonies based on external traditions, resulting in considerable cultural mixing in wedding practices. Many young people adopt external cultural elements because they perceive Gayo traditional wedding customs as overly complicated. Consequently, Gayo wedding traditions are frequently combined with wedding practices from outside cultures. This condition indicates that constraints arising from social and cultural changes significantly affect the implementation of Gayo wedding ceremonies. This situation is reflected in the statement of one informant, Siti Musdaifa:

“Jema mude seni ike kerje gere neh orom edet Gayo, delenen makek edet lueren, contohe baju e nume kerawang Gayo tapi lagu baju edet Sunda waktu akad nikah e.” (Interview, June 2, 2025)

This statement explains that many young people today conduct wedding ceremonies using external traditions rather than Gayo traditional customs. For example, some choose to wear Sundanese traditional attire during the *akad nikah*. Another perspective regarding constraints arising from social and cultural changes was expressed by an informant, M. Rasydin, who stated:

“Jemen seni jema mude ike ngerje gere neh edet Gayo tapi nge edet len, seni ike jema ngerje dele nge campurne edet Gayo orom edet lueren, karna anggapene edet Gayo ni nge jemen nye recok.” (Interview, June 2, 2025)

This statement indicates that many young people today choose to marry using wedding traditions from outside their own culture. In many cases, external wedding customs are combined with Gayo traditional weddings because these traditions are perceived as old-fashioned and overly complicated. This perception reflects a shift in values among the younger generation, which contributes to the gradual decline in

adherence to Gayo traditional wedding practices.

3. Limited Knowledge of the Younger Generation

Many young people today participate in traditional wedding ceremonies without fully understanding the meanings embedded in each stage of the procession. Due to the lack of proper documentation on Gayo traditional weddings, many young people are unfamiliar with the correct procedures for conducting the ceremonies, including the arrangement of customary items during the rituals and the equipment required for the wedding. In addition, the younger generation has a limited understanding of customary traditions, as knowledge is transmitted primarily through oral means and remains confined to certain groups because of the minimal availability of written documentation. The findings of this study, based on an interview with one informant, Muhammad Suhada, revealed that:

“Kekanak seni gere beteh e meh edet ni meh gere roh hana pe, isi batil pe teba beteh e cara e, hana pelen perlengkapenen gere beteh e, sara mi makna e pe memang gere beteh asal oboh plen teba.” (Interview, June 6, 2025)

The statement explains that many young people today lack knowledge of the proper procedures for conducting traditional wedding ceremonies. Even the placement of *isi batil* is often done incorrectly, indicating a limited understanding of customary rules. In addition, many young people are unfamiliar with the ceremonial equipment required during the wedding and do not understand the symbolic meanings of each item used in the procession. This condition reflects a broader decline in the transmission of traditional knowledge among the younger generation.

CONCLUSION

The wedding procession in Gayo tradition consists of several stages, including *bersiengonen* (proposal), *muginte* (engagement), *amal nome nipi jege* (ritual sleep), *betelah* (request or dowry negotiation), *mujele emas* (delivery of gold), *beguru* (giving advice), *jule bai* (escorting the groom), *akad nikah* (Islamic marriage contract), and *jule беру* (escorting the bride). However, the implementation of these traditional ceremonies is often constrained by economic factors, social and cultural changes, and the limited understanding of the younger generation. These constraints frequently result in weddings being conducted without fully following Gayo customary traditions, as traditional ceremonies are perceived as time-consuming and financially demanding.

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CULTURAL AND ISLAMIC VALUES IN INFLUENCING AGRICULTURAL WORK PATTERNS AMID LABOR MOBILITY

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ABSTRACT

This study examines the role of Acehese agricultural culture, integrated with Islamic teachings, in shaping agricultural labor patterns amidst increasing labor mobility in Simpang Mamplam District, Bireuen, Aceh. In Acehese society, agriculture functions not just as an economic activity but also as a cultural and spiritual institution tied to traditional practices such as *peuphon meugo*, *tabu bijeh*, *meu ue*, and *khanduri blang*, as well as the institution of *keujrueng blang*. These traditions are grounded in Islamic values like *amanah* (trust), justice, consultation, and mutual assistance, which have historically fostered collective work and agrarian sustainability. Using an exploratory qualitative design, data were collected from 185 farming households through a structured questionnaire that combined closed and open-ended questions. The findings reveal that while collective traditions remain significant, labor mobility, particularly the migration of younger generations, has reduced the availability of male labor. Women now play a greater role in planting and maintenance, while land preparation and harvesting increasingly rely on hired labor and mechanization. Decision-making has also shifted from communal to more individualistic approaches, leading to emerging inequalities between landowners and smallholders. This study concludes that the integration of Acehese culture and Islamic teachings continues to provide a moral and social framework that sustains agriculture despite structural changes. However, labor mobility has transformed collective labor systems into more individualistic forms.

Keywords: *Acehnese Agriculture; Labor Mobility; Culture; Islam value; Labor Pattern*

INTRODUCTION

Agriculture in Aceh, particularly in rural areas such as Simpang Mamplam District, is not merely an economic activity but an integral part of Acehese cultural heritage in agricultural management, transmitted across generations. The Acehese agricultural system emphasizes mutual cooperation, social solidarity, and adherence to customary norms, which are reflected in traditional practices such as *peuphon meugo*, the ritual marking the opening of the planting season involving the entire community; *keujreung blang*, the customary agricultural institution responsible for labor distribution, irrigation management, and resolution of agrarian disputes; *tabu bijeh*, the seed preparation ritual conducted in accordance with customary cycles and planting seasons; *meu ue*, the collective plowing and land preparation activity; and *khanduri blang*, the harvest ritual that celebrates agricultural success while expressing communal gratitude.

These practices illustrate that agriculture in Aceh functions as a cultural institution in which the social and spiritual lives of the community are inseparably intertwined with agrarian activities (Attas & Anoegrajekti, 2021; Dewi & Julianita, 2021; Hanafiah, Jamil, & Iswandi, 2021).

Islam positions agriculture not merely as an economic activity, but as an essential component of religious teachings with both spiritual and social dimensions. As an organized religion, Islam places significant emphasis on agriculture, recognizing its critical role in sustaining human life. Islamic teachings advocate for the strengthening and prudent management of agricultural practices, as the Qur'an, the primary source of Islamic jurisprudence, repeatedly highlights the vital role of water in supporting life. Allah SWT says to the effect:

“Do not the Unbelievers see that the heavens and the earth were joined together (as one unit of creation), before we clove them asunder? We made from water every living thing. Will they not then believe?” Surah al-Anbiya' [21]:30

Allah SWT declares that every living being originates from water. This statement emphasizes that water is the fundamental source of life, divinely provided to sustain all living creatures. As such, water offers immense benefits to humankind, including supporting plant and animal life that serve as sources of food and medicinal resources (Borhan & Ab. Aziz, 2009). Beyond its technical aspects, Islamic teachings also serve as a normative framework guiding agricultural labor patterns. Values such as *amanah* (trustworthiness) in fulfilling human needs. Allah SWT says to the effect:

“Allah has made the earth a wide expanse for you (19). So that you may tread its spacious paths” Surah Nuh [71]:19–20

This verse emphasizes the obligation to manage agriculture responsibly. It highlights that agricultural practices should not only satisfy immediate human needs. Furthermore, it underscores the importance of ensuring the sustainability of natural resources. (Anwar, Jailani, & Damayanti, 2022).

In this regard, Islam frames agriculture as more than a means of subsistence; it integrates agricultural practice into a broader moral and spiritual responsibility, maintaining a balance between utilization and preservation of the natural environment. Within this framework, the integration of Acehese culture and Islamic teachings plays a crucial role in shaping agricultural work patterns that are harmonious, disciplined, and collective. Such integration not only strengthens social solidarity among farming communities but also contributes to the sustainability of agricultural production.

However, labor mobility in Simpang Mamplam has significantly affected the distribution of traditional agricultural labor. The shift in labor driven by employment outside the agricultural sector and labor migration has necessitated adjustments in work patterns to maintain the effectiveness of the agricultural system (Hadi & Wahyuni, 2024; Mussa, 2019). This context is crucial to examine, as Acehese cultural traditions and Islamic teachings simultaneously guide labor behavior, construct collective work patterns, and influence the sustainability of agricultural systems in Simpang Mamplam District. With a population of 28,562 and 3,180 households managing irrigated rice fields, agriculture remains the primary source of livelihood despite the changes in labor distribution caused by mobility (Badan Pusat Statistik Bireuen, 2025).

This study aims to identify and analyze the role of Acehnese culture in agricultural management and its integration with Islamic teachings in shaping labor work patterns, particularly within the context of social and economic dynamics influencing traditional labor structures. This research emphasizes the interaction between customary practices, religion, and agricultural activities as the foundation of agricultural sustainability in Aceh.

Bourdieu's (1977) Cultural Practice Theory emphasizes that individual actions are shaped by habitus, namely a set of values and dispositions embedded through social experience, as well as by field, the social arena in which individuals interact and compete within prevailing norms and structures of power (Adib, 2012; Daniel, 2022; Nurnazmi & Kholifah, 2023). In the context of Acehnese agriculture, traditional agrarian practices such as *peuphon meugo* (ritual opening of the planting season), *tabu bijeh* (seed preparation based on customary cycles), *meu ue* (collective plowing and land cultivation), and *khanduri blang* (harvest ritual) constitute a cultural field that organizes agricultural labor patterns collectively. Within this framework, every community member fulfills roles based on customary norms and Islamic values that emphasize discipline, responsibility, and mutual assistance (Attas & Anoeграjekti, 2021; Hanafiah et al., 2021; Thaifuri & Gadeng, 2024). Thus, agricultural activities are not solely oriented toward food production but also serve as a medium for social learning and the formation of collective work ethics.

According to Borhan & Ab. Aziz (2009), from a historical perspective, the Qur'an recounts the prosperity of the people of Saba', who excelled in agriculture and advanced farming technologies. Their fertile land, irrigated by the Ma'rib Dam, was renowned for its abundance of fruits and gardens. Allah (s.w.t.) states:

"There was, for Saba', aforetime, a sign in their homeland, two gardens to the right and to the left. Eat of the sustenance (provided) by your Lord, and be grateful to Him: a territory fair and happy, and a Lord Oft-Forgiving!" (QS. Saba' 34:15).

Within Islamic thought, agriculture is regarded as a communal obligation (*fard al-kifayah*). Involvement in food production is, therefore, considered an honorable activity. It is also viewed as a charitable act that benefits the wider community. The Prophet Muhammad (s.a.w.) said:

"There is none amongst the Muslims who plants a tree or sows seeds, and then a bird, or a person or an animal eats from it, but it is regarded as a charitable gift for him." (HR. Bukhari and Muslim).

The Qur'an further emphasizes the multiplied rewards of charity:

"The parable of those who spend their wealth in the way of Allah is that of a grain of corn: it groweth seven ears, and each ear hath a hundred grains. Allah giveth manifold increase to whom He pleaseth: and Allah careth for all and He knoweth all things." (QS. Al-Baqarah 2:261).

According to Jufri, Sahri, Huzaini, & Ismiwati (2025), Islam also stresses consultation and fairness in agricultural labor. Allah (s.w.t.) commands:

"Consult with them in the matter. Then, when you have made up your mind, put your trust in Allah. Indeed, Allah loves those who put their trust." (QS. Ali Imran:159).

The Prophet (s.a.w.) embodied this principle:

“I have never seen someone consult more with his companions than the Messenger of Allah SAW.” (HR. Tirmidhi).

Furthermore, Islamic ethics safeguard workers' rights, as the Prophet (s.a.w.) instructed:

“Give the worker his wages before his sweat dries.” (HR. Ibn Majah: 397; al-Baihaqi, from Abu Hurairah).

These principles are reflected in collective farming practices, such as seed preparation (*tabu bijeh*), land cultivation, and harvesting activities shaped not only by economic needs but also by cultural and religious values. Such integration strengthens social harmony, fosters patience, and cultivates gratitude within farming communities.

Within the framework of labor mobility theory, agricultural work patterns are also influenced by the dynamics of workforce migration across regions and generations (Becker, 1962; Todaro & Smith, 2015). In the Acehese context, labor mobility frequently occurs due to economic motivations, educational aspirations, or better employment opportunities in urban areas. Nevertheless, agricultural labor patterns in rural communities are preserved through customary norms and Islamic teachings, ensuring that labor redistribution in villages remains collectively organized (Faizien, 2025; Hanafiah et al., 2021). In other words, labor mobility does not entirely eliminate traditional work patterns but rather stimulates adaptive strategies that maintain social discipline and spiritual values within agriculture.

Theoretical perspectives on social structure and traditional work organization further highlight the importance of Acehese customary norms, particularly through the institution of *keujreung blang*, which regulates labor division, coordinates agricultural activities, and resolves social conflicts. Previous researches also note that Acehese agricultural traditions not only shape physical farming activities but also construct labor patterns embedded with spiritual and social values (Mardhiah, Syahputri, Marefanda, & Macella, 2024; Shamadiyah & Nasution, 2018). For example, the ritual of *khanduri blang* is not merely a harvest celebration but also a social pedagogy that instills responsibility, cooperation, and gratitude to Allah, reminding farmers that agricultural yields are both a trust and a divine blessing to be managed responsibly.

In the case of Simpang Mamplam District, agricultural labor patterns are organized through these cultural-religious practices, ensuring that the entire farming process from seed preparation to harvesting remains structured, efficient, and aligned with both customary and Islamic norms. Therefore, the literature indicates that Acehese agricultural culture and Islamic teachings interact synergistically in shaping labor patterns. Local culture provides the social framework, norms, and practices that regulate agrarian activities, while Islamic values emphasize moral principles, work ethics, and accountability. The integration of both dimensions not only enhances productivity but also safeguards the social and spiritual sustainability of rural communities. Furthermore, labor mobility theory demonstrates how dynamic workforce redistribution continues to preserve collective labor patterns rooted in Islamic and cultural values, ensuring that Acehese agriculture remains a resilient cultural institution.

Although a number of studies have explored Acehnese agricultural traditions and their cultural-religious significance, there is still limited scholarly attention to how these traditions interact with the dynamics of labor mobility. Previous research has largely emphasized symbolic rituals and cultural identity, yet has not fully examined how migration, changing demographics, and socio-economic shifts reshape collective work patterns in agriculture. This study addresses this gap by analyzing the role of Acehnese agricultural culture, grounded in local wisdom and Islamic teachings in the formation of agricultural labor patterns under the pressures of labor mobility. The findings are expected to enrich theoretical debates on cultural practice and labor migration while also offering practical insights for policymakers, customary institutions, and religious leaders. By highlighting the resilience and adaptability of Acehnese agriculture, this study contributes to discussions on agrarian sustainability within broader contexts of cultural preservation and socio-economic change.

METHOD

This study employed an exploratory qualitative design to gain a comprehensive understanding of the role of Acehnese agricultural culture, rooted in local wisdom and Islamic teachings, in shaping agricultural labor patterns. The qualitative approach was chosen because it is well-suited to capturing the complexity of social realities, in which customary norms, traditional agrarian practices, and Islamic values simultaneously influence behavior, responsibility, and coordination in agricultural work (Bambang, 2018; Riswandi, Andi Patawari, & Fathiah, 2019). Referring to Geertz's (1973) interpretative tradition, the formation of labor patterns is viewed not merely as a technical division of tasks, but as the outcome of interactions between individual agents and the cultural-religious structures deeply embedded in Acehnese society (Indirasari, Susianto, & Gurtaningsih, 2019).

The respondents were selected using purposive sampling, involving 185 farming households actively engaged in traditional agricultural practices. These households were specifically chosen because their members apply both local wisdom and Islamic values in farming, collective work, and land management. This selection ensured that the respondents represented authentic and relevant experiences concerning agricultural work patterns, cooperation, and the integration of cultural and religious values in Aceh.

Data were collected through a structured questionnaire designed to explore both factual information and respondents' personal experiences. The instrument included two types of questions:

1. Closed-ended questions, aimed at gathering basic quantitative data on respondents' demographics, labor distribution, and decision-making patterns; and
2. Open-ended questions, intended to capture narratives, perceptions, and interpretations regarding the ways in which cultural practices and Islamic values shape labor patterns in agriculture.

The data were analyzed using a qualitative thematic approach. Responses from closed-ended questions were analyzed descriptively to present demographic characteristics of the respondents and to identify patterns in labor distribution and collective practices. Open-ended responses were thematically coded and categorized to

uncover underlying meanings, recurring themes, and perspectives regarding the integration of cultural and Islamic values in agricultural work. By combining descriptive quantitative findings with qualitative thematic insights, the study was able to provide a holistic interpretation of how local wisdom and Islamic principles shape work behavior, discipline, coordination, and social solidarity in Acehese agriculture.

RESULT AND DISCUSSION

This study involved 185 respondents from farming households in Simpang Mamplam District. The majority of respondents were male (63.8%) and in the productive age range of 30–50 years (64.9%), indicating that the agricultural sector is still supported by a physically active labor force. In terms of education, most respondents had completed junior high school (33.5%) and senior high school (35.1%), while those with no schooling or only primary education were fewer (20.5%).

This reflects an improvement in education compared to previous generations, which in turn fosters openness to agricultural innovations. More than half of the respondents had worked in the agricultural sector for over 20 years (54.6%), demonstrating a strong attachment to traditional farming practices. Interestingly, most respondents (91.4%) owned agricultural land, either rice fields or plantations, indicating that farming in this region is still predominantly family-based. Only 8.6% of respondents did not own land and worked instead as farm laborers or tenant farmers.

Table 1. Characteristics of Respondents(N=185)

Characteristics	Category	Frequency	Percentage
		(n)	(%)
Gender	Male	118	63.8
	Female	67	36.2
Age (years)	<30	12	6.5
	30–40	54	29.2
	41–50	66	35.7
	51–60	38	20.5
	>60	15	8.1
Education	No schooling/Primary	38	20.5
	Junior High School	62	33.5
	Senior High School	65	35.1
	Higher Education	20	10.8
Main			
Occupation	Land-owning farmer	102	55.1
	Farm laborer	56	30.3
	Tenant farmer	21	11.4
	Others	6	3.2
Land Ownership	Own land	169	91.4
	Landless (laborers & tenants)	16	8.6

Characteristics	Category	Frequency (n)	Percentage (%)
Years in Farming	<10 years	20	10.8
	10–20 years	64	34.6
	>20years	101	54.6

Source: Processed data, 2025

Acehnese agricultural traditions remain deeply embedded in the daily lives of farmers. Rituals such as *peuphon meugo* (season-opening ceremony), *tabu bijeh* (customary seed preparation), *meu ue* (collective plowing), and *khanduri blang* (harvest thanksgiving ritual) continue to be practiced, though the involvement of younger generations has declined. The customary institution *keujreung blang* still functions in regulating labor distribution, irrigation allocation, and resolving agrarian conflicts, although its influence is gradually weakening.

Islamic values are also strongly ingrained in agricultural practices. Nearly all respondents viewed farming as a trust from Allah, which must be carried out with responsibility. Collective labor was also interpreted as a manifestation of the Qur'anic command to cooperate in goodness, while consultation remained an important mechanism in collective decision-making.

Nevertheless, the study also revealed significant changes due to labor mobility. The migration of younger generations into non-agricultural sectors, particularly in urban and service industries, has reduced the availability of male labor in rice fields. This has weakened the practice of collective farming that previously characterized Acehnese agriculture. Women's roles have expanded in planting and maintenance activities, while land preparation and harvesting have increasingly relied on wage laborers, tractor rentals, or simple mechanization. The comparison of work patterns before and after the emergence of labor mobility is summarized in the following Table 2.

Table 2. Comparison of Agricultural Work Patterns in Simpang Mamplam Sub-District.

Work Stage	Before Labor Mobility (Trad ition al)	After Labor Mobility (Ad ap tation)
Preparation (Peuphon Meugo, TabuBijeh)	Collective; women dominant in seed preparation; rituals involved entirecommunity	Still practiced, but youth participation hasdeclined
Land Preparation (Meu Ue)	Male-dominated; strong collective plowing	Reduced male labor → reliance on wage laborers/tractors
Planting & Maintenance	Women dominant; irrigation managed collectively	Women take on larger roles; collective labor weakens
Harvest (Khanduri Blang)	Collective; men performed heavy tasks;ritualsofthanksgiving	Collective practices reduced;wagelabor

		used; rituals simplified
Post-Harvest	Results distributed through consultation (fair, syura)	Distribution more pragmatic; consultation less
		intensive
Culture & Religion	Strongly embedded across all stages	Still present, but collective values weakening, more
		individualistic
General Work Pattern	Collective, disciplined, based on custom & Islam	Adaptive, pragmatic, collective weakened, greater reliance on wage labor/technology

Source: Processed data, 2025

The findings confirm that agricultural labor patterns in Simpang Mamplam District are shaped by the interconnection of local culture, Islamic teachings, and labor mobility dynamics. Acehese agrarian traditions such as *peuphon meugo* (season-opening ritual), *tabu bijeh* (seed preparation), *meu ue* (collective plowing), and *khanduri blang* (harvest thanksgiving) function not only as technical mechanisms in production but also as social instruments that foster solidarity, strengthen community cohesion, and embed spiritual values. In this sense, agriculture is not merely an economic activity, but also a cultural institution organizing social, economic, and spiritual relations within the community.

The customary institution *keujreung blang* plays an essential role in regulating irrigation, distributing workloads, and resolving agrarian conflicts, thereby ensuring order in collective agricultural practices. However, its influence has gradually weakened as younger generations become less engaged, largely due to labor mobility into urban and non-agricultural employment. In daily practice, Islamic values remain the moral foundation guiding farmers' work behavior. The concept of *amanah* is understood as responsibility in land management. The principle of justice is also upheld in labor distribution and harvest sharing.

Collective labor continues to be practiced, though its intensity has declined. For farmers, working together is not merely a strategy of efficiency but also a reflection of *ta'awun* (mutual assistance in goodness). Gratitude is also expressed through rituals such as *khanduri blang*, rooted in the belief that agricultural yield is a blessing from Allah. Yet, labor mobility has brought significant structural changes. Outmigration of young male workers has reduced the availability of physical labor for tasks such as *meu ue* (plowing) and harvesting. Consequently, women now play a greater role in planting and maintenance, while physical tasks rely on hired workers and mechanization. This

indicates a transition from collective, disciplined, and tradition-based labor patterns to more adaptive, pragmatic, and individualistic systems.

Decision-making processes have also shifted. While previously determined through communal decisions, decisions are now often made by individuals or small groups. This shift signals a move from collective principles to pragmatism. Furthermore, mobility has created new forms of social differentiation. Land-owning farmers with sufficient resources can adapt by hiring labor or using tractors, while smallholders with limited capital struggle to adjust. This has led to emerging inequalities in what was once a relatively egalitarian system. The emphasis on fairness thus becomes increasingly vital.

The transformation of agricultural labor patterns in Simpang Mamplam is not limited to the technical aspects of production but extends to social structures, cultural values, and religious practices. Agriculture, once fully collective, is transitioning into a more modern and adaptive system, yet it remains rooted in Acehese traditions and Islamic principles. This highlights that Acehese farming communities are in a phase of transition, striving to balance modernization with cultural and religious legacies that have long underpinned agrarian life.

CONCLUSION

The conclusion should present the answer to the research hypothesis and/or objectives, or the scientific findings obtained. The conclusion should not merely repeat the results and discussion, but rather provide a summary of the findings in relation to the stated objectives or hypothesis. If necessary, the conclusion may also include suggestions for future work or ideas for the continuation of the research. The conclusion must be written in a full paragraph, not in bullet points.

This study demonstrates that agricultural work patterns in Simpang Mamplam District are shaped by the intricate interaction of local culture, Islamic teachings, and the dynamics of labor mobility. Acehese agricultural culture, through its traditional rituals, the customary institution of *keujreung blang*, and the practice of *gotong royong* (mutual cooperation), serves as a social system that maintains solidarity, discipline, and harmony in farming management. Islamic teachings provide a moral foundation that reinforces values such as *amanah* (trust), justice, gratitude, *musyawarah* (consultation), and *ta'awun* (mutual assistance), thus transforming agricultural activities from a purely economic endeavor into a form of worship rich with spiritual values.

However, labor mobility, particularly the migration of the younger generation to non-agricultural sectors, has altered the structure of these work patterns. The once-strong collective practices have weakened, the role of women in planting and maintenance has become more dominant, and the use of hired labor and mechanization has increased. Consequently, work patterns have shifted from a collective system based on custom and religion to an adaptive system that is more pragmatic and individualistic.

Despite these changes, local culture and Islamic values remain as a social and spiritual framework that safeguards the continuity of agriculture. To maintain the sustainability of the traditional Acehese agricultural system, it is necessary to revitalize the role of *keujreung blang* and strengthen the support of religious leaders. This will

encourage the younger generation to once again see farming as both a trust (*amanah*) and a source of blessings.

Based on the research findings, several recommendations can be proposed to strengthen the sustainability of agricultural work patterns in Simpang Mamplam District. First, the revitalization of traditional agricultural institutions, particularly *keujreung blang*, is a critical step to ensure that the division of labor, social solidarity, and agrarian conflict resolution can function effectively again. This revitalization would not only revive customary functions but also create a social mechanism capable of adapting to modern challenges. Second, the reinforcement of Islamic values in farming practices needs to be consistently instilled, both through the role of religious figures and through educational institutions like *pesantren*. This will maintain the awareness that farming is an *amanah* and a form of worship. This is especially vital for building a religious work ethic that can withstand the increasing individualism brought by labor mobility.

Third, the involvement of the younger generation must be increased through incentive policies, training in modern agricultural technology, and innovations that remain aligned with traditional values. These efforts are expected to rekindle young people's interest in farming, not just as an economic job but as a means for cultural and spiritual actualization. Fourth, the increasing use of agricultural mechanization should be designed to fit the local context. Technology should be positioned as a supportive tool rather than a replacement for social solidarity. This way, work efficiency can be achieved without eliminating the collective values that have long been a hallmark of Acehnese agriculture. Finally, multi-stakeholder collaboration is essential. The local government, traditional institutions, religious leaders, and universities need to work together to create policies that reinforce a culturally and Islamically-based agricultural system while allowing for adaptation to the dynamics of labor mobility.

The practical implications of these recommendations are as follows. For the government, these findings provide a basis for designing agricultural policies that not only focus on increasing production but also on strengthening the social and religious institutions that underpin the system's sustainability. For the community, especially farmers, these recommendations encourage an awareness of how to preserve their collective identity while adapting to socio-economic changes. For academics, this research opens up avenues for further studies on the integration of culture, religion, and mobility in agricultural systems, offering a model for agricultural development that is not only technical but also value-based. Through these efforts, Acehnese agriculture can remain a resilient and adaptive cultural and religious institution in the face of modern challenges.

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ASSET-BASED COMMUNITY DEVELOPMENT APPROACH FOR SUSTAINABLE MANAGEMENT OF GOAT MANURE WASTE INTO LIQUID ORGANIC FERTILIZER

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ABSTRACT

The underutilization of livestock waste, particularly goat manure, in Kampung Rantau Panjang is the rationale for this research. The objective of this research is to process goat manure waste into liquid organic fertilizer (LOF) using a simple fermentation technology that employs Effective Microorganisms 4 (EM4) and molasses. This study employed an Asset-Based Community Development (ABCD) approach across six stages: discovery, dream, design, determination, destiny, and reflection. The fermentation process lasted 14-20 days under semi-anaerobic conditions, with the following material composition: 2 kg of goat manure, 10 liters of clean water, 10 ml of EM4, and 200 ml of molasses. The fermentation results demonstrated physical changes, including a distinctive fermentation odor, color changes, and a more homogeneous solution. The community was actively involved in producing LOF, thereby reinforcing their skills in independently managing livestock waste. In addition to producing an environmentally friendly product, this activity demonstrates the effectiveness of the ABCD approach in exploring local potential, increasing community participation, and promoting farmer self-reliance in fertilizer provision. Therefore, utilizing goat manure waste to produce liquid organic fertilizer offers a sustainable solution to support local agriculture and reduce dependence on chemical fertilizers.

Keywords: *EM4; fermentation; fertilizer; molasses.*

INTRODUCTION

Kampung Rantau Panjang is a settlement located in the Karang Baru sub-district, Aceh Tamiang Regency, Aceh Province. This area holds significant potential for agricultural Development due to its tropical climate and high rainfall over recent years (BPS Aceh Tamiang Regency, 2023). Water conditions and road access are important factors influencing the general and agricultural activities of the local population. Given its land potential, Kampung Rantau Panjang is an ideal location for implementing liquid organic fertilizer (LOF) transformation technology. LOF is an essential component in modern agriculture as it can enhance soil nutrients and fertility in an environmentally friendly manner (Wijaya et al., 2017). LOF is produced through the fermentation of organic materials, such as kitchen waste, green waste, or animal manure, using effective microorganisms, yielding a fertilizer rich in macro- and micronutrients essential for plant

growth. The use of LOF not only reduces dependence on chemical fertilizers but also supports sustainable agriculture by utilizing readily available organic waste in the surrounding environment.

In addition to farming, the majority of the community in Kampung Rantau Panjang also engages in cattle and goat farming. This results in an abundant availability of goat manure that has not yet been optimally utilized. In fact, goat manure possesses great potential to be processed into high-value liquid organic fertilizer. Utilizing livestock waste not only increases agricultural productivity but also helps reduce environmental pollution from the accumulation of unmanaged organic waste (Satrio et al., 2024). Livestock waste, such as goat manure, contains nitrogen, organic carbon, and potassium, which are beneficial for plant growth; however, its nutrient content is strongly influenced by the type of animal feed given (Kalbani et al., 2016).

Based on initial observations, the local community's management of livestock waste remains suboptimal. Most farmers dispose of or accumulate livestock waste without adequate processing. This condition not only poses environmental problems, such as soil and water pollution, but also overlooks the significant value in the waste, which can be processed into value-added products, including liquid organic fertilizer (Arwani et al., 2023). Additionally, farmers in the area generally still rely on chemical fertilizers for crop cultivation. Excessive dependence on chemical fertilizers is concerning because it may accelerate soil degradation, reduce soil fertility, and disrupt the balance of agricultural ecosystems in the long term, thereby negatively affecting research outcomes and land productivity (Zhuang et al., 2024).

One environmentally friendly solution to address organic waste problems is fermentation using EM4 and molasses. Research conducted by Junaedi et al. (2023) demonstrated that fermentation of cattle manure with this method can produce high-quality liquid organic fertilizer, which can serve as an alternative to chemical fertilizers and support sustainable agriculture. Furthermore, a study by Cintiyah et al. (2024) showed that integrating livestock and *Indigofera zollingeriana* with EM4 and molasses can produce liquid organic fertilizer that effectively enhances hydroponic growth and reduces national standards. On the other hand, Pyakurel et al. (2019) reported that combining molasses with organic fertilizer increased soil organic carbon content (4.51%), nitrogen (0.24%), and potassium (557.80 mg/kg), thereby significantly improving spinach yield. Based on these relevant studies, the objective of this activity is to provide education and practical training to the community of Kampung Rantau Panjang to optimize the utilization of goat manure waste formulated with molasses as a fermentation substrate. This approach is expected to be an effective means of improving soil quality and agricultural productivity in an environmentally friendly manner.

METHOD

This activity was conducted in Kampung Rantau Panjang, Karang Baru sub-district, Aceh Tamiang Regency. The location was chosen because the community's primary occupation is livestock farming. The activity was carried out from January to February 2024 at one of the local farmers' barns (Figure 1).



Figure 1. The goat-farming environment in Kampung Rantau Panjang serves as the primary site for fermenting liquid organic fertilizer.

The approach used in this activity is called Asset-Based Community Development (ABCD). The ABCD approach reflects the potentials already recognized by the community, such as willingness, skills, and existing local resources. This approach is highly relevant to Kampung Rantau Panjang, where the majority of the population are goat farmers and agriculturalists. Although not yet fully utilized, goats represent a substantial potential as a raw material for liquid organic fertilizer. The implementation of the ABCD approach was carried out through the following six stages:

1. Discovery

An inventory of community assets and potentials was conducted through direct field observations and in-depth interviews with community leaders, livestock farmers, and members of farmer groups. Based on the data collected, it was found that goat manure produced by farmers had not yet been fully utilized as an economic or environmental resource. Consequently, the manure tended to accumulate around residential areas and even on roads, potentially causing sanitation and environmental health problems. These findings highlight the need for innovative approaches to managing livestock waste, for instance, by utilizing goat manure as an organic fertilizer or a biogas feedstock, thereby providing additional benefits to the community and supporting sustainable Development in the area.

2. Dream

At this stage, active community participation was emphasized through group discussions to formulate collective hopes and goals. During these forums, farmers expressed aspirations to manage livestock waste, particularly goat manure independently. Interviews revealed that the community hoped to process the waste into high-quality organic fertilizer, thereby reducing reliance on chemical fertilizers, a major constraint on crop cultivation. Furthermore, this initiative was expected to improve production cost efficiency, maintain soil fertility, and contribute to sustainable environmental conservation efforts.

3. Design

The KKM student team, together with the local community, designed and implemented a production process for liquid organic fertilizer that leverages local

resources, with goat manure as the primary raw material. The fertilizer production employed fermentation technology using effective microorganism cultures (EM4) and molasses as bioactivators. EM4 is a mixture of beneficial microorganisms that accelerate the decomposition of organic matter and enhance soil fertility, while molasses serves as an energy source for microorganisms during fermentation. This method of goat manure fermentation produces environmentally friendly, nutrient-rich liquid organic fertilizer, thereby optimizing the sustainable use of livestock waste.

4. Define

The objective of this activity is to enhance the community's skills and knowledge in managing livestock waste. The community is expected to be capable of processing livestock waste into liquid organic fertilizer with high utility value and environmental friendliness. This organic fertilizer is expected to serve as an alternative to synthetic chemical fertilizers, thereby reducing adverse environmental and health impacts. Additionally, utilizing livestock waste as organic fertilizer can improve the economic efficiency of farming, lower agricultural production costs, and create new business opportunities for the community.

5. Destiny

The production process for liquid organic fertilizer was implemented directly through practical approaches. Community members were actively involved in every fermentation stage, from preparing and mixing primary raw materials—goat manure, EM4, and molasses—to the fermentation process itself. They also participated in packaging the final fertilizer product ready for use. This activity aimed not only to impart technical knowledge but also to enhance the community's capacity to independently manage livestock waste.

6. Reflection

The reflection and evaluation process was conducted through discussions and joint assessments with the community. The success of the activity was measured by the achievement of its goals, including improvements in community skills for managing livestock waste to produce liquid organic fertilizer. Additionally, this evaluation identified challenges and obstacles encountered during implementation and provided constructive feedback for future improvements.

The activities conducted with the community employed several tools and materials, including a 20-liter plastic bucket, a stirring stick, a wheelbarrow, 2 kg of goat manure, 10 liters of clean water, 200 ml of molasses, and 10 ml of liquid EM4. This formula is based on a study by Adi et al. (2023), which demonstrated that the addition of EM4 and molasses during the fermentation of organic waste can significantly improve the quality of the resulting liquid. The use of locally available materials in this process also supports the asset-based community Development approach by optimizing existing community resources.

The procedure for producing liquid organic fertilizer from goat manure began by placing 2 kg of goat manure into a bioreactor bucket filled with 10 liters of water. The bucket was equipped with ventilation holes to ensure sufficient air circulation. Subsequently, 10 ml of effective microorganism solution (EM4) and 200 ml of molasses, serving as a nutrient source for the decomposing microbes, were added. The mixture was

stirred until homogeneous, and the container was tightly sealed to initiate semi-anaerobic fermentation, which lasted 14 to 20 days. During fermentation, the solution was stirred every 2 to 3 days to maintain process stability and accelerate the decomposition of organic material. After fermentation was complete, the solution was filtered to separate the residue, and the resulting liquid fertilizer was stored in plastic bottles for use as a nutrient-rich liquid organic fertilizer.

Data were collected through participatory observation, with intensive monitoring conducted throughout the fermentation process to obtain direct information on the changes occurring. In addition, in-depth interviews were conducted with farmers to explore their understanding and interest in utilizing livestock waste as a valuable resource. The evaluation of fermentation outcomes was conducted using visual indicators, including changes in color, odor, and texture of the fermentation solution.

RESULT AND DISCUSSION

This activity was conducted in Kampung Rantau Panjang, Karang Baru sub-district, Aceh Tamiang Regency. This activity adopted an asset-based community Development approach, optimizing local assets to support sustainable livestock waste management, as evidenced by the community's involvement in processing goat manure into liquid organic fertilizer. The objective of this activity was to evaluate the effectiveness of a simple fermentation technology for processing goat manure into liquid organic fertilizer (LOF), thereby simultaneously addressing livestock waste management and enhancing sustainable agricultural productivity at the local level (Figure 2).



Figure 2. The process of collecting goat manure waste and preparing it for the production of liquid organic fertilizer.

The abundant goat manure in this area has significant value as a raw material for liquid organic fertilizer (LOF) due to its high nitrogen, organic carbon, and potassium content. To maximize this potential, fermentation was conducted for 14-20 days using a mixture of bioactivators, EM4, and molasses as the carbon source. Observations indicated physical changes during fermentation, including a transition from dark brown to black, a reduction in the characteristic fermentation odor after day 10, and a progressively more homogeneous liquid consistency.

The resulting fermented product was then packaged in plastic jerrycans and made ready for use by local farmers. This project demonstrates that producing LOF from goat manure waste can be readily achieved at the village level, thereby reducing dependence on chemical fertilizers. On the other hand, the use of fertilizers derived from organic materials, such as goat manure, has been shown to increase soil nitrogen, phosphorus, and potassium levels and to improve soil texture and fertility (Zhuang et al., 2024). Indrawati et al. (2020) reported that fermentation of organic waste using EM4 and molasses yielded nitrogen, phosphate, and potassium contents of 0.076%, 0.32%, and 0.47%, respectively, demonstrating the effectiveness of these fermentation agents in enhancing the quality of liquid fertilizer.

The implementation of this activity is critical because goat manure in Kampung Rantau Panjang has not been utilized optimally (Arwani et al., 2023; Junaedi et al., 2023). By increasing nutrient content through fermentation, this activity is expected to serve as a long-term strategy to support farmer independence, reduce production costs, and decrease reliance on chemical fertilizers. These outcomes also align with sustainable agricultural practices by promoting environmentally friendly waste management and resource efficiency at the community level.

CONCLUSION

This study demonstrates that the abundant goat manure waste available in Kampung Rantau Panjang has significant potential to be transformed into liquid organic fertilizer through a simple fermentation process using EM4 and molasses. The fermentation results exhibited physical changes and a distinctive aroma, indicating that the process proceeded successfully, and the involved community responded positively to this practice. This activity not only encourages the utilization of livestock waste as a productive resource but also strengthens farmer autonomy and reduces dependence on chemical fertilizers. The application of the Asset-Based Community Development (ABCD) approach in this activity proved effective in exploring local potential and enhancing community participation to support more sustainable agriculture.

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